

Virtual University of Pakistan

BIF501

INTRODUCTION TO BIOINFORMATICS

-2

Created by Team

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Mates
sign of bright future

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Topic # 1

Applications of Bioinformatics

1. Drug Development
2. Crop Improvement
3. Microbial Genome
4. Gene Therapy
5. Biotechnology
6. Comparative Study
7. Evolutionary Studies
8. Veterinary Science
9. Molecular Medicine
10. Personalized Medicine
11. Preventive Medicine
12. Waste Cleanup
13. Antibiotic Resistance
14. Alternate Energy Science
15. Insect Resistance
16. Climate Change Studies
17. Nutritional Quality

Drug development

- Drugs target only about 500 proteins
- Disease mechanisms and using computational tools identify and validate new drug targets

Crop improvement

- Comparative genetics of the plant genomes
- Information obtained from the model crop systems can be used to suggest improvements to other food crops.
- At present the complete genomes of *Arabidopsis thaliana* (water cress) and *Oryza sativa* (rice) are available.

Microbial genome applications

- Complete genome sequences
- Environment, health, energy and industrial applications
- Isolation of the genes that give them their unique abilities to survive under extreme conditions.

Gene Therapy

- Gene therapy-used to treat, cure or even prevent disease
- Clinical trials

Biotechnology

- *Archaeoglobus fulgidus* and *Thermotoga maritima*
- *Corynebacterium glutamicum*
- *Xanthomonas campestris*
- *Lactococcus lactis*

Evolutionary studies

- The sequencing of genomes from all three domains of life; eukaryota, bacteria and archaea

Topic # 2

Applications of Bioinformatics

Veterinary Science

- Farm animals including cows and sheep have been sequenced

Molecular Medicine

- The human genome project
- 3000-4000 hereditary disease including Cystic Fibrosis and Huntingtons disease
- Response to an environmental stress causes cancers, heart disease, diabetes
- Human Genome Project Data Base

Personalized medicine

- Pharmacogenomics
- Sequence variants in DNA
- Trial and error to find the best drug
- Patient's genetic profile
- With the specific details of the genetic mechanisms of diseases being unraveled, the development of diagnostic tests to measure a persons susceptibility to different diseases may become a distinct reality.
- Preventative actions such as change of lifestyle or having treatment at the earliest possible stages when they are more likely to be successful, could result in huge advances in our struggle to conquer disease.

Waste cleanup

- *Deinococcus radiodurans*

- Potential usefulness in cleaning up waste sites that contain radiation and toxic chemicals

Antibiotic Resistance

- *Enterococcus faecalis*
- Virulence region-resistant genes
- The discovery of the region, known as a pathogenicity island

Alternative energy sources

- *Chlorobium tepidum*
- Capacity for generating energy from light

Insect resistance

- *Bacillus thuringiensis*
- Control serious pests of cotton, maize and potatoes
- Insecticides can be reduced and hence the nutritional quality of the crops is increased

Climate change Studies

- Increasing levels of carbon dioxide emission-global climate change.
- Study the genomes of microbes that use carbon dioxide as their sole carbon source.

Improve nutritional quality

- Genes transferred into rice to increase levels of Vitamin A, iron and other micronutrients
- Reducing occurrences of blindness and anaemia

Topic # 3

Nucleotide Sequence Databases

Biological Databases

Biological databases in general store biological data and their main goals are

- **Data storage**
- **Information retrieval**
- **Knowledge discovery**

Classification

Biological databases can be classified as

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- **Primary databases**
- **Secondary databases**
- **Specialized Databases**

Biological databases can also be classified on the bases of types of data as

- **Nucleotide databases**
- **Protein databases**
- **RNA databases**
- **Genome databases**
- **Expression databases**

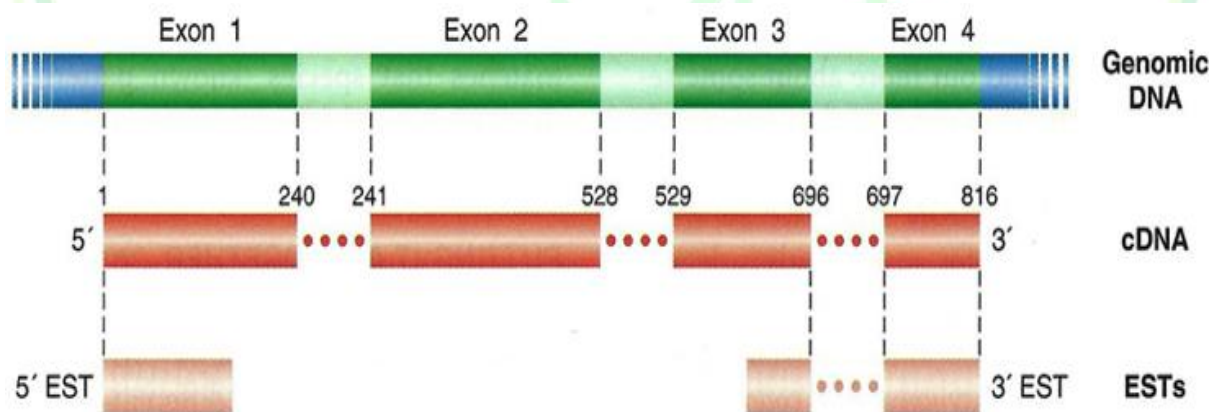
Issues

Due to limited Q/A

- **Redundancy**
- **Inconsistency**
- **Incompatibility (format, terminology, data types, etc.)**

Nucleotide Sequence databases

- Nucleotide Sequence Databases store DNA and cDNA or EST sequences.



Origin

- First assembled into Genbank (1982) at Los Alamos National Laboratory (LANL), New Mexico by Walter Goad et al
- **GeneBank** is now under NCBI
- **National Center for Biotechnology Information**

NCBI Resources How To Sign in to NCBI

NCBI National Center for Biotechnology Information

All Databases Recent All Databases Assembly BioProject BioSample BioSystems Books ClinVar Clone Conserved Domains dbGaP dbVar Epigenomics EST Gene Genome GEO DataSets GEO Profiles

NCBI Home Resource List (A-Z) All Resources Chemicals & Bioassays Data & Software DNA & RNA Domains & Structures Genes & Expression Genetics & Medicine Genomes & Maps Homology Literature Proteins Sequence Analysis Taxonomy Training & Tutorials Variation

NCBI National Center for Biotechnology Information advances science and health by providing biological and genomic information.

NCBI | Mission | Organization | Research | NCBI News

Popular Resources PubMed Bookshelf PubMed Central PubMed Health BLAST Nucleotide Genome SNP Gene Protein PubChem

NCBI Announcements NCBI's next webinar is The Statistics of Local Pairwise Sequence Alignment, Parts 1 and 2 Jan 13, 2015

3D Structures Explore three-dimensional structures of proteins, DNA, and RNA molecules. Examine sequence-structure relationships, active sites, molecular interactions, biological activities of bound chemicals, and associated biosystems.

GeneBank

NCBI Resources How To Sign in to NCBI

GenBank Nucleotide Search

GenBank Submit Genomes WGS HTGs EST/GSS Metagenomes TPA TSA INSDC

GenBank Overview

What is GenBank?

GenBank® is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences (*Nucleic Acids Research*, 2013 Jan;41(D1):D36-42). GenBank is part of the [International Nucleotide Sequence Database Collaboration](#), which comprises the DNA DataBank of Japan (DDBJ), the European Molecular Biology Laboratory (EMBL), and GenBank at NCBI. These three organizations exchange data on a daily basis.

The complete [release notes](#) for the current version of GenBank are available on the NCBI ftp site. A new release is made every two months. GenBank growth [statistics](#) for both the traditional GenBank divisions and the WGS division are available from each release.

An [annotated sample GenBank record](#) for a *Saccharomyces cerevisiae* gene demonstrates many of the features of the GenBank flat file format.

Access to GenBank

There are several ways to search and retrieve data from GenBank.

- Search GenBank for sequence identifiers and annotations with [Entrez Nucleotide](#), which is divided into three divisions: [CoreNucleotide](#) (the main collection), [dbEST](#) (Expressed Sequence Tags), and [dbGSS](#) (Genome Survey Sequences).
- Search and align GenBank sequences to a query sequence using [BLAST](#) (Basic Local Alignment Search Tool). BLAST searches CoreNucleotide, dbEST, and dbGSS independently; see [BLAST info](#) for more information about the numerous BLAST databases.

GenBank Resources

- [GenBank Home](#)
- [Submission Types](#)
- [Submission Tools](#)
- [Search GenBank](#)
- [Update GenBank Records](#)

EMBL and DDBJ

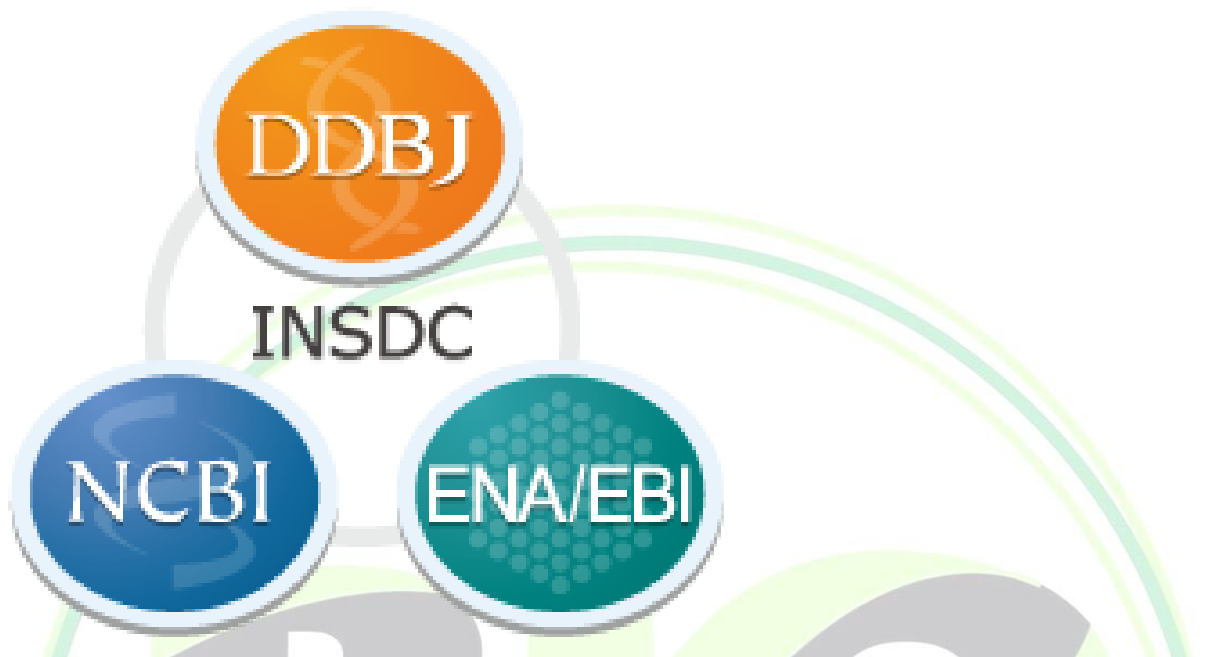
- European Molecular biology Lab (**EMBL** established 1980)
- DNA databank of Japan (**DDBJ**) established in Mishima Japan (1984)

INSDC

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- Genebank, DDBJ and EMBL joined together in **International Nucleotide Sequence Database Collaboration (INSDC)**



INSDC International Nucleotide Sequence Database Collaboration

ABOUT INSDC POLICY ADVISORS DOCUMENTS

International Nucleotide Sequence Database Collaboration

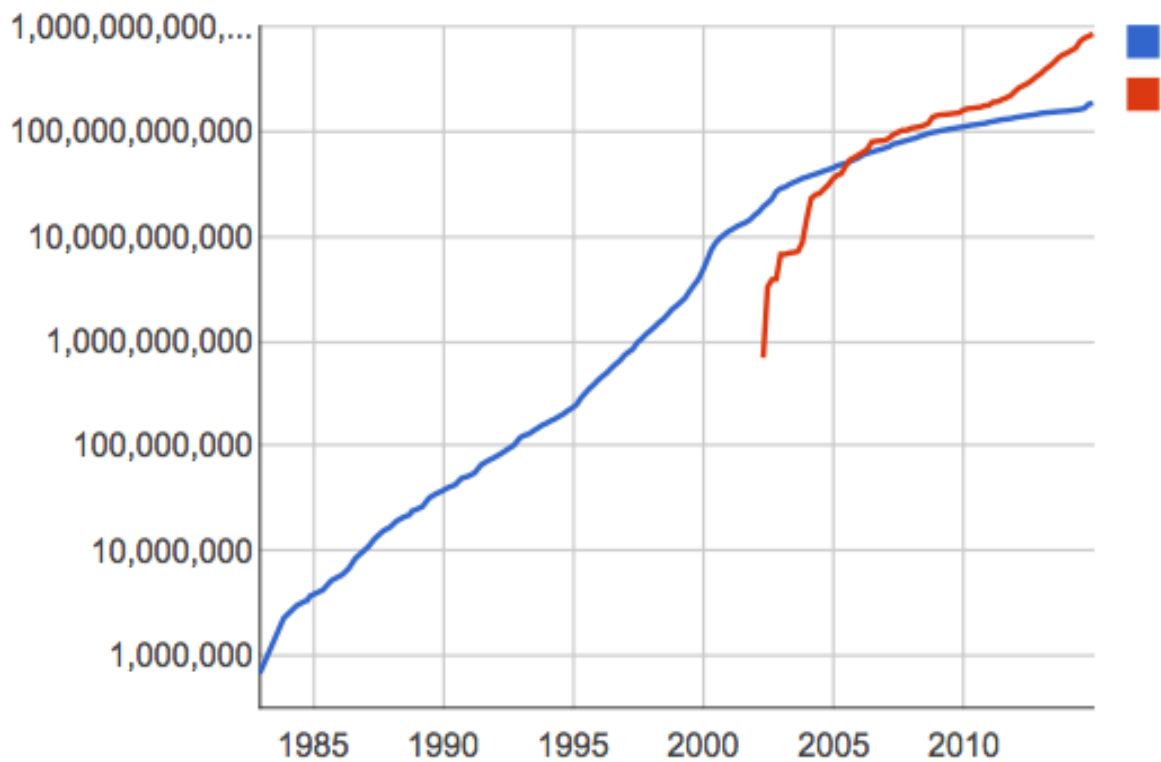
- The International Nucleotide Sequence Database Collaboration (INSDC) is a long-standing foundational initiative that operates between [DDBJ](#), [EMBL-EBI](#) and [NCBI](#). INSDC covers the spectrum of data raw reads, though alignments and assemblies to functional annotation, enriched with contextual information relating to samples and experimental configurations.

Data type	DDBJ	EMBL-EBI	NCBI
Next generation reads	Sequence Read Archive	European Nucleotide Archive (ENA)	Sequence Read Archive
Capillary reads	Trace Archive		Trace Archive
Annotated sequences	DDBJ		GenBank
Samples	BioSample		BioSample
Studies	BioProject		BioProject

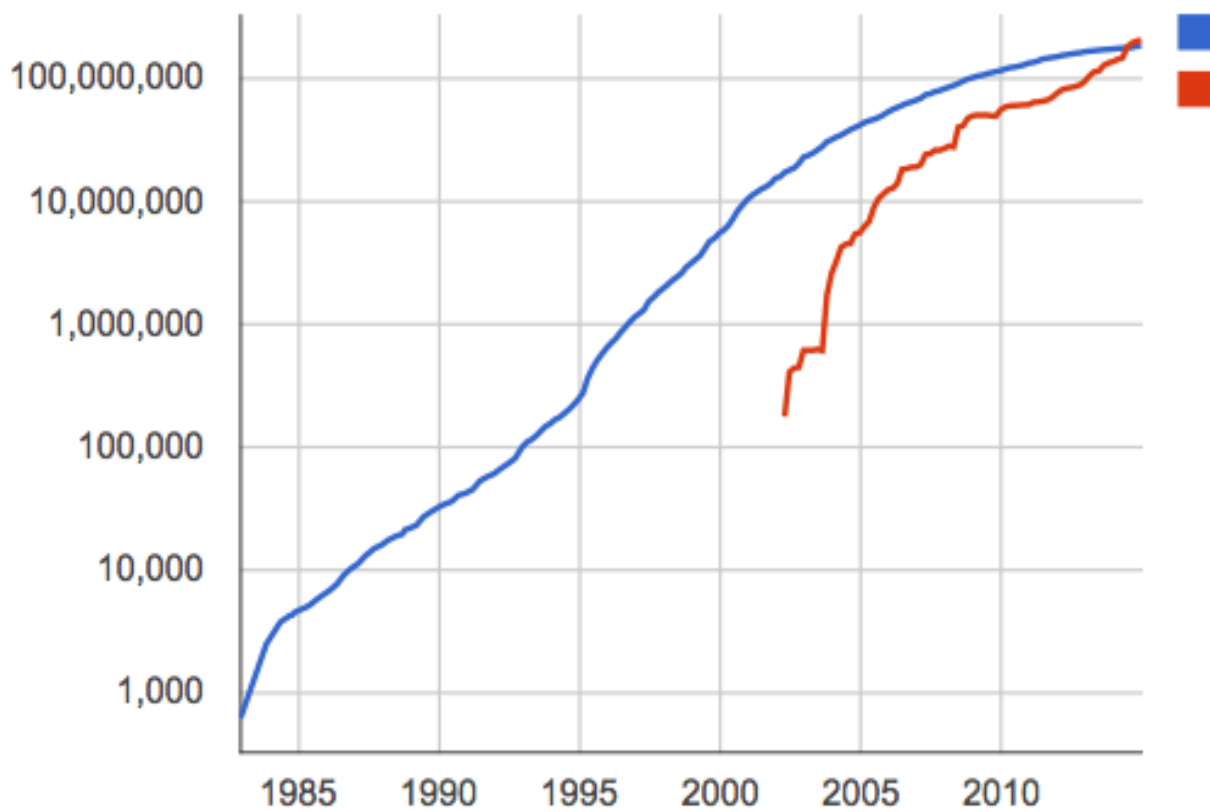
Growth of Genebank

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<http://www.ncbi.nlm.nih.gov/genbank/statistics>



<http://www.ncbi.nlm.nih.gov/genbank/statistics>

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Conclusions

- Biological databases store biological data
- **INSDC** is joint venture of NCBI, EMBL and DDBJ
- Growth of bases in Genbank is exponential, doubling every 18 months

Topic # 4

Protein Databases

Introduction

Protein databases store

- **Protein sequences**
- **Motif**
- **Structure**
- **Structure alignments**

Origin

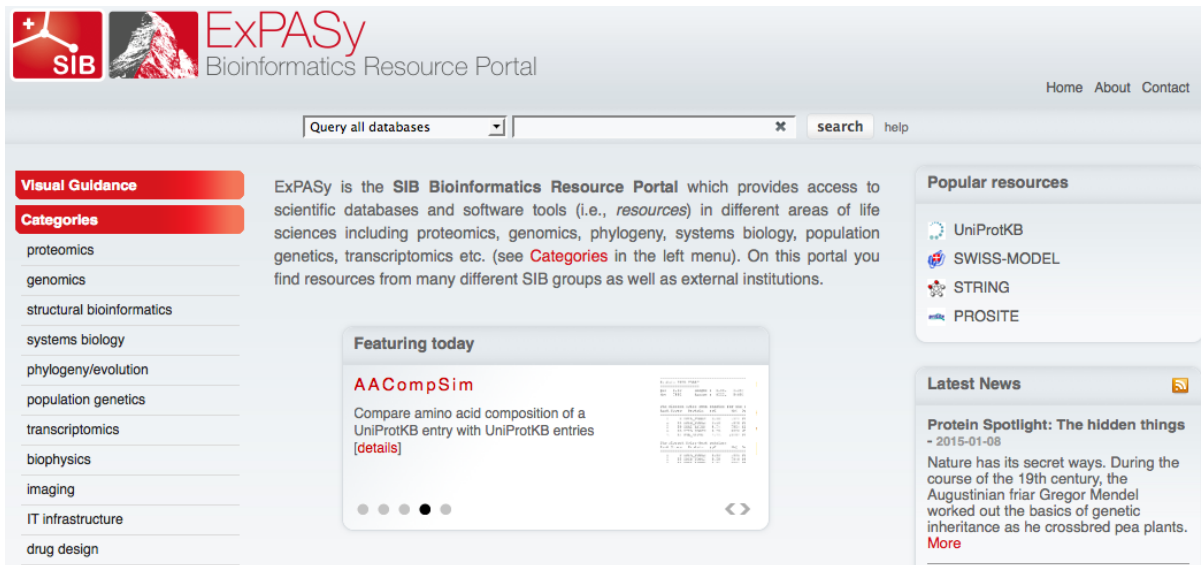
- First sequences to be collected were Proteins using **Sanger and Tuppy's** methods (1951) where Common protein families like cytochromes were sequenced
- Atlas of protein sequences (mainly cytochromes) was assembled by Margret Dayhoff and her collaborators at **National Biomedical Research Foundation (NBRF)** in 1960s

PIR ((Protein Information Resource)

- The collection of Dayhoff and co became **PIR**
- which is now a collaboration of **NBRF**, **Munich Center for Protein Sequences (MIPS)** and **Japan International Protein Information Database (JIPID)**

Protein Sequences

- **Swiss-Prot** is Collaboration between the SIB and EBI
- Weekly releases from about 50 servers across the world, the main source being ExPASy in Geneva



<http://www.expasy.org/>

Protein Sequences

- International partnership between PIR, EBI and SIB
- Created **UniProt**, by unifying the PIR-PSD, Swiss-Prot, and TrEMBL databases

<http://www.uniprot.org/>

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PIR A UniProt CONORTIUM MEMBER
Protein Information Resource

Protein Search Site Search

About PIR Databases Search/Analysis Download Support

PRO Hierarchy (Note that the implicit relationship is *is_a*, whereas ^d indicates *derives_from* relationship.)

6 shown of 223047 records

expand	sort (no)	sort (size)	find	Category
☐	GO:0032991	macromolecular complex		
☐	PR:000025493	LPS:GPI-anchored CD14		complex
☐	PR:000025494	LPS:secreted CD14		complex
☐	GO:0043234	protein complex		
☐	PR:000018263	amino acid chain		
☐	PR:000000001	protein		

Home | About PIR | Databases | Search/Analysis | Download | Support

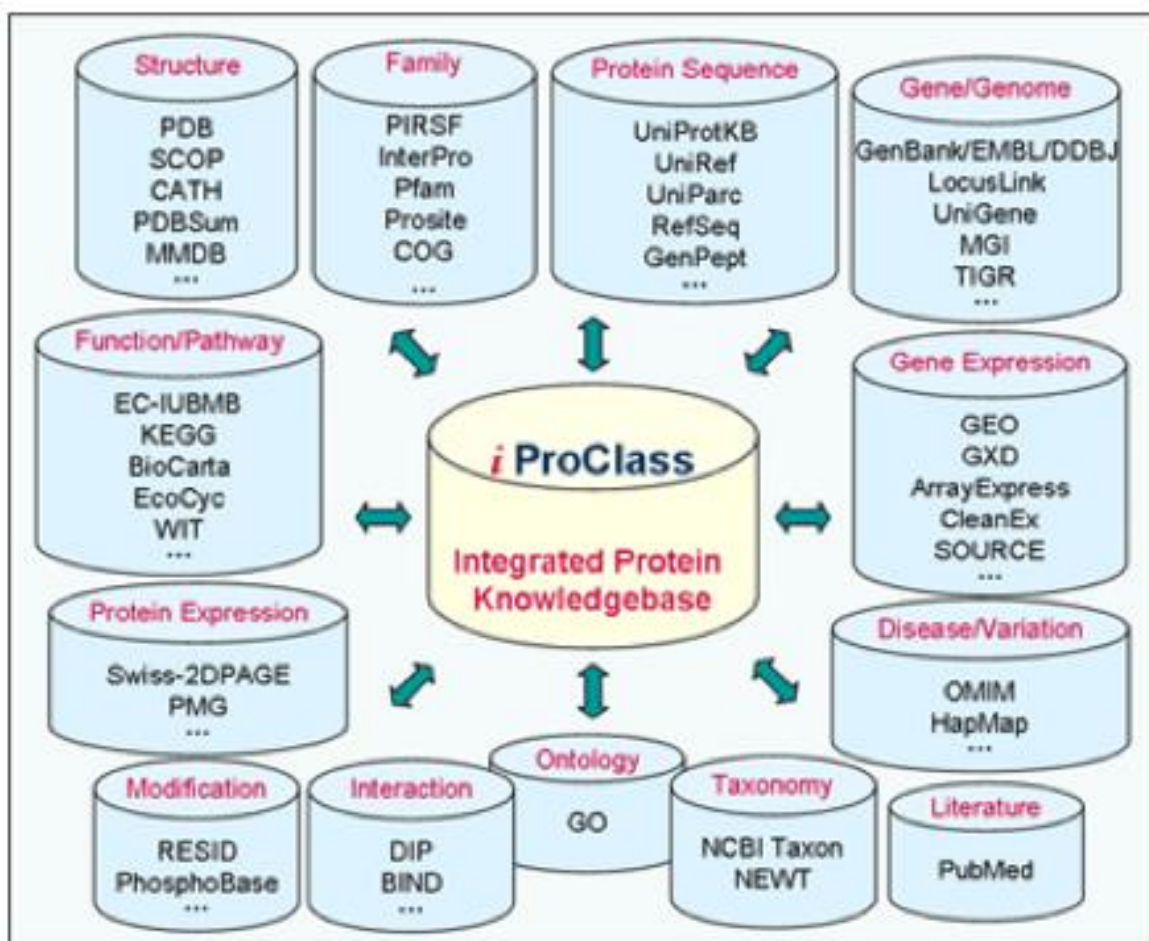
SITE MAP | TERMS OF USE

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University of Delaware
15 Innovation Way, Suite 205
Newark, DE 19711, USA

Georgetown University Medical Center
3300 Whitehaven Street, NW, Suite 1200
Washington, DC 20007, USA

<http://pir.georgetown.edu/>

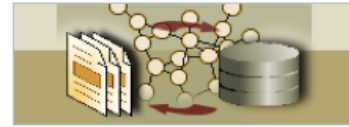
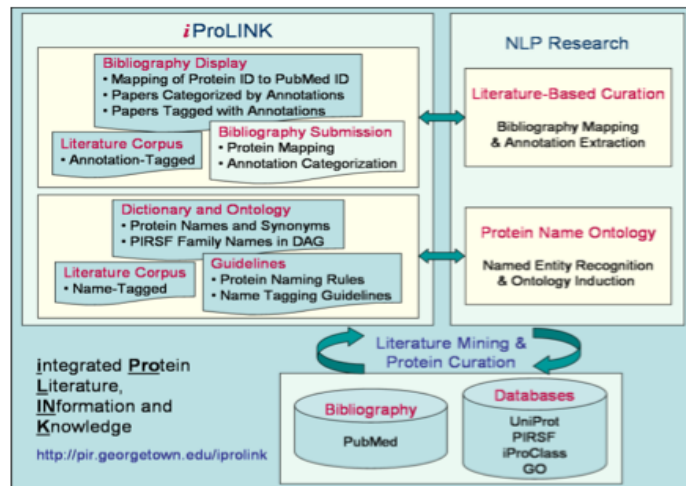


<http://www.uniprot.org/>

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iProLINK (**i**ntegrated **P**rotein **L**iterature, **I**Nformation and **K**nowledge) has been developed as a resource to facilitate text mining in the area of literature-based database curation, named entity recognition, and protein ontology development. The collection of data sources can be utilized by computational and biological researchers to explore literature information on proteins and their features or properties (Hu *et al.*, 2004).



Bibliography Mapping/Annotation Extraction

- [Bibliography Display/Submission](#)
- [Annotation-Tagged Corpora](#)
- [RLIMS-P Text Mining Tool](#)
- [eFIP Text Mining Tool](#)
- [eGIFT Text Mining Tool](#)
- [iSimp Sentence Simplification System](#)

Entity Recognition/Ontology Development

- [Name Tagging Guidelines/Corpora](#)
- [Protein Ontology Development](#)

Collaborators

iProLINK Paper

<http://www.uniprot.org/>

RCSB PDB Deposit Search Visualize Analyze Download Learn More MyPDB Login

RCSB PDB An Information Portal to 105499 Biological Macromolecular Structures

Search by PDB ID, author, macromolecule, sequence, or ligands **Go**

Advanced Search | Browse by Annotations

PDBe-101 PDB EMDatabank Protein Data Bank StructuralBiology Knowledgebase

Welcome

Deposit Search Visualize Analyze Download

A Structural View of Biology

This resource is powered by the Protein Data Bank archive—information about the 3D shapes of proteins, nucleic acids, and complex assemblies that helps students and researchers understand all aspects of biomedicine and agriculture, from protein synthesis to health and disease.

As a member of the wwPDB, the RCSB PDB curates and annotates PDB data.

The RCSB PDB builds upon the data by creating tools and resources for research and education in molecular biology, structural biology, computational biology, and beyond.

Structure and Health Focus: Ebola Virus Proteins

January Molecule of the Month

<http://www.rcsb.org/pdb/home/home.do>

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[Browse](#) [Stats & History](#) [ASTRAL Subsets](#) [Downloads](#) [Related Resources](#) [References](#) [Help](#) [About](#)

Welcome to SCOPe!

SCOPe is a database developed at the Berkeley Lab and UC Berkeley to extend the development and maintenance of SCOP.

SCOP was conceived at the MRC Laboratory of Molecular Biology, and developed in collaboration with researchers in Berkeley.

Work on SCOP (version 1) concluded in June 2009 with the release of SCOP 1.75.

SCOPe classifies many newer structures through a combination of automation and manual curation, and corrects some errors in SCOP,

aiming to have the same accuracy as the hand-curated SCOP releases. SCOPe also incorporates and updates the ASTRAL database.

For prior releases, click on the [Stats & History](#) tab above. For more info, click on the [About](#) tab above.

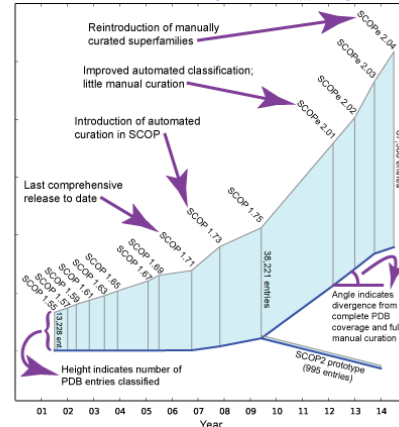
New PDB entries were last added on **2014-12-18**; for more info on periodic updates click on the [Help](#) tab above.

Search SCOPe (example):

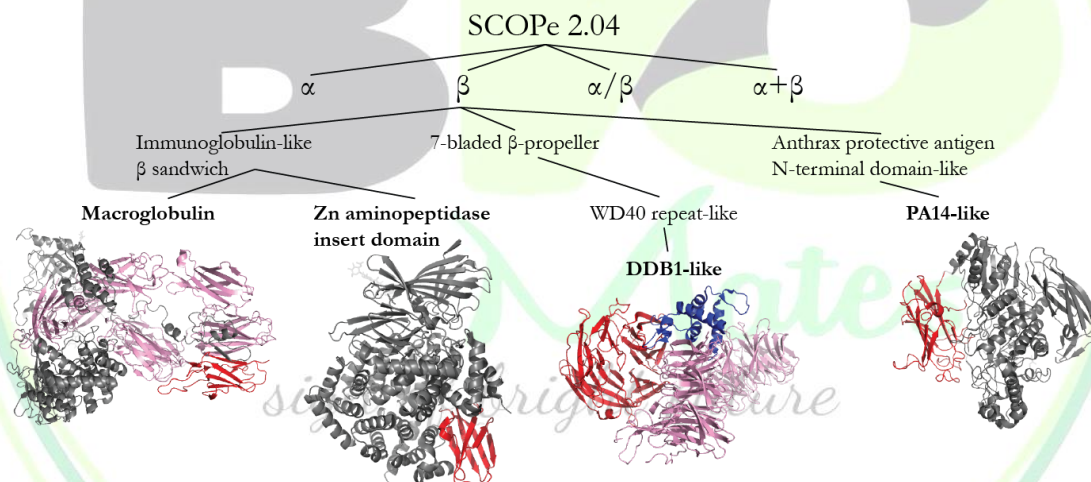
Classes in SCOPe 2.04:

1. [a: All alpha proteins](#) [46456] (285 folds)
2. [b: All beta proteins](#) [48724] (176 folds)
3. [c: Alpha and beta proteins \(a/b\)](#) [51349] (148 folds)
4. [d: Alpha and beta proteins \(a+b\)](#) [53931] (380 folds)
5. [e: Multi-domain proteins \(alpha and beta\)](#) [56572] (68 folds)
6. [f: Membrane and cell surface proteins and peptides](#) [56835] (57 folds)
7. [g: Small proteins](#) [56992] (91 folds)
8. [h: Coiled coil proteins](#) [57942] (7 folds)
9. [i: Low resolution protein structures](#) [58117] (25 folds)

[Click for information about changes to SCOP\(e\) design and size.](#)



<http://scop.mrc-lmb.cam.ac.uk/scop/>



<http://scop.mrc-lmb.cam.ac.uk/scop/>

Conclusions

- First sequences to be collected were Protein sequences
- Protein databases are classified on the basis of sequences, motifs and structures
- Growth of Sequence in Databases is exponential

Topic # 5

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Genome & Organism Specific Databases

Origin

- First attempt to sequence free living organism was launched in late 1990's
 - (Blattner et al. 1997)
- Viruses had already been sequenced
- *Haemophilus influenzae* was the first published genome
 - (Fleischmann et al. 1995)
- The project initiated at The Institute of Genome Research (TIGR) under leadership of Craig Venter
- Use of shotgun sequencing method speeded up the process
- 1.8 million bp
- Took 9-months and cost was 1 million
- Paved the way for sequencing of many other organisms

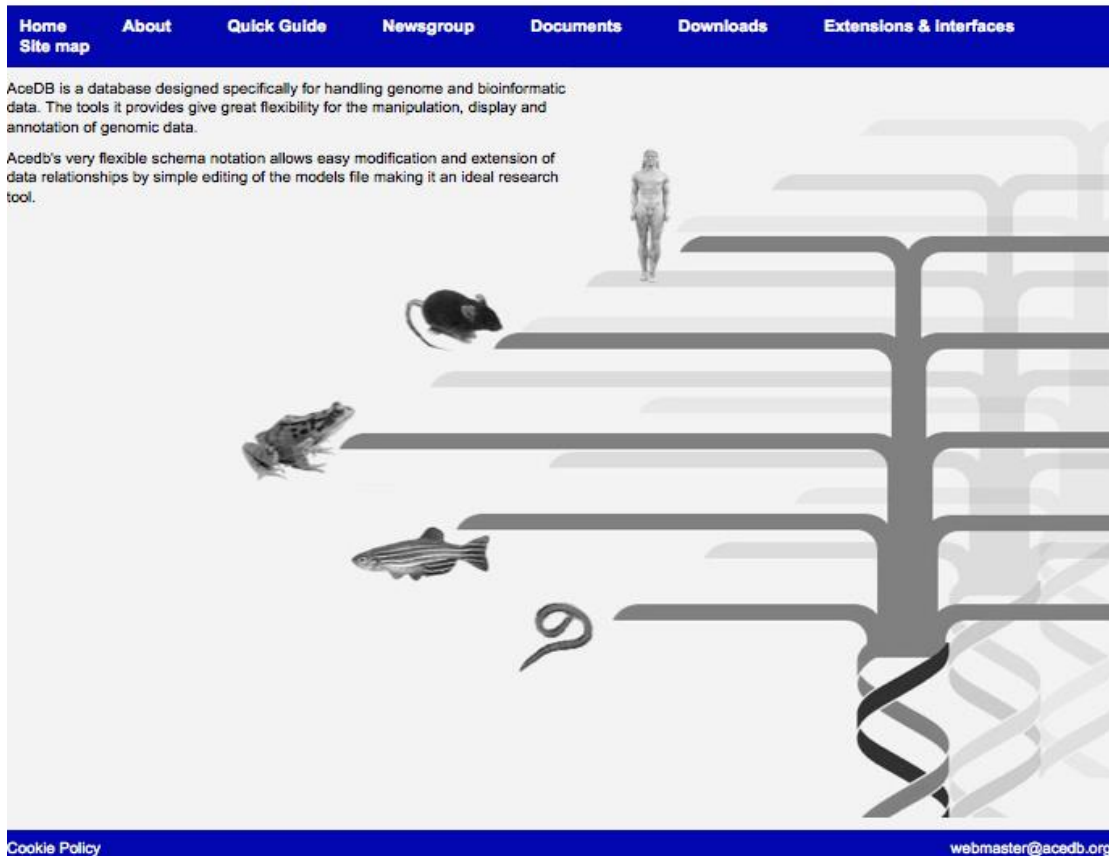
Examples

- **AceDB** (**A** **C.** **elegans** **DataBase**) was the first genome database developed in 1989
 - by **Richard durbin** and **Thierry-Miegi**

Mates
sign of bright future

AceDB is a database designed specifically for handling genome and bioinformatic data. The tools it provides give great flexibility for the manipulation, display and annotation of genomic data.

AceDB's very flexible schema notation allows easy modification and extension of data relationships by simple editing of the models file making it an ideal research tool.



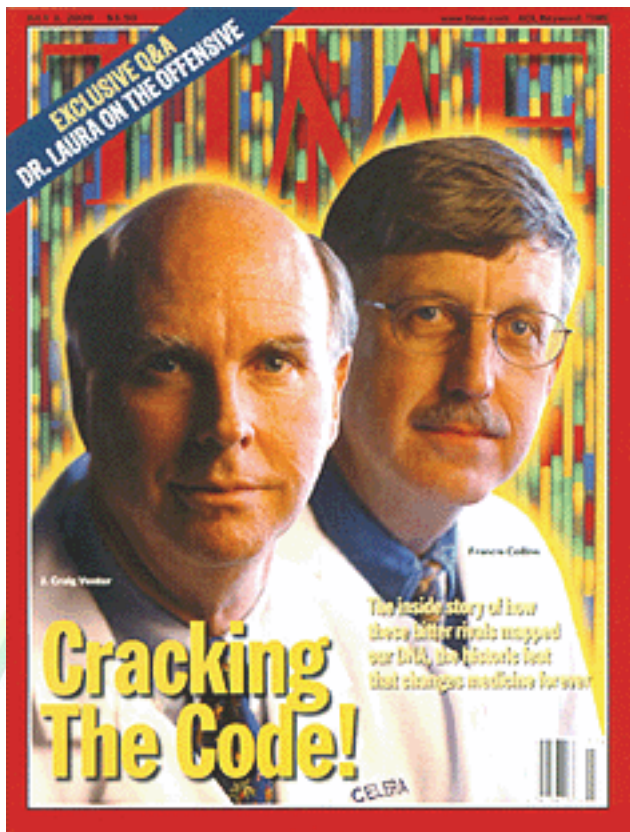
<http://www.acedb.org/>

Examples

- **TAIR** (The Arabidopsis Information Resource) <http://www.arabidopsis.org/> and **SGB** (Saccharomyces Genome Database) <http://www.yeastgenome.org/> utilized **AceDB** system

Human Genome Project

- Pilot project, the **Human Genome Initiative** begun by Department Of Energy (DOE) in 1986
- National **Human Genome Research Initiative (NHGRI)**, a federally funded organization (NIH) started in 1988 (**Francis Collin**)
- Celera, a commercial vendor (**Craig Venter**) joined the venture in 1998



-
- Both simultaneously announced the completion of the project in 2000
- Total 3.4 billion bases sequenced at a cost of \$1/base

UCSC Genome Bioinformatics

Genomes - Blat - Tables - Gene Sorter - PCR - VigiGene - Session - FAQ - Help

Genome Browser

Ebola

Blat

Table Browser

Gene Sorter

In Silico PCR

Genome Graphs

Galaxy

VigiGene

Utilities

Downloads

Release Log

Custom

About the UCSC Genome Bioinformatics Site

Welcome to the UCSC Genome Browser website. This site contains the reference sequence and working draft assemblies for a large collection of genomes. It also provides portals to [ENCODE](#) data at UCSC (2003 to 2012) and to the [Neanderthal](#) project. Download or purchase the Genome Browser source code, or the Genome Browser in a Box (GBiB) at our [online store](#).

We encourage you to explore these sequences with our tools. The [Genome Browser](#) zooms and scrolls over chromosomes, showing the work of annotators worldwide. The [Gene Sorter](#) shows expression, homology and other information on groups of genes that can be related in many ways. [Blat](#) quickly maps your sequence to the genome. The [Table Browser](#) provides convenient access to the underlying database. [VigiGene](#) lets you browse through a large collection of *in situ* mouse and frog images to examine expression patterns. [Genome Graphs](#) allows you to upload and display genome-wide data sets.

The UCSC Genome Browser is developed and maintained by the Genome Bioinformatics Group, a cross-departmental team within the [UC Santa Cruz Genomics Institute](#) and the Center for Biomolecular Science and Engineering (CBSE) at the University of California Santa Cruz (UCSC). If you have feedback or questions concerning the tools or data on this website, feel free to contact us on our [public mailing list](#).

The Genome Browser project team relies on public funding to support our work. Donations are welcome -- we have many more ideas than our funding supports! If you have ideas, drop a comment in our [suggestion box](#).

[DONATE NOW](#)

News

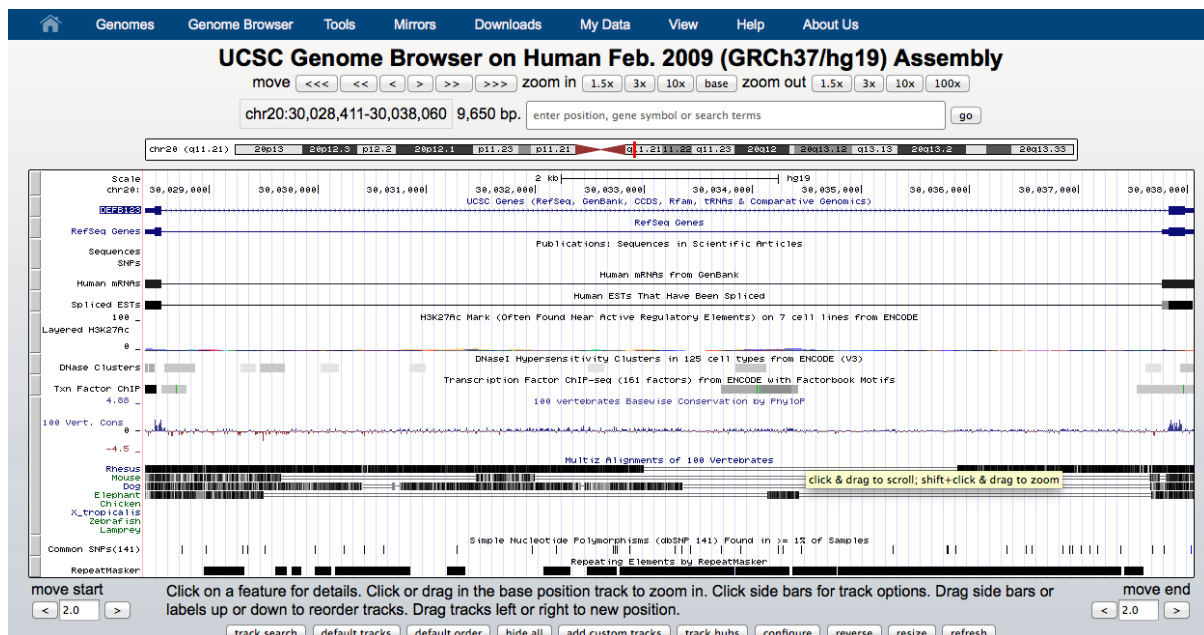
To receive announcements of new genome assembly releases, new software features, updates and training seminars by email, subscribe to the [genome-announce](#) mailing list. Please see our [blog](#) for posts about Genome Browser tools, features, projects and more.

News Archives ►

<http://genome.ucsc.edu/>

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<http://genome.ucsc.edu/>

Conclusion

- Success of *Haemophilus influenzae* paved the way for other genome sequencing projects
- Human Genome Project was accomplished by NHGRI and Celera
- Genome browsers help viewing Genome features

Topic #6

Gene Expression Databases

Gene Expression Omnibus (GEO)

- A public repository for the archiving and distribution of gene expression data submitted by the scientific community
- MIAME compliant data
- Minimum Information About a Microarray Experiment
<http://www.mged.org/Workgroups/MIAME/miame.html>
- Convenient for deposition of gene expression data, as required by funding agencies and journals
- Curated resource for gene expression data
- Browsing, querying, analysis and retrieval

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The screenshot shows the NCBI GEO website with the 'Query & Browse' dropdown menu open. The menu options are: Search GEO DataSets, Search GEO Profiles, Analyze with GEO2R, GEO BLAST, Programmatic Access, Repository Browser, and FTP Site. The main content area includes a 'Gene Expression' section, a search bar, and three columns: 'Getting Started', 'Tools', and 'Browse Content'.

Gene Expression

GEO is a public functional genomics data repository supporting MIAME-compliant data submissions. Array- and sequence-based data are accepted. Tools are provided to help users query and download experiments and curated gene expression profiles.

Getting Started

- Overview
- FAQ
- About GEO DataSets
- About GEO Profiles
- About GEO2R Analysis
- How to Construct a Query
- How to Download Data

Tools

- Search for Studies at GEO DataSets
- Search for Gene Expression at GEO Profiles
- Search GEO Documentation
- Analyze a Study with GEO2R
- GEO BLAST
- Programmatic Access
- FTP Site

Browse Content

Repository Browser

DataSets:	3848
Series:	51804
Platforms:	13522
Samples:	1259935

- <http://www.ncbi.nlm.nih.gov/geo/>

Gene Architecture

GEO has four kinds of records

- **Sample (GSM)** preparation and description of the samples
- **Platform (GPL)** technology used and the features detected
microarray or RNASeq
- **Series (GSE)**
defines a set of samples and how they are related
- **Datasets (GDS)**
sample data collections assembled by GEO

The screenshot shows the NCBI GEO website with the 'Browse Content' section highlighted with a red circle. The main content area includes a 'Gene Expression Omnibus' section, a search bar, and three columns: 'Getting Started', 'Tools', and 'Browse Content'.

Gene Expression Omnibus

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Gene Expression Omnibus

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Keyword or GEO Accession

Getting Started

Overview

FAQ

About GEO DataSets

About GEO Profiles

About GEO2R Analysis

How to Construct a Query

How to Download Data

Tools

Search for Studies at GEO DataSets

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Search GEO Documentation

Analyze a Study with GEO2R

GEO BLAST

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FTP Site

Browse Content

Repository Browser

DataSets:	3848
Series:	51804
Platforms:	13522
Samples:	1259935

Public holdings

Series	Platforms	Samples	Organisms	History
Series type	Count			
Expression profiling by array	36,895			
Expression profiling by genome tiling array	612			
Expression profiling by high throughput sequencing	3,446			
Expression profiling by SAGE	241			
Expression profiling by MPSS	20			
Expression profiling by RT-PCR	269			
Expression profiling by SNP array	12			
Genome variation profiling by array	557			
Genome variation profiling by genome tiling array	978			
Genome variation profiling by high throughput sequencing	56			
Genome variation profiling by SNP array	726			
Genome binding/occupancy profiling by array	156			
Genome binding/occupancy profiling by genome tiling array	2,042			
Genome binding/occupancy profiling by high throughput sequencing	3,317			
Genome binding/occupancy profiling by SNP array	11			
Methylation profiling by array	476			
Methylation profiling by genome tiling array	579			
Methylation profiling by high throughput sequencing	563			
Methylation profiling by SNP array	9			

Total holdings

	Public	Unreleased	Total
Series	51,806	7,896	59,702
Platforms	13,522	438	13,960
Samples	1,259,985	192,394	1,452,379

Show additional filters

Entry type

Series (4)

Organism

Select ...

Study type

Expression profiling by

array

More ...

Author

Select ...

Attribute name

tissue

strain

More ...

Publication

dates

30 days

1 year

Custom range...

Display Settings: Summary, Sorted by Default order

Did you mean: [colon cancer RNA seq](#) (60 items)

Results: 4

1. [Tumor cell-specific inhibition of MYC function using small molecule inhibitors of the HUWE1 ubiquitin ligase](#)

(Submitter supplied) Deregulated expression of MYC is a driver of colorectal carcinogenesis, necessitating novel strategies to inhibit MYC function. The ubiquitin ligase HUWE1 (HECTH9, ARFBP1, MULE) associates with both MYC and the MYC-associated protein MIZ1. We show here that HUWE1 is required for growth of colorectal cancer cells in culture and in orthotopic xenograft models. Using high throughput screening, we identify small molecule inhibitors of HUWE1, which inhibit MYC-dependent transactivation in colorectal cancer cells, but not in stem and normal colon epithelial cells. [more...](#)

Organism: Homo sapiens
Type: Expression profiling by high throughput sequencing; Genome binding/occupancy profiling by high throughput sequencing

Platform: GPL10999 22 Samples

Download data: GEO (TXT, WIG), SRA SRP044171

Series Accession: GSE59223 ID: 200059223

[PubMed](#) [Similar studies](#)

2. [Sequential ChIP-bisulfite sequencing enables direct genome-scale investigation of chromatin and DNA methylation cross-talk](#)

(Submitter supplied) Cross-talk between DNA methylation and histone modifications drives the establishment of composite epigenetic signatures and is traditionally studied using correlative rather than direct approaches. Here we present sequential ChIP-bisulfite-sequencing (ChIP-BS-seq) as an approach to

Send to:

Filters: [Manage Filters](#)

Top Organisms [\[Tree\]](#)

Homo sapiens (4)

Mus musculus (2)

Find related data

Database: [Select](#)

Find items

Search details

("colonic neoplasms"[MeSH Terms] OR colon cancer[All Fields]) AND RNASeq[All Fields]

Search

See more...

Recent activity

[Turn Off](#) [Clear](#)

colon cancer RNA Seq (60)

GEO DataSets

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[GEO Publications](#) | [FAQ](#) | [MIAME](#) | [Email GEO](#)

[NCBI > GEO > Accession Display](#)

Not logged in | [Login](#)

Scope: Format: Amount: GEO accession:

Series GSE57043
[Query DataSets for GSE57043](#)

Status	Public on Apr 25, 2014
Title	Dicer knockout NSCLC RNAseq and miRseq
Organism	Mus musculus
Experiment type	Expression profiling by high throughput sequencing Non-coding RNA profiling by high throughput sequencing
Summary	Dicer knockout NSCLC mRNAseq profiles the transcriptome, Dicer knockout NSCLC miRseq profiles the miRnome
Overall design	DicerHet and DicerKO NSCLC, 2 biological reps each genotype for mRNAseq, 1 biological rep each for miRseq
Contributor(s)	Sharp PA, Chen S
Citation(s)	Chen S, Xue Y, Wu X, Le C et al. Global microRNA depletion suppresses tumor angiogenesis. <i>Genes Dev</i> 2014 May 15;28(10):1054-67. PMID: 24788094
Submission date	Apr 24, 2014
Last update date	Oct 14, 2014
Contact name	Sidi Chen
E-mail	chensidi@mit.edu
Phone	7734144158
Organization name	MIT
Department	Biology
Lab	Sharp
Street address	77 Mass Ave, 76-461
City	Cambridge
State/province	MA
ZIP/Postal code	02139
Country	USA

Data is submitted to GEO as a Series, which represents the experiment design

Mates
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Platforms (1) [GPL13112](#) Illumina HiSeq 2000 (Mus musculus)

Samples (6) [GSM1373652](#) DcrHet_1_mRNA

≡ [Less...](#)

[GSM1373653](#) DcrHet_2_mRNA

[GSM1373654](#) DcrKO_1_mRNA

[GSM1373655](#) DcrKO_2_mRNA

[GSM1373656](#) DcrHet_1_miR

[GSM1373657](#) DcrKO_1_miR

Relations

BioProject [PRJNA245291](#)

SRA [SRP041414](#)

Download family	Format
SOFT formatted family file(s)	SOFT ?
MINiML formatted family file(s)	MINiML ?
Series Matrix File(s)	TXT ?

Supplementary file	Size	Download	File type/resource
GSE57043_dicerko_fpkkm.txt.gz	875.5 Kb	(ftp) (http)	TXT
GSE57043_dicerko_hairpin_rpm.txt.gz	4.1 Kb	(ftp) (http)	TXT
GSE57043_dicerko_mature_rpm.txt.gz	1.6 Kb	(ftp) (http)	TXT
SRP/SRP041/SRP041414		(ftp)	SRA Study

Raw data provided as supplementary file

Processed data is available on Series record



An expression signature for p53 status in human breast cancer predicts mutation status, transcriptional effects, and patient survival

Lance D. Miller[†], Johanna Smeds[‡], Joshy George^{*}, Vinsensius B. Vega^{*}, Liza Vergara^{*}, Alexander Ploner[§], Yudi Pawitan[§], Per Hall[§], Sigrid Klaar[‡], Edison T. Liu^{†*}, and Jonas Bergh[‡]

^{*}Genome Institute of Singapore, 60 Biopolis Street, #02-01, Singapore 138672; [‡]Department of Oncology and Pathology, Radiumhemmet, Karolinska Institute and Hospital, S-17176 Stockholm, Sweden; and [§]Department of Medical Epidemiology and Biostatistics, Karolinska Institutet, 17177 Stockholm, Sweden

GENETICS. For the article "An expression signature for p53 status in human breast cancer predicts mutation status, transcriptional effects, and patient survival," by Lance D. Miller, Johanna Smeds, Joshy George, Vinsensius B. Vega, Liza Vergara, Alexander Ploner, Yudi Pawitan, Per Hall, Sigrid Klaar, Edison T. Liu, and Jonas Bergh, which appeared in issue 38, September 20, 2005, of *Proc. Natl. Acad. Sci. USA* (102, 13550-13555; first published September 2, 2005; 10.1073/pnas.0506230102), the breast cancer microarray data discussed in this publication have been deposited in the National Center for Biotechnology Information's Gene Expression Omnibus database (GEO, www.ncbi.nlm.nih.gov/geo/) and are accessible through GEO Series accession no. GSE3494 [NCBI GEO] .

Conclusion

- **GEO** is a public repository for the archiving and distribution of gene expression data

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- Best resource to get microarray and Next Generation Sequencing (RNASeq) data

Topic #7

Medical Databases

Introduction

Informatics in health care may be called as health informatics

- It deals with the resources, devices, and methods required to optimize the acquisition, storage, retrieval, and use of information in health and biomedicine.

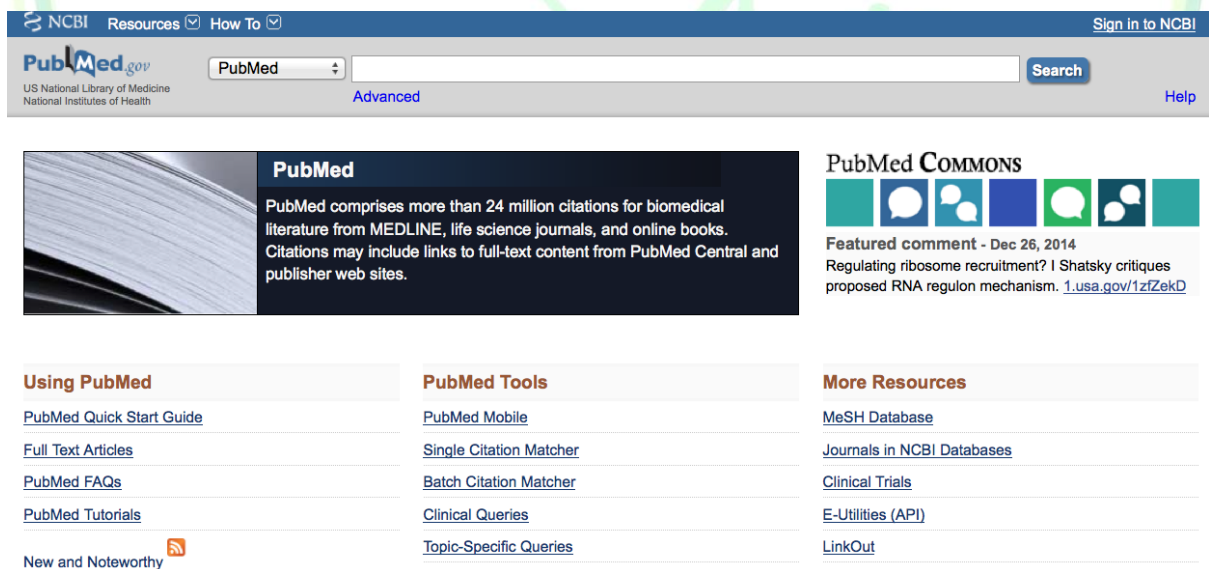
(wiki)

Medical databases store and provide medical information

- The premier database for biomedical literature is the National Library of Medicine (NLM)'s MEDLINE, accessible through PubMed

PUBMED

- Comprises of more than 24 million citations for biomedical literature from MEDLINE, life science journals, and online books
- Citations may include links to full-text content from PubMed Central and publisher web sites



PubMed

PubMed comprises more than 24 million citations for biomedical literature from MEDLINE, life science journals, and online books. Citations may include links to full-text content from PubMed Central and publisher web sites.

PubMed Commons

Featured comment - Dec 26, 2014
Regulating ribosome recruitment? | Shatsky critiques proposed RNA regulon mechanism. 1.usa.gov/1zfZekD

Using PubMed	PubMed Tools	More Resources
PubMed Quick Start Guide	PubMed Mobile	MeSH Database
Full Text Articles	Single Citation Matcher	Journals in NCBI Databases
PubMed FAQs	Batch Citation Matcher	Clinical Trials
PubMed Tutorials	Clinical Queries	E-Utilities (API)
New and Noteworthy	Topic-Specific Queries	LinkOut

MEDLINE

- MEDLINE is the primary resource for biomedical journal articles
- Millions of citations to articles in biomedical journals

VU Bio Mates

<https://www.facebook.com/groups/vubiomates/>

- MEDLINE uses the MeSH vocabulary

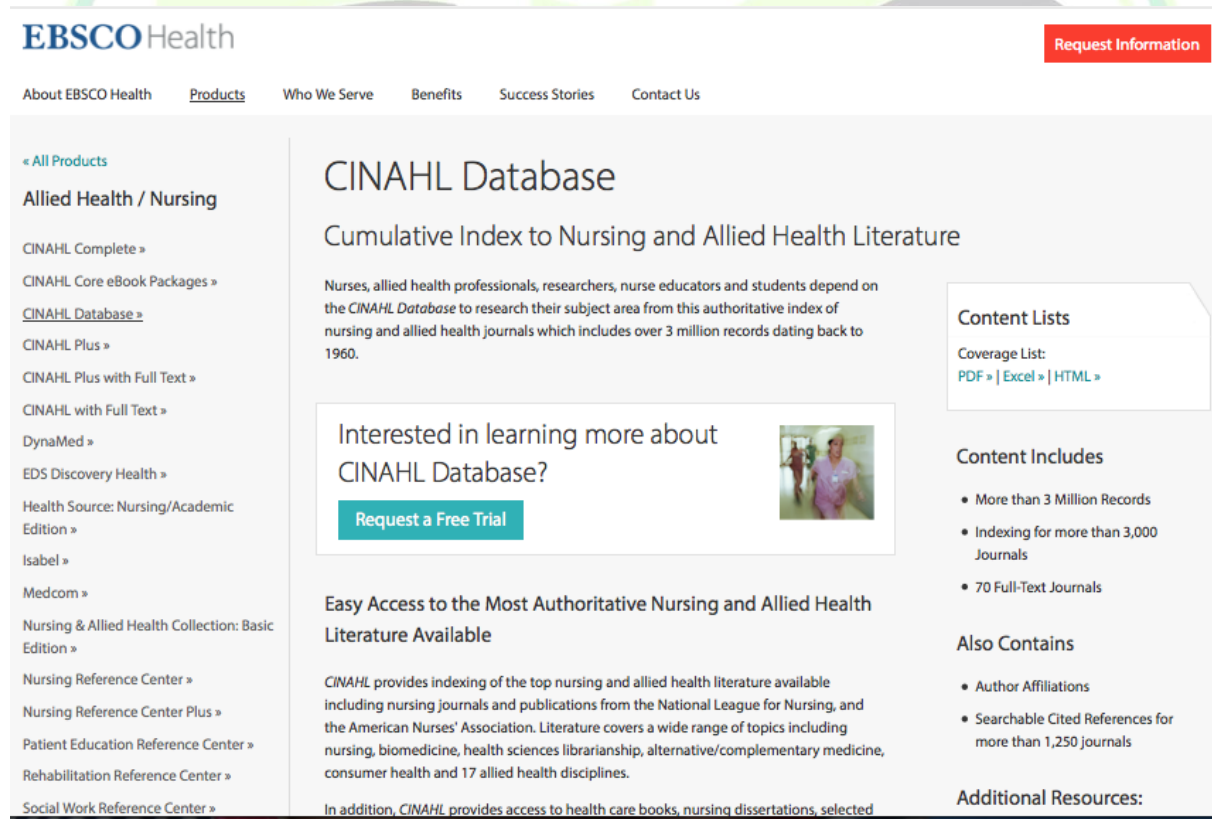
Other Databases

MEDLINE is the primary resource, but other databases may also be helpful

- Academic OneFile
- CINAHL (Cumulated Index of Nursing and Allied Health Literature)
- PsycINFO
- Web of Knowledge

Academic OneFile

- Academic OneFile lists articles from journals covering a broad range of subjects
- While it does not primarily focus on medical topics, useful articles can still be found here



The image shows the EBSCO Health website interface for the CINAHL Database. The header includes the EBSCO Health logo and a 'Request Information' button. A navigation bar lists links: About EBSCO Health, Products, Who We Serve, Benefits, Success Stories, and Contact Us. The main content area is titled 'CINAHL Database' and 'Cumulative Index to Nursing and Allied Health Literature'. It describes the database as an authoritative index of nursing and allied health journals with over 3 million records dating back to 1960. A sidebar on the left lists various products like CINAHL Complete, CINAHL Core eBook Packages, CINAHL Database, CINAHL Plus, CINAHL Plus with Full Text, CINAHL with Full Text, DynaMed, EDS Discovery Health, Health Source: Nursing/Academic Edition, Isabel, Medcom, Nursing & Allied Health Collection: Basic Edition, Nursing Reference Center, Nursing Reference Center Plus, Patient Education Reference Center, Rehabilitation Reference Center, and Social Work Reference Center. A central box asks 'Interested in learning more about CINAHL Database?' with a 'Request a Free Trial' button. To the right, there are sections for 'Content Lists' (Coverage List: PDF, Excel, HTML), 'Content Includes' (More than 3 Million Records, Indexing for more than 3,000 Journals, 70 Full-Text Journals), 'Also Contains' (Author Affiliations, Searchable Cited References for more than 1,250 journals), and 'Additional Resources'.

PsycINFO

- PsycINFO searches the psychological literature
- While it does not primarily focus on medical topics, useful articles can still be found here
- <http://www.apa.org/pubs/databases/psycinfo/coverage.aspx>

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<https://www.facebook.com/groups/vubiomates/>

PsycINFO® Journal Coverage List

October 2014 Update

Currently, there are 2,562 journals covered in the PsycINFO® database. The list changes continuously as journals are added and discontinued throughout the year, so it is updated online monthly.

- [Download the journal coverage list in Excel format \(2,358KB\)](#)
- [View a list of the currently covered neuroscience titles \(PDF, 625KB\)](#)
Updated February 2010
- [View journal coverage facts](#)
- [View journal coverage policy](#)
- [View journals added in 2013](#)

Other Information

Visit the [journals added](#) page to see a list of the journals we've added to the Journal Coverage List since the last update.

With the exception of [journals indexed cover-to-cover](#), not all articles from each journal are included in the database. PsycINFO staff examine each article and select only those that have psychological relevance.

More About the Database

- [PsycINFO® Home](#)
- [FAQs](#)
- [Coverage List](#)
- [Sample Records](#)
- [Publisher Relations](#)
- [Cited Reference Facts](#)
- [Subsets & Special Collections](#)
- [Archive of Sample Searches Podcasts](#)

Librarians

- [Institutional Access](#)
- [Institutional Pricing](#)
- [Free Trial for Institutions](#)

<http://www.apa.org/pubs/databases/psycinfo/coverage.aspx>

Web of Science

- Major source for articles in a wide range of fields, including the sciences, social sciences, and humanities.
- Excellent place to find articles from scientific journals that may not be included in MEDLINE

Conclusions

Informatics in health care may be called as health informatics

- Medical databases deal with the acquisition, storage, retrieval, and use of information in health and biomedicine.

Topic #8

Sequence Submission

Introduction

- Sequences are submitted to the databases in order to share them with the scientific community
- Generally sequences are submitted at the time of publication and are reviewed by peers

Caution

- It is important to ensure that sequence files do not contain any special characters

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- Control characters in addition to standard ASCII characters must be removed else they might mess up the analysis or data transfer
- Nomenclature committee of International Union of Biochemistry (IUB) has established standard codes to represent ambiguous bases or aminoacids

Table 2.1. *Base-nucleic acid codes*

Symbol	Meaning	Explanation
G	G	Guanine
A	A	Adenine
T	T	Thymine
C	C	Cytosine
R	A or G	puRine
Y	C or T	pYrimidine
M	A or C	aMino
K	G or T	Keto
S	C or G	Strong interactions 3 h bonds
W	A or T	Weak interactions 2 h bonds
H	A, C or T not G	H follows G in alphabet
B	C, G or T not A	B follows A in alphabet
V	A, C or G not T (not U)	V follows U in alphabet
D	A, G or T not C	D follows C in alphabet
N	A,C,G or T	Any base

Adapted from NC-IUB (1984).

Mount, pg 28

Table 2.2. *Table of standard amino acid code letters*

1-letter code	3-letter code	Amino acid
A ^a	Ala	alanine
C	Cys	cysteine
D	Asp	aspartic acid
E	Glu	glutamic acid
F	Phe	phenylalanine
G	Gly	glycine
H	His	histidine
I	Ile	isoleucine
K	Lys	lysine
L	Leu	leucine
M	Met	methionine
N	Asn	asparagine
P	Pro	proline
Q	Gln	glutamine
R	Arg	arginine
S	Ser	serine
T	Thr	threonine
V	Val	valine
W	Trp	tryptophan
X	Xxx	undetermined amino acid
Y	Tyr	tyrosine
Z ^b	Glx	either glutamic acid or glutamine

Adapted from IUPAC-IUB (1969, 1972, 1983).

^a Letters not shown are not commonly used.

^b Note that sometimes when computer programs translate DNA sequences, they will put a "Z" at the end to indicate the termination codon. This character should be deleted from the sequence.

Mount, pg 28

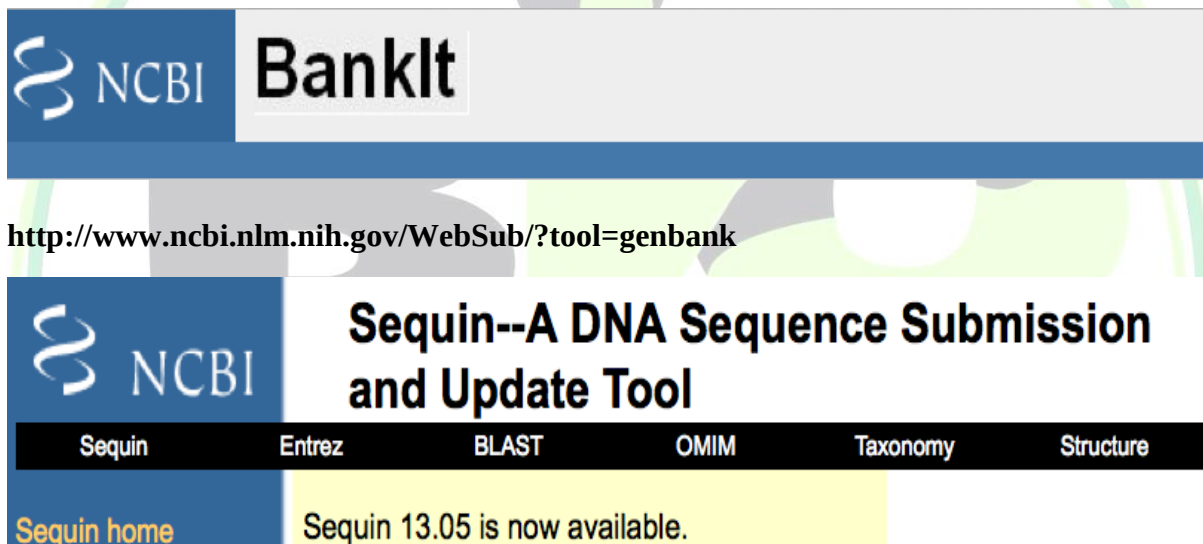
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NCBI

NCBI has two options

- **BANKIt**
 - For simple sequences and annotations
 - For submission through web
 - Do not require advanced tools
- **Sequin**
 - For Complex sequences and annotations
 - For off-line submissions
 - Do require advanced tools and graphical reports



UniProt

SPIN web-based tool for submitting

- directly sequenced protein sequences
- biological annotations to the knowledgebase

Welcome to the new SPIN website!

Please use this new website for any new submissions you would like to make.

To finish and submit any submissions you may have started using the old website please go [here](#) if. Please note that submissions created using the new website will not be seen in the old one and vice versa.

About SPIN

SPIN is the web-based tool for submitting directly sequenced protein sequences and their biological annotations to the UniProt Knowledgebase. The information required to create a database entry will be collected during the submission process.

SPIN should not be used for the submission of translations of nucleic acid sequences, please use [Webin](#) if for these submissions.

Please sign in

*

*

[Forgot your password?](#)

New user?

 [Create an account](#)

<https://www.ebi.ac.uk/swissprot/Submissions/spin>

Conclusion

- Sequences are stored in databases in specific format

Topic #9

DNA Sequence Retrieval

Introduction

- Databases not merely collect and organize data but allow intelligent data retrieval
- And/or analysis

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Gene [Save search](#) [Advanced](#)

Display Settings: ☒ Tabular, 20 per page, Sorted by Relevance

Send to: ☐

Did you mean p53 as a gene symbol?
Search Gene for [p53](#) as a symbol.

Results: 1 to 20 of 9031

<< First < Prev Page **1** of 452 [Next >](#) [Last >>](#)

Filters activated: Current only. [Clear all](#) to show 9271 items.

Name/Gene ID	Description	Location	Aliases	MIM
☐ p53 ID: 2768677	CG33336 gene product from transcript CG33336-RB [<i>Drosophila melanogaster</i> (fruit fly)]	Chromosome 3R, NT_033777.3 (23049657..23054082, complement)	Dmel_CG33336, CG10873, CG31325, CG33336, D- DMP53, Dm-P53, DmP53, Dmel\CG33336, Dmp53, Dp53, dmp53, dp53, prac	
☐ TP53 ID: 7157	tumor protein p53 [<i>Homo sapiens</i> (human)]	Chromosome 17, NC_000017.11 (7668402..7687550, complement)	BCC7, LFS1, P53, TRP53	191170

clear

<http://www.ncbi.nlm.nih.gov/>

p53 [*Drosophila melanogaster* (fruit fly)]

Gene ID: 2768677, updated on 4-Jan-2015

☒ Summary ☐ ☐

Official Symbol p53 provided by [FlyBase](#)
Primary source [FLYBASE:FBgn0039044](#)
Locus tag [Dmel_CG33336](#)
Gene type protein coding
RefSeq status REVIEWED
Organism [Drosophila melanogaster](#) (old-lineage: Eukaryota; Metazoa; Arthropoda; Hexapoda; Insecta; Pterygota; Neoptera; Endopterygota; Diptera; Brachycera; Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora)
Lineage Eukaryota; Metazoa; Ecdysozoa; Arthropoda; Hexapoda; Insecta; Pterygota; Neoptera; Endopterygota; Diptera; Brachycera; Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora
Also known as CG10873; CG31325; CG33336; D-p53; Dm-P53; Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53; prac

<http://www.ncbi.nlm.nih.gov/>

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<https://www.facebook.com/groups/vubiomates/>

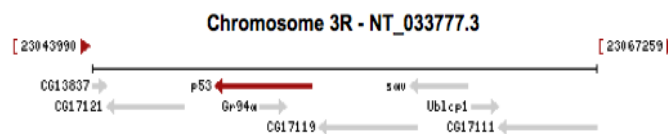
Genomic context

Location: 94D10-94D10

See p53 in [Epigenomics](#), [MapViewer](#)

Exon count: 10

Annotation release	Status	Assembly	Chr	Location
Release 6.01	current	Release 6 plus ISO1 MT (GCF_000001215.4)	3R	NT_033777.3 (23049657..23054082, complement)
Release 5.57	previous assembly	Release 5 (GCF_000001215.2)	3R	NT_033777.2 (18875379..18879804, complement)



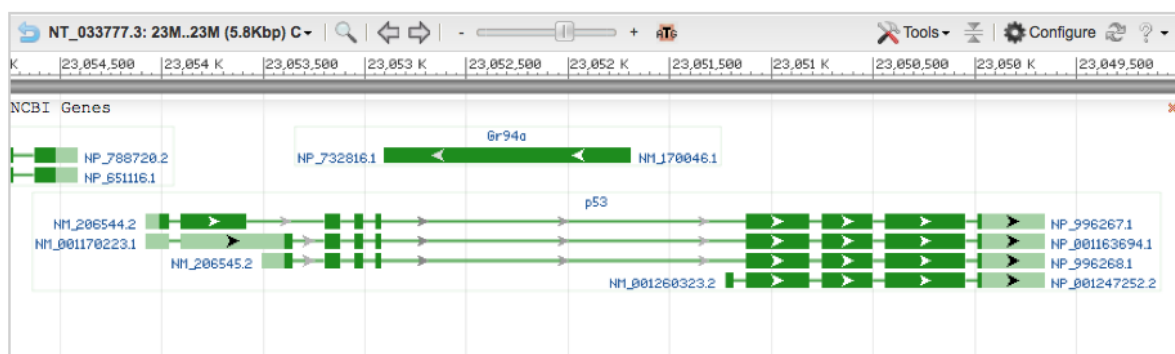
<http://www.ncbi.nlm.nih.gov/>

Genomic regions, transcripts, and products

Go to [reference sequence details](#)

Genomic Sequence: NT_033777.3 Chromosome 3R Reference Release 6 plus ISO1 MT Primary Assembly

Go to nucleotide: [Graphics](#) [FASTA](#) [GenBank](#)



<http://www.ncbi.nlm.nih.gov/>

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Drosophila melanogaster chromosome 3R

NCBI Reference Sequence: NT_033777.3

[FASTA](#) [Graphics](#)

LOCUS NT_033777 4426 bp DNA linear INV 05-AUG-2014
DEFINITION Drosophila melanogaster chromosome 3R.
ACCESSION [NT_033777](#) REGION: complement(23049657..23054082)
VERSION NT_033777.3 GI:671162122
DBLINK BioProject: [PRJNA164](#)
BioSample: [SAMN02803731](#)
KEYWORDS RefSeq.
SOURCE Drosophila melanogaster (fruit fly)
ORGANISM [Drosophila melanogaster](#)
Eukaryota; Metazoa; Ecdysozoa; Arthropoda; Hexapoda; Insecta;
Pterygota; Neoptera; Endopterygota; Diptera; Brachycera;
Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora.
REFERENCE 1 (bases 1 to 4426)
AUTHORS Hoskins,R.A., Carlson,J.W., Kennedy,C., Acevedo,D., Evans-Holm,M.,
Frise,E., Wan,K.H., Park,S., Mendez-Lago,M., Rossi,F.,
Villasante,A., Dimitri,P., Karpen,G.H. and Celniker,S.E.
TITLE Sequence finishing and mapping of Drosophila melanogaster
heterochromatin
JOURNAL Science 316 (5831), 1625-1628 (2007)
PUBMED [17569867](#)

<http://www.ncbi.nlm.nih.gov/>

FEATURES	Location/Qualifiers
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<http://www.ncbi.nlm.nih.gov/>

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          prac"
          /note="CG33336 gene product from transcript CG33336-RB;
          CG33336-PB; p53-PB; p53-like regulator of apoptosis and
          cell cycle; Dmp53; protein 53; drosophila p53"
          /codon_start=1
          /product="p53, isoform B"
          /protein_id="NP_996267.1"
          /db_xref="GI:45553461"
          /db_xref="FLYBASE:FBpp0083753"
          /db_xref="FLYBASE:FBgn0039044"
          /db_xref="GeneID:2768677"
          /translation="MSLHKSASFSLTFNQNTSIVSRNSRTIFEAFKEFLDFWDIGNE
          VSAESAVRVSSNGAFNLPQSFGNESNEYAHLATPVDPAYGGNNTNNMQFTNNLEILA
          NNNSDGNNKINACNKFVCHKGTDEDDSTEVDIKEDIPKTVEVSGSELTEPMAFLQG
          LNSGNLMQFSQQSVLREMMLQDIQIQANTLPKLEHNIGGYCFSMVLDEPPKSLWMYS
          IPLNKLYIRMNKA FNVDVQFKSKMPIQPLNLRVFLCFSNDVSA PVRCQNHLSVEPLT
          ANNAKMRESLLRSENPN SVYCGNAQGGKISERFSVVVPLNMSRSVTRSGLTRQTLAFK
          FVCQNSCIGRKETSLVFCLEKACGDIVGQHVHVKICTCPKRDRIQDERQLNSKKRKS
          VPEAAEEDPSKVRRCIAIKTEDTESNDSRDCDDSAAEWNVSRTPDGDYRLAITCPNK
          EWLLQSIEGMIKEAAAEVLRNPNQENLRRHANKLLSLKKRAYELP"

```

<http://www.ncbi.nlm.nih.gov/>

ORIGIN

```

1  cctggagcac ggaagattct tgcggacaca aatcgcaact gctaaataaa atttatttat
61 ttgagtgcac agccatgagt cttcacaagt ccgcgtcggt tagcttgact tttaccaggt
121 gagcggagat attttattcg gtcttaccga acaaaataat gttgcgcctt tttgcagaaa
181 cacttcgatt gtttcgcgta gcaatagtcg cacaattttt gaagctttca aggagttcct
241 ggatttttgg gatatacgga acgaagtttc tgcagagtca gcagttcggg tctccagcaa
301 cggagctttc aacttgccgc agagttttgg caacgaatcc aacgaatatg cccacctggc
361 tacgcctgtg gatccagcct acggaggcaa caacacgaac aacatgatgc agttcacgaa
421 caatctggaa attttggcca acaataattc cgatggcaat aacaaaatta atgcatgcaa
481 caaattcgtc tgccacaagg ggtgagcaaa ttcaaaacac gcgctccaat cgataaacat
541 tggctacggc gattgttcgc gctgcgtggc gaatggcaaa atccaaatag tcggtggcca
601 ctacgattct gtagtttttt gttagcgaat ttttaatat tagcctcctt cccaacaag
661 atcgcttgat cagatatagc cgactaagat gtatatatca cagccaatgt cgtggcaca
721 agaaaggtac agtgcgga ccaattgatg atcgaaacag agaaaccttg catgtagcaa

```

```

4261 ggcattgttc atggccgaaa agaaaacatt tttatatttt tgatagtata ctgttggtta
4321 ctgcagttct atgtgactac gtaacttttg tctaccacaa caacataact ctgtacaaaa
4381 aagccaaaag tgaattttatt aaagagttgt catattttgc aaacat

```

//

<http://www.ncbi.nlm.nih.gov/>

Conclusions

- DNA Sequences are stored in DNA sequence databases in specified formats
- Genebank format is a standard format

Lecture no.10

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Protein Sequence Retrieval

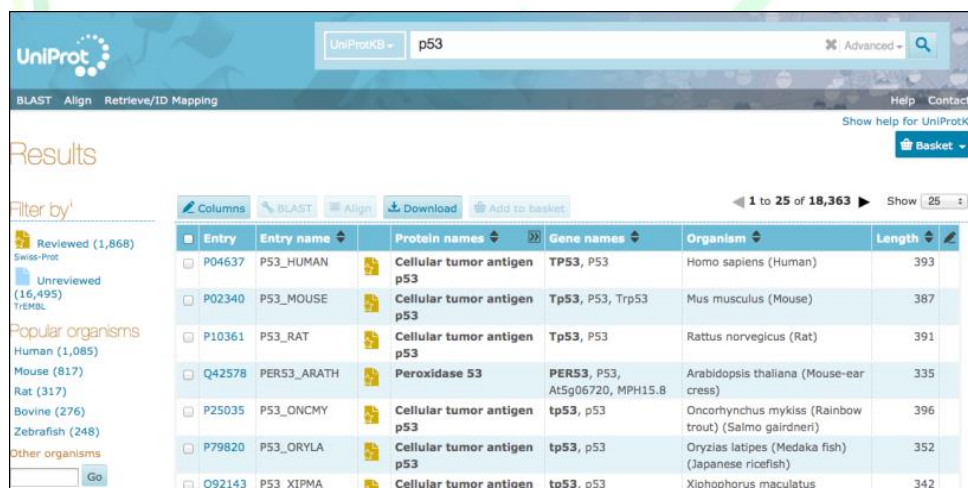
Protein Sequences:

Now we talk about the data retrieval and first we'll talk about the protein sequence retrievals and structures. Protein data is of the following types:

- Actual sequences (from the proteomic data or some other experimental techniques) or translated sequences (sometimes, we go to nucleotides databases, we get those nucleotides and then we translate them by using some softwares, so these are kind of predicted protein sequences) .
- Structures (we can also make structures from those proteins that maybe predicted or the real structures coming from various X-ray Crystallography Techniques).
- Annotations (sometimes, we are interested in the functions of the proteins so those are stored as annotations).

UniProt (It is an international partnership between PIR, EBI and SIB):

Now as far as the resources are concerned, we have multiple resources for protein sequences but **UniProt** claims to be the biggest and integrated resource whereas for the structures **PDB** seems like a good resource.



The screenshot shows the UniProt website interface. At the top, there's a search bar with 'p53' entered. Below the search bar, there's a navigation menu with 'BLAST', 'Align', and 'Retrieve/ID Mapping'. The main content area is titled 'Results' and shows a table of search results. The table has columns for 'Entry', 'Entry name', 'Protein names', 'Gene names', 'Organism', and 'Length'. The first few rows of the table are visible, showing results for 'p53' from various organisms like Human, Mouse, Rat, and Arabidopsis.

Entry	Entry name	Protein names	Gene names	Organism	Length
P04637	P53_HUMAN	Cellular tumor antigen p53	TP53, P53	Homo sapiens (Human)	393
P02340	P53_MOUSE	Cellular tumor antigen p53	TP53, P53, Trp53	Mus musculus (Mouse)	387
P10361	P53_RAT	Cellular tumor antigen p53	TP53, P53	Rattus norvegicus (Rat)	391
Q42578	PER53_ARATH	Peroxidase 53	PER53, P53, At5g06720, MPH15.8	Arabidopsis thaliana (Mouse-ear cress)	335
P25035	P53_ONCMY	Cellular tumor antigen p53	tp53, p53	Oncorhynchus mykiss (Rainbow trout) (Salmo gairdneri)	396
P79820	P53_ORYLA	Cellular tumor antigen p53	tp53, p53	Oryzias latipes (Medaka fish) (Japanese ricefish)	352
Q92143	P53_XIPMA	Cellular tumor antigen	tp53, p53	Xiphophorus maculatus	342

As shown in his figure, is the webpage for data retrieval from UniProt, so we want to search a protein, say p53, where we put it into the search box and press enter which gives us the output. And

we see that there are 18,000 different records and it is showing us the first 25 out of them.

We can have different columns for the output on this webpage so we can have *entry*; *it's ID*, *entry name* (the Suffix Human is written so it's coming from Human, it can be from mouse, rat and Arabidopsis), the *protein name* is Cellular tumour antigen, then *gene name* which is TP53 (where TP stands for Tumour Protein), the *organism* is obviously the human (here) and in the end we have it's length i.e. 393 bp (base pairs).

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The link to this webpage is <http://www.uniprot.org/uniprot/>.

P04637 - P53_HUMAN

Protein Cellular tumor antigen p53
Gene TP53
Organism Homo sapiens (Human)
Status Reviewed - Experimental evidence at protein level¹

Display None **BLAST** **Align** **Format** **Add to basket** **History** **Feedback** **Help video**

FUNCTION **Function¹**
 Acts as a tumor suppressor in many tumor types; induces growth arrest or apoptosis depending on the physiological circumstances and cell type. Involved in cell cycle regulation as a trans-activator that acts to negatively regulate cell division by controlling a set of genes required for this process. One of the activated genes is an inhibitor of cyclin-dependent kinases. Apoptosis induction seems to be mediated either by stimulation of BAX and FAS antigen expression, or by repression of Bcl-2 expression. In cooperation with mitochondrial PRIF is involved in activating oxidative stress-induced necrosis; the function is largely independent of transcription. Induces the transcription of long intergenic non-coding RNA p21 (lincRNA-p21) and lincRNA-Mkn1. lincRNA-p21 participates in TP53-dependent transcriptional repression leading to apoptosis and seem to have to effect on cell-cycle regulation. Implicated in Notch signaling cross-over. Prevents CDK7 kinase activity when associated to CAK complex in response to DNA damage, thus stopping cell cycle progression. Isoform 2 enhances the transactivation activity of isoform 1 from some but not all TP53-inducible promoters. Isoform 4 suppresses transactivation activity and impairs growth suppression mediated by isoform 1. Isoform 7 inhibits isoform 1-mediated apoptosis. ^{11 Publications}

SEQUENCES (9) **CROSS-REFERENCES** **PUBLICATIONS** **ENTRY INFORMATION** **MISCELLANEOUS** **SIMILAR PROTEINS**

Feature key	Position(s)	Length	Description	Graphical view	Feature identifier	Actions
Site ¹	120 - 120	1	Interaction with DNA			
Metal binding ¹	176 - 176	1	Zinc			
Metal binding ¹	179 - 179	1	Zinc			
Metal binding ¹	238 - 238	1	Zinc			
Metal binding ¹	242 - 242	1	Zinc			

Regions

Feature key	Position(s)	Length	Description	Graphical view	Feature identifier	Actions
DNA binding ¹	102 - 292	191				Add BLAST

GO - Molecular function¹

- ATP binding ^{Source: UniProtKB}
- chromatin binding ^{Source: UniProtKB}
- damaged DNA binding ^{Source: RefGenome}
- enzyme binding ^{Source: UniProtKB}
- histone deacetylase regulator activity ^{Source: Ensembl}
- p53 binding ^{Source: RefGenome}
- protein heterodimerization activity ^{Source: UniProtKB}
- protein N-terminus binding ^{Source: UniProtKB}
- protein phosphatase binding ^{Source: UniProtKB}
- chaperone binding ^{Source: UniProtKB}
- copper ion binding ^{Source: UniProtKB}
- DNA binding ^{Source: UniProtKB}
- histone acetyltransferase binding ^{Source: UniProtKB}
- identical protein binding ^{Source: IntAct}
- protease binding ^{Source: UniProtKB}
- protein kinase binding ^{Source: UniProtKB}
- protein phosphatase 2A binding ^{Source: UniProtKB}
- receptor tyrosine kinase binding ^{Source: BHF-UCL}

So, let's check the first one and here we reach on the record for this protein (shown in the figure on the left) which is Cellular Tumour Antigen p53 protein, commonly known as TP53.

We can have different tabs, showing us the outputs. We can look

into the *functions*, its *taxonomy*, and lot many other characteristics so if we look into the function so it gives us some description about what it's doing.

After scrolling the same webpage (shown in the figure on the left), we can see the *feature key* and in some *site* written (there are unique sites in different proteins which are having some

specific properties in them so this is just one amino-acid present in this protein that *interacts with the DNA*). Similarly, there are different *metal binding sites* and we can see that it's mainly binding to the *Zinc* metal. The number of amino-acids is shown here so these are the regions where it interacts with the metal.

Down below, we can also see the *DNA binding* region, for example here, the amino acids are from 102 to 292 and that is also shown in the *Graphical view* as well.

GO-Molecular function or GO-Gene Ontologies, so gene ontologies are the different functional annotation term, there they define different functions, so amongst them we have molecular functions, biological processes, and we have cellular components. So here we just see a *Molecular function*, so it tells us that it performs the functions as shown in the figure, mainly it's a *ATP binding*, it's *p53 binding* with various other functions like *DNA binding*. So all those functions related to these proteins are present in the heading of GO-Molecular Function.

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Keywords - Molecular functionⁱ

Activator

Keywords - Biological processⁱ

Apoptosis, Cell cycle, Host-virus interaction, Necrosis, Transcription, Transcription regulation

Keywords - Ligandⁱ

DNA-binding, Metal-binding, Zinc

Enzyme and pathway databases

Reactome ⁱ	REACT_118568. Pre-NOTCH Transcription and Translation. REACT_1194. Activation of NOXA and translocation to mitochondria. REACT_121. Activation of PUMA and translocation to mitochondria. REACT_169121. Formation of Senescence-Associated Heterochromatin Foci (SAHF). REACT_169185. DNA Damage/Telomere Stress Induced Senescence. REACT_169325. Oncogene Induced Senescence. REACT_169436. Oxidative Stress Induced Senescence. REACT_20549. Autodegradation of the E3 ubiquitin ligase COP1. REACT_24970. Factors involved in megakaryocyte development and platelet production. REACT_309. Stabilization of p53.
SignalLink ⁱ	P04637.

Next, we move on to some other functions, in the *Biological process* category (shown in the figure) we see that it is

related to *Apoptosis* (which is a cell death and it is related to cell-cycle and some other components).

In the below section, we see some *enzymes and pathway databases*, and *Reactome* is a database in which we have a group of reactions which are categorized so these are the list of those reactions with which it is related.

Protein family/group databasesTCDB¹ 1.C.110.1.1.1. the pore-forming pnc-27 peptide of 32 aas from the p53 tumor suppressor protein (pnc-27) family.**Names & Taxonomy¹**

Protein names ¹	Recommended name: Cellular tumor antigen p53 Alternative name(s): - Antigen NY-CO-13 - Phosphoprotein p53 - Tumor suppressor p53
Gene names ¹	Name: TP53 Synonyms: P53
Organism ¹	Homo sapiens (Human)
Taxonomic identifier ¹	9606 [NCBI]
Taxonomic lineage ¹	Eukaryota > Metazoa > Chordata > Craniata > Vertebrata > Euteleostomi > Mammalia > Eutheria > Euarchontoglires > Primates > Haplorrhini > Catarrhini > Hominidae > Homo [88]
Proteomes ¹	UP000005640: Chromosome 17

When we move further (as shown in the figure on the left) till we reach its *Taxonomy*.

On the top, we can see something written as *Protein family or group databases* which is TCDB. Basically, there is another classification in which the proteins are classified on the basis of being as transporter proteins so it is associated with the transportation across the membranes and there is 5-digit number, so there is a specific classification code which is given to each protein, and this protein has the specific code as shown in the figure.

So then we have the *names and taxonomies*, where there are *protein names*, and the taxonomy of the individual can be seen in the *Taxonomic lineage* row. Let's see how we reach to its sequence and is shown in the figure below:

Isoform 1 (Identifier: **P04637-1**) [UniParc] [FASTA](#) [Add to Basket](#)

Also known as: p53, p53alpha

This isoform has been chosen as the 'canonical' sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

« Hide

10	20	30	40	50
MEEPQSDPSV	EPPLSQETFS	DLWKLPPENN	VLSPLPSQAM	DDLMLSPDDI
60	70	80	90	100
EQWFTEDPGP	DEAPRMPEAA	PPVAPAPAAP	TPAAPAPAPS	WPLSSSVPSQ
110	120	130	140	150
KTYQGSYGFR	LGFLHSGTAK	SVTCTYSPAL	NKMFCQLAKT	CPVQLWVDST
160	170	180	190	200
PPPGTRVRAM	AIYKQSQHMT	EVVRRCPHHE	RCSDSGLAP	PQHLIRVEGN
210	220	230	240	250
LRVEYLDNRN	TFRHSVVVPY	EPPEVGSDDT	TIHYNMCMNS	SCMGGMNRRP
260	270	280	290	300
ILTIITLEDSS	SGNLLGRNSF	EVRVCACPGR	DRRTEENLR	KKGEPHHELP
310	320	330	340	350
PGSTKRALPN	NTSSSPQPKK	KPLDGEYFTL	QIRGRERFEM	FRELNEALEL
360	370	380	390	
KDAQAGKEPG	GSAHSSHLK	SKKGQTSRHH	KKLMFKTEGP	DSD

In this figure, we can see the sequence of the protein which is found to be at the end of the page.

<https://www.facebook.com/groups/vubiomates/>

Here, it says *Isoform 1*, so different proteins have different isoforms, different alternative splice variants so this is Isoform 1 as exhibited by its name which is P04637-1, and is the kind of first isoform. We can see the sequence of the protein and starts with a methionine (always a first amino acid in those proteins) and ending at 390TH amino acid. So, it's a 393 aa long protein and the sequence is right here. You can click on the FASTA button on the top and then you can get this output in FASTA format (we'll discuss it later).

NCBI:

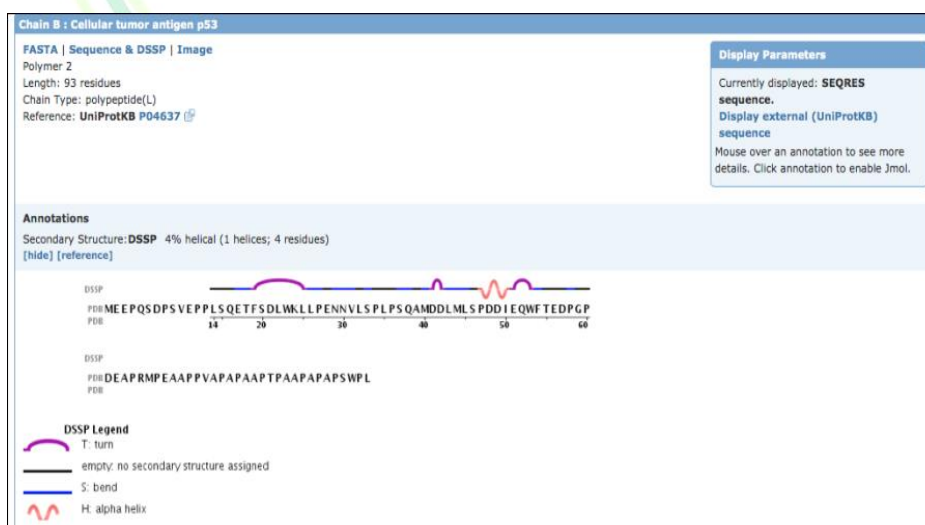
```
ORIGIN
      1 meepqsdpsv epplsqetfs dlwklpenn vlsplpsqam ddmlspddi eqwftedpgp
     61 deaprmpeaa ppvapapaap tpaapapaps wplsssvpsq ktyqgsygfr lgflhsgtak
    121 svtctyspal nkmfcqlakt cpvqlwvdst ppgtrvrarn aiykqsgqht evvrrcphhe
    181 rcsdsdglap pqhlirvegn lrveylddrn tfrhsvvpy eppevgdct tihynmcns
    241 scmggmnrrp iltiitleds sgnllgrnsf evrvacpgr drtteenlr kkgephhelp
    301 pgstkralpn ntssspqpkk kpldgeyftl qirgrerfem frelnealel kdaqagkepg
    361 gsrhsshhlk skkgqstsrh kklmfktegp dsd
//
```

We can also get the same protein from NCBI (as shown in the figure on the left)

In NCBI, obviously the sequence is pretty similar and the arrangement is slightly

different so it is *ORIGIN*, where the sequence starts and sequence ends at those two slashes (/). So, we can get the protein sequence from NCBI as well and the link to this website is <http://www.ncbi.nlm.nih.gov/>.

PDB:



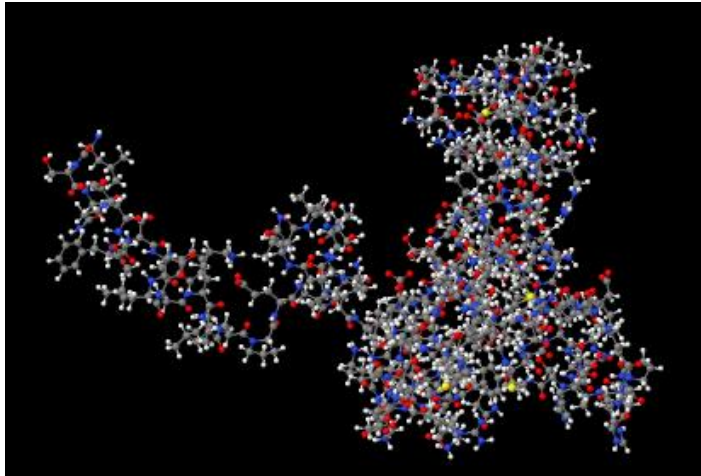
PDB gives us the structures, so we can go to PDB webpage (as shown in the figure on the left) and search for the same ID i.e. P04637 and it gives us the sections or the regions from where it can make up some specific

structures.

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You can see the *turns* in Annotations section, the black ones are the empty lines where no secondary structure can be formed, blue ones show those bends and the orange ones are designated as alpha helices regions. So in PDB, we can have structures in this format as well as the 3D-Structures as shown in the figure below:



The link towards PDB is:

<http://www.rcsb.org/pdb/explore/remediatedSequence.do?structureId=2LY4&bionumber=1>

Conclusions:

We conclude that:

- UniProt is the integrated resource between PIR, EBI and SIB and
- PDB is a good resource to get the protein structure.

TOPIC # 11

Sequence Formats

FASTA Sequence Format

- DNA sequences are stored in specified formats
- Different softwares need sequences in different formats
- FASTA is the most frequently used format to present DNA and Protein sequences
- It is recommended that all lines of text be shorter than 80 characters in length

```
>gi|568815581:c7687550-7668402 Homo sapiens chromosome 17, GRCh38 Primary Assembly
GATGGGATTGGGGTTTTCCCCTCCCATGTGCTCATCTAGAGCCACCGTCCAGGGAGCAGGTAGCTGCTGG
GCTCTCCACGACGGTGACACGC-----
```

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>gi|120407068|ref|NP_000537.3| cellular tumor antigen p53 isoform a [Homo sapiens]
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLAPPVGLFLHSGTAKSVTCTYSPALNKMFCQLAKT--*

Sometimes a * might be present in the end

Genebank Format

- A sequence file in GenBank format can contain several sequences
- One sequence starts with a line containing the word LOCUS and a number of annotation lines
- The start of the sequence is marked by a line containing "ORIGIN" and the end of the sequence is marked by two slashes ("/")

```
LOCUS          AAU03518   237 bp   DNA   PRI   04-FEB-1995
DEFINITION      Aspergillus awamori internal transcribed spacer 1 (ITS1) and
                18S rRNA and 5.8S rRNA genes, partial sequence.
ACCESSION U03518
BASE COUNT 41 a  77 c  67 g 52 t
ORIGIN
1 aacctgcgga aggatcatta ccgagtgagg gtccttggg cccaacctcc catccgtgtc
61 tattgtaccc tgtgtcttcg gcgggcccgc cgcttgcgg ccgccggggg ggcgcctctg
121 cccccgggc ccgtgcccgc cggagacccc aacacgaaca ctgtctgaaa gcgtgcagtc
181 tgagttgatt gaatgcaatc agttaaaact ttcaacaatg gatctcttgg ttccggc
//
```

EMBLFormat

An example sequence in EMBL format is:

```
ID  AA03518   standard; DNA; FUN; 237 BP.
XX
AC  U03518;
XX
DE  Aspergillus awamori internal transcribed spacer 1 (ITS1) and 18S
DE  rRNA and 5.8S rRNA genes, partial sequence.
```

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XX

SQ Sequence 237 BP; 41 A; 77 C; 67 G; 52 T; 0 other;

```
aacctgcgga aggatcatta ccgagtgcgg gtcctttggg cccaacctcc catccgtgtc    60
tattgtaccc tgttgcttcg gcgggcccgc cgcttgtcgg ccgccggggg ggcgcctctg    120
ccccccgggc ccgtgcccgc cggagacccc aacacgaaca ctgtctgaaa gcgtgcagtc    180
tgagttgatt gaatgcaatc agttaaactt ttaacaatg gatctcttgg ttccggc      237
```

//

SwissProt Format

- SwissProt protein sequence format is similar to EMBL format
- There is considerably more information about physical and biochemical properties of a protein is provided
- ID - Identification.
- AC - Accession number(s).
- DT - Date.
- DE - Description.
- GN - Gene name(s).
- OS - Organism species.
- OG - Organelle.
- OC - Organism classification.
- RN - Reference number.
- RP - Reference position.
- RC - Reference comments.
- RX - Reference cross-references.
- RA - Reference authors.
- RL - Reference location.
- CC - Comments or notes.
- DR - Database cross-references.

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- KW - Keywords.
- FT - Feature table data.
- SQ - Sequence header.
- // - Termination line.

XML Format

- Extensible Markup Language
- Readable by both man and machine
- Becoming standard data format for transferring genome data

<xsd:annotation>

<xsd:documentation>

XML Schema for SBOL core data model compatible with RDF/XML serialization.

<dc:date>2012-01-19</dc:date>

<dc:creator>Evren Sirin</dc:creator>

<dc:contributor>Michal Galdzicki</dc:contributor>

</xsd:documentation>

</xsd:annotation>

Sequence converters

- READSEQ is a useful sequence converter developed by D.G.Gilbert at Indiana University, USA
- Recognizes DNA or Protein sequence file and interconvert them

Conclusions

- Databases store sequences in specified formats
- Genebank, DDBJ and EMBL has similar formats
- Different softwares need sequences in different formats

Topic #12

Data Retrieval

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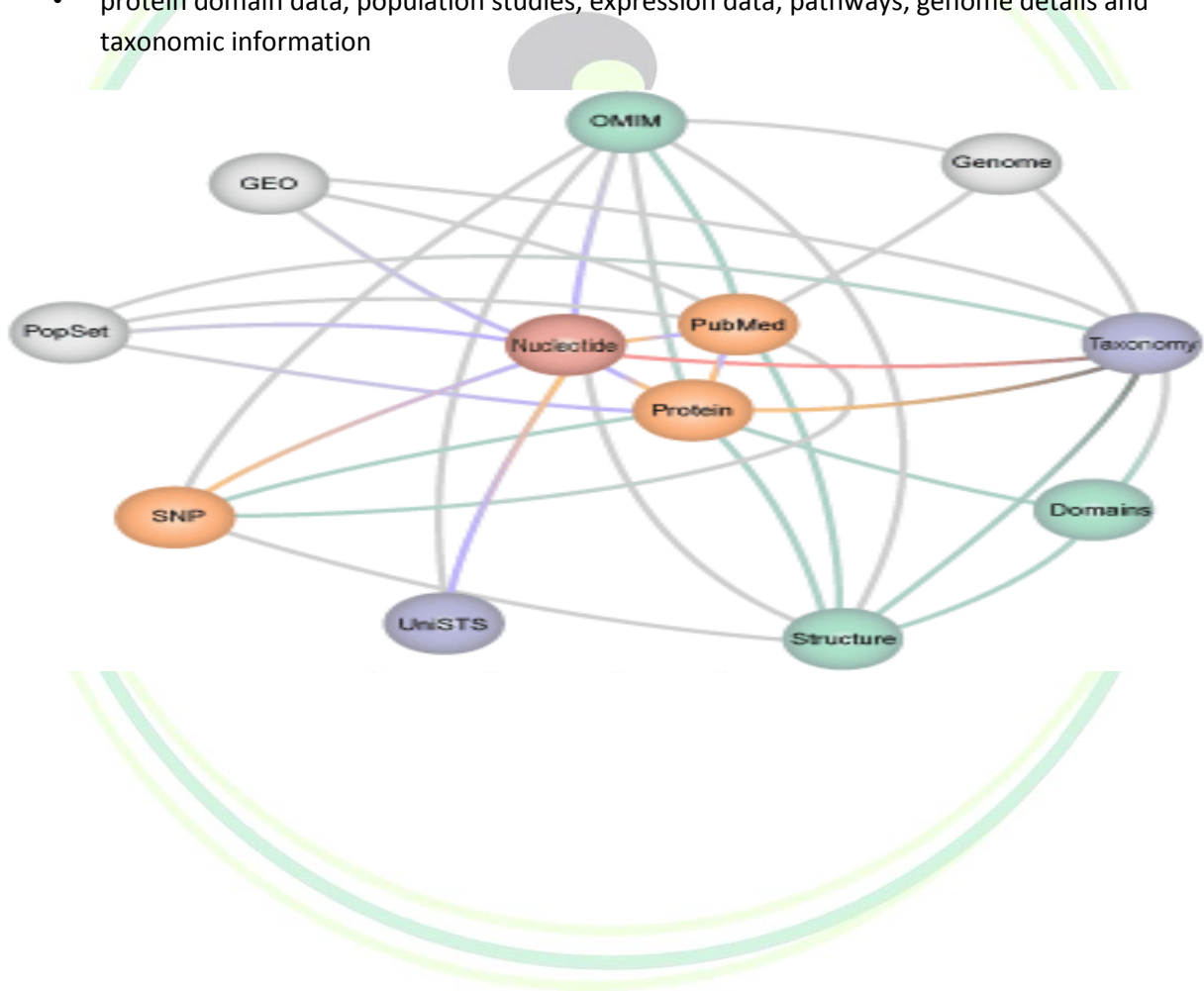
<https://www.facebook.com/groups/vubiomates/>

Data Retrieval

- Nearly all biological databases are available for download as simple text (flat) files
- A local version of the database allows one greater freedom in processing the data

Entrez

- is an integrated search engine which allows users to search and retrieve different data from the NCBI
- It can be accessed from the site www.ncbi.nlm.nih.gov/Entrez/
- integrates PubMed and 39 other scientific literatures, nucleotide and protein databases
- protein domain data, population studies, expression data, pathways, genome details and taxonomic information



Search NCBI databases

[Help](#)

Literature

Books	books and reports
MeSH	ontology used for PubMed indexing
NLM Catalog	books, journals and more in the NLM Collections
PubMed	scientific & medical abstracts/citations
PubMed Central	full-text journal articles

Health

ClinVar	human variations of clinical significance
dbGaP	genotype/phenotype interaction studies
GTR	genetic testing registry
MedGen	medical genetics literature and links
OMIM	online mendelian inheritance in man
PubMed Health	clinical effectiveness, disease and drug reports

Genomes

Assembly	genomic assembly information
BioProject	biological projects providing data to NCBI
BioSample	descriptions of biological source materials

Genes

EST	expressed sequence tag sequences
Gene	collected information about gene loci
GEO DataSets	functional genomics studies
GEO Profiles	gene expression and molecular abundance profiles
HomoloGene	homologous gene sets for selected organisms
PopSet	sequence sets from phylogenetic and population studies
UniGene	clusters of expressed transcripts

Proteins

Conserved Domains	conserved protein domains
Protein	protein sequences
Protein Clusters	sequence similarity-based protein clusters
Structure	experimentally-determined biomolecular structures

Chemicals

BioSystems	molecular pathways with links to genes, proteins and chemicals
PubChem BioAssay	bioactivity screening studies

Bulk Data Retrieval

- The best option is to use ftp (File transfer protocol)
- The File Transfer Protocol (FTP) is a standard network protocol used to transfer files
- Via command line or application programs like FTP clients
- Data needs to be transformed or processed using programming languages
- PERL and Python are good for processing Biological data

Conclusions

- Data is transferred over the internet
- Data needs to be transformed or processed

Topic # 13

Genome Informatics

Introduction

- Genome sequencing provides the sequences of all the genes of an organism
- A major application of Bioinformatics is analysis of full genomes that have been sequenced
- Challenge is to identify those genes that are predicted to have a particular biological function

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Genomics

- Study of all of a person's genes (the **Genome**), including interactions of those genes with each other and with the person's environment (**NHGRI**)

Genome Informatics

- Genome informatics is the field in which computational and statistical techniques are applied to derive biological information from genome sequences
- Genome informatics includes methods to analyze DNA sequence information and to predict protein sequence and structure

(Iossifov, et al. 2014)

Genome Sequences

Availability of genome sequences facilitates;

- The discovery and utilization of sequence polymorphisms
- Opportunity to explore genetic variability both between and within the organisms

Genome Analysis

Following tasks;

- Sequencing
- Assembly
- Repeat identification and masking out
- Gene prediction
- Looking for EST and cDNA sequences
- Genome annotation
- Expression analysis
- Metabolic pathways and regulation studies
- Functional genomics
- Gene location/gene map identification
- Comparative genomics
- Identify clusters of functionally related genes
- Evolutionary modeling

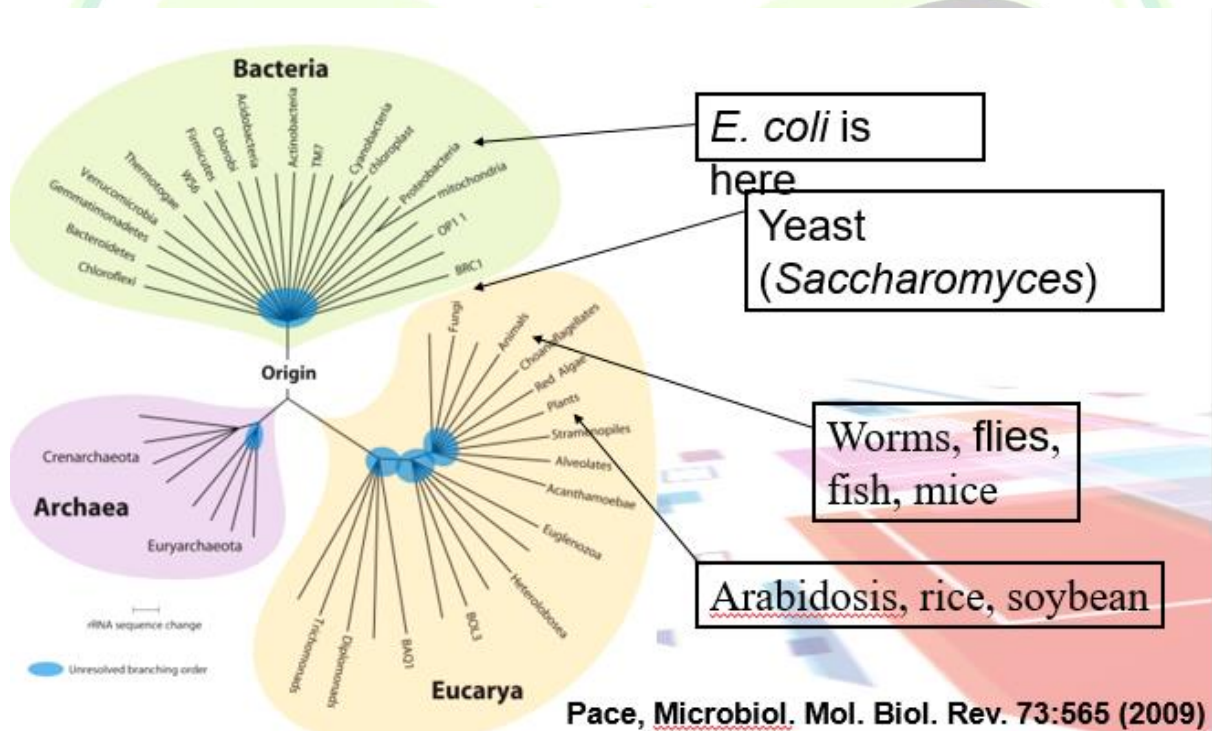
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- Self-comparison of proteome

Model organisms

- E. coli – bacteria
- S. cerevisiae – yeast
- C. elegans – worm
- D. melanogaster – fly
- Danio rerio – zebrafish
- Mus musculus - mouse
- Homo sapiens – you and me
- Arabidopsis - plant



Conclusions

- Sequencing and analysis of full genomes paves the way for future discoveries
- Different model organisms can help explore our Genome and what matters most for us

Topic #14

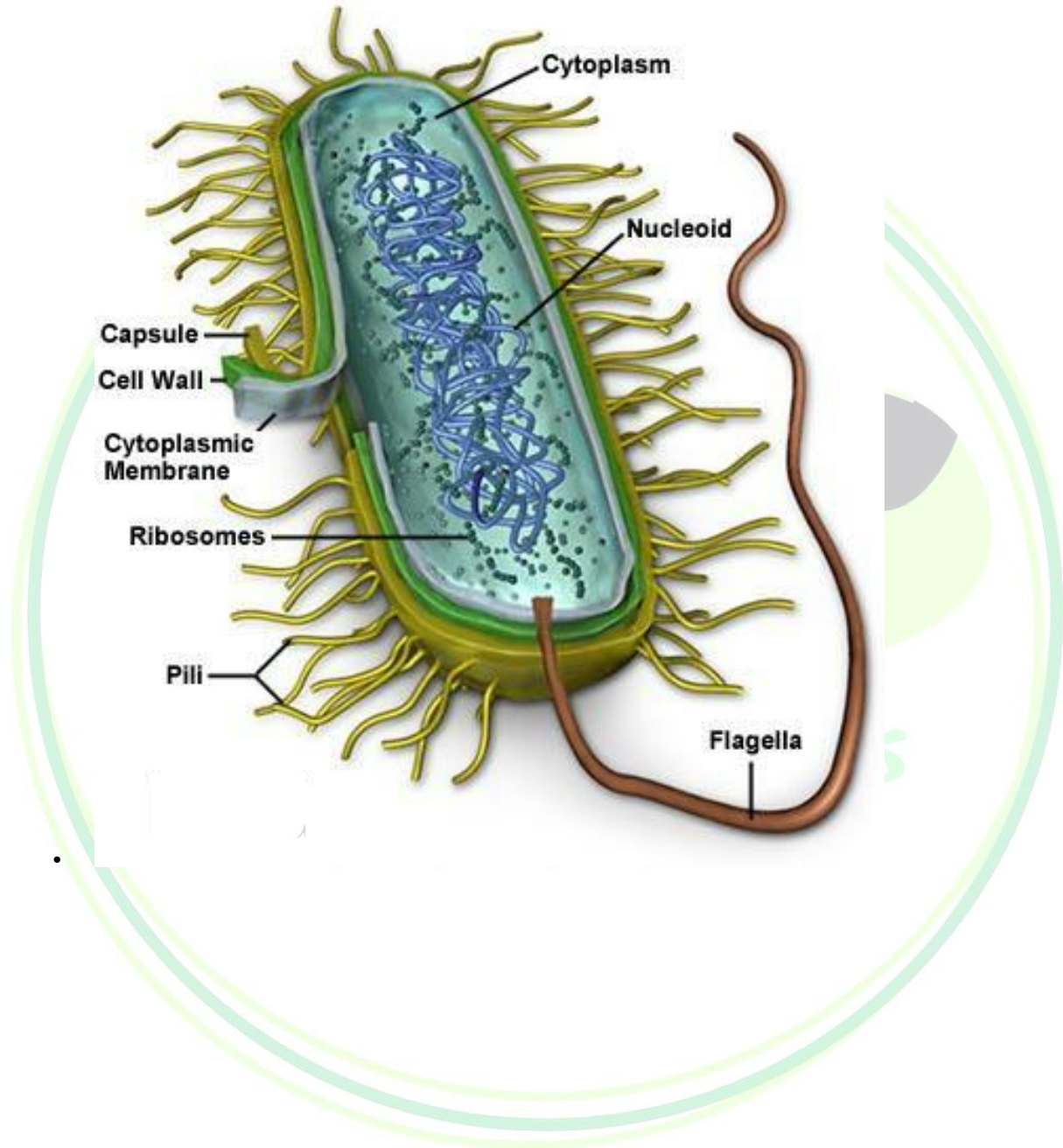
Prokaryotic Genome

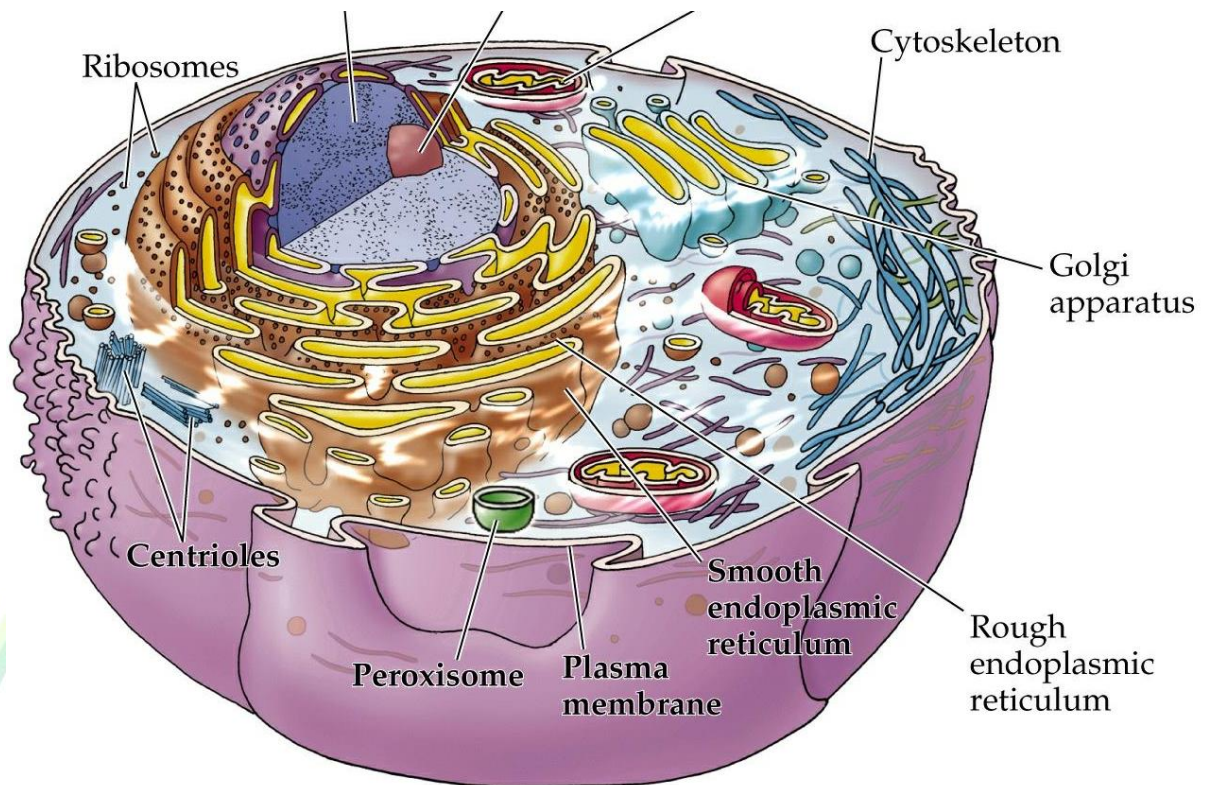
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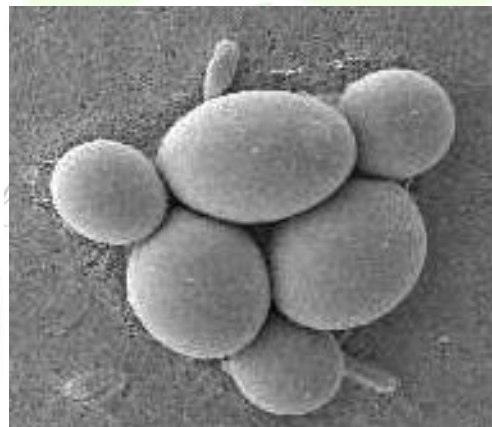
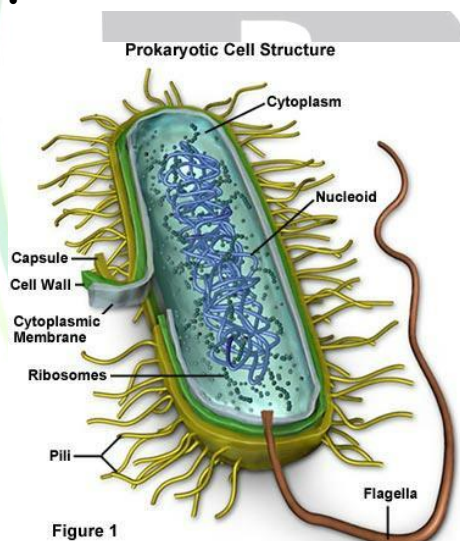
- Prokaryotes are the organisms whose Genetic material (DNA) is not enclosed in a nuclear membrane
- No membrane bound organelles

Prokaryotic Cell Structure





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10 um

Mitochondria evolved from a bacterial endosymbiont

- First prokaryotic Genome sequenced was that of *Hemophilus influenzae*
- Paved the way for sequencing of many other organisms

Selection criteria

Following are the criteria for organisms selected for sequencing;

- They had been subjected to a detailed biological analysis and thus were model organisms

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- important human pathogens
- They were of phylogenetic interest
- Sequences were annotated as they were sequenced

Table 10.2. Features of representative prokaryotic genomes

Organism (reference)	Phylogenetic group	Genome size (Mbp) (no. protein- encoding genes)	Novel functions
<i>Escherichia coli</i> (Blattner et al. 1997)	Bacteria	4.6 (4288)	model organism
<i>Methanococcus jannaschii</i> (Bult et al. 1996)	Archaea	1.66 (1682) ^a	grows at high temperature and pressure and produces methane
<i>Hemophilus influenzae</i> (Fleischmann et al. 1995)	Bacteria	1.83 (1743)	human pathogen
<i>Mycoplasma pneumoniae</i> (Himmelreich et al. 1996)	Bacteria	0.82 (676)	human pathogen that grows inside cells; metabolically weak
<i>Bacillus subtilis</i> (Kunst et al. 1997)	Bacteria	4.2 (4098)	model organism
<i>Aquifex aeolicus</i> (Deckert et al. 1998)	Bacteria	1.55 (1512) ^b	ancient species, grows at high temperature and can grow in a hydrogen, oxygen, carbon dioxide atmosphere in the presence of only mineral salts
<i>Synechocystis</i> sp. (Kaneko et al. 1996a,b)	Bacteria	3.57 (3168)	ancient organism that produces oxygen by light-harvesting; may have oxygenated atmosphere

Conclusions

- Prokaryotes are simple Genomes
- Easy models to study Biochemistry and Molecular biology of life processes
- Sequencing is done on economically important organisms

sign of bright future

Topic #15

Eukaryotic Genome

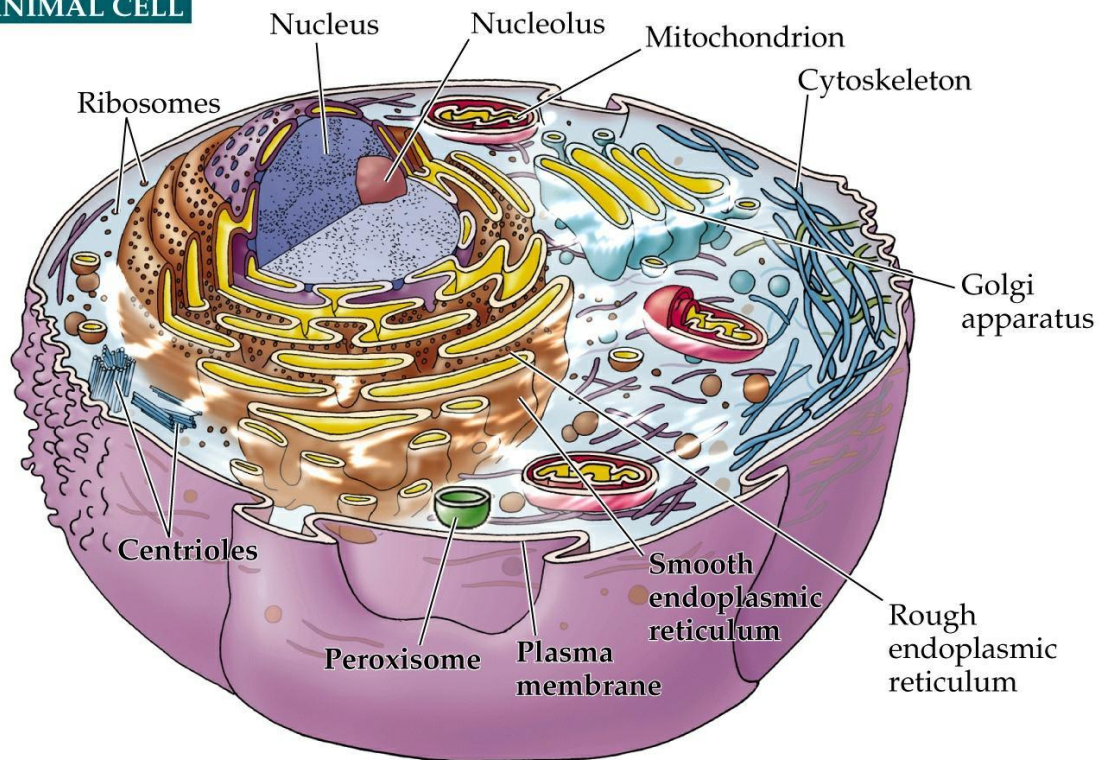
Introduction

- Eukaryotes have larger genomes
- Have tandem repeats
- Have introns in their protein-coding genes
- Heterochromatin and euchromatin region

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AN ANIMAL CELL



Peroxisomes are organelles in all eukaryotes that participate in the metabolism of fatty acids and other metabolites. Peroxisomes have enzymes that rid the cell of toxic peroxides. They have a single lipid bilayer membrane that separates their contents from the cytosol (the internal fluid of the cell) and contain membrane proteins critical for various functions, such as importing proteins into the organelles and aiding in proliferation. Like lysosomes, peroxisomes are part of the secretory pathway of a cell, but they are more dynamic and can replicate by enlarging and then dividing. Peroxisomes were identified as organelles by the Belgian cytologist Christian de Duve in 1967[1] after they had been first described in a Swedish PhD thesis a decade earlier.[2]

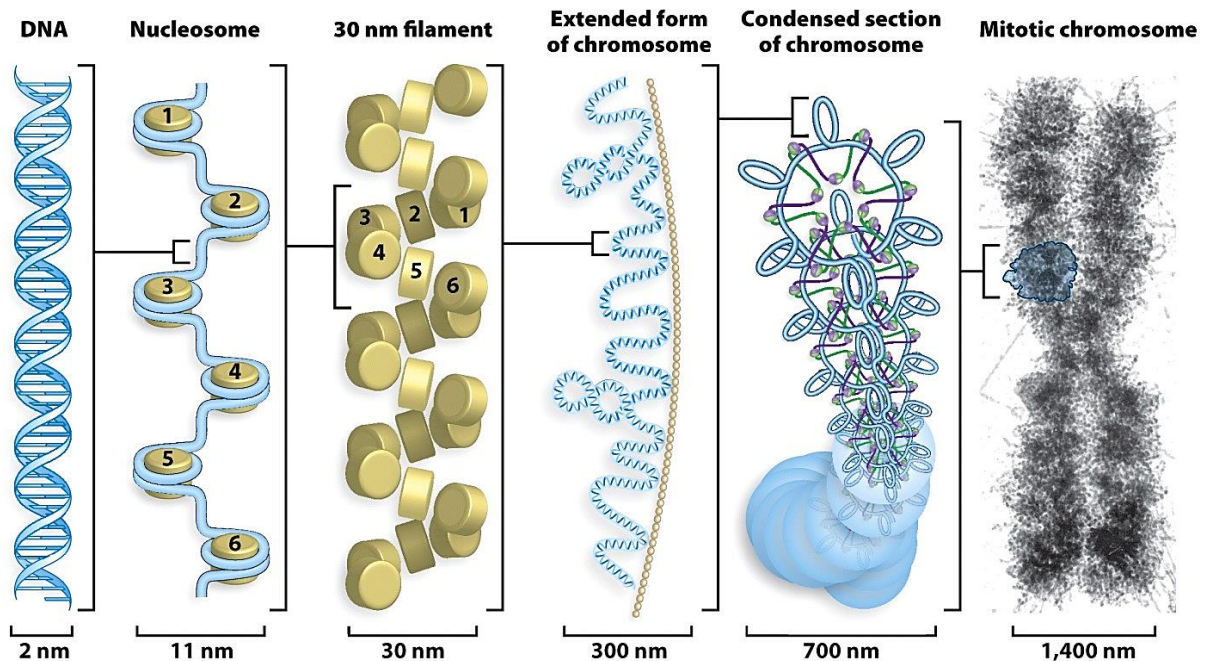


Figure 10-16
Molecular Biology: Principles and Practice

CM: Centromere

TM: Telomere

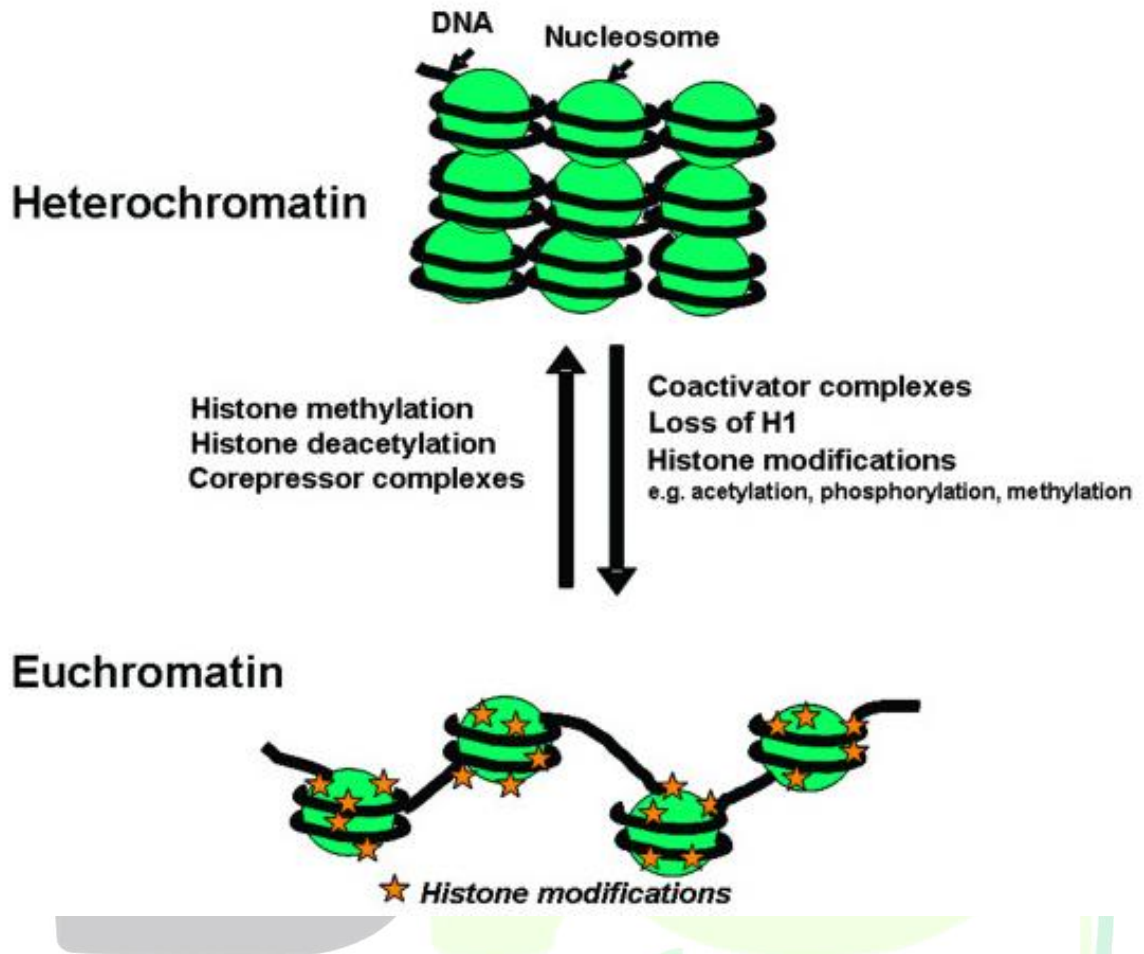
Peroxisomes are organelles in all eukaryotes that participate in the metabolism of fatty acids and other metabolites. Peroxisomes have enzymes that rid the cell of toxic peroxides. They have a single lipid bilayer membrane that separates their contents from the cytosol (the internal fluid of the cell) and contain membrane proteins critical for various functions, such as importing proteins into the organelles and aiding in proliferation. Like lysosomes, peroxisomes are part of the secretory pathway of a cell, but they are more dynamic and can replicate by enlarging and then dividing. Peroxisomes were identified as organelles by the Belgian cytologist Christian de Duve in 1967[1] after they had been first described in a Swedish PhD thesis a decade earlier.[2]

Staining with dyes

- Dense heterochromatin
- Light Euchromatin
- Heterochromatic regions are poorly transcribed (expressed)
- Euchromatic regions are highly expressed

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<http://pixshark.com/euchromatin-and-heterochromatin.htm>

Conclusions

- Eukaryotes are distinguished by the presence of prominent nuclei
- have larger genomes, tandem repeats and introns in their protein-coding genes

Topic # 16

Epichromosomal Elements (EEs)

Introduction

- Genome is the total collection of genetic material and is made up of
 - Chromosomes

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– Epichromosomal Elements

Prokaryotic EEs

- Plasmids
- Self replicating
- Additional rings
- Bacteriophages
- Host colonization
- Transposons
- Parasitic DNA elements

Eukaryotic EEs

- Organellar DNA
- Mitochondria
- Chloroplasts
- Plasmids
- yeast
- Transposons
- Viral genomes
- retroviruses
- membrane-bound organelles
- hundreds to thousands of them
- Site of respiration (mitochondria) or photosynthesis (chloroplast)
- mtDNA
- cpDNA

Endosymbiont hypothesis

- originally proposed in 1883 by Andreas Schimper, but extended by Lynn Margulis in the 1980s
- Mitochondria and Chloroplast are derived from endosymbiotic bacteria

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Organelle Genome

- Circular
- Double stranded
- Supercoiled
- No histones
- Multiple copies
- of mtDNA/cell or cpDNA/cell

Genome	Size & Organization
Plant plastid	150 kb circle
Plant mitochondria	150 – 2000 kb multipartite
Human mitochondria	17 kb circle
saccharomyces mitochondria	75 kb circle

Organelle Genome

- Encode only a subset of genes required to elaborate a functional organelle

rRNAs, tRNAs, ribosomal proteins, membrane-associated respiratory or photosynthetic components

- Other components encoded by nuclear genome, translated in the cytosol and imported into the organelle

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- 10% of nuclear genes devoted to mitochondrial function; 15% to plastid function
-

Conclusions

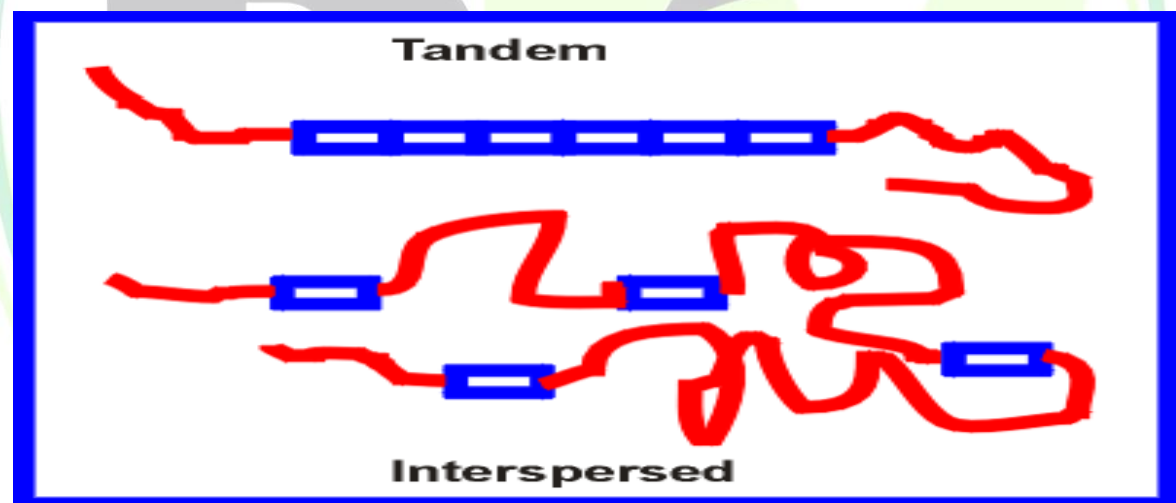
- Organelle genome is similar to prokaryotes
- High copy number
- mtDNA and cpDNA depends on Nuclear DNA

Topic # 17

Genome Repeats

Sequence Repeats

- Repeats skew the base composition
- Contribute to differences in buoyant densities
- Repeat containing DNA can be separated as satellite DNA on the bases of these densities



<http://mcb1.ims.abdn.ac.uk/djs/web/lectures/repeats1.html#anchor10305>

Satellite DNA

- One to several thousand bp long
- Tandem array of 100 million bases long
- Near centromere and telomere
- Minisatellite and Nicrosatellite

Minisatellite

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- 15 bases long in array of several hundred to thousands kb
- Typically in euchromatin
- VNTR e.g, used to identify human individuals in forensics

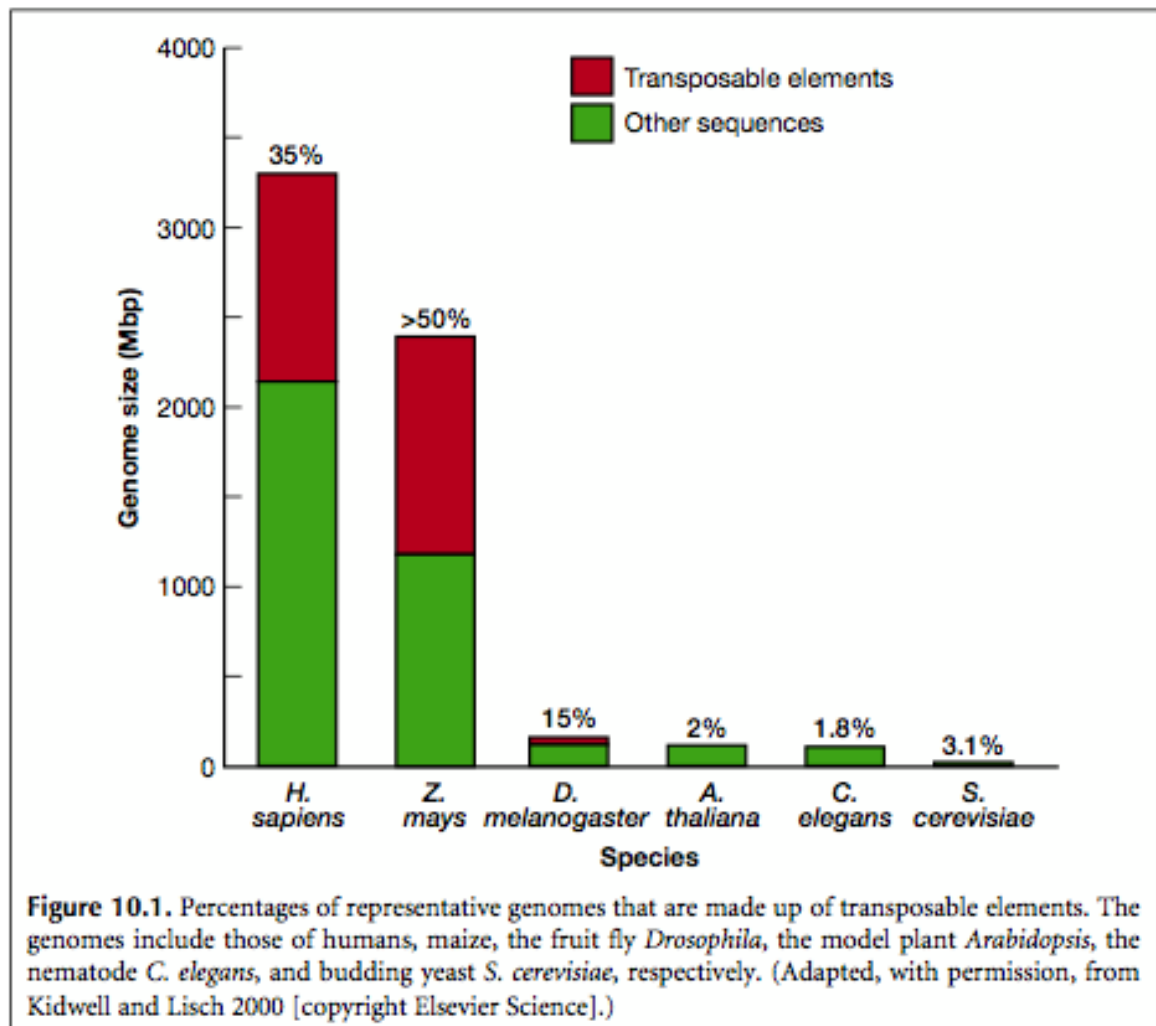
Microsatellite

- 2-6 bases long in arrays of 10-100 bases
- Inherited to offsprings
- Useful markers for genetic analysis and evolutionary studies
- Found in telomeres TTAGGG
- SSRs and STRs

Transposable Elements (TEs)

- Large portion of eukaryotic genome
- Thought to play important role in evolution of these genomes
- Move (Jump) from one location to another faster than chromosome replicate
- Have a potential to increase in number
- Make up a large proportion of eukaryotic genome
- Detectable until blend into genome due to mutations

sign of bright future



Conclusions

- Large proportion of eukaryotic genome is composed of repeats
- Different repeats act as markers to detect genetic variation and are also used to study evolution

Topic # 18

Transposable Elements (TEs)

Introduction

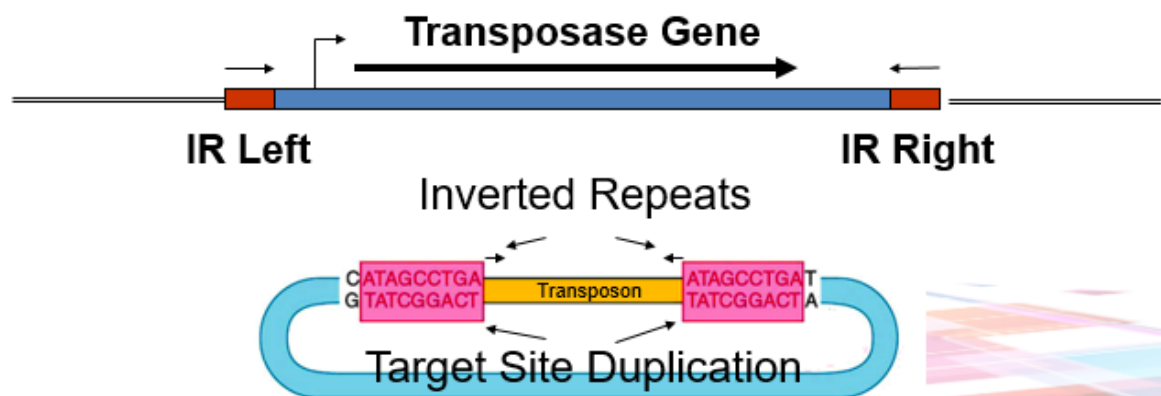
Any segment of DNA, able to transpose

- **Insertion Sequences (IS elements)**

- The simplest transposable elements
- Code only for the ability to transpose
- prokaryotes

IS elements

- usually very small (< 1 Kb to 2 Kb)
- Flanked by inverted repeat sequences (IRs)
- Encode at least one gene that provides their own transposition functions
- Do not code for noticeable (phenotypic) traits
- Can cause mutations by transposition into genes



- **Transposons** *sign of bright future*
 - More complex transposable elements
 - Code for the ability to transpose
 - Code for other detectable traits as well as ability to transpose

Eukaryotic TEs

- 2 Classes; Class I and Class II
- Class I
 - Encode reverse transcriptase
 - Use RNA-mediated mechanisms of transcription
- Class I

Three classes

- LTR (Long Terminal Repeats) retrotransposons
- Retrotransposons
- Retrovirus like Elements

Eukaryotic TEs

- **SINES** (Short Interspersed Nuclear Elements)
 - 80-300 bp
 - **Alu repeats**
- **LINES** (Long Interspersed Nuclear Elements)
 - 6-8 KB long

Eukaryotic TEs

- **Class II TEs**
 - Ac-Ds in maize
 - P elements in Drosophila
- **MITES** (Miniature Inverted repeat TEs)
 - Features of both Class I and II
 - 400 bp

sign of bright future

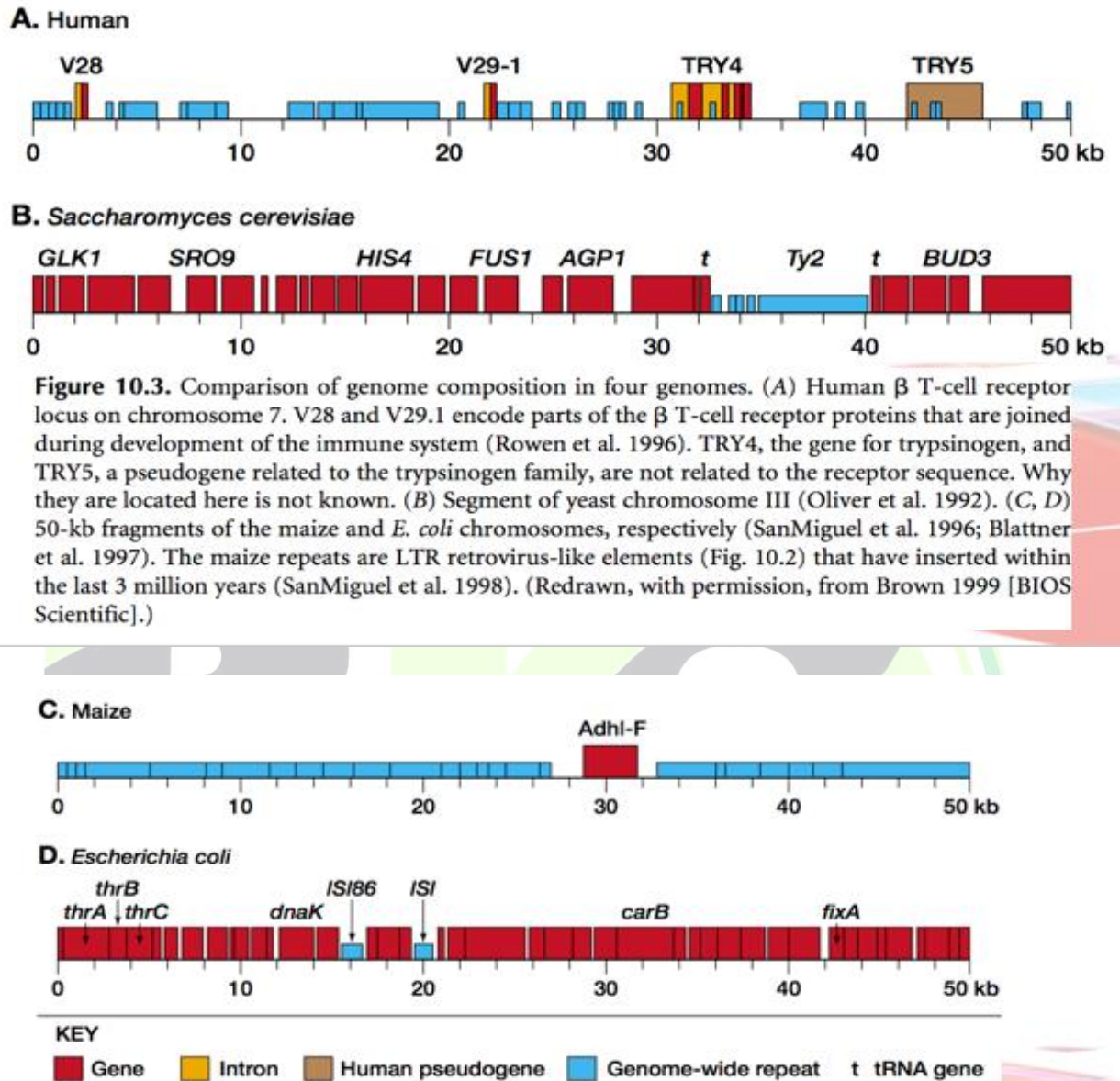


Figure 10.3. Comparison of genome composition in four genomes. (A) Human β T-cell receptor locus on chromosome 7. V28 and V29.1 encode parts of the β T-cell receptor proteins that are joined during development of the immune system (Rowen et al. 1996). TRY4, the gene for trypsinogen, and TRY5, a pseudogene related to the trypsinogen family, are not related to the receptor sequence. Why they are located here is not known. (B) Segment of yeast chromosome III (Oliver et al. 1992). (C, D) 50-kb fragments of the maize and *E. coli* chromosomes, respectively (SanMiguel et al. 1996; Blattner et al. 1997). The maize repeats are LTR retrovirus-like elements (Fig. 10.2) that have inserted within the last 3 million years (SanMiguel et al. 1998). (Redrawn, with permission, from Brown 1999 [BIOS Scientific].)

Conclusions

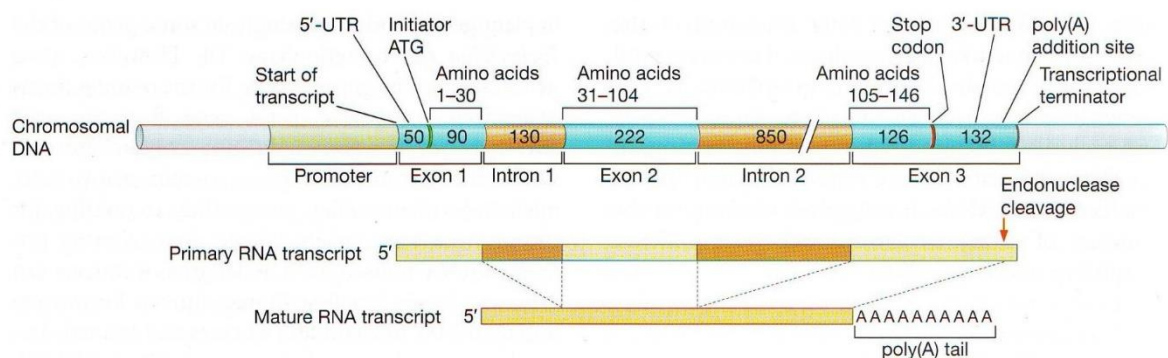
- Transposable elements make up a significant part of eukaryotic genome
- Move within and across genomes
- Cause genome expansion

Topic # 19

Eukaryotic Gene Structure

Eukaryotic genes

- Eukaryotic genes are complicated
- They possess exons which are protein coding regions and are interrupted with introns
- Whole gene is transcribed as a large transcript
- Introns are spliced leaving an ORF that is translated



Human β -globin gene

- Introns start on the 5' ends as GT and end as AG towards 3' end **GT-----AG**
- This trend is highly conserved

Introns proportion

- A small proportion in yeast, only 239 introns in genome
- Human genes have hundreds, often 95% of genes
- Might have embedded genes

Intron origin

- **Intron-early**
 - used to assemble first genes from ancient conserved introns
- **Intron-late**
 - Broke up previously continuous genes by inserting into them

Number of Genes

- Degree of compactness contributes to variation in gene density

- Compact genomes have higher genome density

Organism	Genome size (haploid MB)	Predicted genes
<i>A. thaliana</i> (plant)	130	~25,000
<i>C. elegans</i> (worm)	100	18,424
<i>Drosophila melanogaster</i>	180	13,601
<i>Escherichia coli</i>	4.7	4,288
<i>Homo sapiens</i> (human)	3000	45,000 – 120,000
<i>S. cerevisiae</i> (yeast)	13.5	6,241

Mount: Table 10.3, 11.3 2nd edn

Pseudogenes

- Non functional genes
- Derived from functional genes through mutations following genome duplication
- Processed pseudogenes lack introns and promoters

Gene families

- Set of genes having similar sequences and functions
- Gene families arise from gene duplication and subsequent divergence

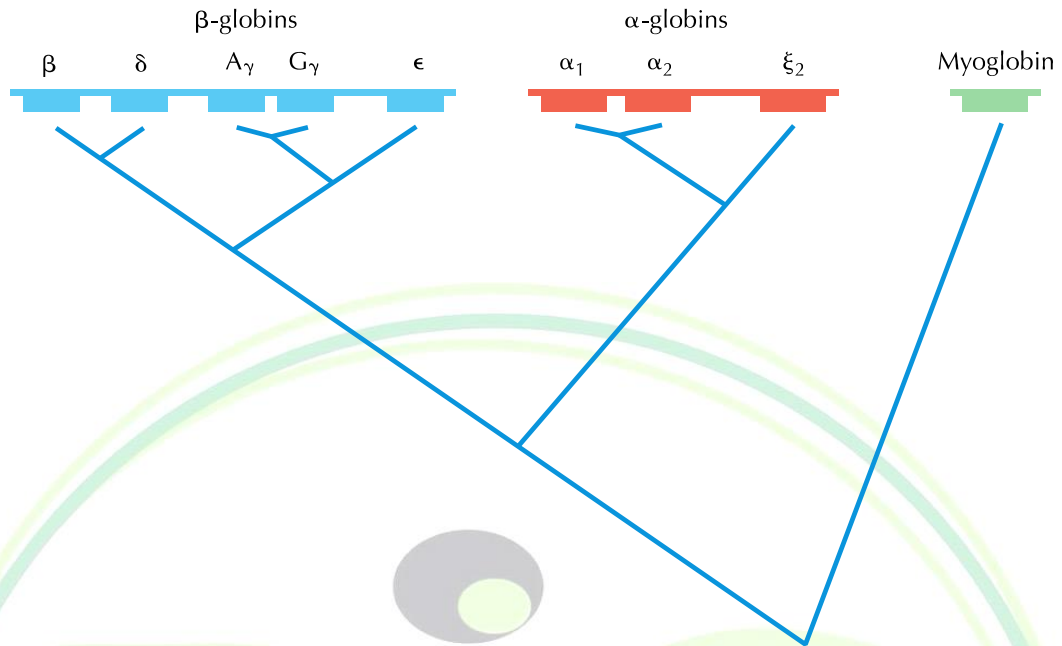


FIGURE 27.29. Gene duplications during the evolution of the human globin gene families. The initial split gave rise to two lineages, one leading to the modern gene for myoglobin and the other to the globin genes. Subsequently, the proto- α -globin and proto- β -globin lineages split following a duplication. Other duplications took place within the α and β lineages. (Modified from Strachan T. and Read A.P. *Human Molecular Genetics 2*, Fig. 14.16, © 1999 Garland Science.)

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Conclusions

- Eukaryotic genes have exons and introns
- Introns make up a significant portion of Human genome
- Pseudogenes are non functional genes
- Similar genes make gene families

Topic # 20

Comparative Genomics

Introduction

- Comparison of gene number, gene content and gene location in both prokaryotic and eukaryotic groups of organisms
- Availability of genome makes possible a comparison of all the proteins (proteome) encoded by one organism with those of another

Orthologs

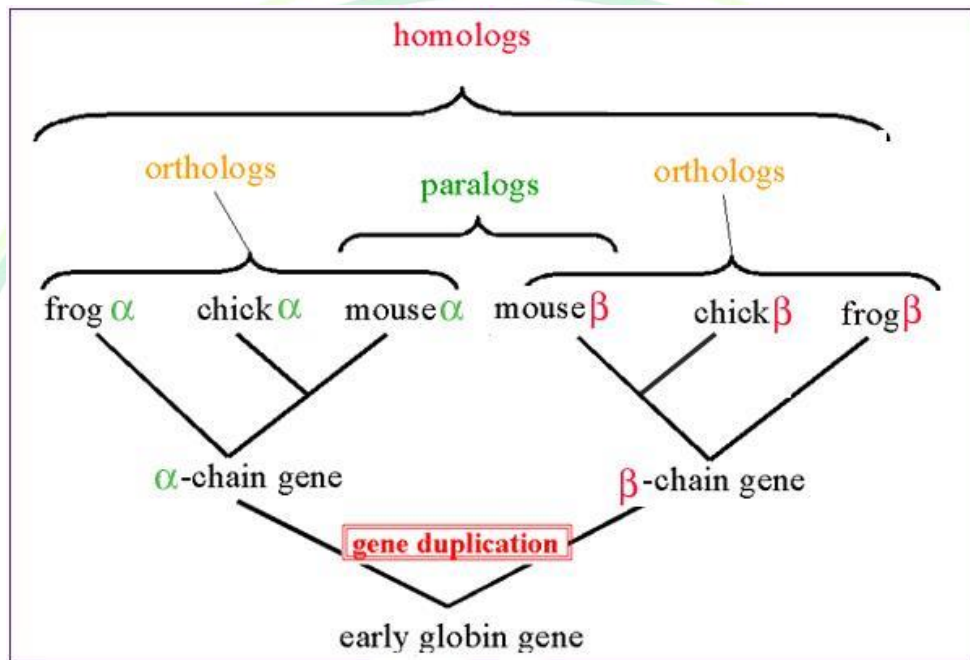
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- Genes in two organisms that are so similar that they must have the same function and evolutionary history are called orthologs (Fitch, 1970)

Paralogs

- Gene families originating from rare gene duplication events over the evolutionary time are called paralogs
- Unlike pseudogenes, 2nd copy of gene remains functional



<http://www.ncbi.nlm.nih.gov/Education/BLASTinfo/Orthology.html>

Drosophila and yeast

- Drosophila* has core proteome only twice the size of that of yeast
- Drosophila* proteome is more similar to mammalian proteomes than worm or yeast

Drosophila and *C. elegans*

- Despite the large differences between fly and worm, they use a core proteome of similar size
- Nearly 30% of fly genes have putative orthologs in the worm

Drosophila and Human

- Some human disease genes absent in *Drosophila*
- A number of previously unknown counterparts to human cancer and neurological disorders genes

Conclusions

- Comparative genomics reveals the relationship among different organisms
- Fruit fly has more similarities with mammals

Topic #21

Comparative Proteomics

Introduction

- For a new Genome, predicted genes are translated
- The collection of protein sequences encoded by the genome makes up the proteome of that individual

A. Types of proteome analysis

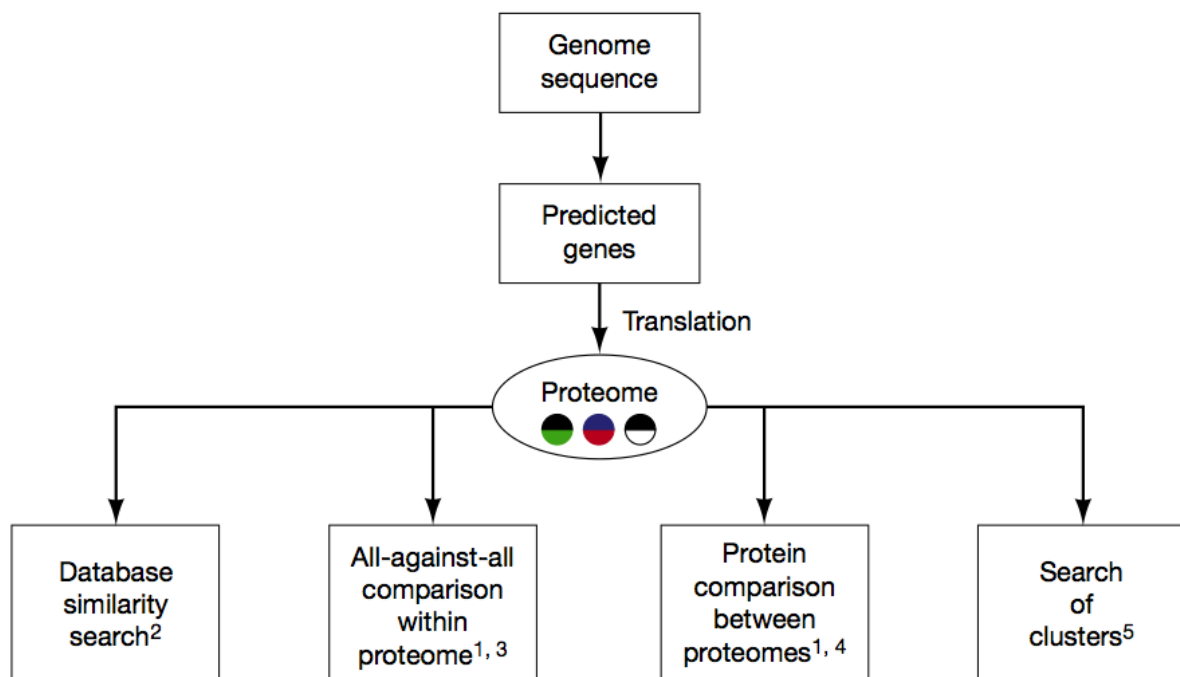


Figure 10.4. Analysis of the proteome encoded by genomes. (A) Types of proteome analyses. (B) Examples of database hits resulting from domain structure of proteins. (C) Cluster analysis of similar sequences. (D) Domain identification.

All against all, self comparison

- Comparison of all proteins with each other within proteome
- Identify unique proteins from the ones having paralogs
- Identify Gene families

B. Examples of database hits resulting from domain structure of proteins⁶

	Amino acid alignment	Sequence number	Typical range of P/E value
(i)		1 (query)	—
		2	$<10^{-20}$
		3	$10^{-8} - 10^{-20}$
		4	$10^{-8} - 10^{-20}$
		5	$10^{-6} - 10^{-8}$

Figure 10.4. Analysis of the proteome encoded by genomes. (A) Types of proteome analyses. (B) Examples of database hits resulting from domain structure of proteins. (C) Cluster analysis of similar sequences. (D) Domain identification.

Cluster Analysis

- To sort out the relationships of all related proteins
- Clustering classify the proteins based on some objective criteria e.g.
 - E value cut off
 - Distance in alignment

Clustering by subgraph

- Each sequence is a vertex
- Significant alignment score is an edge
- Trimming by removing weak edges (High P/E)

(i) Graphic representation

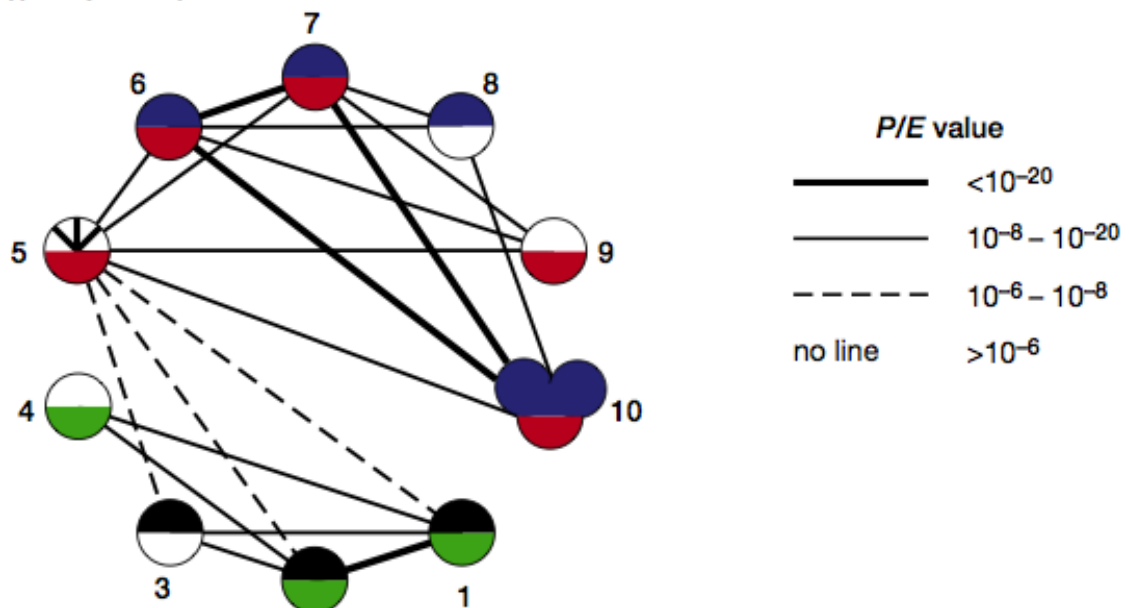


Figure 10.4. Analysis of the proteome encoded by genomes. (A) Types of proteome analyses. (B) Examples of database hits resulting from domain structure of proteins. (C) Cluster analysis of similar sequences. (D) Domain identification.

Clustering by Linkage

- Each sequence is a vertex
- Significant alignment score is an edge
- Trimming by removing weak edges (High P/E)
- Or remove $> e^{-6}$
- Remaining subgraph should share 2/3rd of edges

Single Linkage

- A group of sequences in all-against-all comparison is subjected to MSA
- Create distance matrix
- Neighbour joining is then used to do clustering

(ii) Single linkage cluster analysis

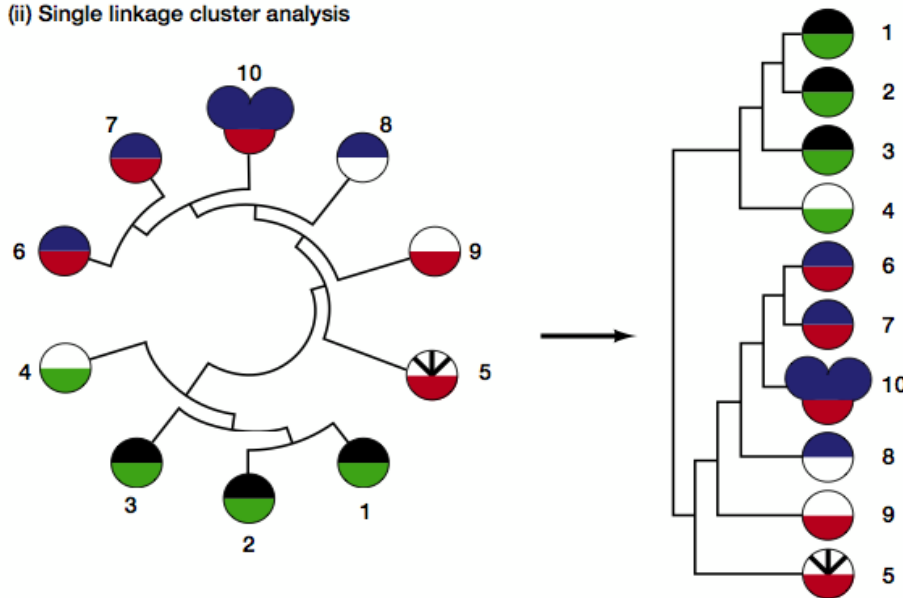


Figure 10.4. Analysis of the proteome encoded by genomes. (A) Types of proteome analyses. (B) Examples of database hits resulting from domain structure of proteins. (C) Cluster analysis of similar sequences. (D) Domain identification.

Core Proteome

- All-against-all comparison provides an indication of Gene/Protein families
- Unique set of proteins is core proteome

Table 10.4. Numbers of gene families and duplicated genes in model organisms (Rubin et al. 2000)

Organism	Total number of genes	Number of gene families ^a	Number of duplicated genes ^b
<i>Hemophilus influenzae</i> (bacteria)	1709	1425 ^c	284
<i>Saccharomyces cerevisiae</i> (yeast)	6241	4383	1858
<i>Caenorhabditis elegans</i> (worm)	18,424	9453	8971
<i>Drosophila melanogaster</i> (fly)	13,600	8065	5536

^a The number of clustered groups in the all-against-all analysis using the algorithm described in the text. This number represents the core proteome of the organism.

^b Count of number of duplicated genes within the protein family clusters.

^c 178 families have paralogs.

Conclusions

- Genome is translated into proteome
- Self comparison of proteome yields gene families and duplications
- Unique set of proteins is core proteome

Topic # 22

Between-Proteome comparisons

Introduction

- Each proteome is used as a query in a database similarity search against another proteome or a set of proteomes
- If proteome is not available, EST database may be searched

Significance

- Helps finding orthologs, Gene families and domains
- Proteins with highly significant alignment score are likely to be orthologs
- Mostly the proteins related to core biological functions

Finding true orthologs

- Method 1
 - Reciprocal Hits
 - $E < 0.01$
 - 60% coverage
- Keep matched pairs with a very conservative P value 10^{-10} to 10^{-100}

Clusters of Orthologous Group (COG)

- Orthologs are assumed to be derived from common ancestor
- Each orthologs might also have paralogs
- Orthologs are clustered to form COG

Table 10.4. Numbers of gene families and duplicated genes in model organisms (Rubin et al. 2000)

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^b Count of number of duplicated genes within the protein family clusters.

^c 178 families have paralogs.

Proteomes to EST databases

- Expressed Sequence Tags (EST) are cDNA copies of cell's mRNA sequences
- For organisms whose genome sequence is not available

- EST are single DNA reads
- Mostly 3' biased
- May be Incomplete being dependent on gene expression
- **TBLASTN** is frequently used

Family and Domain Analysis

- Proteins are organized into domains that represent modules of structure or function
- Domain comparison reveals their biological roles

D. Domain identification⁹

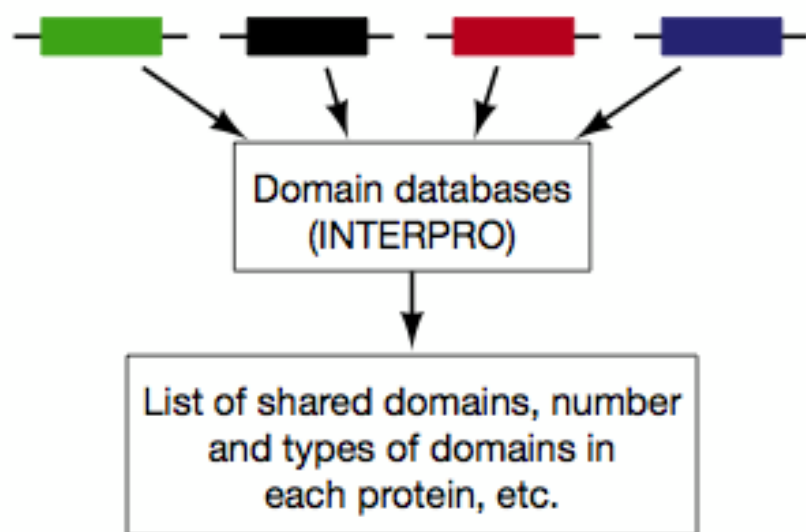


Figure 10.4. Analysis of the proteome encoded by genomes. (A) Types of proteome analyses. (B) Examples of database hits resulting from domain structure of proteins. (C) Cluster analysis of similar sequences. (D) Domain identification.

Conclusions

- Proteome comparison helps finding orthologs, gene families and protein domains
- Domain comparison reveals their biological roles

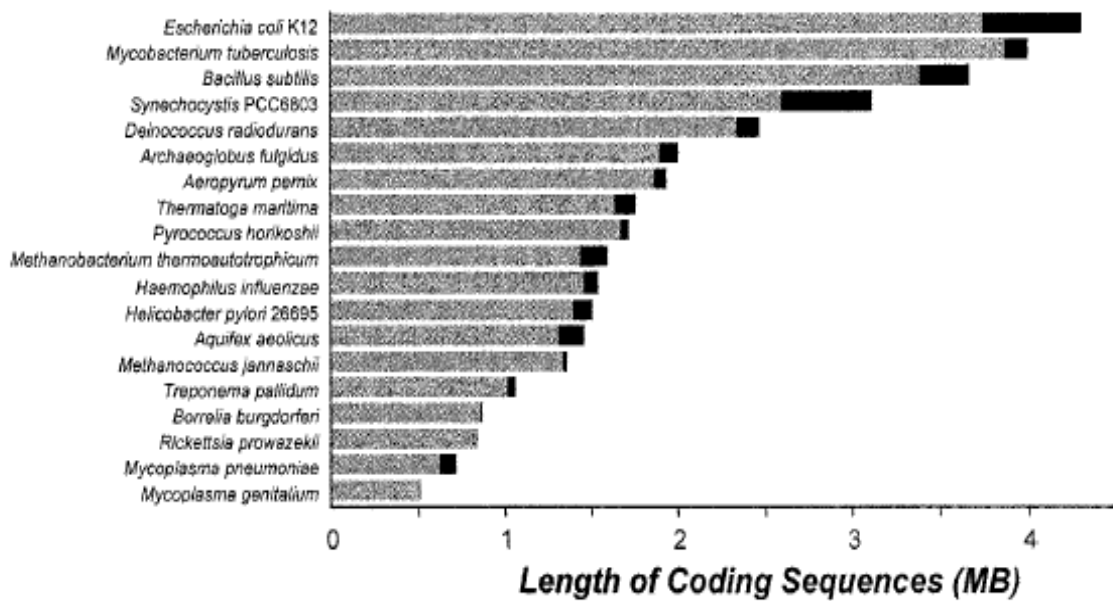
Topic #23

Horizontal Gene Transfer

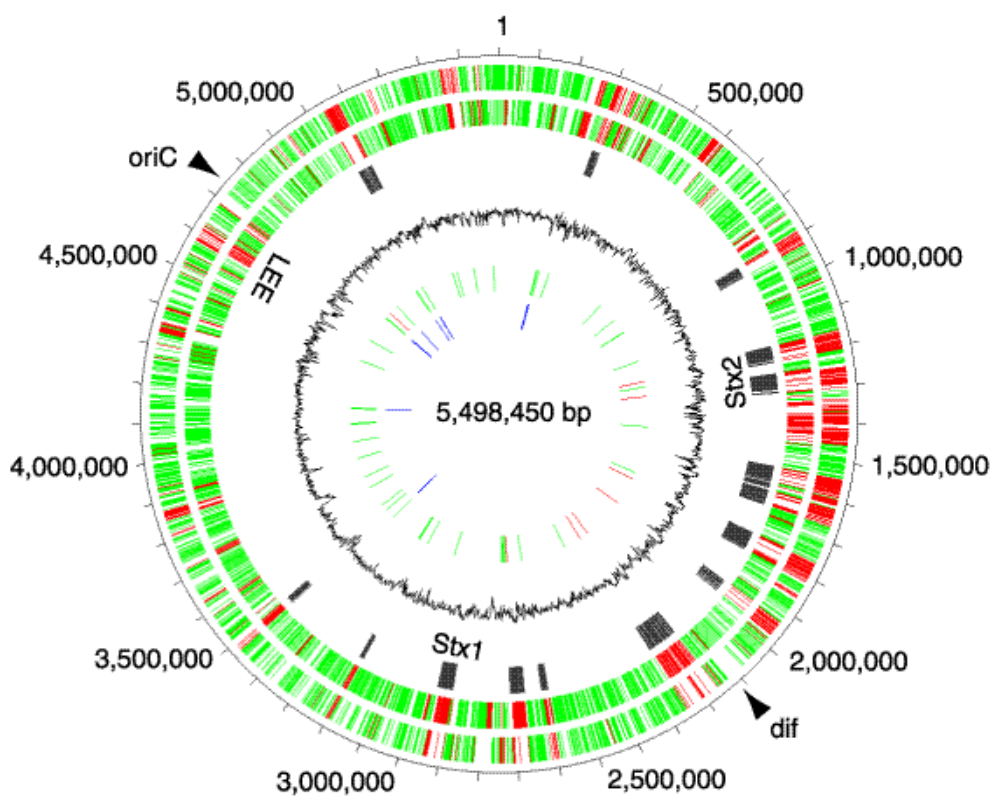
Introduction

- Horizontal Transfer of genes between different organisms
- Vertical Transfer is from parents to offsprings

- Relatively slow process in evolution



(Genome size)



0157-H7 has about 1,400 genes not present in K12

K12 has about 500 genes not present in 0157-H7

Green: Genes conserved in both bacteria

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Red: Genes present in 0157:H7 but not in K12

Other Examples of Genes of Foreign Origin in Complex Eukaryotes

Eukaryote	Foreign Genes	Source
Various Plants	Hormone synthases	Bacterial
Aphids	Carotenoids	Fungal
Sturgeons	Various (15 genes)	<u>Trematodes</u>
Sea Slug	<u>psbB</u> , encodes a nuclear factor	Alga

Conclusion

- Horizontal Transfer of genes between different organisms is a relatively slow process that leads to acquisition of new traits

Topic #24

Gene Order (Synteny)

Gene Mapping

- Gene mapping is determining the location of and relative distances between genes on a chromosome

Genetic vs physical Mapping

- Genetic map distances are based on the genetic linkage information measured in Centi-Morgans (CM)

Genetic vs physical Mapping

- Physical maps use actual physical distances usually measured in number of base pairs

Synteny

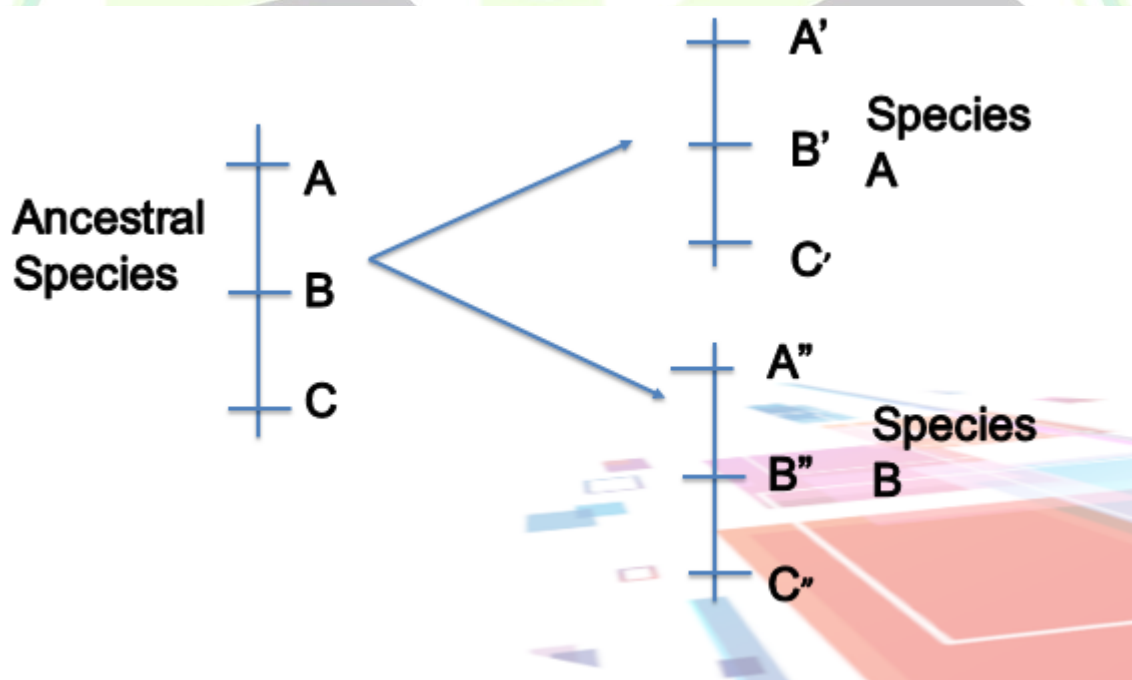
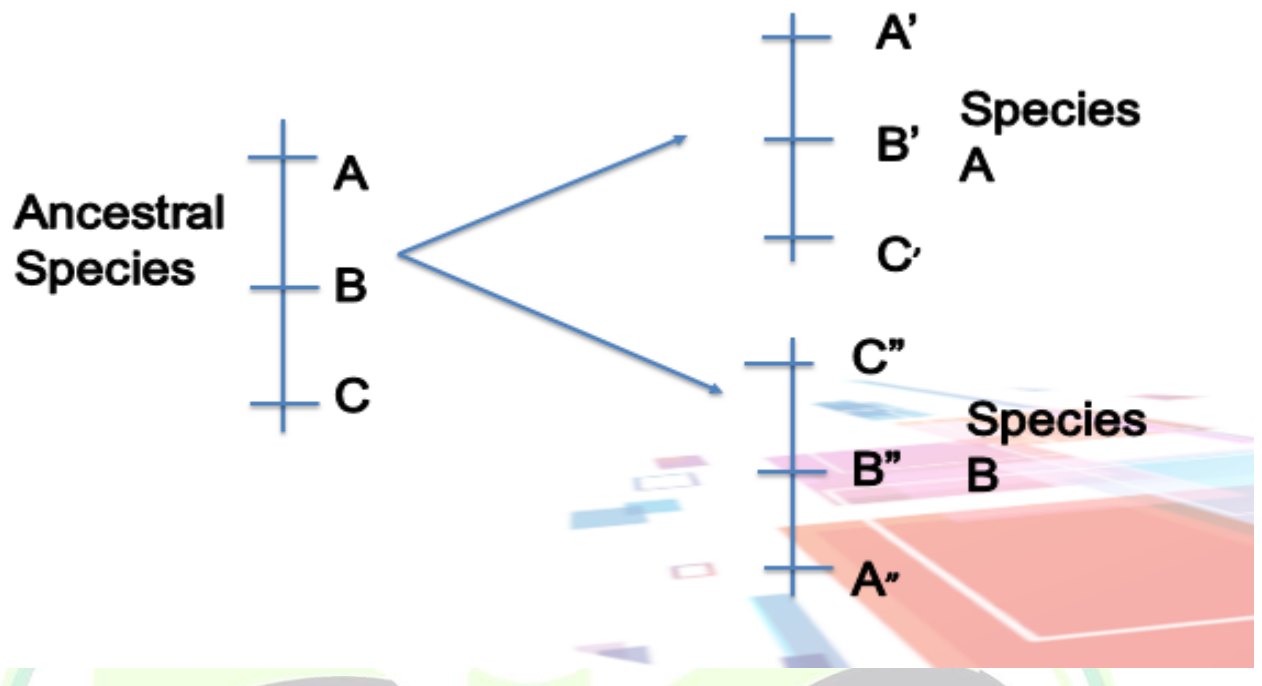
- Arrangement of genes on the chromosomes
- Comparing gene orders provides a good measure of similarities and differences among different organisms

Conserved Synteny

- Species diverged from common ancestor have similar chromosomes and gene order
- Gene duplication and rearrangements changes synteny
- As a results species diverge

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Clusters of metabolically related genes

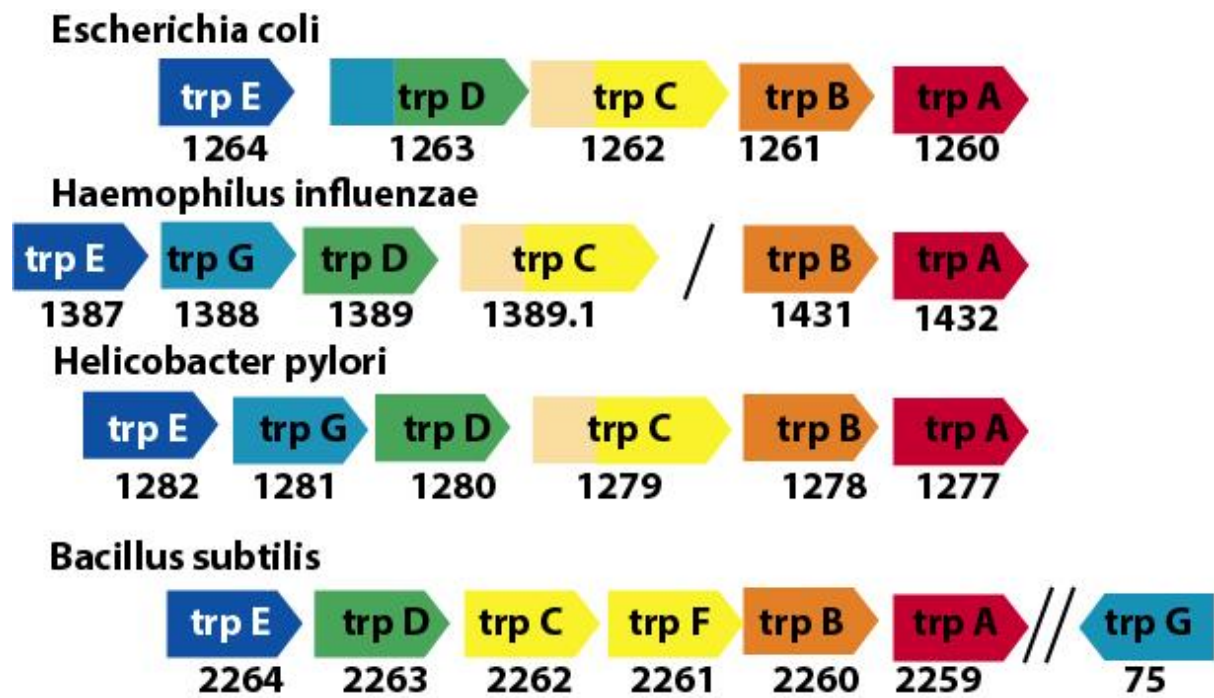


Figure 10.9 (Mount)

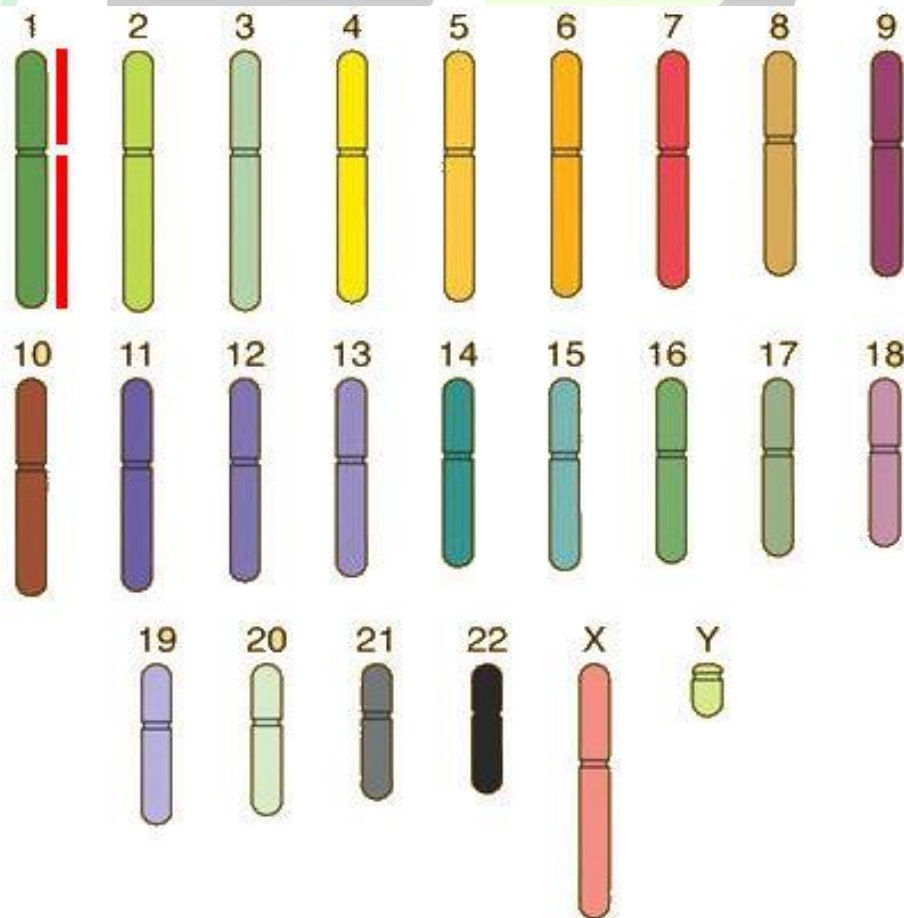


Figure 10.7 (Mount)

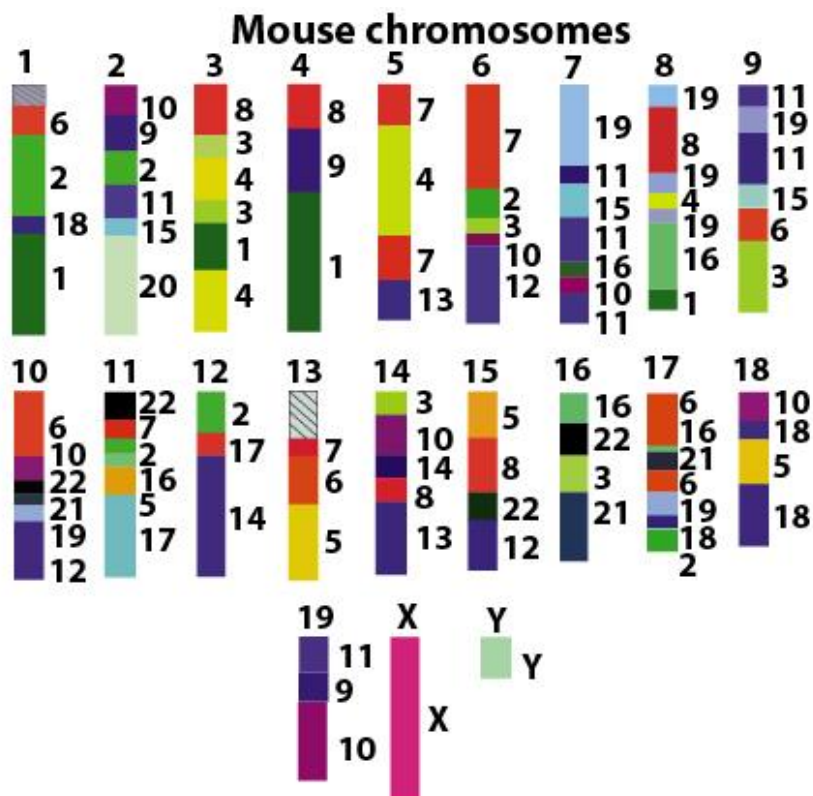


Figure 10.7 (Mount)

Analysis of Chromosomal Rearrangements

Sequence Alignment

A U U A C A G G U U C G U C
 // // // - | | | / / - /
 U U A G G A G G C G A C

Genome Alignment



Figure 10.8: Mount

Alignment Reduction

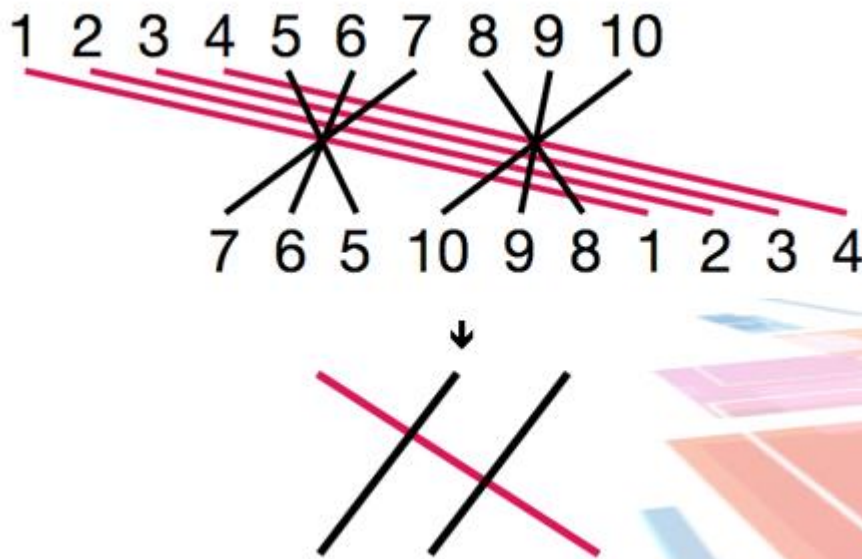


Figure 10.8: Mount

Alignment Reduction

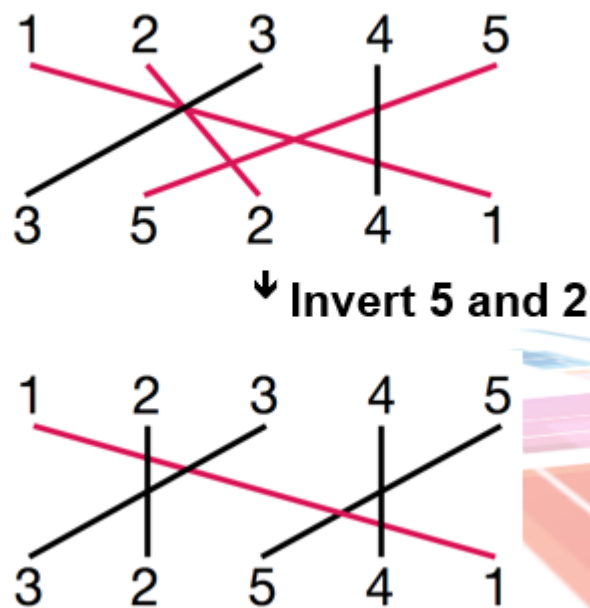
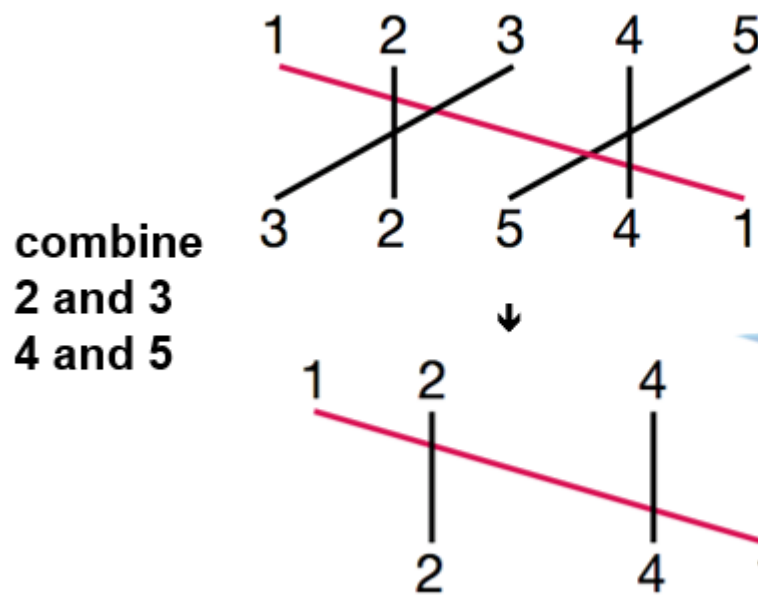


Figure 10.8: Mount

Alignment Reduction



combine
2 and 3
4 and 5

Figure 10.8: Mount

Conclusions

- Species diverged from a common, expected to have similar synteny
- Functionally related genes stay close as clusters

Topic # 25

Genome Annotations

Genome annotation is the process of getting information about a genome including;

- Identification of protein coding genes and their products
- Locating RNA-encoding genes
- Recognizing the non-coding regions in a genome
- Function of genes
- Biochemistry and structure of gene products
- Literature links
- Links to genetic maps
- Location of repeats
- Location of STS (sequence tag sites)
- Location of sequence polymorphisms

Genome annotation is an important first step in genome analysis, based on;

- Finding significant alignment to sequences of known function in databases

Annotations steps

There are two steps in genome annotation

- Structural annotation
- Functional annotation

Structural Annotation

Identification of gene features like;

- Promoters
- Terminators
- Shine-Dalgarno sites
- DNA motifs
- Co-transcription units
- Operons in microbes

Annotations Tools

- **MAGPIE** and **GENEQUIZ** are the tools, designed to assist with genome annotations
- **MAGPIE** (Multipurpose Automated Genome Project Investigation Environment)
- It's an automated genome analysis tool that is used for structural annotation

GeneQuiz

- Focuses on deriving a predicted protein function based upon available evidence, including evaluation of similarity to the closest homologue in database

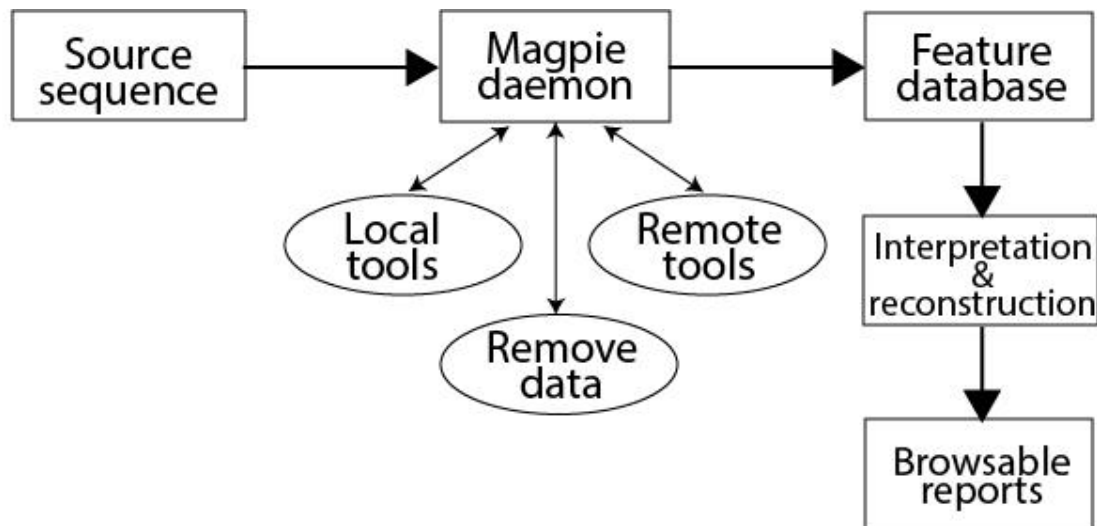


Figure 10.10 (Mount)

Functional Annotations

Attaching biological information to genes like;

- Biological function
- Biochemical function
- Gene expression
- Regulation and interactions

8 group classification

- An **early classification** of in e.coli where groups included;
- Enzymes
- Transporters
- Regulators
- Membranes
- Structural elements

An **early classification** of in e.coli where groups included;

Protein factors

Leader peptides

(control transcription and translation)

Carriers

90% of e.coli genes fell into these categories

Enzyme Commission (EC) numbers

Enzyme Commission (EC) under **IUBMB**

Enzymes are classified on the basis of reactions they catalyze

EC a.b.c.d

a one of 6 classes of biochemical reactions

b group of substrate

c acceptor molecule

d details of biochemical reaction

tripeptide aminopeptidases

EC 3.4.11.4

3 Hydrolases (use water to break substrate)

3.4 Hydrolases that act on peptide bonds

3.4.11 Hydrolases that cleave amino terminal amino acids of polypeptide

tripeptide aminopeptidases

EC 3.4.11.4

Hydrolases that cleave amino terminal amino acids of a tripeptide

70% of e.coli genes shared a and b, catalyze same biochemical reaction

Three groups scheme

Energy

Information

Communication

Plant devote half of their genome to **energy metabolism**

Animals devote half of their genome to **communication**

Conclusions

Finding genes and their coding regions is an important task in Genome annotations

Functional annotations correlate genes to different classes of functions

Topic # 26

Genome Sequencing

Introduction

- Genome sequencing involves recognition of nucleotides in a Genome and determining their precise order
- Advances in sequencing technologies have revolutionized the pace of scientific discovery
- DNA sequencing began in 1970s with the development of Maxam-Gilbert's method and later got the pace with Sanger's method
- Most of the modern day sequencing employs variants of Sanger's methods

First published Nucleotides

Proc. Nat. Acad. Sci. USA
Vol. 70, No. 12, Part I, pp. 3581–3584, December 1973

The Nucleotide Sequence of the *lac* Operator

(regulation/protein–nucleic acid interaction/DNA–RNA sequencing/oligonucleotide priming)

WALTER GILBERT AND ALLAN MAXAM

Department of Biochemistry and Molecular Biology, Harvard University, Cambridge, Massachusetts 02138

Communicated by J. D. Watson, August 9, 1973

ABSTRACT The *lac* repressor protects the *lac* operator against digestion with deoxyribonuclease. The protected fragment is double-stranded and about 27 base-pairs long. We determined the sequence of RNA transcription copies of this fragment and present a sequence for 24 base pairs. It is:

5'--TGGAATTGTGAGCGGATAACAATT3'
3'--ACCTTAACA CTCGCTATTGTTAA5'

The sequence has 2-fold symmetry regions; the two longest are separated by one turn of the DNA double helix.

bind again to the repressor, and is also a part of the operator. Here we shall describe its sequence.

METHODS

Sonicated DNA Fragments. Sonicated DNA fragments were made by growing a temperature-sensitive mutant λ c1857 ϕ lac5S7 at 34° in a glucose–50 (pH 7.4) medium in 3 mM phosphate buffer. Cells were grown to a cell density of 4×10^8 /ml, then harvested by centrifuging the cells at a density of 8×10^8 /ml.

Location of protein-coding genes within bacteriophage MS2 RNA. Note that the *lys* gene overlaps segments of both the *cp* and *rep* genes.

First genome sequenced

- First complete genome of a virus (phage) MS2 was sequenced by Walter Fiers and colleagues at University of Ghent, Belgium (1972-76)

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- RNA was first sequenced among nucleic acids
- Among free living organisms, *Haemophilus influenzae* was the first published genome
 - (Fleischmann et al. 1995)

MS2

- Infects E.Coli and Enterobacteriaceae
- Single stranded sense RNA
- Encodes MS2 coat protein

MS2 sequencing

- The entire nucleotide sequence was established by nuclease digestion and characterization of fragments

MS2 genome

- 49 different codons in the genetic code specify the sequence of the 129 amino-acids long coat polypeptide

MS2 genome



Location of protein-coding genes within bacteriophage MS2 RNA. Note that the lys gene overlaps segments of both the cp and rep genes.

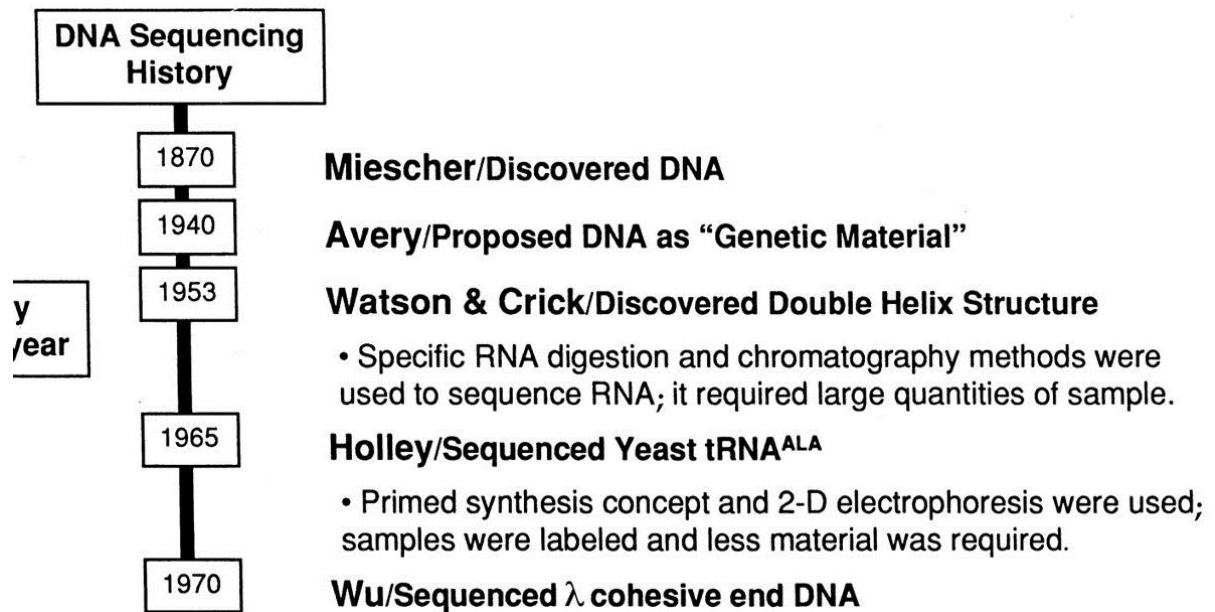
Lys lysis protein

Rep replicase

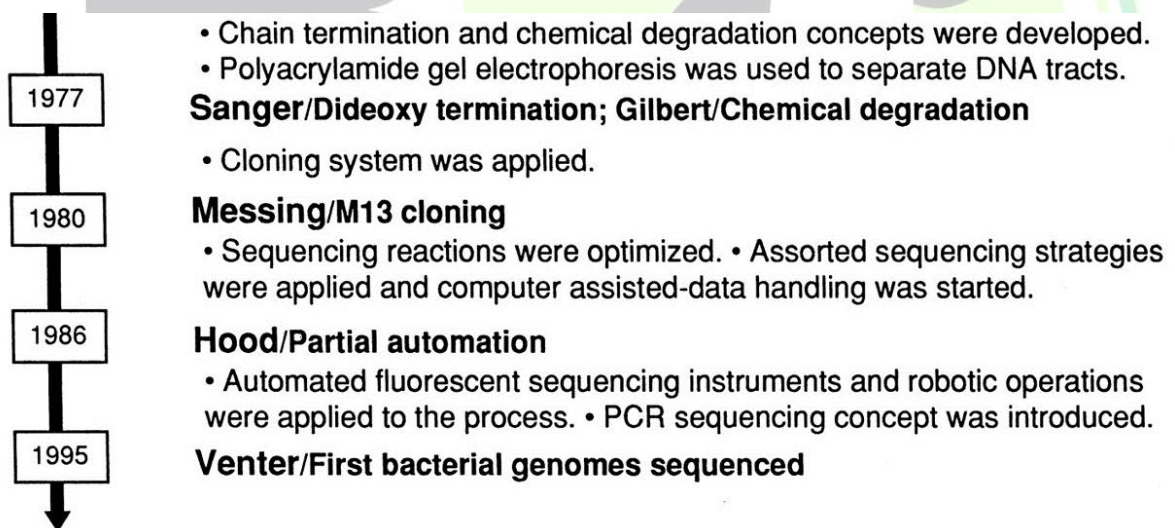
Cp coat protein

Mat assembly protein

Significant Milestones in the history of DNA Sequencing

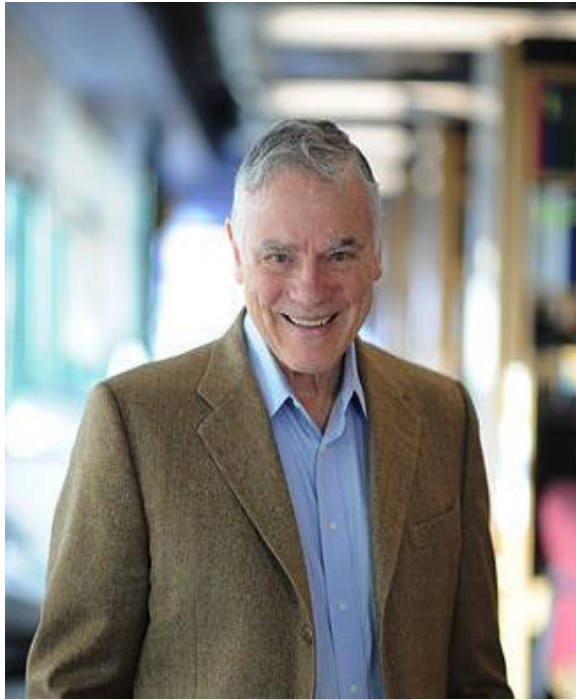


Significant Milestones in the history of DNA Sequencing



Automated sequencing

- Leroy E. Hood's laboratory at the California Institute of Technology invented the first semi-automated DNA sequencer in 1986
- Key technology used in **Human Genome Project**



Leroy E. Hood

Institute of system biology

Seattle Washington

Institute of system biology, Seattle Washington

Conclusions

- Genome sequencing involves recognition and determining the precise order of nucleotides in a Genome
- Advances in sequencing technologies have revolutionized the pace of scientific discovery

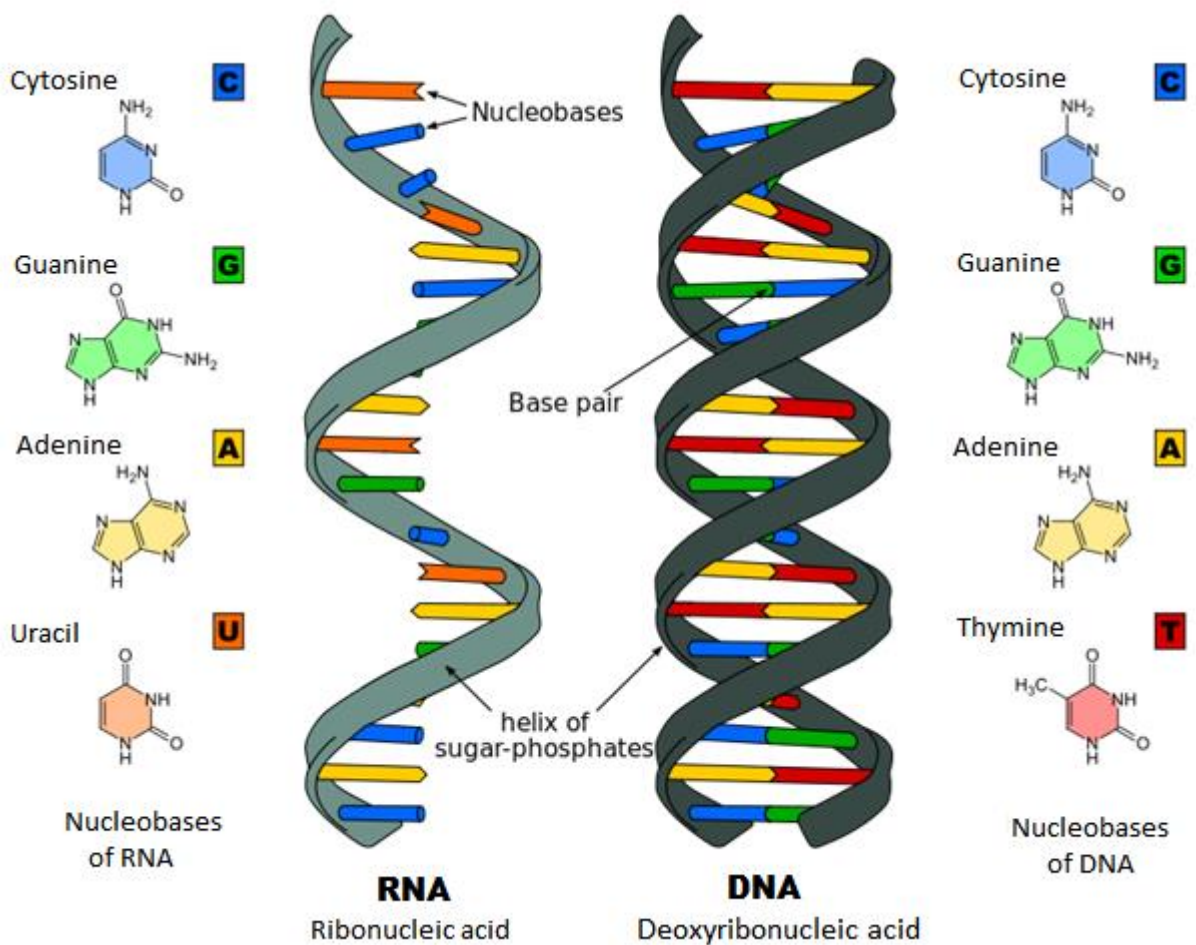
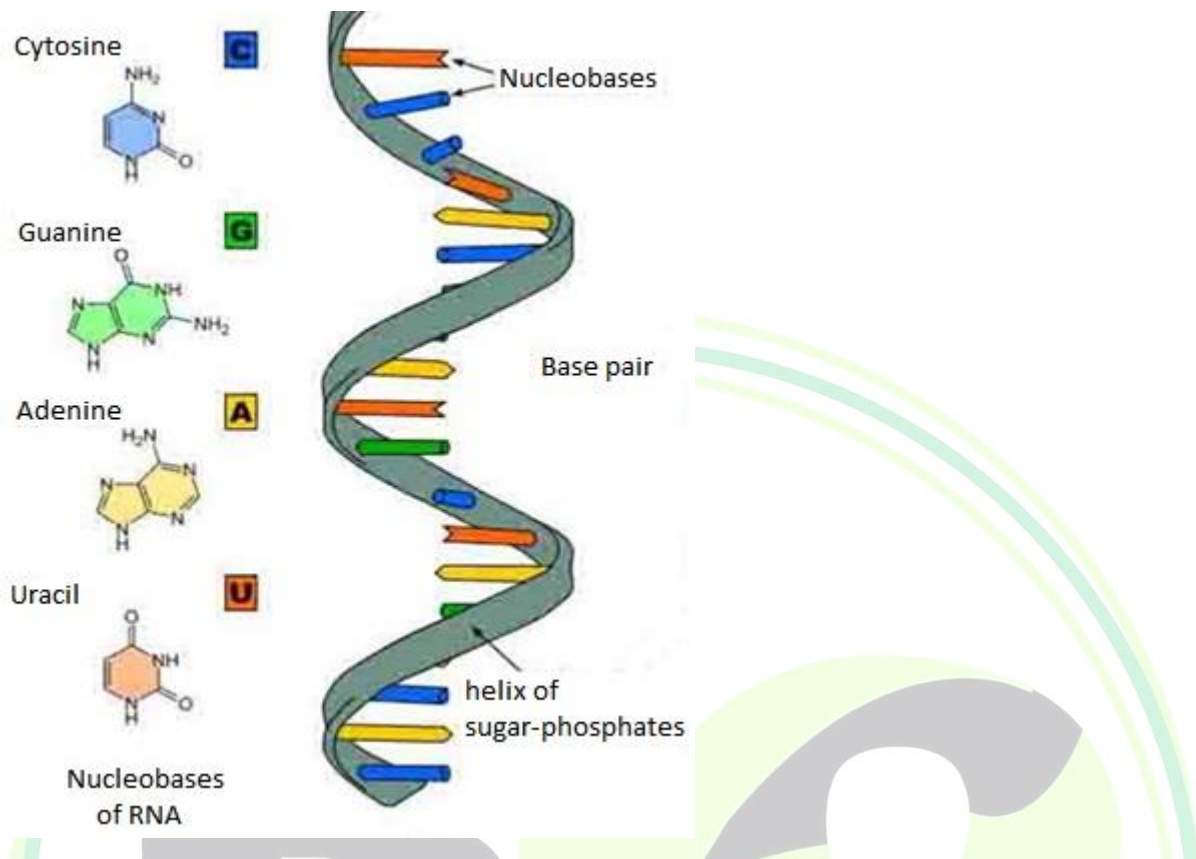
Topic # 27

Advanced Computing Approaches

Structure of RNA

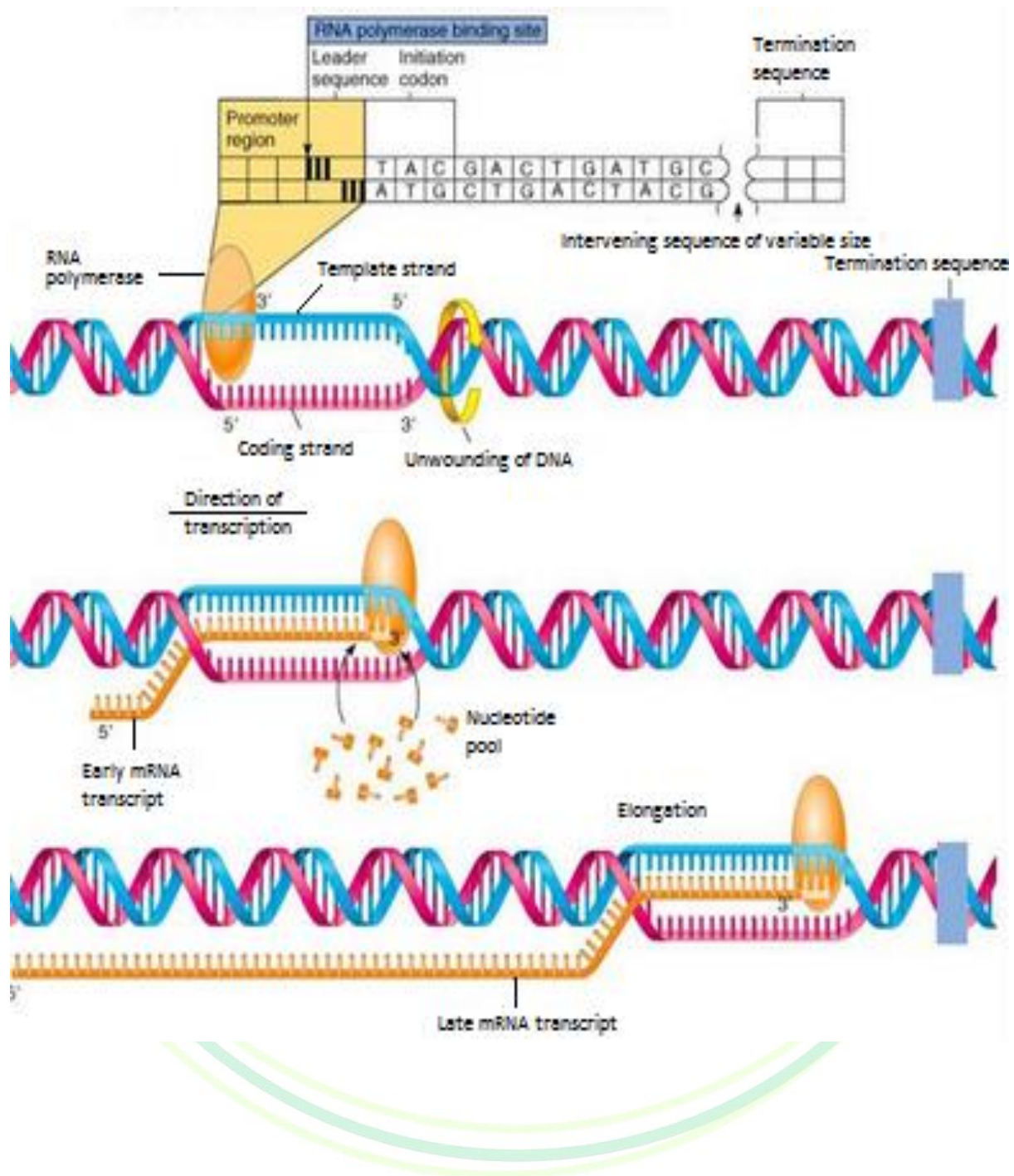
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Topic # 28

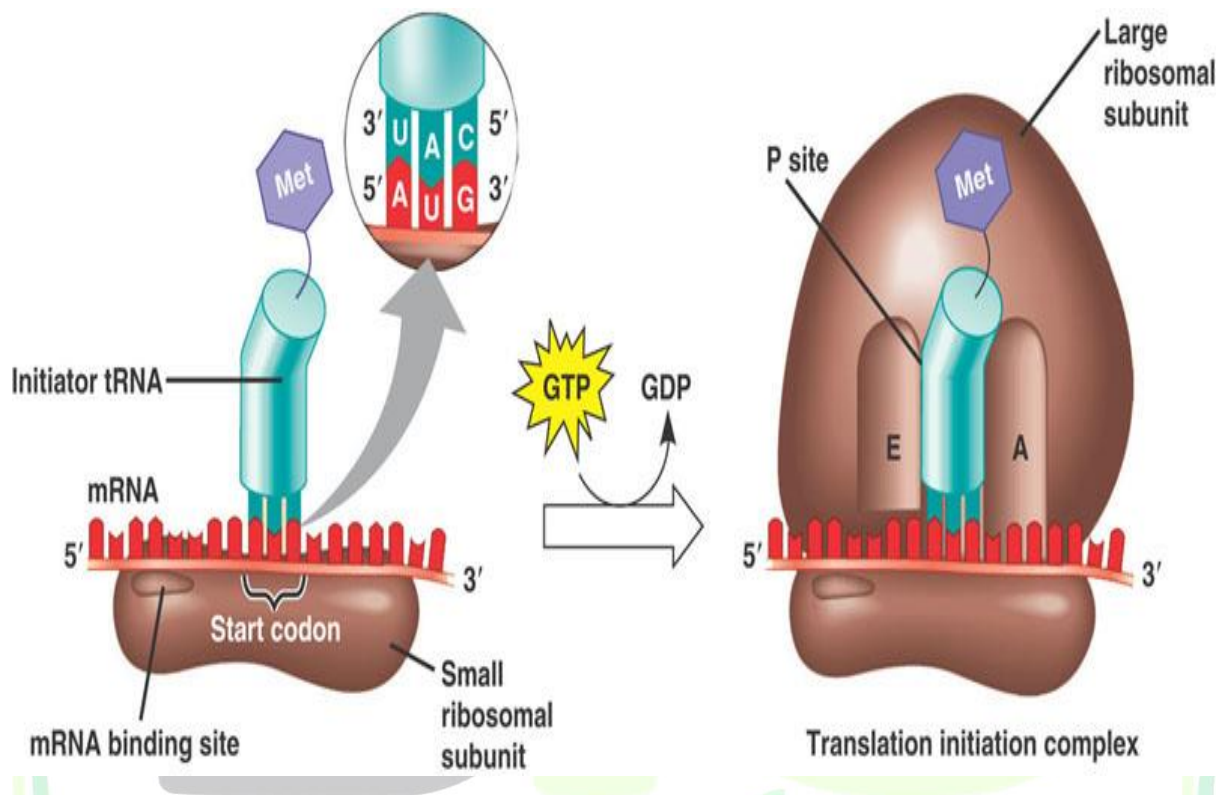
DNA Transcription



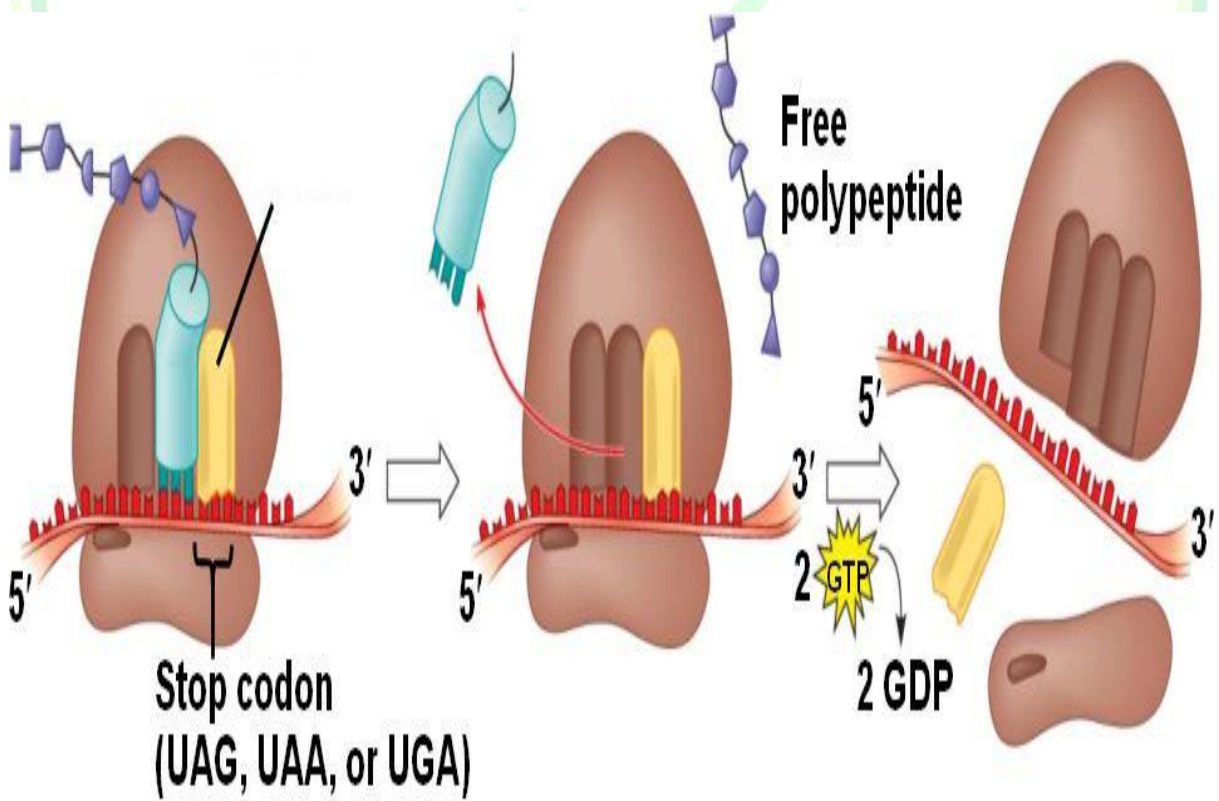
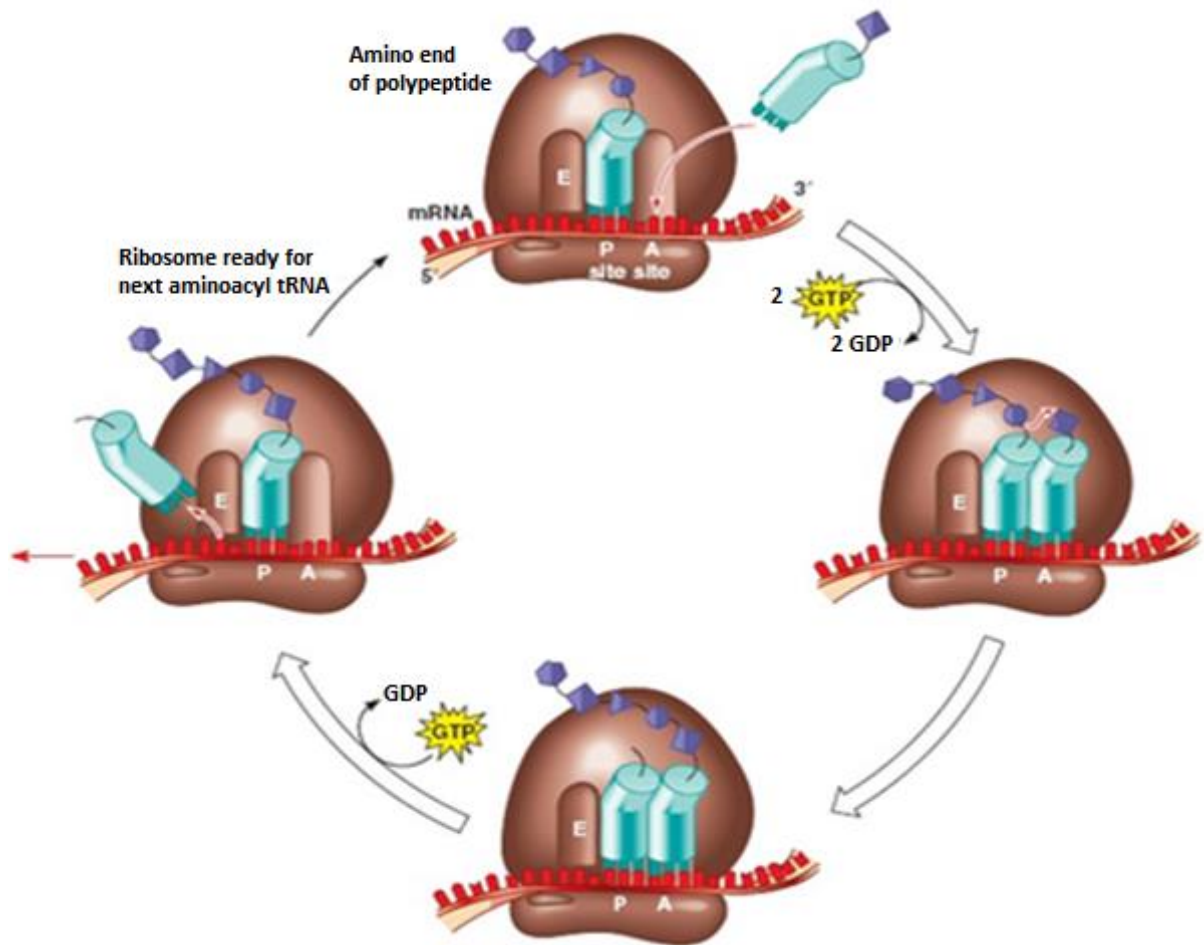
Topic # 29

Protein Translation

Initiation



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		Second base					
		U	C	A	G		
First base (5' terminus)	U	UUU } Phe	UCU }	UAU } Tyr	UGU } Cys	Third base (3' terminus)	U
		UUC }	UCC } Ser	UAC }	UGC }		C
		UUA } Leu	UCA }	UAA } Stop	UGA } Stop		A
		UUG }	UCG }	UAG } Stop	UGG } Trp		G
	C	CUU }	CCU }	CAU } His	CGU }		U
		CUC } Leu	CCC } Pro	CAC }	CGC } Arg		C
		CUA }	CCA }	CAA } Gln	CGA }		A
		CUG }	CCG }	CAG }	CGG }		G
	A	AUU }	ACU }	AAU } Asn	AGU } Ser		U
		AUC } Ile	ACC }	AAC }	AGC }		C
		AUA }	ACA } Thr	AAA } Lys	AGA } Arg		A
		AUG } Met	ACG }	AAG }	AGG }		G
	G	GUU }	GCU }	GAU } Asp	GGU }		U
		GUC } Val	GCC } Ala	GAC }	GGC } Gly		C
		GUA }	GCA }	GAA } Glu	GGA }		A
		GUG }	GCG }	GAG }	GGG }		G

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Topic # 30

Dynamic Programming

Block Game

		m blocks					
		0	1	2	3	4	5
n blocks	0						
	1						
	2						
	3						
	4						
	5						

Table "R"

$$\begin{aligned}
 &3+4, \\
 &R_{3,4} = W \\
 &R_{3,4} = L \\
 &\underline{R_{n,m}}
 \end{aligned}$$

	0	1	2	3	4	5	6	7	8	9	10
0		W									
1	W	W									
2											
3											
4											
5											
6											
7											
8											
9											
10											

$R_{0,1}$ $R_{1,0}$ $R_{1,1}$

$(1,0), (1,0), (1,1)$

$(2,0) \rightarrow (1,0) = W$

$(0,2) \rightarrow (0,1) = W$

$(2,1) \rightarrow$

$(1,1), (2,0), (1,0)$

$(2,0) = L$

$(1,2) = L$

$R_{2,2} (2,2) \rightarrow$

$(2,1), (1,2), (1,1) = W$

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Topic #31

Dynamic Programming

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Block Game

$(i-1, j)$, $(i-1, j-1)$, $(i, j-1)$

	0	1	2	3	4	5	6	7	8	9	10
0	L	W	L	W	L	W	L	W	L	W	L
1	W	W	W	W	W	W	W	W	W	W	W
2	L	W	L	W	L	W	L	W	L	W	L
3	W	W	W	W	W	W	W	W	W	W	W
4	L	W	L	W	L	W	L	W	L	W	L
5	W	W	W	W	W	W	W	W	W	W	W
6	L	W	L	W	L	W	L	W	L	W	L
7	W	W	W	W	W	W	W	W	W	W	W
8	L	W	L	W	L	W	L	W	L	W	L
9	W	W	W	W	W	W	W	W	W	W	W
10	L	W	L	W	L	W	L	W	L	W	L

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BLOCKS (n, m)

```
1.  $R_{0,0} = L$ 
2. for  $i \leftarrow 1$  to  $n$ 
3.   if  $R_{i-1,0} = W$ 
4.      $R_{i,0} \leftarrow L$ 
5.   else
6.      $R_{i,0} \leftarrow W$ 
7.   for  $j \leftarrow 1$  to  $m$ 
8.     if  $R_{0,j-1} = W$ 
9.        $R_{0,j} \leftarrow L$ 
10.    else
11.       $R_{0,j} \leftarrow W$ 
12.    for  $i \leftarrow 1$  to  $n$ 
13.      for  $j \leftarrow 1$  to  $m$ 
14.        if  $R_{i-1,j-1} = W$  and  $R_{i,j-1} = W$  and  $R_{i-1,j} = W$ 
15.           $R_{i,j} \leftarrow L$ 
16.        else
17.           $R_{i,j} \leftarrow W$ 
18. return  $R_{n,m}$ 
```

FASTBLOCK(n, m)

```
1. if  $n$  and  $m$  are both even
2.   return  $L$ 
3. else
4.   return  $W$ 
```

Topic # 32

Algorithm for Dynamic Programming

Block Game Algorithm

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BLOCKS (n, m)

```

1.  $R_{0,0} = L$ 
2. for  $i \leftarrow 1$  to  $n$ 
3.   if  $R_{i-1,0} = W$ 
4.      $R_{i,0} \leftarrow L$ 
5.   else
6.      $R_{i,0} \leftarrow W$ 
7.   for  $j \leftarrow 1$  to  $m$ 
8.     if  $R_{0,j-1} = W$ 
9.        $R_{0,j} \leftarrow L$ 
10.    else
11.       $R_{0,j} \leftarrow W$ 

```

$i = 2$

$R_{2-1,0} = R_{1,0} = W$

$j = 2$

$R_{0,2-1} = R_{0,1} = W$

BLOCKS (n, m)

```

12. for  $i \leftarrow 1$  to  $n$ 
13. for  $j \leftarrow 1$  to  $m$ 
14. if  $R_{i-1,j-1} = W$  and  $R_{i,j-1} = W$  and  $R_{i-1,j} = W$ 
15.    $R_{i,j} \leftarrow L$ 
16. else
17.    $R_{i,j} \leftarrow W$ 
18. return  $R_{n,m}$ 

```

$i = 2, j = 2$

$R_{2-1,2-1}, R_{2,2-1}, R_{2-1,2}$

$R_{1,1} = W$

$R_{2,1} = W$

$R_{1,2} = W$

FASTBLOCK(n, m)

```

1. if  $n$  and  $m$  are both even
2.   return  $L$ 
3. else
4.   return  $W$ 

```

$R_{n,m}$

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$$R_{2,2} = L$$

$$R_{4,4} = L$$

$$R_{4,5} = W$$

Topic # 33

Restriction Mapping

EcoRI (Escherichia coli)

GAATTC

ATGTTTGCATTACGATAGAATTCCGTCAAAGTGCTAG
TACAAACGTAATGCTATCTTAAGGCAGTTTCACGATC
GCCGTTATACGCTGGATTAAATTGCTGTGAAATGGT
CGGCAATATGCGACCTAAATTTAACGACACTTTACCA
TACTGCCAAGACCGAATTCCTGCGAGTGCTGAAACG
ATGACGGTTCTGGCTTAAGGACGCTCACGACTTTGC
GCGATATTACGAATGTGCTTACAGCACCGAATTCATC
CGCTATAAAGCTTACACGAATGTCGTGGCTTAAGTAG

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ATGTTTGCATTACGATAGAATTCCGTCAAAGTGCTAG
TACAAACGTAATGCTATCTTAAGGCAGTTTCACGATC
GCCGTTATACGCTGGATTAAATTGCTGTGAAATGGT
CGGCAATATGCGACCTAAATTTAACGACACTTTACCA
TACTGCCAAGACCGAATTCCTGCGAGTGCTGAAACG
ATGACGGTTCTGGCTTAAGGACGCTCACGACTTTGC
GCGATATTACGAATGTGCTTACAGCACCGAATTCATC
CGCTATAAAGCTTACACGAATGTCGTGGCTTAAGTAG

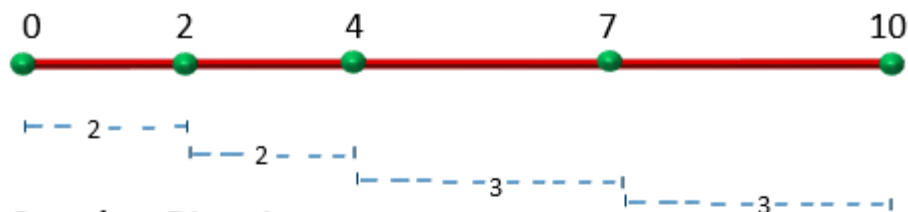
CUT BY RESTRICTION ENDONUCLEASE

X---X---X---G---A---A---T---T---C---X---X---X
Y---Y---Y---C---T---T---A---A---G---Y---Y---Y

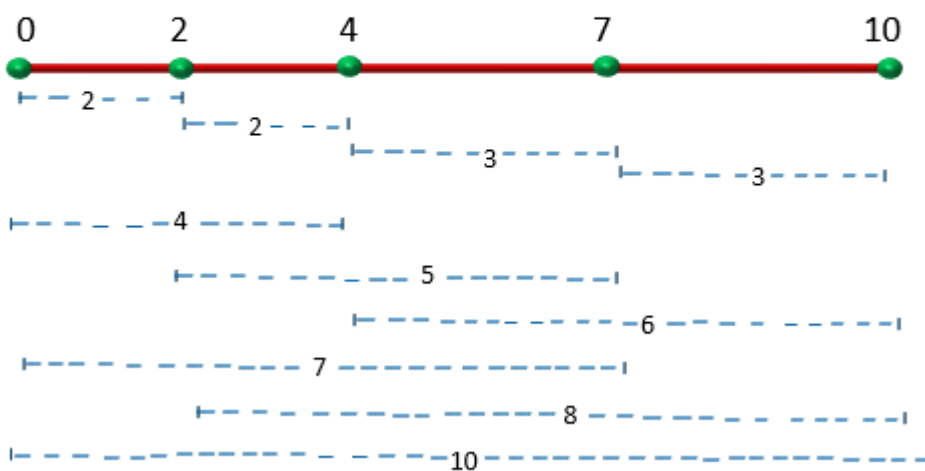


X---X---X---G A---A---T---T---C---X---X---X
Y---Y---Y---C---T---T---A---A G---Y---Y---Y

“STICKY ENDS”



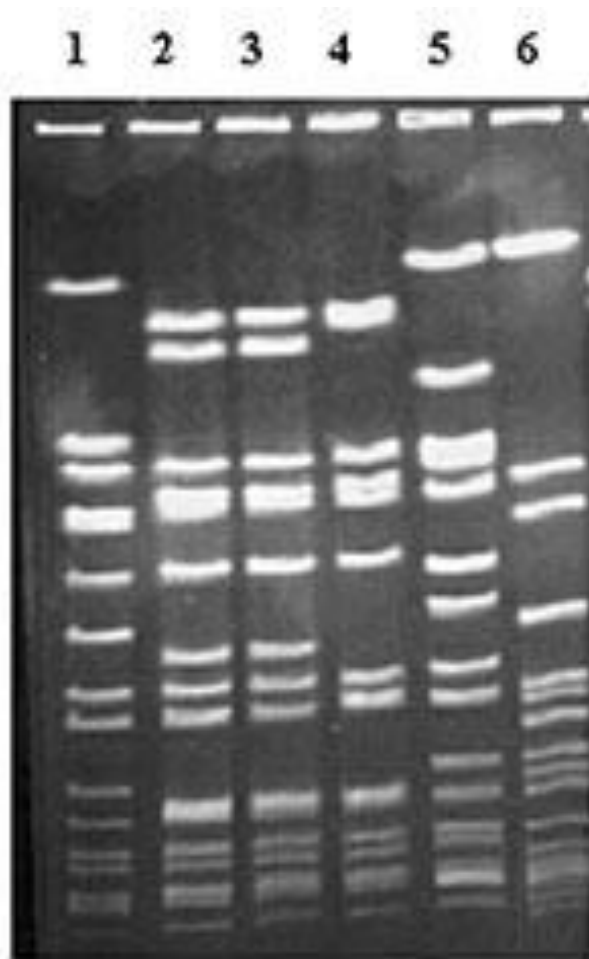
Complete Digestion



Partial Digestion

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$\{2, 2, 2, 3, 3, 4, 5\}$

If $X = \{x_1 = 0, x_2, \dots, x_n\}$

$\Delta X = \{x_j - x_i : 1 \leq i < j \leq n\}$

$X = \{0, 2, 4, 7, 10\}$, then $\Delta X = \{2, 2, 3, 3, 4, 5, 6, 7, 8, 10\}$,

Representation of ΔX

	0	2	4	7	10
0		2	4	7	10
2			2	5	8
4				3	6

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10

The element at (i, j) in the table is the value $x_j - x_i$ for $1 \leq i < j \leq n$.

Topic # 34

Restriction Mapping

$\{2, 2, 2, 3, 3, 4, 5\}$

If $X = \{x_1 = 0, x_2, \dots, x_n\}$

$\Delta X = \{x_j - x_i : 1 \leq i < j \leq n\}$

$X = \{0, 2, 4, 7, 10\}$, then $\Delta X = \{2, 2, 3, 3, 4, 5, 6, 7, 8, 10\}$,

Representation of ΔX

	0	2	4	7	10
0		2	4	7	10
2			2	5	8
4				3	6
7					3
10					

The element at (i, j) in the table is the value $x_j - x_i$ for $1 \leq i < j \leq n$.

Topic # 35

Partial Digest Problem

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Partial Digest Problem:

Given all pairwise distances between points on a line, reconstruct the positions of those points.

Input: The multiset of pairwise distances L , containing $\binom{n}{2}$ integers.

Output: A set X , of n integers, such that $\Delta X = L$

ΔA is equal to $\Delta(A \oplus \{v\})$,

where $A \oplus \{v\}$ is defined to be

$\{a + v : a \in A\}$,

Also $\Delta A = \Delta(-A)$, where $-A = \{-a : a \in A\}$

$A = \{0, 2, 4, 7, 10\}$, $\Delta(A \oplus \{100\}) =$

$\{100, 102, 104, 107, 110\}$,

and $-A = \{-10, -7, -4, -2, 0\}$

$\{0, 1, 3, 8, 9, 11, 12, 13, 15\}$ and $\{0, 1, 3, 4, 5, 7, 12, 13, 15\}$

	0	1	3	4	5	7	12	13	15
0		1	3	4	5	7	12	13	15
1			2	3	4	6	11	12	14
3				1	2	4	9	10	12
4					1	3	8	9	11
5						2	7	8	10
7							5	6	8
12								1	3
13									2
15									

	0	1	3	8	9	11	12	13	15
0		1	3	8	9	11	12	13	15
1			2	7	8	10	11	12	14
3				5	6	8	9	10	12
8					1	3	4	5	7
9						2	3	4	6
11							1	2	4
12								1	3
13									2
15									

$\{1_4, 2_4, 3_4, 4_3, 5_2, 6_2, 7_2, 8_3, 9_2, 10_2, 11_2, 12_3, 13, 14, 15\}$

$$U \oplus V = \{u + v : u \in U, v \in V\}$$

$$U \ominus V = \{u - v : u \in U, v \in V\}$$

$$U = \{6, 7, 9\} \text{ and } V = \{-6, 2, 6\}$$

$U \oplus V$	-6	2	6
6	0	8	12
7	1	9	13
9	3	11	15

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$U \ominus V$	-6	2	6
6	12	4	0
7	13	5	1
9	15	7	3

BRUTEFORCEPDP(L, n)

1. M maximum element in L
2. for every set of $n - 2$ integers $0 < x_2 < \dots < x_{n-1} < M$
3. $X = \{0, x_2, \dots, x_{n-1}, M\}$
4. Form ΔX from X
5. **if** $X = L$
6. **return** X
7. **output** "No Solution"

ANOTHERBRUTEFORCEPDP(L, n)

1. M maximum element in L
2. **for** every set of $n-2$ integers $0 < x_2 < \dots < x_{n-1} < M$
from L
3. $X = \{0, x_2, \dots, x_{n-1}, M\}$
4. Form ΔX from X
5. **if** $X = L$

6. **return** X
7. **output** “No Solution”

Topic # 36

Practical Restriction Mapping Algorithm

Brute Force Algorithm

Largest distance in “ L ”

Outermost points of “ X ”

Remaining distance “ δ ”

$$L = \{2, 2, 3, 3, 4, 5, 6, 7, 8, 10\}$$

Size of L is $\binom{n}{2} = \frac{n(n-1)}{2} = 10$ where is “ n ” is number of points in the solutions. Here “ n ” is 5 and positions of ‘ X ’ as $x_1 = 0, x_2, x_3, x_4$ and x_5 .



$$\binom{n}{k} = \binom{n}{2} =$$

$$\frac{n!}{(n-k)!k!} = \frac{n!}{(n-2)!2!}$$

$$= \frac{n(n-1)(n-2)!}{(n-2)!2!}$$

$$= \frac{n(n-1)}{2}$$

$$\frac{n(n-1)}{2} = 10$$

$$n^2 - n = 2 \times 10$$

$$n^2 - n = 20$$

$$n^2 - 5n + 4n - 20 = 0$$

$$n(n-5) + 4(n-5) = 0$$

$$(n-5)(n+4) = 0$$

$$n = 5; n = -4$$

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$$\binom{5}{2} \binom{-4}{2}$$



$$L = \{2, 2, 3, 3, 4, 5, 6, 7, 8, \mathbf{10}\}$$

$$X = \{0, 10\} \quad L = \{\mathbf{2}, 2, 3, 3, 4, 5, 6, 7, \mathbf{8}\}$$



$$x_5 - x_2 = 8 \text{ and } x_2 - x_1 = 2$$

$$X = \{0, 2, 10\} \quad L = \{2, 3, 3, 4, 5, 6, 7\}$$

$$X = \{0, 2, 10\} \quad L = \{2, \mathbf{3}, 3, 4, 5, 6, 7\}$$

$$x_4 = 7 \text{ or } x_3 = 3$$

If $x_3 = 3$ then $x_3 - x_2 = 1$ must be in L , but not, so x_4 must be 7, $x_5 - x_4 = 3$, $x_4 - x_2 = 5$, and $x_4 - x_1 = 7$ from L

$$X = \{0, 2, 7, 10\} \quad L = \{2, 3, \mathbf{4}, 6\}$$



Two choices, $x_3 = 4$ or $x_3 = 6$, If $x_3 = 6$, then $x_4 - x_3 = 1$ must be in L but not. Then $x_3 = 4$

$$X = \{0, 2, 4, 7, 10\}$$



Topic # 37

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Partial Digest Algorithm

Brute Force Algorithm

List of pairwise distances, L , and uses the function DELETE(y, L)

Value y from L , notation (y, X) to denote the multiset of distances

For example,

$$\Delta(2, \{1, 3, 4, 5\}) = \{1, 1, 2, 3\}$$

PARTIALDIGEST (L)

1. $width$ Maximum element in L
2. DELETE ($width, L$)
3. $X = \{0, width\}$
4. PLACE (L, X)

PLACE (L, X)

1. if L is empty
2. output X
3. return
4. y Maximum element in L
5. if $\Delta(y, X) \subseteq L$
6. Add y to X and remove lengths $\Delta(y, X)$ from L
7. PLACE (L, X)
8. Remove y from X and add lengths (y, X) to L
9. if $(width - y, X) \subseteq L$
10. Add $width - y$ to X and remove lengths $(width - y, X)$ from L
11. PLACE (L, X)
12. Remove $width - y$ from X and add lengths $(width - y, X)$ to L
13. return

Topic # 38

Partial Digest Algorithm

PARTIALDIGEST (L)

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PARTIALDIGEST (L)

1. $width \leftarrow$ Max element in L 10
2. DELETE ($width, L$) 10, L
3. $X \leftarrow \{0, width\}$ 0, 10
4. PLACE (L, X)

$X = \{0, 10\}$

PLACE (L, X)

1. if L is empty
2. **output** X
3. **return**
4. $y \leftarrow$ Maximum element in L
5. if $\Delta(y, X) \subseteq L$
6. Add y to X and remove lengths $\Delta(y, X)$ from L
7. PLACE (L, X)
8. Remove y from X and add lengths (y, X) to L

$L = \{2, 2, 3, 3, 4, 5, 6, 7, 8, \mathbf{10}\}$

$L = \{2, 2, 3, 3, 4, 5, 6, 7, 8\}$

$X = \{0, 10\}$

8

$\Delta(y, X) = 8(0, 10) = (8, 10)$

9. if $(width - y, X) \subseteq L$
10. Add $width - y$ to X and remove lengths $(width - y, X)$ from L
11. PLACE (L, X)
12. Remove $width - y$ from X and add lengths $(width - y, X)$ to L
13. **return**

$L = \{2, 2, 3, 3, 4, 5, 6, 7, 8\}$

$X = \{0, 10\}$

$width - y, X = 10 - 8(0, 10)$

$= 2(0, 10)$

$= (2, 8)$

$L = \{2, 3, 3, 4, 5, 6, 7\}$

$X = \{0, 2, 10\}$

PLACE (L, X)

1. if L is empty
2. **output** X
3. **return**
4. $y \leftarrow$ Maximum element in L
5. **if** $\Delta(y, X) \subseteq L$
6. Add y to X and remove
 lengths $\Delta(y, X)$ from L
7. PLACE (L, X)
8. Remove y from X and add
 lengths (y, X) to L

$L = \{2, 3, 3, 4, 5, 6, 7\}$

$X = \{0, 2, 10\}$

7
 $\Delta(y, X) = 7(0, 2, 10) = (7, 5, 3)$

$L = \{2, 3, 3, 4, 5, 6\}$

$X = \{0, 2, 7, 10\}$

PLACE (L, X)

1. if L is empty
2. **output** X
3. **return**
4. $y \leftarrow$ Maximum element in L
5. **if** $\Delta(y, X) \subseteq L$
6. Add y to X and remove
 lengths $\Delta(y, X)$ from L
7. PLACE (L, X)
8. Remove y from X and add
 lengths (y, X) to L

$L = \{2, 3, 3, 4, 5, 6\}$

$X = \{0, 2, 7, 10\}$

6
 $\Delta(y, X) = 6(0, 2, 7, 10) = (6, 4, 1, 4)$

PLACE (L, X)

1. if L is empty
2. **output** X
3. **return**
4. $y \leftarrow$ Maximum element in L
5. if $\Delta(y, X) \subseteq L$
6. Add y to X and remove
 lengths $\Delta(y, X)$ from L
7. PLACE (L, X)
8. Remove y from X and add
 lengths (y, X) to L

$L = \{2, 3, 3, 4, 5, 6\}$

$X = \{0, 2, 7, 10\}$

6

$\Delta(y, X) = 6(0, 2, 7, 10) = (6, 4, 1, 4)$

Topic # 39

Regulatory Motifs in DNA Sequences

Sequence Motifs

Biological function

Nucleases and transcription factors

Processes at RNA level

Specific sequence located upstream of genes TCGGGGATTTC

NF-kB binding sites (nuclear factor kappa-light-chain-enhancer of activated B cells)

Regulatory motifs, transcription factors

Set of upstream regions in genes in the genome, each region containing one NF-kB sites

Suppose we do not know either location or sequence of NF-kB sites

“The Gold Bug” by Edgar Allan provided some clue of finding DNA motifs, one of the character find parchment written below

53++!305))6*;4826)4+.)4+);806*;48!8'60))85;]8*:+*8!83(88)5*!;46(;88*96*?;8)*+(:485);5
!2:+(:4956*2(5*-)8'8*;4069285) ;6!8) 4++;1(+9;48081;8:8+1;48!85;4)485!
528806*81(+9;48;(88;4(+?34;48)4+;161;:188;+?;

“;48” codes for “THE”

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53++!305))6*THE26)H+.)H+)TE06*THE!E'60))E5T]E*.*E!E3(EE)5*!TH6(TEE*96*?TE
)* +(THE5) T5*!2.*+(TH956*2(5*-H)E'E*T
 H0692E5)T)6!E)H++T1(+9THE0E1TE:E+1THE!E5TH)HE5!52EE06*E1(+9THET(EETH(
 +?3HTHE) H+T161T:1EET+?T

“;” for “T”

“4” for “H”

“8” for “E”

Topic # 40

Profiles

Conserved Pattern

32 nucleotide

7 sequences

- 1- CGGGGCTGGGTCGTCACATTCCCCTTTTCGATA
- 2- TTTGAGGGTGCCCAATAACCAAAGCGGACAAA
- 3- GGGATGCCGTTTGACGACCTAAATCAACGGCC
- 4- AAGGCCAGGAGCGCCTTTGCTGGTTCTACCTG
- 5- AATTTTCTAAAAAGATTATAATGTCGGTCCTC
- 6- CTGCTGTACAACTGAGATCATGCTGCTTCAAC
- 7- TACATGATCTTTTGTGGATGAGGGAATGATGC

Figure 1

P = ATGCAACT

l = 8

- 1- CGGGGCTATGCAACTGGGTCGTCACATTCCCCTTTTCGATA
- 2- TTTGAGGGTGCCCAATAAATGCAACTCCAAAGCGGACAAA
- 3- GGATGCAACTGATGCCGTTTGACGACCTAAATCAACGGCC
- 4- AAGGATGCAACTCCAGGAGCGCCTTTGCTGGTTCTACCTG
- 5- AATTTTCTAAAAAGATTATAATGTCGGTCCATGCAACTTC
- 6- CTGCTGTACAACTGAGATCATGCTGCATGCAACTTTCAAC
- 7- TACATGATCTTTTGAATGCAACTTGGATGAGGGAATGATGC

Figure 2

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P = ATGCAACT

$l = 8$

- 1- CGGGGCTATGCAACTGGGTCGTCACATTCCCCTTTTCGATA
- 2- TTTGAGGGTGCCCAATAAATGCAACTCCAAAGCGGACAAA
- 3- GGATGCAACTGATGCCGTTTGACGACCTAAATCAACGGCC
- 4- AAGGATGCAACTCCAGGAGCGCCTTTGCTGGTTCTACCTG
- 5- AATTTTCTAAAAAGATTATAATGTCGGTCCATGCAACTTC
- 6- CTGCTGTACAACTGAGATCATGCTGCATGCAACTTTCAAC
- 7- TACATGATCTTTTGATGCAACTTGGATGAGGGAATGATGC

Figure 3

P = ATGCAACT

$l = 8$

$7 \times (32 + 8) = 280$ nucleotides

Probability = $280/4^8 = 0.004$

- 1- CGGGGCTATcCAgCTGGGTCGTCACATTCCCCTTTTCGATA
- 2- TTTGAGGGTGCCCAATAAaggGCAACTCCAAAGCGGACAAA
- 3- GGATGgAtCTGATGCCGTTTGACGACCTAAATCAACGGCC
- 4- AAGGAAaGCAACcCCAGGAGCGCCTTTGCTGGTTCTACCTG
- 5- AATTTTCTAAAAAGATTATAATGTCGGTCCtTGgAACTTC
- 6- CTGCTGTACAACTGAGATCATGCTGCATGCcAtTTTCAAC
- 7- TACATGATCTTTTGATGgcACTTGGATGAGGGAATGATGC

Figure 4

Topic # 41

Profiles

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Conserved Pattern

18 sequences
NF-kB
TCGGGGATTTC
/ = 12

Alignment

Profile

Consensus

```

1- TCGGGGATTTC A
2- ACGGGGATTTT
3- TCGGTACTTTAC
4- TTGGGGACTTTT
5- CCGGTGATTCCC
6- GCGGGGAATTTC
7- TCGGGGATTCCT
8- TCGGGGATTCCT
9- TAGGGGA ACTAC
10- TCGGGTATAAAC
11- TCGGGGGTTTTT
12- CCGGTGACTTAC
13- CCAGGGACTCCC
14- AAGGGGACTTCC
15- TTGGGGACTTTT
16- TTTGGGAGTCCC
17- TCGGTGATTTC
18- TAGGGGAAGACC

```

```

A: 2 3 1 0 0 1 16 3 1 2 4 1
T: 12 3 1 0 4 1 0 9 15 11 5 6
G: 1 0 16 18 14 16 1 1 1 0 0 0
C: 3 12 0 0 0 0 1 5 1 5 9 11

```

TCGGGGATTTC

score = 12, 12, 16, 18, 14, 16, 16, 9, 15, 11, 9, 11 = 159

- 1- position 8 - Sequence 1
- 2- position 19 - Sequence 2
- 3- position 3- Sequence 3
- 4- position 5- Sequence 4
- 5- position 31- Sequence 5
- 6- position 27- Sequence 6
- 7- position 15- Sequence 7

1-
CGGGGCTATcCAgCTGGGTCGTCACATTCCCCTT

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2- TTTGAGGGTGCCCAATAAaggGCAACTCCAAAGCGGACAAA

3- GGATGgAtCTGATGCCGTTTGACGACCTA

4- AAGGAaGCAACcCCAGGAGCGCCTTTGCTGG

5- AATTTTCTAAAAAGATTATAATGTCGGTCCtTGgAACTTC

6- CTGCTGTACAACTGAGATCATGCTGCATGCcAtTTTCAAC

7- TACATGATCTTTTGATGgcACTTGGATGAGGGAATGATGC

Figure 6

Topic # 42

Profiles

Conserved Pattern

Alignment		1	2	3	4	5	6	7	8
		A	T	C	C	A	G	C	T
		G	G	G	C	A	A	C	T
		A	T	G	G	A	T	C	T
		A	A	G	C	A	A	C	C
		T	T	G	G	A	A	C	T
		A	T	G	C	C	A	T	T
		A	T	G	G	C	A	C	T
Profile	A	5	1	0	0	5	5	0	0
	T	1	5	0	0	0	1	1	6
	G	1	1	6	3	0	1	0	0
	C	0	0	1	4	2	0	6	1
	Consensus	A	T	G	C	A	A	C	T

Figure 7

If $s = (8, 19, 3, 5, 31, 27, 15)$

then $\text{Score}(s) = 5 + 5 + 6 + 4 + 5 + 5 + 6 + 6 = 42$

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Set of t DNA sequences

“ n ” nucleotides

one position in each of these “ t ” sequences

$s = (s_1, s_2, \dots, s_t)$, with $1 \leq s_i \leq n - l + 1$

l -mers can be compiled into $t \times l$ alignment matrix

whose (i, j) th element is the nucleotide in the $s_i + j - 1$ th element in the i th sequence of figure 7

Topic # 43

Motif Finding Problem

Profile Matrix

If $P(s)$ denotes profile matrix, starting from s

$M_{P(s)}(j)$ -largest count in column j of $P(s)$

Alignment		A	T	C	C	A	G	C	T	
		G	G	G	C	A	A	C	T	
		A	T	G	G	A	T	C	T	$M_{P(s)}(1) = 5$
		A	A	G	C	A	A	C	C	$M_{P(s)}(2) = 5$
		T	T	G	G	A	A	C	T	$M_{P(s)}(3) = 6$
		A	T	G	C	C	A	T	T	$M_{P(s)}(4) = 4$
Profile		A	T	C	C	A	G	C	T	$M_{P(s)}(5) = 5$
	A	5	1	0	0	5	5	0	0	$M_{P(s)}(6) = 5$
	T	1	5	0	0	0	1	1	6	$M_{P(s)}(7) = 6$
	G	1	1	6	3	0	1	0	0	$M_{P(s)}(8) = 6$
	C	0	0	1	4	2	0	6	1	
Consensus		A	T	G	C	A	A	C	T	

For the starting positions in above figure, $\text{Score}(s, \text{DNA}) =$

$5 + 5 + 6 + 4 + 5 + 5 + 6 + 6 = 42$. $\text{Score}(s, \text{DNA})$ can be used to measure

The strength of a profile corresponding to the starting positions s . consensus score of $l \cdot t$ corresponds to the best possible alignment $\text{Score}(s, \text{DNA})$

consensus score is defined to be $Score(s, DNA) = \sum_{j=1}^l MP_{(s)}(j)$

$Score(s, DNA) = 5 + 5 + 6 + 4 + 5 + 5 + 6 + 6 = 42$

consensus score of $l \cdot t$ corresponds to the best possible alignment

A consensus score of $l \cdot t/4$, is worst possible alignment

Motif Finding Problem:

Given a set of DNA sequences, find a set of l -mers, one from each sequence, that maximizes the consensus score.

Input: A $t \times n$ matrix of DNA, and, the length of the pattern to find.

Output: An array of t starting positions $s = l(s_1, s_2, \dots, s_t)$ maximizing $Score(s, DNA)$.

Topic # 44

Motif Finding Problem

Median String

Median string

Given two l -mers v and w , can compute the *Hamming distance* between them, $d_H(v, w)$, as the number of positions that differ in the two strings

A T T G T C

: x : x : :

A C T C T C

$s = (s_1, s_1, \dots, s_t)$

v is some l -mer

$d_H(v, s)$ to denote the total Hamming distance between v and the l -mers starting at positions s : $d_H(v, s) = \sum_{i=1}^t d_H(v, s_i)$ where $d_H(v, s_i)$ is the Hamming distance between v and the l -mer that starts at s_i in the i th DNA sequence

$$\text{TotalDistance}(v, \text{DNA}) = \min_s (d_H(v, s))$$

Finding $\text{TotalDistance}(v, \text{DNA})$ is a simple problem:

find the best match for v in the first DNA sequence (i.e., a position minimizing $d_H(v, s_1)$ for $1 \leq s_1 \leq n-l+1$), then the best match in the second sequence and so on

Median string for DNA as the string v that minimizes $\text{TotalDistance}(v, \text{DNA})$; this minimization is performed

over all 4^l strings v of length l .

Median String Problem:

Given a set of DNA sequences, find a median string.

Input: A $t \times n$ matrix DNA, and l , the length of the pattern to find

Output: A string v of l nucleotides that minimizes $\text{TotalDistance}(v, \text{DNA})$ over all strings of that length

Double minimization: finding a string v that minimizes $\text{TotalDistance}(v, \text{DNA})$, which is in turn the smallest distance

among all choices of starting points s in the DNA sequences.

Min $\min d_H(v, s)$

all choices of all choices

l -mers v starting positions s

The Median String problem- Minimization problem

The Motif Finding problem- Maximization problem

Computationally equal

Let s be a set of starting positions with consensus score $\text{Score}(s, \text{DNA})$, and let w be the consensus string of the corresponding profile. Then

$$d_H(w, s) = lt - \text{Score}(s, \text{DNA})$$

1. **A T C C A G C T**
 2. G G **G C A A C T**
 3. **A T G G A T C T**
 4. **A A G C A A C C**
 5. T **T G G A A C T**
 6. **A T G C C A T T**
 7. **A T G G C A C T**
A T G C A A C T

Hamming distance between the consensus

string w and each of the seven implanted patterns is 2, and $d_H(w, s) = 2 \times 7 = 14$

$$= 7 \times 8 - 42 = 14$$

Topic # 44

Motif Finding Problem

Median String

Median string

Given two l -mers v and w , can compute the *Hamming distance* between them, $d_H(v, w)$, as the number of positions that differ in the two strings

A T T G T C

: x : x ::

A C T C T C

$s = (s_1, s_1, \dots, s_t)$

v is some l -mer

$d_H(v, s)$ to denote the total Hamming distance between v and the l -mers starting at positions s :

$d_H(v, s) = \sum_{i=1}^t d_H(v, s_i)$ where $d_H(v, s_i)$ is the Hamming distance between v and the l -mer that starts at s_i in the i th DNA sequence

$\text{TotalDistance}(v, \text{DNA}) = \min_s (d_H(v, s))$

Finding $\text{TotalDistance}(v, \text{DNA})$ is a simple problem:

find the best match for v in the first DNA sequence (i.e., a position minimizing $d_H(v, s_1)$ for $1 \leq s_1 \leq n-l+1$), then the best match in the second sequence and so on

Median string for DNA as the string v that minimizes $\text{TotalDistance}(v, \text{DNA})$; this minimization is performed

over all 4^l strings v of length l .

Median String Problem:

Given a set of DNA sequences, find a median string.

Input: A $t \times n$ matrix DNA, and l , the length of the pattern to find

Output: A string v of l nucleotides that minimizes $\text{TotalDistance}(v, \text{DNA})$ over all strings of that length

Double minimization: finding a string v that minimizes $\text{TotalDistance}(v, \text{DNA})$, which is in turn the smallest distance

among all choices of starting points s in the DNA sequences.

Min $\min d_H(v, s)$

all choices of all choices

l -mers v starting positions s

The Median String problem- Minimization problem

The Motif Finding problem- Maximization problem

Computationally equal

Let s be a set of starting positions with consensus score $\text{Score}(s, \text{DNA})$, and let w be the consensus string of the corresponding profile. Then

$$d_H(w, s) = lt - \text{Score}(s, \text{DNA})$$

```
1. AT CC AGCT
2. GG GC AACT
3. AT GG ATCT
4. AAGC AAC C
5. TT GG AACT
6. ATG C CATT
7. ATG G CACT
   ATGC AA CT
```

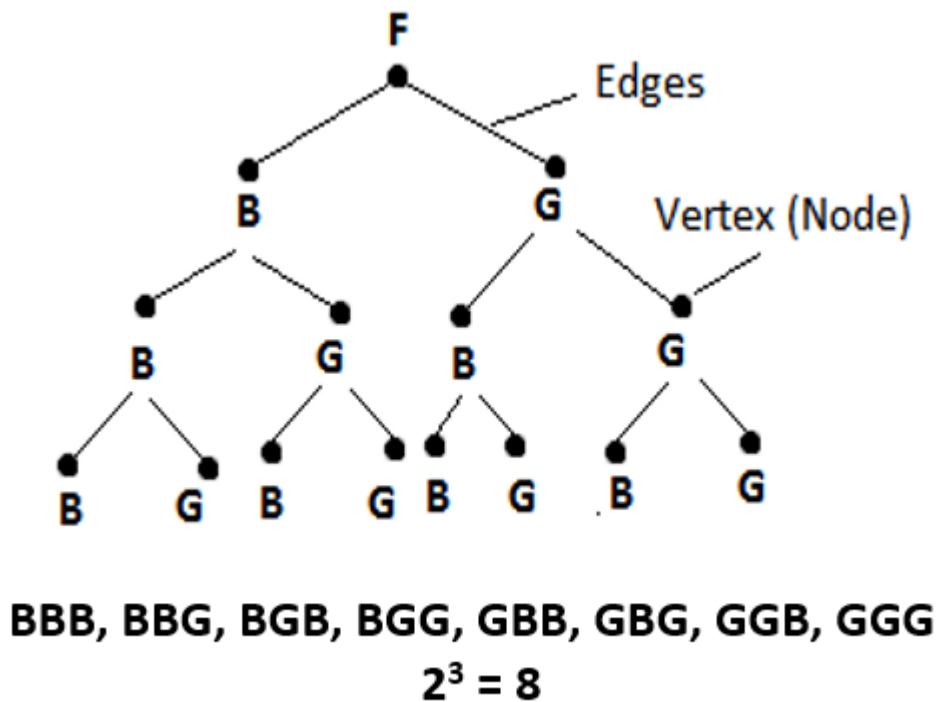
Hamming distance between the consensus

string w and each of the seven implanted patterns is 2, and $d_H(w, s) = 2 \times 7 = 14$

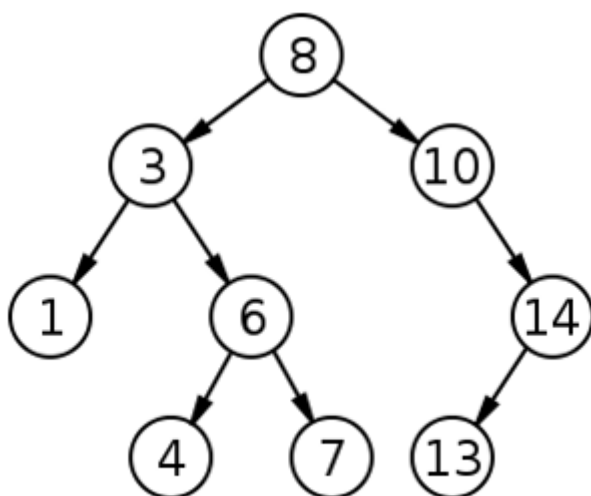
$$= 7 \times 8 - 42 = 14$$

Topic # 45

Introduction



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Topic # 46

Search Trees

Best Alternative

Median String and

Motif Finding Problems

Number of alternatives

To find best one

$$(n - l + 1)^t$$

(1, 1, ..., 1, 1)

(1, 1, ..., 1, 2)

(1, 1, ..., 1, 3)

(1, 1, ..., 1, n - l + 1)

(1, 1, ..., 2, 1)

(1, 1, ..., 2, 2)

(1, 1, ..., 2, 3)

(1, 1, ..., 2, n - l + 1)

(n - l + 1, n - l + 1, ..., n - l + 1, 1)

(n - l + 1, n - l + 1, ..., n - l + 1, 2)

(n - l + 1, n - l + 1, ..., ..., n - l + 1, 3)

(n - l + 1, n - l + 1, ..., ..., n - l + 1, n - l + 1)

Figure 1

AA...AA

AA...AT

AA...AG

AA...AC

AA...TA

AA...TT

AA...TG

AA · · · TC

...

CC · · · GG

CC · · · GC

CC · · · CA

CC · · · CT

CC · · · CG

CC · · · CC

All 4^l

Figure 2

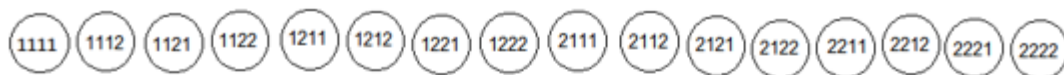


Figure 3

All 4-mers in the alphabet of $\{1, 2\}$

1 for A

2 for T

3 for G

4 for C

$(1, 1, \dots, 1, 1)$

$(1, 1, \dots, 1, 2)$

$(1, 1, \dots, 1, 3)$

$(1, 1, \dots, 1, 4)$

$(1, 1, \dots, 2, 1)$

$(1, 1, \dots, 2, 2)$

$(1, 1, \dots, 2, 3)$

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$(1, 1, \dots, 2, 4)$

...

$(4, 4, \dots, 3, 3)$

$(4, 4, \dots, 3, 4)$

$(4, 4, \dots, 4, 1)$

$(4, 4, \dots, 4, 2)$

$(4, 4, \dots, 4, 3)$

$(4, 4, \dots, 4, 4)$

1 for A

2 for T

3 for G

4 for C

Consider all k^L L -mers in a k -letter alphabet

For Motif Finding problem, $k = n-l+1$,

For Median String problem, $k = 4$.

All 2^4 4-mers in the two-letter alphabet of 1 and 2.

Topic # 47

Algorithm for Search Trees

Search tree Algorithms

Next leave

All leaves

Preorder

L -mer $a = (a_1 a_2 \dots a_L)$

NEXTLEAF(a, L, k)

1. **for** $i \leftarrow L$ to 1
2. **if** $a_i < k$
3. $a_i \leftarrow a_i + 1$
4. **return** a
5. $a_i \leftarrow 1$
6. **return** a

a = elements of L -mer

L = l -mer

k = number of elements

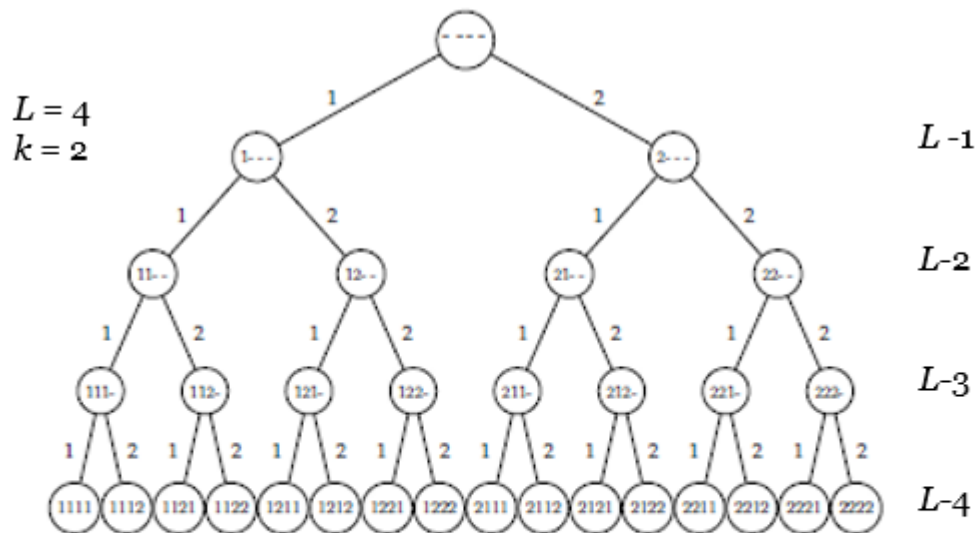
ALLEAVES(L, k)

1. $a \leftarrow (1, \dots, 1)$
2. **while** forever
3. **output** a
4. $a \leftarrow \text{NEXTLEAF}(a, L, k)$
5. **if** $a = (1, 1, \dots, 1)$
6. **return**



ates

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Motif Finding Problem , $L = t$ levels $k = n - l + 1$

Median String Problem , $L = l$ and $k = 4$

t = number of DNA sequences

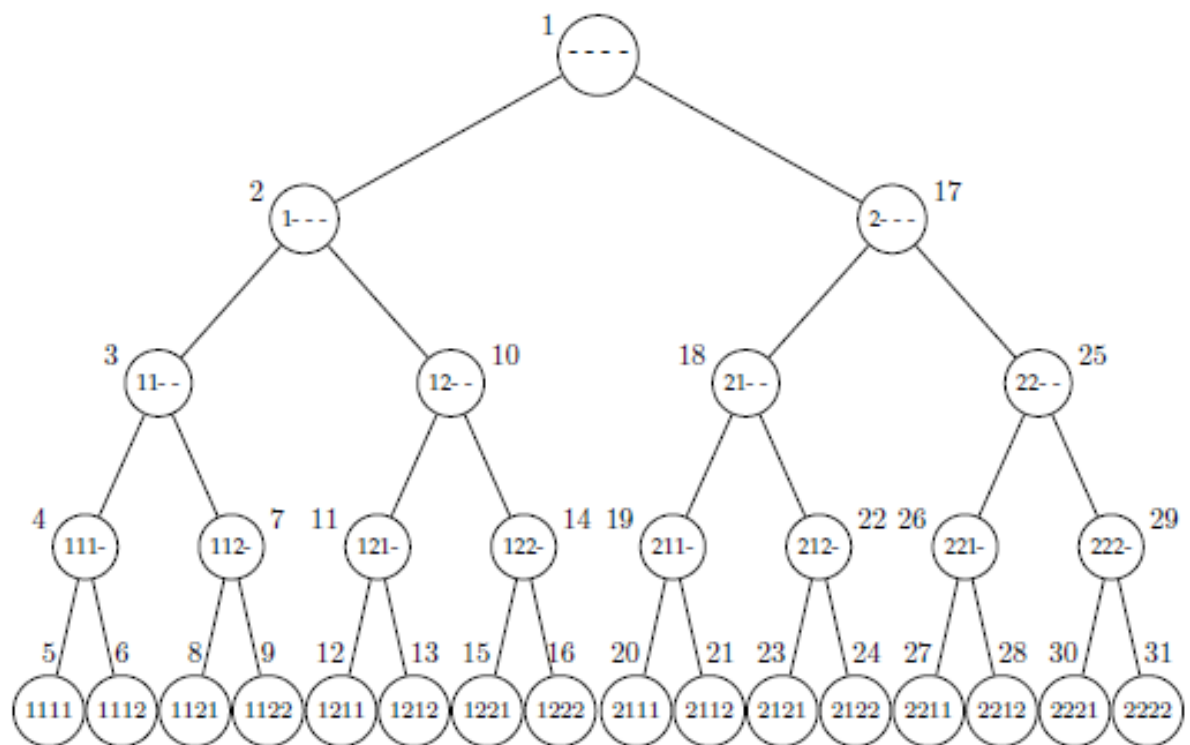
n = length of each sequence, l = length of the profile

Topic # 48

Algorithm for Search Trees

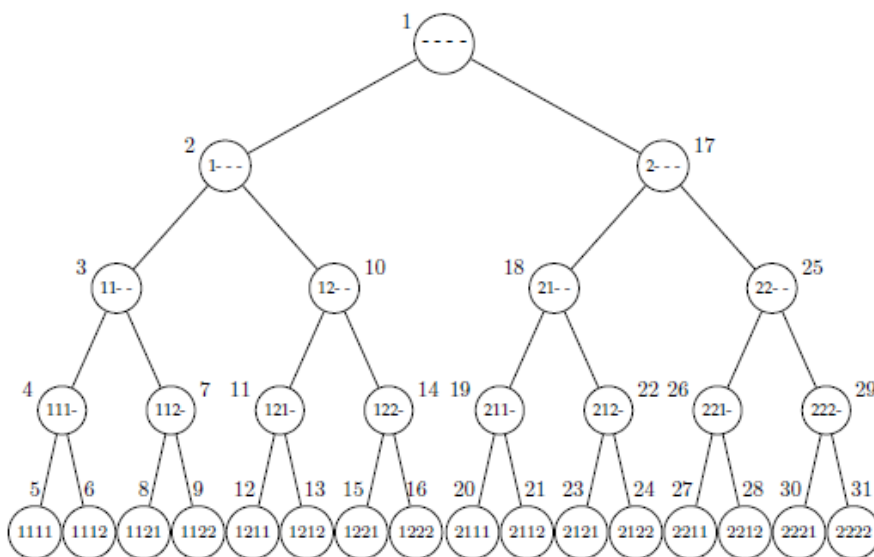
PREORDER(v)

1. output v
2. if v has children
3. PREORDER(left child of v)
4. PREORDER(right child of v)



$L = 4$

$k = 2$



1. (-,-,-,-)

2. (1,-,-,-)

3. (1,1,-,-)

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4. (1,1,1,-)
5. (1,1,1,1)
6. (1,1,1,2)
7. (1,1,2,-)
8. (1,1,2,1)
9. (1,1,2,2)
10. (1,2,-,-)
11. (1,2,1,-)
12. (1,2,1,1)
13. (1,2,1,2)
14. (1,2,2,-)
15. (1,2,2,1)
16. (1,2,2,2)
17. (2,-,-,-)
18. (2,1,-,-)
19. (2,1,1,-)
20. (2,1,1,1)
21. (2,1,1,2)
22. (2,1,2,-)
23. (2,1,2,1)
24. (2,1,2,2)
25. (2,2,-,-)
26. (2,2,1,-)
27. (2,2,1,1)
28. (2,2,1,2)
29. (2,2,2,-)
30. (2,2,2,1)
31. (2,2,2,2)



Topic # 49

Next Vertex Algorithm

Traversing vertices

Input: $a = (a_1, \dots, a_L)$ at level i

Output: Next vertex in the tree

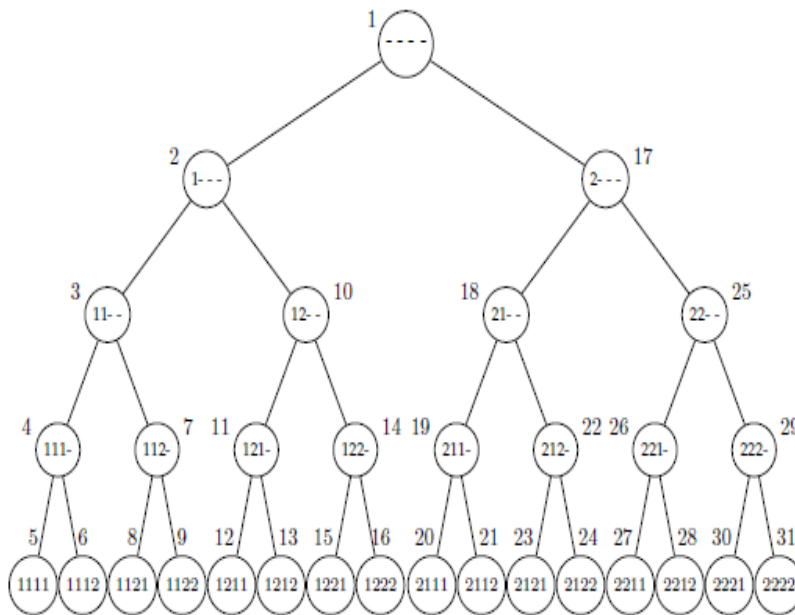
values $a_1, \dots, a_i$ and ignores $a_{i+1}, \dots, a_L$

NEXTVERTEX takes inputs that are similar to NEXTLEAF, with the exception that the “current

leaf” is now the “current vertex,” and uses parameter i for vertices

NEXTVERTEX(a, i, L, k)

1. **if** $i < L$
2. $a_{i+1} \leftarrow 1$
3. **return** ($a, i + 1$)
4. **else**
5. **for** $j \leftarrow L$ to 1
6. **if** $a_j < k$
7. $a_j \leftarrow a_j + 1$
8. **return** (a, j)
9. **return** ($a, 0$)



When $i < L$, NEXTVERTEX ($\mathbf{a}$, i , L , k) moves down to the next lower level and explores that subtree of $\mathbf{a}$. If $i = L$, NEXTVERTEX either moves along the lowest level as long as $a_L < k$ or jumps back up in the tree.

Topic # 50

Advanced Computing Approaches

Algorithm

- Some entity needs to carry out the steps specified by the algorithm
- Humans are generally slow
- A computer is less intelligent but can perform simple steps quickly and reliably
- Algorithm must be rephrased in programming language
- Pseudocode: language often used to describe algorithm
- Complex operations are grouped together into mini-algorithms called subgroups
- **Variable** is written as x or *total*
- An **array** of n elements is an ordered collection of n variables $a_1, a_2, \dots, a_n$
- An algorithm or pseudocode is denoted by a name, followed by the list of arguments

Topic #51

ByPass Algorithm

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Branch and Bound Algorithm

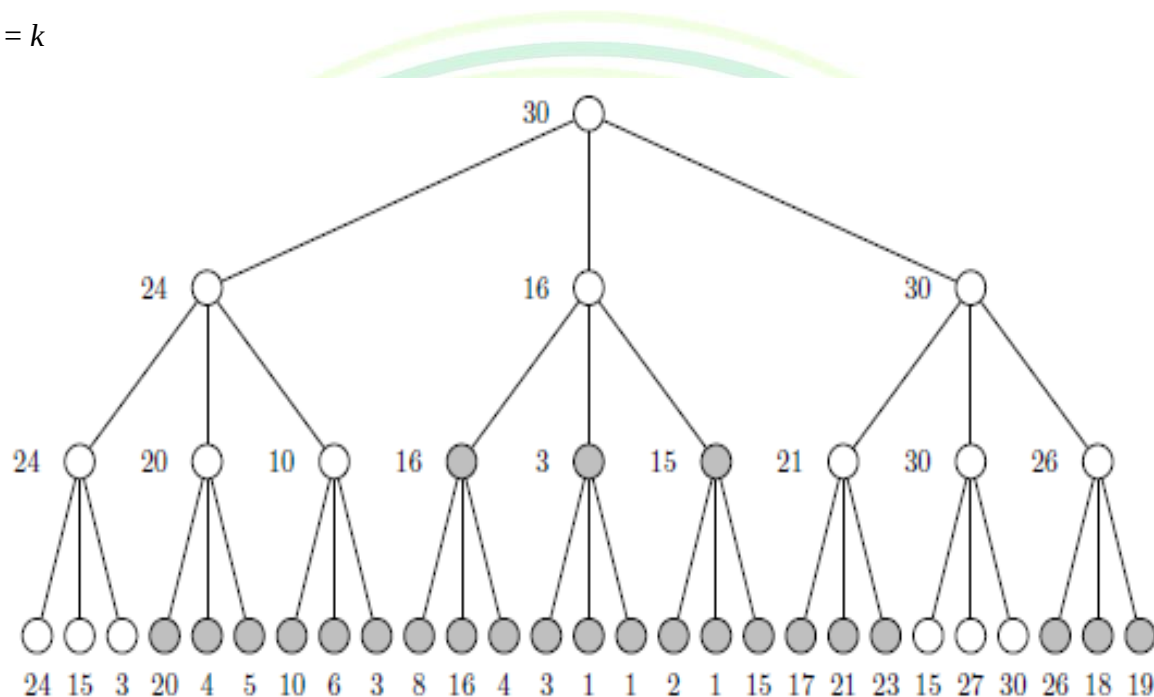
NEXTVERTEX Algorithm

BYPASS Algorithm

Skip the subtree rooted at vertex (a, i)

Increment a_i (unless

$a_i = k$



A tree that has uninteresting subtrees. The numbers next to a leaf represent the “score” for that L-mer. Scores at internal vertices represent the maximum score in the subtree rooted at that vertex. To improve the brute force algorithm, we can “prune” subtrees that do not contain high-scoring leaves. For example, since the score of the very first leaf is 24, it does not make sense to analyze the 4th, 5th, or 6th leaves whose scores are 20, 4, and 5, respectively. Therefore, the subtree containing these vertices can be ignored.

BYPASS($\mathbf{a}, i, L, k$)

1. **for** $j \leftarrow i$ to 1
2. **if** $a_j < k$
3. $a_j \leftarrow a_j + 1$
4. **return** $(\mathbf{a}, j)$
5. **return** $(\mathbf{a}, 0)$

Topic # 52

Finding Motifs

Brute force approach

BRUTEFORCEMOTIFSEARCH(DNA, t, n, l)

1. $bestScore \leftarrow 0$
2. for each $(s_1, \dots, s_t)$ from $(1, \dots, 1)$ to $(n - l + 1, \dots, n - l + 1)$
3. if $Score(s, DNA) > bestScore$
4. $bestScore \leftarrow Score(s, DNA)$
5. $bestMotif \leftarrow (s_1, s_2, \dots, s_t)$
6. return $bestMotif$

$n-l+1$ choices for the first index (s_1), then for s_2, s_3

number of positions is $(n - l + 1)^t$

$Score(s, DNA)$, which requires $O(l)$ operations- $O(ln^t)$.

BRUTEFORCEMOTIFSEARCHAGAIN(DNA, t, n, l)

1. $s \leftarrow (1, 1, \dots, 1)$
2. $bestScore \leftarrow Score(s, DNA)$
3. while forever
4. $s \leftarrow NEXTLEAF(s, t, n - l + 1)$
5. if $Score(s, DNA) > bestScore$
6. $bestScore \leftarrow Score(s, DNA)$
7. $bestMotif \leftarrow (s_1, s_2, \dots, s_t)$
8. if $s = (1, 1, \dots, 1)$
9. return $bestMotif$

Topic # 53

Simple Motif Search Algorithm

Simple Motif Search Algorithm

SIMPLEMOTIFSEARCH(DNA, t, n, l)

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1. $s \leftarrow (1, \dots, 1)$
2. $bestScore \leftarrow 0$
3. $i \leftarrow 1$
4. while $i > 0$
5. if $i < t$
6. $(s, i) \leftarrow \text{NEXTVERTEX}(s, i, t, n - l + 1)$
7. else
8. if $\text{Score}(s, \text{DNA}) > bestScore$
9. $bestScore \leftarrow \text{Score}(s, \text{DNA})$
10. $bestMotif \leftarrow (s_1, s_2, \dots, s_t)$
11. $(s, i) \leftarrow \text{NEXTVERTEX}(s, i, t, n - l + 1)$
12. return $bestMotif$

Simple Motif Search Algorithm

Some sets of starting positions can be ruled out

If the first i of t starting positions [i.e., $(s_1, s_2, \dots, s_i)$]

Sequences $i+1, i+2, \dots, t$,

$s = (s_1, s_2, \dots, s_t)$, define the partial consensus score, $\text{Score}(s, i, \text{DNA}) - i \times l$ alignment matrix

Partial consensus score for $s_1, \dots, s_i$, remaining $t-i$ rows can only improve the consensus score by $(t-i) \cdot l$

First i starting positions $(s_1, \dots, s_i)$ could be at most $\text{Score}(s, i, \text{DNA}) + (t-i) \cdot l$

$\text{Score}(s, i, \text{DNA}) + (t-i) \cdot l$

is less than the currently best score, $bestScore$

$t-i$ sequences in the sample

$\text{Score}(s, i, \text{DNA}) + (t-i) \cdot l$

$(n-l+1)^{t-i}$

Topic # 54

Branch and Bound Algorithm

Branch and Bound

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Motif Search

BRANCHANDBOUNDMOTIFSEARCH(*DNA, t, n, l*)

```
1. s      (1, ..., 1)
2. bestScore      0
3. i      1
4. while i > 0
5.     if i < t
6.         optimisticScore      Score (s, i, DNA) +
            (t - i) · l
7.         if optimisticScore < bestScore
8.             (s, i)      BYPASS(s, i, t, n - l + 1)
9.         else
10.            (s, i)      NEXTVERTEX(s, i, t, n - l + 1)
11.        else
12.            if Score(s, DNA) > bestScore
13.                bestScore      Score(s)
14.                bestMotif      (s1, s2, ..., st)
15.            (s, i)      NEXTVERTEX(s, i, t, n - l + 1)
16. return bestMotif
```

Topic # 55

Brute Force Algorithm

Brute Force

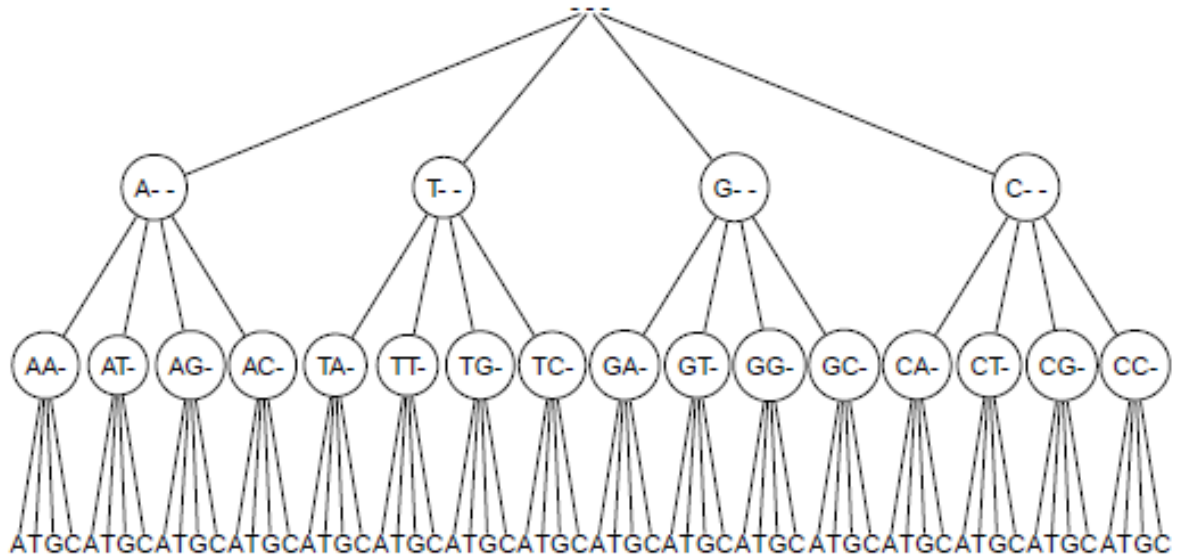
Median Search

BRUTEFORCEMEDIANSEARCH(*DNA, t, n, l*)

```
1. bestWord      AAA · · · AA
2. bestDistance
3. for      each l-mer word from AAA...A to TTT...T
4.     if TOTALDISTANCE(word, DNA) < bestDistance
```

6. *bestWord* word

```
7. return bestWord
```



A search tree for the Median String problem. Each branching point can give rise to only four children, as opposed to the $n-l+1$ children in the Motif Finding problem.

SIMPLEMEDIANSEARCH(DNA, t, n, l)

1. $s = (1, 1, \dots, 1)$

2. *bestDistance*

3. $i \rightarrow 1$

4. while i > 0

5. if $i < l$

6. (s, i) NEXTVERTEX($s, i, l, 4$)

7. else

8 *word* nucleotide string corresponding to $(s_1, s_2, \dots, s_l)$

9. if TOTALDISTANCE(*word*,*DNA*) < *bestDistance*

10. bestDistance TOTALDISTANCE(word,DNA)

```
11.      bestWord      word
```

12. (s, i) NEXTVERTEX($s, i, l, 4$)
13. return *bestWord*

BRANCHANDBOUNDMEDIANSEARCH(DNA, t, n, l)

1. $s \leftarrow (1, 1, \dots, 1)$
2. *bestDistance*
3. $i \leftarrow 1$
4. while $i > 0$
5. if $i < l$
6. *prefix* nucleotide string corresponding to $(s_1, s_2, \dots, s_i)$
7. *optimisticDistance* TOTALDISTANCE(*prefix*,DNA)
8. if *optimisticDistance* > *bestDistance*
9. (s, i) BYPASS($s, i, l, 4$)
10. else
11. (s, i) NEXTVERTEX($s, i, l, 4$)
12. else
13. *word* nucleotide string corresponding to $(s_1, s_2, \dots, s_l)$
14. if TOTALDISTANCE(*word*,DNA) < *bestDistance*
15. *bestDistance* TOTALDISTANCE(*word*,DNA)
16. *bestWord* *word*
17. (s, i) NEXTVERTEX($s, i, l, 4$)
18. return *bestWord*

Topic # 56

Genomic Rearrangements

Waardenburg's syndrome

Hearing loss

Two Different colored eyes

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Gene present on chromosome 2

Spotch gene in mice

Human genome-mouse genome

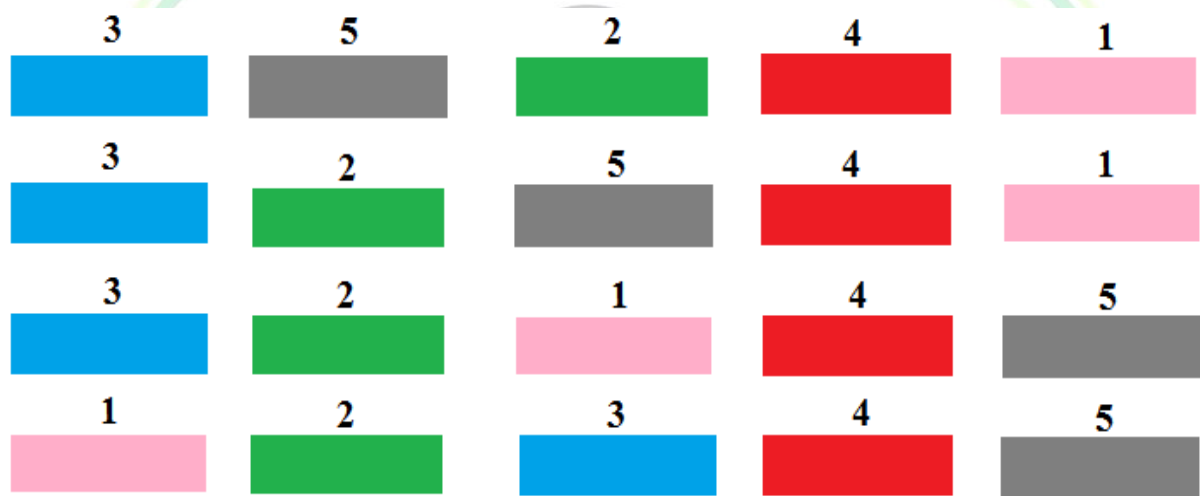
Cut into 300 genomic fragments-Synteny blocks

Chromosome 2 in humans-mouse chromosomes 1, 2, 3, 5, 6, 7, 10, 11, 12, 14, and 17

Genome rearrangement results in a change of gene ordering

Analysis of human and mouse genomes-250 genomic rearrangements

Mouse X chromosome



Human X chromosome

Transformation of the mouse gene order into the human gene order on the X chromosome

Topic # 57

Greedy Approach for Motif Search

Approximation algorithm

Brute force algorithm to solve the Motif Finding problem

running time of $O(l \cdot n^l)$

Cannot run it on biological samples

Faster greedy technique-not

correct, good performance

Approximation algorithm

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CONSENSUS- as good or better

GREEDYMOTIFSEARCH scans each DNA sequence only once. Once we have scanned a particular sequence i , we decide which of its l -mer has the

best contribution to the partial alignment score $\text{Score}(s, i, \text{DNA})$ for the first i sequences and immediately claim that this l -mer is part of the alignment.

GREEDYMOTIFSEARCH (DNA, t, n, l)

```
1. bestMotif (1, 1, ..., 1)
2. s (1, 1, ..., 1)
3. for  $s_1$  1 to  $n - l + 1$ 
4.   for  $s_2$  1 to  $n - l + 1$ 
5.     if  $\text{Score}(s, 2, \text{DNA}) > \text{Score}(\text{bestMotif},$ 
         $2, \text{DNA})$ 
6.        $\text{BestMotif}_1 \leftarrow s_1$ 
7.        $\text{BestMotif}_2 \leftarrow s_2$ 
8.    $s_1 \leftarrow \text{BestMotif}_1$ 
9.    $s_2 \leftarrow \text{BestMotif}_2$ 
10. for  $i$  3 to  $t$ 
11.   for  $s_i$  1 to  $n - l + 1$ 
12.     if  $\text{Score}(s, i, \text{DNA}) > \text{Score}(\text{bestMotif},$ 
         $i, \text{DNA})$ 
13.        $\text{BestMotif}_i \leftarrow s_i$ 
14.        $s_i \leftarrow \text{bestMotif}_i$ 
15. return bestMotif
```

Approximation algorithm

Two closest l -mers

$2 \times l$ seed matrix

$l(n - l + 1)^2$ operations

$t - 2$ iterations-by scanning the i th sequence (for $3 \leq i \leq t$)

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$t-2$ sequences and selecting the one l -mer that has the maximumScore(s, i)

$l \cdot (n - l + 1)$ operations

Approximation algorithm

Running time of this algorithm is $O(ln^2 + lnt)$, which is vastly better than the $O(lnt)$ of IMLEMOTIFSEARCH or even the $O(4lnt)$ of BRUTEFORCEMEDIAN-STRING

Topic # 58

The Power of DNA Sequence Comparison

Introduction

Discovery of new gene-no idea of functions

Find similarities with genes of known function

Newly discovered cancer-causing -sis oncogene matched a normal gene involved in growth and development called platelet-derived growth factor

Oncogene v-sis is the simian sarcoma virus

Scientists became suspicious that cancer might be caused by a normal growth gene

Discovery of cystic fibrosis gene

Abnormal secretions, and is diagnosed in children
produce abnormally thick mucus that clogs the lungs

10 million Americans are carriers of the cystic fibrosis gene

There is a 25% chance that the child will have cystic fibrosis

In 1989, 1 million nucleotides on the chromosome 7

The area around the cystic fibrosis gene was sequenced

Database of all known genes

Similarities between a gene that had already been discovered, code for *adenosine triphosphate (ATP) binding proteins*

These proteins constitute the ion transport channel

The disease involves sweat secretions with abnormally high sodium content

Link between cancer-causing genes and normal growth genes and elucidating the nature of cystic fibrosis

Topic # 59

Brute Force vs Greedy Algorithm

Dynamic Programming

Manhattan Tourist Problem:

Find a longest path in a weighted grid

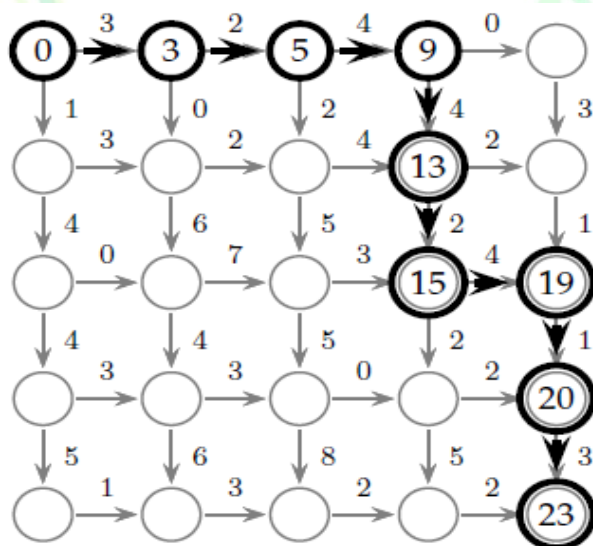
Input: A weighted grid G with
two distinguished vertices: a
source and a *sink*

Output: A longest path in G
from *source* to *sink*

Tourists only move south and east, any grid positions
west or north of the source are unusable

Any grid positions south or east of the sink are unusable, the source vertex is at $(0, 0)$

Sink vertex at (n, m) defines the southeastern most corner of the grid



The brute force approach search for the longest path

Not an option for medium to large grid

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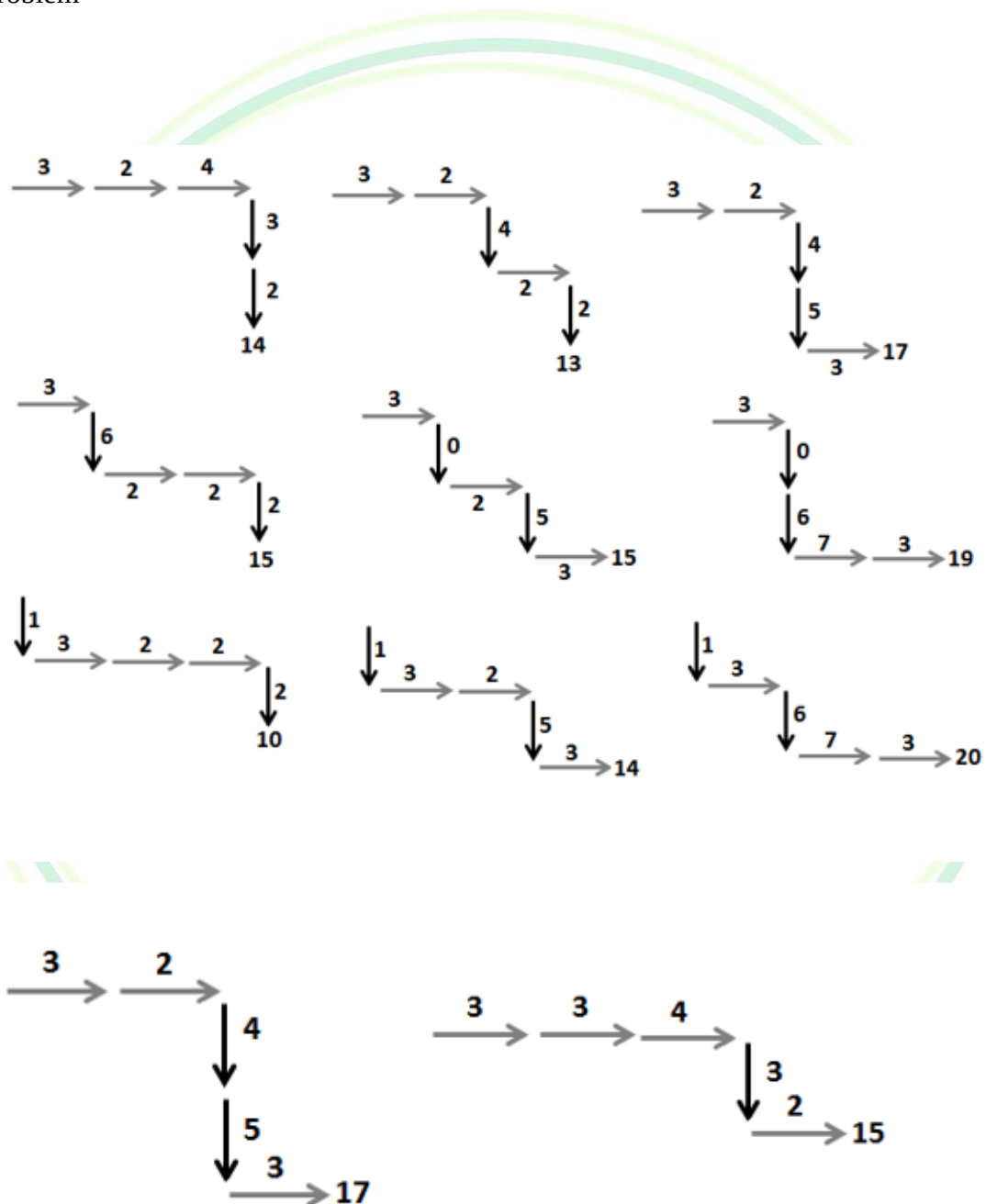
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Use a greedy strategy, choose between two possible directions (south or east) by comparing them and selecting one with large increase for one step (local maximum)

Good at beginning

May lead to area with few attractions

No known greedy strategy for the Manhattan Tourist problem provides an optimal solution to the problem



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Instead of solving the Manhattan Tourist problem directly, that is, finding the longest path from source $(0, 0)$ to sink (n, m) , we solve a more general problem: find the longest path from source to an arbitrary vertex (i, j) with

$0 \leq i \leq n, 0 \leq j \leq m$. We will denote the length of such a best path as $s_{i,j}$, noticing that $s_{n,m}$ is the weight of the path that represents the solution to the Manhattan Tourist problem

Topic #60

Sequence Similarity

Meaning of “sequence similarity or “distance” between DNA sequences. Hamming distance is not typically used to compare DNA or protein sequences. Calculation rigidly assumes that the i th symbol of one sequence is already aligned against the i th symbol of the other

It is common case that the i th symbol in one sequence corresponds to a symbol at different position in other. Mutation in DNA-evolutionary process: DNA replication- substitutions, insertions, and deletions of nucleotides, leads to “edited” DNA texts. Whether the i th symbol in one DNA sequence corresponds to the i th symbol in the other

ATATATAT and TATATATA

A T A T A T A T -
- T A T A T A T A

align the $(i+1)$ -st letter in ATATATAT against the i th letter in TATATATA for $1 \leq i \leq 7$

ATATATAT and TATAAT- subtle similarities

1	2	3	4	5	6	7	8
A	T	A	T	A	T	A	T
-	T	A	T	A	-	A	T
	1	2	3	4		5	6

Topic # 61

Edit Distance

Edit distance between two strings as the minimum number of editing operations needed to transform one string into another, where the edit operations are insertion, deletion, and substitution of one symbol for another

It is often the case that the i th symbol in one sequence corresponds to a symbol at different position in other. Mutation in DNA-evolutionary process: DNA replication- substitutions, insertions, and deletions of nucleotides, leads to “edited” DNA texts. Whether the i th symbol in one DNA sequence corresponds to the i th symbol in the other



A T A T A T A T -
- T A T A T A T A

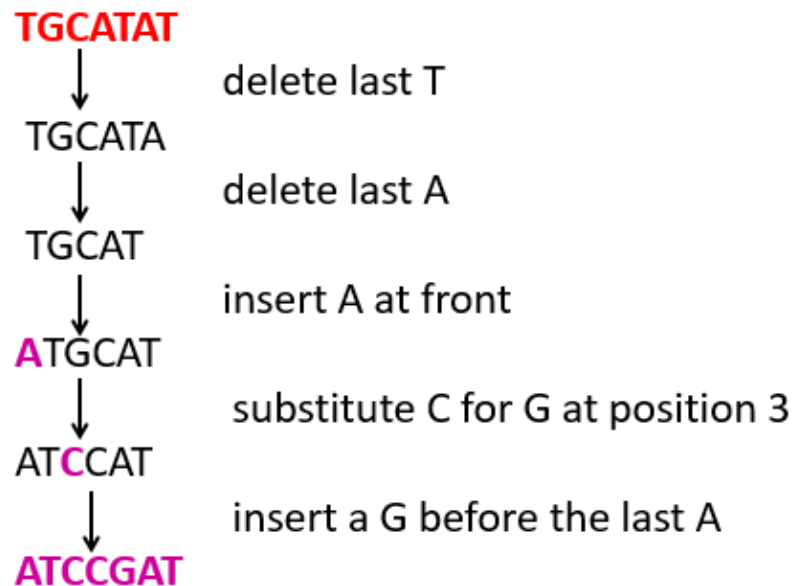
align the $(i+1)$ -st letter in ATATATAT against the i th letter in TATATATA for $1 \leq i \leq 7$



1 2 3 4 5 6 7 8
A T A T A T A T
- T A T A - A T
1 2 3 4 5 6

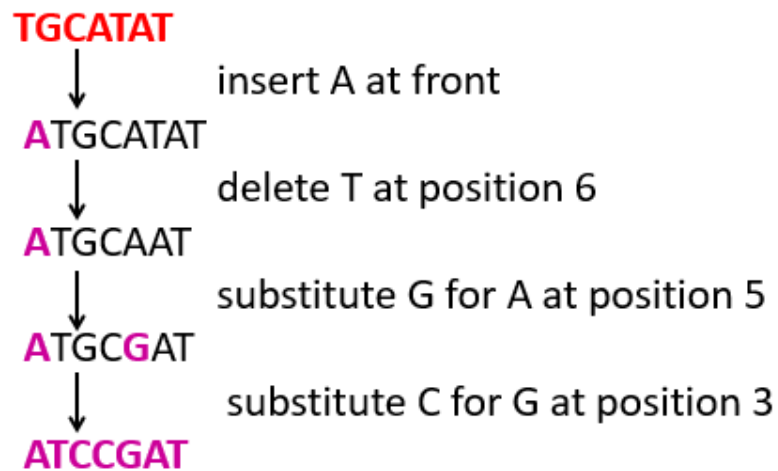
sign of bright future

TGCATAT can be transformed into **ATCCGAT** with five editing operations



Edit distance between TGCATAT and ATCCGAT is at most 5

TGCATAT can also be transformed into **ATCCGAT** with four editing operations



Edit distance between TGCATAT and ATCCGAT is 4

Topic # 62

Alignment

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The *alignment* of the strings v (of n characters) and w (of m characters, with m not necessarily the same as n) is a two-row matrix such that the first row contains the characters of v in order while the second row contains the characters of w in order, where spaces may be interspersed throughout the strings in different places

As a result, the characters in each string appear in order, though not necessarily adjacently.

No column of the alignment matrix contains spaces in both rows, so that the alignment may have at most $n + m$ columns.

A	T	--	G	T	T	A	T	--
A	T	C	G	T	--	A	--	C

Columns that contain the same letter in both rows are called *matches*, while columns containing different letters are called *mismatches*. The columns of the alignment containing one space are called *indels*, with the columns containing a space in the top row called *insertions* and the columns with a space in the bottom row *deletions*. Five matches, zero mismatches, and four indels. The number of matches plus the number of mismatches plus the number of indels is equal to the length of the alignment matrix and must be smaller than $n + m$

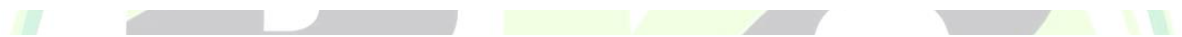
A	T	--	G	T	T	A	T	--
A	T	C	G	T	--	A	--	C

Each of the two rows in the alignment matrix is represented as a string interspersed by space symbols “--”; for example AT--GTTAT-- is a representation of the row corresponding to $v = \text{ATGTTAT}$, while ATCGT--A--C is a representation of the row corresponding to $w = \text{ATCGTAC}$

0	1	2	2	3	4	5	6	7	7
	A	T	--	G	T	T	A	T	--
	A	T	C	G	T	--	A	--	C
0	1	2	3	4	5	5	6	6	7

AT--GTAT-- is **1 2 2 3 4 5 6 7 7**, which shows the number of symbols of **v** present up to a given position. Similarly, ATCGT-A-C is represented as **1 2 3 4 5 5 6 6 7**. When both rows of an alignment are represented in this way the resulting matrix is

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix} \begin{pmatrix} 5 \\ 5 \end{pmatrix} \begin{pmatrix} 6 \\ 6 \end{pmatrix} \begin{pmatrix} 7 \\ 6 \end{pmatrix} \begin{pmatrix} 7 \\ 7 \end{pmatrix}$$



$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix} \begin{pmatrix} 5 \\ 5 \end{pmatrix} \begin{pmatrix} 6 \\ 6 \end{pmatrix} \begin{pmatrix} 7 \\ 6 \end{pmatrix} \begin{pmatrix} 7 \\ 7 \end{pmatrix}$$

$$\begin{aligned} (0, 0) &\rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (3, 4) \\ (4, 5) &\rightarrow (5, 5) \rightarrow (6, 6) \rightarrow (7, 6) \rightarrow (7, 7) \end{aligned}$$

from (0,0) to (n, m) in that grid

Topic # 63

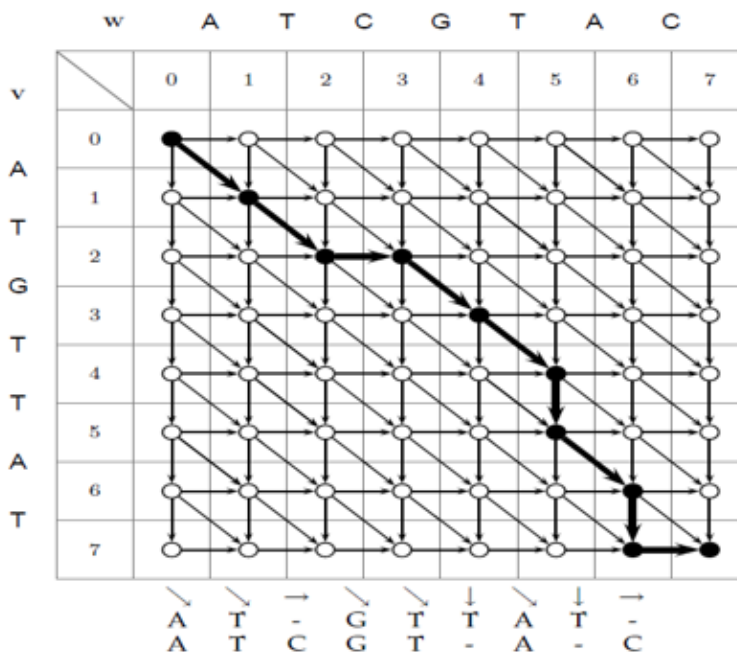
Edit Graph

The grid that is achieved after alignment is similar to the Manhattan grid where each entry in the grid looks like a city block

The graph called Edit Graph

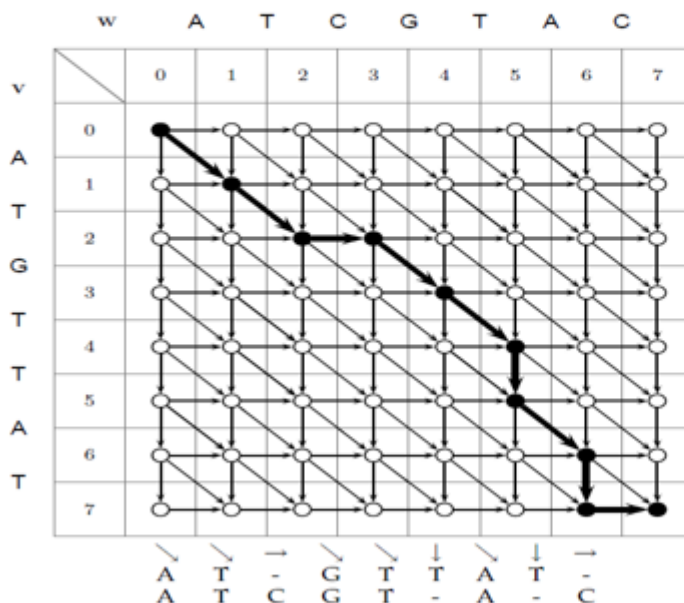
The main difference between the Manhattan and Edit Graph is that we can move along the diagonals in the Edit Graph

$(0, 0) \rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (3, 4)$
 $(4, 5) \rightarrow (5, 5) \rightarrow (6, 6) \rightarrow (7, 6) \rightarrow (7, 7)$

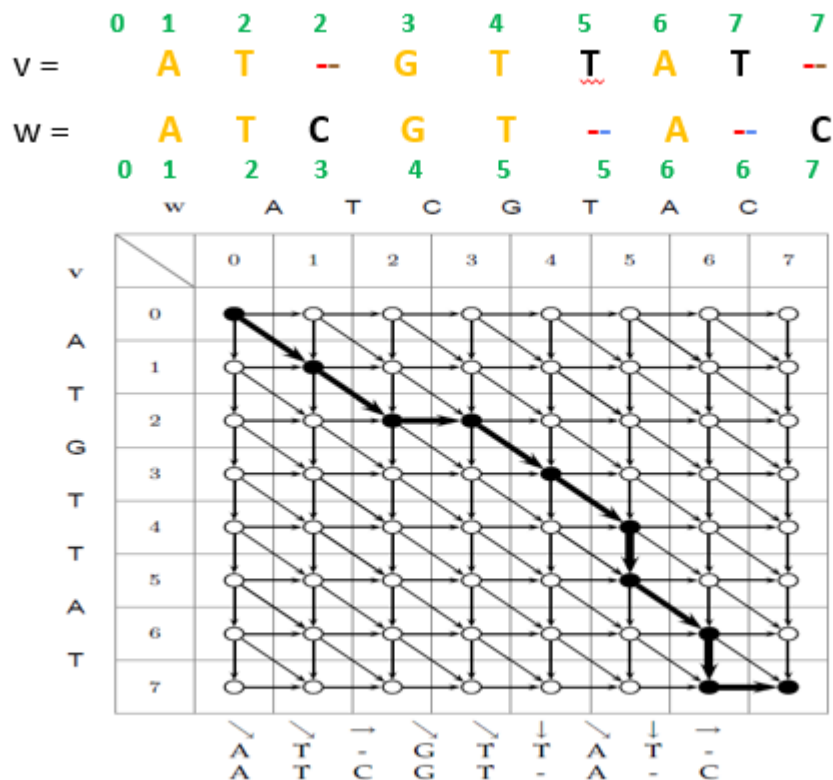


Edit Graph by introducing a vertex for every intersection of streets in the grid.
 This graph will help in calculating edit distance
 Alignment-path
 Path-Alignment
 Edge-one column

A T - G T T A T -
 $\begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix} \begin{pmatrix} 5 \\ 5 \end{pmatrix} \begin{pmatrix} 6 \\ 6 \end{pmatrix} \begin{pmatrix} 7 \\ 6 \end{pmatrix} \begin{pmatrix} 7 \\ 7 \end{pmatrix}$
 A T G C T - A - C



Diagonal edges in the path that end at vertex (i, j) in the graph correspond to the column $\begin{pmatrix} v_i \\ w_j \end{pmatrix}$, horizontal edges corresponds to $\begin{pmatrix} - \\ w_j \end{pmatrix}$, Vertical edges correspond to $\begin{pmatrix} v_i \\ - \end{pmatrix}$



Topic # 64

Edit Graph

Analyzing the merit of an alignment-corresponding path in the edit graph. For any two strings- different alignment matrices and corresponding paths.

Surplus of mismatches and indels and a small number of matches, while others have many matches and few indels and mismatches.

Relative merits of one alignment over another-scoring function- input an alignment matrix and produces a score that determines the “goodness” of the alignment.

Variety of scoring functions-higher scores to alignments with more matches.

Column as a positive number if both letters are the same, and as a negative number if the two letters are different. The score for the whole alignment is the sum of the individual column scores.

Topic # 65

Longest Common Subsequences

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The simplest form of a sequence similarity analysis is the Longest Common Subsequence (LCS) problem, where we eliminate the operation of substitution and allow only insertions and deletions. A subsequence of a string v is simply an (ordered) sequence of characters (not necessarily consecutive) from v .

$v = \text{ATTGCTA}$ AGCA and ATTA are subsequences

TGTT and TCG are not subsequences

A common subsequence of two strings is a subsequence of both of them. Common subsequence of strings

$v = v_1 \dots v_n$ and $w = w_1 \dots w_m$

as a sequence of positions in v ,

$1 \leq i_1 < i_2 < \dots < i_k \leq n$

and a sequence of positions in w ,

$1 \leq j_1 < j_2 < \dots < j_k \leq m$

such that the symbols at the corresponding positions in v and w coincide:

$v_{i_t} = w_{j_t}$ for $1 \leq t \leq k$

TCTA is a common to both ATCTGAT and TGCATA

Typically many common subsequences between two

strings v and w —how to find the longest one

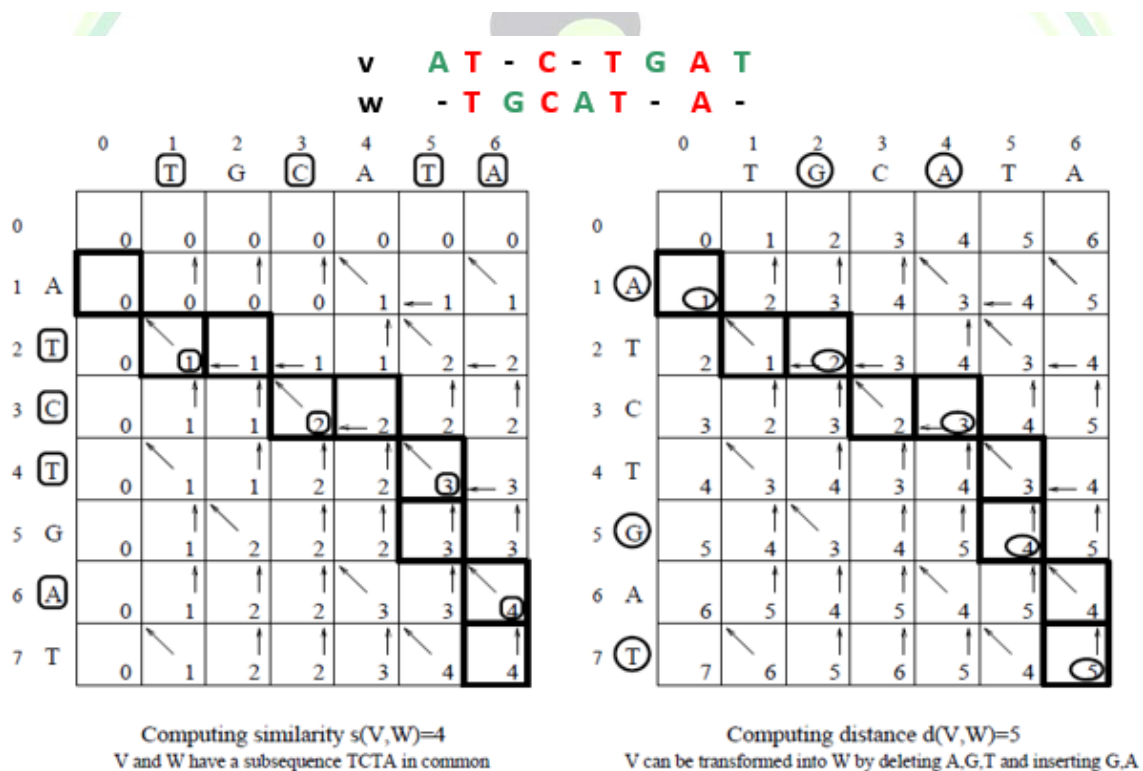
$s(v,w)$ —be the length of the longest common subsequence of v and w

edit distance between v and w —under the assumption that only insertions and deletions are allowed—is $d(v,w) = n + m - 2s(v,w)$

and corresponds to the minimum number of insertions and deletions needed to transform v into w

Alignment **v** — **A** **T** - **C** - **T** **G** **A** **T**
 w — - **T** **G** **C** **A** **T** - **A** -

Above figure presents an LCS of length 4 for the strings $v = \text{ATCTGAT}$ and $w = \text{TGCATA}$ and a shortest sequence of two insertions and three deletions transforming v into w



Longest Common Subsequence Problem:

Find the longest subsequence common to two strings

Input: Two strings, v and w

Output: The longest common subsequence of v and w

Topic # 66

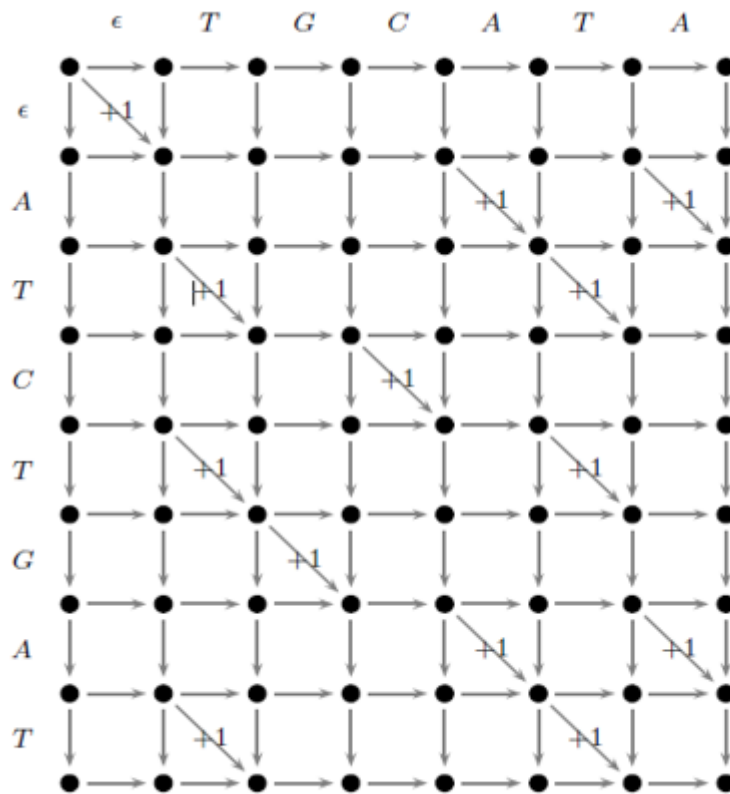
Recurrence for LCS problem

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Every common subsequence corresponds to an alignment with no mismatches.

This can be obtained simply by removing all diagonal edges from the edit graph whose characters do not match, thus transforming it into a graph like that shown in the figure



An LCS Edit Graph

The relationship between

the Manhattan Tourist problem and the LCS Problem is further illustrate by showing that these two problems lead to very similar recurrences

Define $s_{i,j}$ to be the length of an LCS between $v_1 \dots v_i$, the i -prefix of v and $w_1 \dots w_j$, the j -prefix of w . Clearly, $s_{i,0} = s_{0,j} = 0$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$

$s_{i,j}$ satisfies the following recurrence

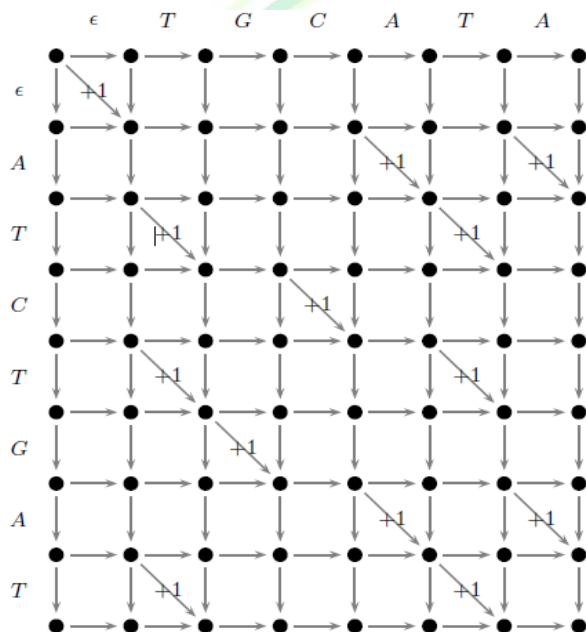
$$s_{i,j} = \max \begin{cases} s_{i-1,j} \\ s_{i,j-1} \\ s_{i-1,j-1} + 1, \text{ if } v_i = w_j \end{cases}$$

The first term- when v_i is not present in the LCS

of the i -prefix of v and j -prefix of w (deletion of v_i); the second term- when w_j is not present in this LCS (an insertion of w_i); and the third term-when both v_i and w_j are present in the LCS (v_i matches w_j).

These recurrences can be rewritten by adding some zeros here and there as

$$s_{i,j} = \max \begin{matrix} s_{i-1,j} + 0 \\ s_{i,j-1} + 0 \\ s_{i-1,j-1} + 1, \text{ if } v_i = w_j \end{matrix}$$



‘s’ is used to represent dynamic programming table, the data structure

The length of an LCS between v and w can be read from the element (n,m) of the dynamic programming table, but to reconstruct the LCS from the dynamic programming table, one must keep some additional information about which of the three quantities, $s_{i-1,j}$, $s_{i,j-1}$, or $s_{i-1,j-1} + 1$, corresponds to the maximum in the recurrence for $s_{i,j}$.

Topic #67

Algorithms for LCS

The length of an LCS

Some additional information about which of the three quantities, $s_{i-1,j}$, $s_{i,j-1}$, or $s_{i-1,j-1} + 1$

The following algorithm achieves this goal by introducing backtracking pointers that take one of the three values, or .

LCS(v,w)

1. for i 0 to n
2. $s_{i,0}$ 0
3. for j 1 to m
4. $s_{0,j}$ 0
5. for i 1 to n
6. for j 1 to m
7. $s_{i,j} = \max \begin{matrix} s_{i-1,j} \\ s_{i,j-1} \\ s_{i-1,j-1} + 1, \text{ if } v_i = w_j \end{matrix}$
8. if $s_{i,j} = s_{i-1,j}$ $b_{i,j} = 0$
9. if $s_{i,j} = s_{i,j-1}$ $b_{i,j} = 1$
10. if $s_{i,j} = s_{i-1,j-1} + 1$ $b_{i,j} = 2$
11. return $(s_{n,m}, b)$

PRINTLCS(v,w, i,j)

1. for $i = 0$ or $j = 0$
2. return
3. $b_{i,j} =$
4. PRINTLCS (b, v, $i-1, j-1$)
5. print v_i
6. Else
7. if $b_{i,j} =$
8. PRINTLCS (b, v, $i-1, j$)
9. Else
10. PRINTLCS (b, v, $i, j-1$)

	0	1	2	3	4	5	6
		T	G	C	A	T	A
0		0	0	0	0	0	0
1 A	0	0	0	0	1	1	1
2 T	0	1	1	1	1	2	2
3 C	0	1	1	2	2	2	2
4 T	0	1	1	2	2	3	3
5 G	0	1	2	2	2	3	3
6 A	0	1	2	2	3	3	4
7 T	0	1	2	2	3	4	4

Computing similarity $s(V,W)=4$
V and W have a subsequence TCTA in common

	0	1	2	3	4	5	6	
		T	G	C	A	T	A	
0		0	1	2	3	4	5	6
1 A	0	1	2	3	4	3	4	5
2 T	2	1	2	3	4	3	4	
3 C	3	2	3	2	3	4	5	
4 T	4	3	4	3	4	3	4	
5 G	5	4	3	4	5	4	5	
6 A	6	5	4	5	4	5	4	
7 T	7	6	5	6	5	4	5	

Computing distance $d(V,W)=5$
V can be transformed into W by deleting A,G,T and inserting G,A

Dynamic programming table-computation of the similarity score $s(v,w)$ between v and w , while the table on the right-computation of the edit distance between v and w -insertions and deletions are the only allowed operations. The edit distance $d(v,w)$ is computed according to the initial conditions $d_{i,0} = i$, $d_{0,j} = j$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$ and the following recurrence:

The edit distance $d(v,w)$ is computed according to the initial conditions $d_{i,0} = i$, $d_{0,j} = j$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$ and the following recurrence:

$$d_{i,j} = \min \begin{cases} d_{i-1,j} + 1 \\ d_{i,j-1} + 1 \\ d_{i-1,j-1}, & \text{if } v_i = w_j \end{cases}$$

Topic #68

Scoring Alignments

Scoring matrices for DNA sequence- μ and σ

Scoring matrices for protein sequences are complicated – *pointed accepted mutation* (PAM) and *block substitution* (BLOSUM), frequency amino acid 'x' replaces amino acid 'y' in evolutionary related sequences.

Random mutagenesis-change amino acid sequence

Some mutations- do not alter but other do

Some amino acid substitutions are commonly found through the process of molecular evolution-

Asparagine (Asp), Serine (Ser)

Glutamate (Glu), Aspartate (Asp)

More mutable

Cysteine (Cys), Tryptophan (Trp)

Probability

Ser Phe 3 times more Try Phe

Types of changes- most and least common- amino acid scoring matrix, sequence alignment

Amino acids sequences- very few matches-scoring matrix $\delta(i, j)$ – how often a. a ‘i’ substitutes a. a ‘j’

Topic # 69

Scoring Alignments

Large set of alignments of related sequences

Computing $\delta(i, j)$

Amounts to counting how many times the amino acid ‘i’ is aligned with amino acid ‘j’

Needs to know scoring matrix

Met Ala Phe Ser Gly Asp Glu Ser.

Met Ala Phe Ser -- Asp Glu Ser.

If proteins are 90% identical, premium +1 for matches and -1 for mismatches and indels will do the job. Then “obvious” alignments are constructed that are used to compute scoring matrix δ .

sign of bright future

The simplified description hides subtle details are important in the construction of scoring matrix

Ser Phe Try Phe

(related proteins in mouse and rat) LESS (related proteins in mouse and human)

15 million years

80 million years

The best scoring matrix to compare two proteins depends on similarity of these organisms

Topic #70

PAM matrix

Problem is rectified- analyzing similar proteins e.g, one mutation/100 a.a. Proteins- human and chimpanzee, Such sequences, *one PAM unit diverged*. PAM unit as the amount of time in

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which an “average” protein mutates 1% of it’s a.a. *PAM1* scoring matrix- many alignments of extremely similar **proteins**

For a given set of base alignments

define $f(i,j)$ as the total number of times amino acids i and j are aligned against each other, divided by the total number of aligned positions.

Also define $g(i,j)$ as $\frac{f(i,j)}{f(i)}$ where $f(i)$ is the frequency of the amino acid i in all proteins from data set.

$g(i,j)$ defines the probability that an amino acid i mutates into amino acids j within 1 PAM unit. The (i,j) entry of the PAM 1 matrix is defined as $\delta(i,j) =$

$\log \frac{f(i,j)}{f(i) \cdot f(j)} = \log \frac{g(i,j)}{f(j)}$ ($f(i)$, $f(j)$, the frequency of aligning amino acid i against amino acid j that one expects simply by chance). The *PAM n* matrix can be defined as the result of applying the PAM 1 matrix n times

If g is the 20 x 20 matrix of frequencies $g(i,j)$, then g^n (multiplying this matrix by itself n times) gives the probability that amino acid i mutates into amino acid j during n PAM units.

The (i,j) entry of the *PAM n* matrix is defined as $\log \frac{g^n(i,j)}{f(j)}$

For large n , the resulting PAM matrices often allow one to find related proteins even when the practically no matches in the alignment. In this case the underlying nucleotide sequences are so diverged that their comparison usually fails to find any statistically significant similarities.

Similarity between the

Cancer –causing v-sis oncogene

Growth factor PDGF (Platelet derived growth factor)

Remained undetected

Russell Doolittle and colleagues has not transformed the nucleotide sequences into amino acids sequences prior to performing the comparison.

Topic #71

Local Sequence Alignment

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Global Alignment- similarity 2 strings- extends over entire length, eg, same protein family- very conserved, same length – fruit flies to humans Score of alignment- 2 substrings ‘v’, ‘w’- larger than score of alignment between entire length of ‘v’ and ‘w’.

Homeobox genes, regulate embryonic development- variety of species. Different in different species- one region- *homeodomain*- highly conserved. How to find conserved area- ignore areas-little similarity. Temple Smith and Micheal Waterman proposed modification of the global sequence alignment dynamic programming algorithm-LAP



```

v  -- T--CC-C- AGT -- TATGT - CAGGGGACACG -- A - GCATGCAGA-GAC
      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
w  AATTGCCGCC-GTCGT-T- T T CAG - - - - CA- GT T A T G -T- CAGAT-- C
  
```

CAGTTATGTCCAG

```

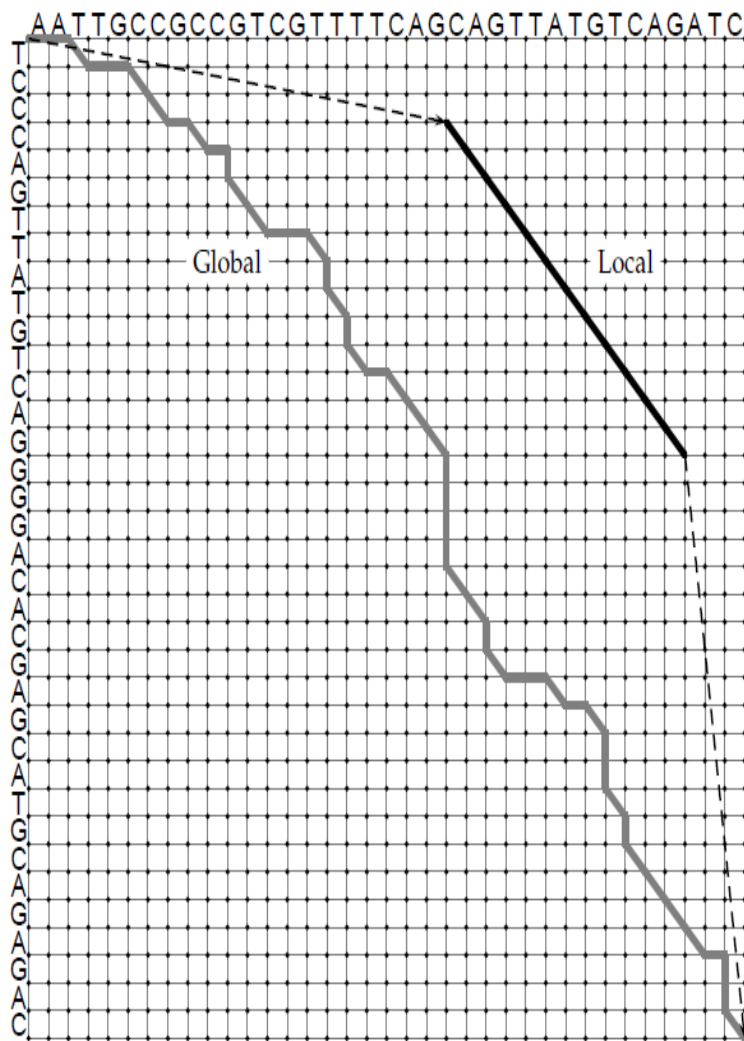
v                                tccCAGTTATGTCAGgggacacgagcatgcagagac
      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
w  aattgccgccgctcgttttcagCAGTTATGTCAGatc
  
```



Mates
sign of bright future

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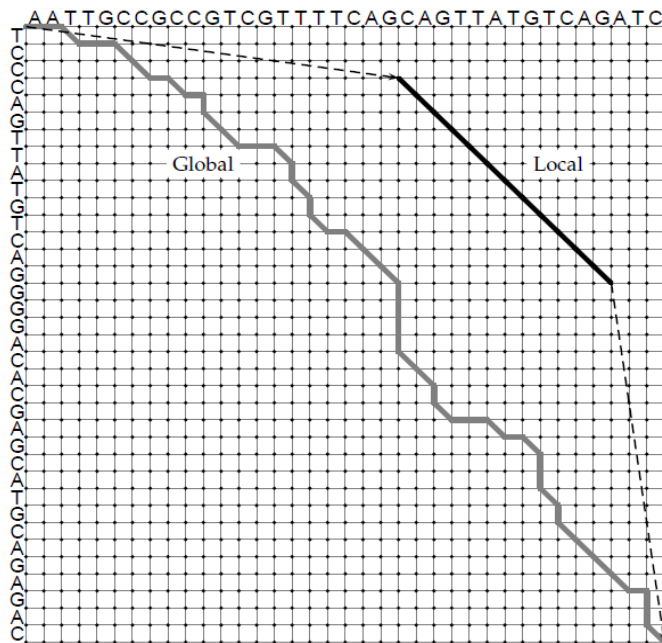
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The conserved domain are identical and cover one third of the entire length, n , of these genes. Path from *source* to *sink* will include approx $\frac{2}{3}n$ horizontal edges, $\frac{1}{3}n$ diagonal match edges (corresponding to homeodomains) and $\frac{2}{3}n$ vertical edges

The score of this path is

$$-\frac{2}{3}n\sigma + \frac{1}{3}n - \frac{2}{3}n\sigma = n \left(\frac{1}{3} - \frac{4}{3}\sigma \right)$$



The score of this path is

$$-\frac{2}{3}n\sigma + \frac{1}{3}n - \frac{2}{3}n\sigma = n\left(\frac{1}{3} - \frac{4}{3}\sigma\right)$$

This path contains so many indels that it is unlikely to be the highest scoring alignment. Biologically irrelevant diagonal paths – likely have score- mismatches are

penalized less than indels. Score of diagonal path is $n(\frac{1}{4} - \frac{3}{4}\mu)$, since every diagonal edge $\frac{1}{4}$ and mismatch with probability $\frac{3}{4}\mu$. Since $(\frac{1}{3} - \frac{4}{3}\sigma) < (\frac{1}{4} - \frac{3}{4}\mu)$ for indels and mismatch penalties, global alignment- miss correct solution of real biological problem, biologically irrelevant near-diagonal path.

Topic # 72 ppts is same as topic # 71 in LMS.

Topic #73

Local Alignment Problem

Biological significant in certain parts of DNA fragments, maximize the alignment score $s(v_i \dots v_{i'}, w_j \dots w_{j'})$ over all substrings $v_i \dots v_{i'}$ of v and $w_j \dots w_{j'}$ of w . Local Alignment Problem- not extend over entire length as in Global Alignment Problem

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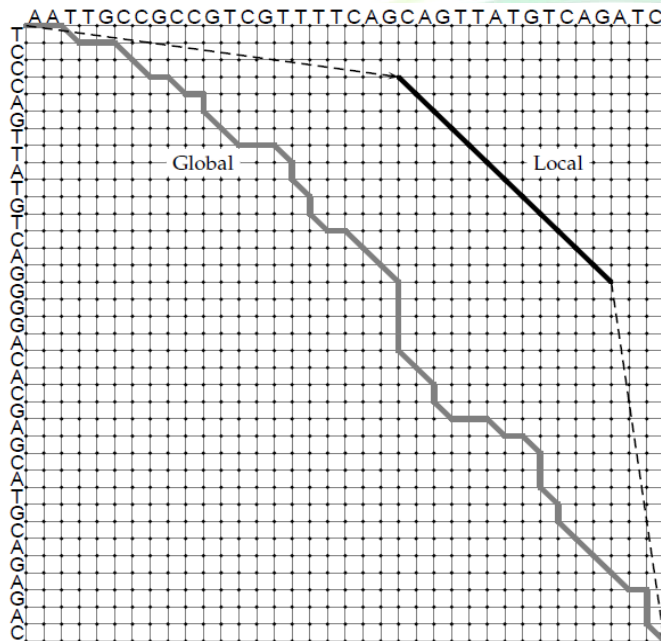
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Local Alignment Problem:

Find the best local alignment between two strings

Input: Strings v and w and a scoring matrix δ

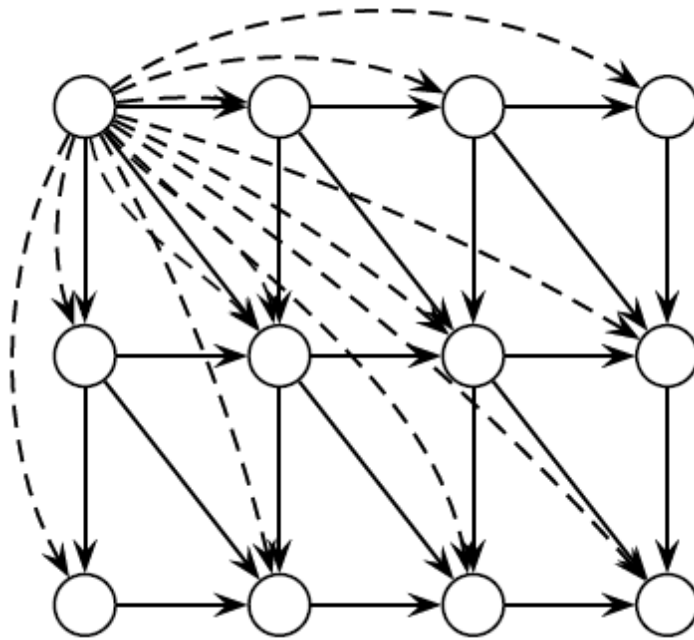
Output: Substring of v and w whose global alignment, as defined by δ , is maximal among all global alignment of all substrings of v and w .



Global Sequence Alignment- finding the longest path between vertices $(0,0)$ and (n,m) in the edit graph

Local Alignment-finding the longest path among the paths between arbitrary vertices (i, j) and (i', j') in the edit graph.

Find the longest path between every pair of vertices (i, j) and (i', j') - then select longest of these computed paths. Instead of finding the longest path from (i, j) to (i', j') , LAP- finding the longest path from the source $(0,0)$ to every other vertex by adding edges to weight 0



These edges make the source vertex (0,0) a predecessor provide a “free ride” from the source to any other vertex (i, j) . Following recurrence reflects the transformation of the edit graph

$$s_{i,j} = \max \begin{cases} 0 \\ s_{i-1,j} + \delta(v_i, -) \\ s_{i,j-1} + \delta(-, w_j) \\ s_{i-1,j-1} + \delta(v_i, w_j) \end{cases}$$

Topic #74

Progressive Multiple Alignment

Another approach-strong pairwise alignment Greedy progressive multiple alignment heuristic-pair of strings with greatest similarity-new string- “once a gap, always a gap.” Multiple alignment of k sequences is reduced to the multiple alignment of $k-1$ sequences.

The motivation for the choice of the closest strings at the early steps of the algorithm is that close strings often provide the most reliable information about a real alignment

Many popular iterative multiple alignment algorithms including the tool CLUSTAL, use similar strategies

ATGTCATATTCGGAC
 ATGTCATATTCGGAC
 ATGTCATATTCGGAC

ATG- CATA

ATGTCATA

Progressive multiple alignment algorithms- problem with CLUSTAL-may be misled by some spuriously strong pairwise alignment effect, a bad seed. The error in initial pairwise alignment will propagate all the way through to the whole multiple alignment. Many algorithms have been proposed, even with systematic deficiencies are quite useful in computational biology

Multiple alignment for k sequences

Generalization of the Pairwise Alignment problem Existence of a k -dimensional scoring matrix

k -dimensional scoring matrices are not very practical

Describe two other scoring approaches that are

more biologically relevant. The choice of the scoring function can drastically affect the quality of the resulting alignment, and no single scoring approach

is perfect in all circumstances.

Multiple alignment of k sequences-a path of edges in a k -dimensional-Manhattan gridlike edit graph.

The weights of the edges-scoring function

Intuitively, assign higher scores to the columns with a low variation in letters-high scores-highly conserved sequences

Multiple Longest Common Subsequence problem, the score of a column is set to 1 if all the characters in the column are the same, and 0 if even one character disagrees

Topic # 75

Gene Prediction

Sydney Brenner and Francis Crick

Every triplet codes for one amino acid

Introduce deletions in DNA-dramatically alters its protein product. Deleting three consecutive nucleotides results in minor changes in the protein

The phrase

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THE SLY FOX AND THE SHY DOG

(written in triplets)

Turns into nonsense after deleting one letter Y from SLY

THE SYF OXA NDT HES HYD OG

or two letters LY from SLY

THE SFO XAN DTH ESH YDO G

but makes some sense after deleting three nucleotides SLY

THE SOX AND THE SHY DOG

Charles Yanofsky proved that a gene and its protein product are collinear. Yanofsky's experiment was so influential that nobody even questioned about codons and for almost 2 decades biologists believed that a protein was encoded by a long string of adjacent triplets.

Discovery of split human genes-collection of substrings

Raised the computational problem-prediction of genes

Human genome is larger and complex than bacterial genome

Salamander genome is ten times larger than the human genome
amounts of so-called junk DNA

Large

Human genes - exons that are separated by this junk DNA. The difference in the sizes of the salamander and human genomes thus presumably reflects larger amounts of junk DNA and repeats in the salamander genome.

Split genes are analogous to a magazine article- page 1, 13, 43, 51, 74, 80, and 91, with pages of advertising appearing in between. Junk DNA represents "advertising" that separates exons.

Topic # 76

Gene Prediction

The jump is inconsistent from species to species.

A gene in insect genome- different in worm

Number of exons

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The information in one part in human-broken up into two in the mouse.

While the genes themselves are related, they may be quite different in terms of the parts' structure

Split genes-1977

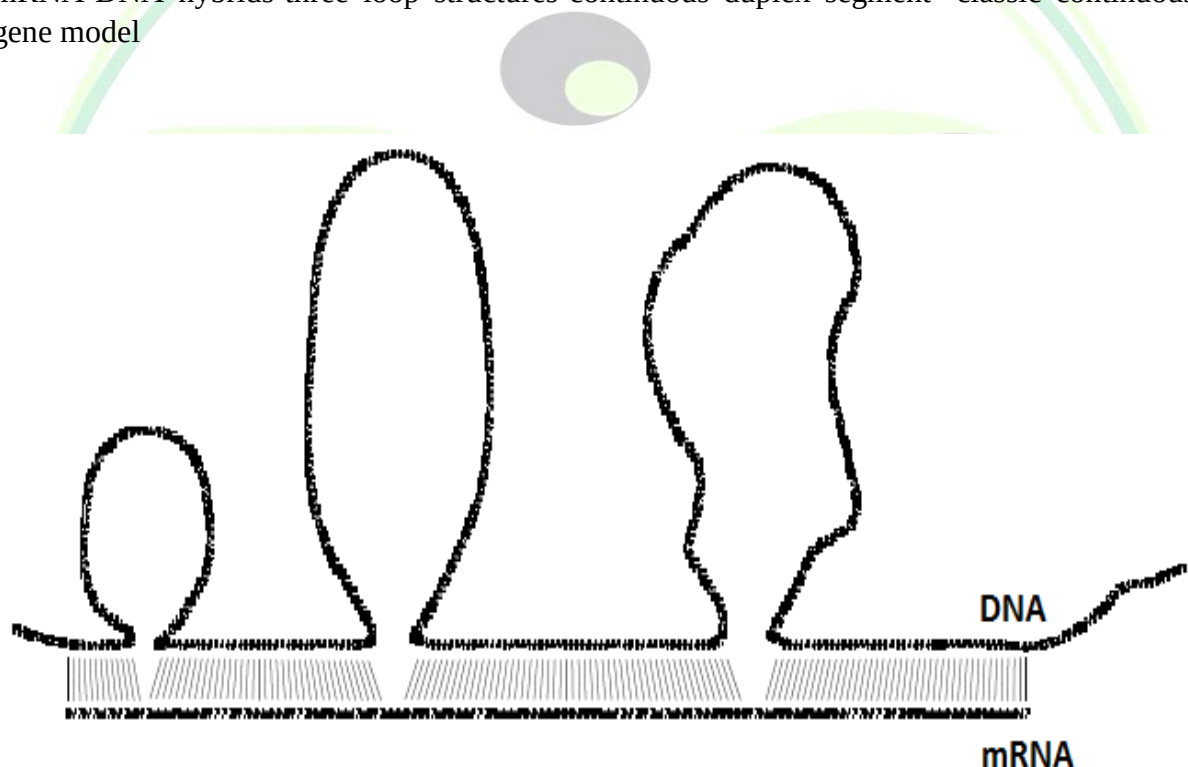
Phillip Sharp and Richard Roberts- Adenovirus

“An Amazing Sequence Arrangement at the 5' End of Adenovirus 2 Messenger RNA.”

Sharp's group- hexon.

Hexon mRNA, mRNA was hybridized to adenovirus DNA- the hybrid molecules-electron microscopy.

mRNA-DNA hybrids-three loop structures-continuous duplex segment- classic continuous gene model



An electron microscopy experiment led to the discovery of split genes

Hexon mRNA is built from four separate fragments of the adenovirus genome

These four continuous segments, exons, in the adenovirus genome are separated by three “junk” fragments called introns.

Topic # 77

Gene Prediction

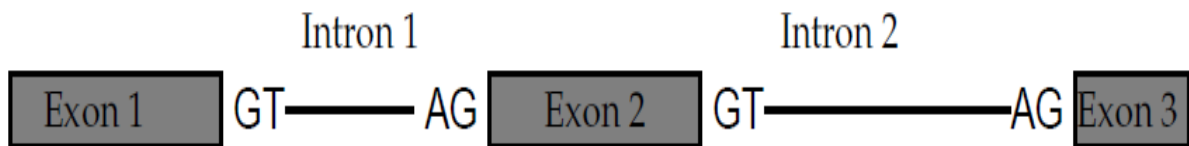
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Human genes-3% of the human genome- no in silico gene recognition algorithm The intron-exon model-in eukaryotic organisms Prokaryotic organisms Gene prediction algorithms for prokaryotes tend to be somewhat simpler than those for eukaryotes

Two categories-predicting gene location. The statistical approach to gene prediction *Splicing signals*- exon-intron junctions

Dinucleotides AG and GT on the left- and right-hand sides of an exon are highly conserved



Less conserved positions on both sides of the exons

The simplest way to represent such binding sites is by a profile describing the propensities of different nucleotides to occur at different positions

Using profiles-detect splice sites

Profiles are quite weak-match frequently in the genome at non splice sites.

Attempts to improve the accuracy of gene prediction- second category-based on similarity.

Topic #78

Second Approach for Gene Prediction

The similarity-based approach to gene prediction relies on the observation that a newly sequenced gene has a good chance of being related to one that is already known. For example, 99% of mouse genes have human analogs.

Simply look for a similar sequence-based on the genes known in another-exon sequence and the exon structure are different

The commonality-produce similar proteins

Similarity-based methods attempt to solve a combinatorial puzzle-putative exons in a genomic sequence – mouse-human

Know a human protein, and we want to discover the exon structure of the related gene in the mouse genome. The more sequence data we collect, the more accurate and reliable similarity based

methods become. Consequently, the trend in gene prediction has recently shifted from statistically motivated approaches to similarity-based algorithms

Topic # 79

Statistical Approach to Gene Prediction

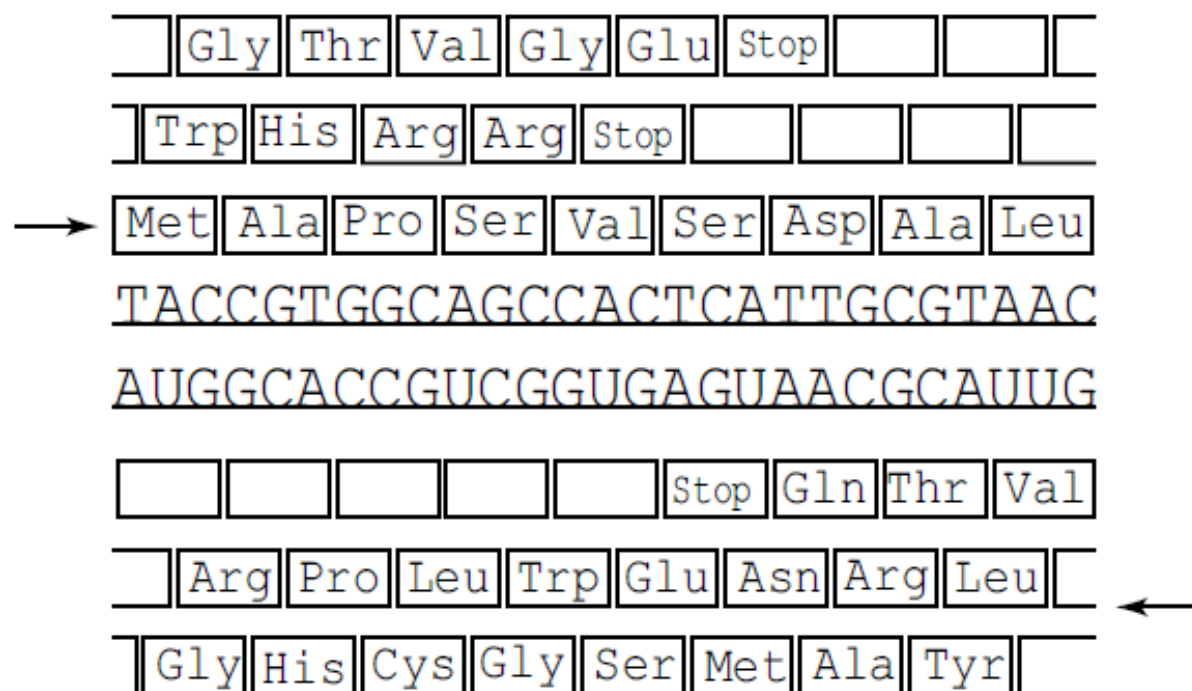
Statistical approaches to finding genes-statistical variations between coding (exons) and non-coding regions

Open reading frames (ORFs) Fenome of length n as a sequence of $n/3$ codons

The three “stop” codons, (TAA, TAG, and TGA)

The subsegments of these that start from a start codon, ATG, are ORFs.

ORFs within a single genomic sequence may overlap since there are six possible “reading frames”: three on one strand starting at positions 1, 2, and 3, and three on the reverse strand



The six reading frames for the DNA sequence

One would expect to find frequent stop codons in noncoding DNA, since the average number of codons between two consecutive stop codons in “random”

DNA should be $64/3 \sim 21$. This is much smaller than the number of codons in an average protein, which is roughly 300. Therefore, ORFs longer than some threshold length indicate potential genes. However, gene prediction algorithms based on selecting significantly long ORFs may fail to detect short genes or genes with short exons

Many statistical gene prediction algorithms rely on statistical features in protein-coding regions, such as biases in *codon usage*. We can enter the frequency of occurrence of each codon within a given sequence into a 64-element *codon usage array*

	U	C	A	G
U	UUU Phe 57	UCU Ser 16	UAU Tyr 58	UGU cys 45
	UUC Phe 43	UCC Ser 15	UAC Tyr 42	UGC cys 55
	UUA Leu 13	UCA Ser 13	UAA stp 62	UGA stp 30
	UUG Leu 13	UCG Ser 15	UAG stp 8	UGG Trp 100
C	CUU Leu 11	CCU Pro 17	CAU His 57	CGU Arg 37
	CUC Leu 10	CCC Pro 17	CAC His 43	CGC Arg 38
	CUA Leu 4	CCA Pro 20	CAA Gln 45	CGA Arg 7
	CUG Leu 49	CCG Pro 51	CAG Gln 66	CGG Arg 10
A	AUU Ile 50	ACU Thr 18	AAU Asn 46	AGU Ser 15
	AUC Ile 41	ACC Thr 42	AAC Asn 54	AGC Ser 26
	AUA Ile 9	ACA Thr 15	AAA Lys 75	AGA Arg 5
	AUG Met 100	ACG Thr 26	AAG Lys 25	AGG Arg 3
G	GUU Val 27	GCU Ala 17	GAU Asp 63	GGU Gly 34
	GUC Val 21	GCC Ala 27	GAC Asp 37	GGC Gly 39
	GUA Val 16	GCA Ala 22	GAA Glu 68	GGA Gly 12
	GUG Val 36	GCG Ala 34	GAG Glu 32	GGG Gly 15

Topic # 80

Statistical Approach to Gene Prediction

Some more facts about genetic code and codon usage in humans

	U	C	A	G
U	UUU Phe 57	UCU Ser 16	UAU Tyr 58	UGU cys 45
	UUC Phe 43	UCC Ser 15	UAC Tyr 42	UGC cys 55
	UUA Leu 13	UCA Ser 13	UAA stp 62	UGA stp 30
	UUG Leu 13	UCG Ser 15	UAG stp 8	UGG Trp 100
C	CUU Leu 11	CCU Pro 17	CAU His 57	CGU Arg 37
	CUC Leu 10	CCC Pro 17	CAC His 43	CGC Arg 38
	CUA Leu 4	CCA Pro 20	CAA Gln 45	CGA Arg 7
	CUG Leu 49	CCG Pro 51	CAG Gln 66	CGG Arg 10
A	AUU Ile 50	ACU Thr 18	AAU Asn 46	AGU Ser 15
	AUC Ile 41	ACC Thr 42	AAC Asn 54	AGC Ser 26
	AUA Ile 9	ACA Thr 15	AAA Lys 75	AGA Arg 5
	AUG Met 100	ACG Thr 26	AAG Lys 25	AGG Arg 3
G	GUU Val 27	GCU Ala 17	GAU Asp 63	GGU Gly 34
	GUC Val 21	GCC Ala 27	GAC Asp 37	GGC Gly 39
	GUA Val 16	GCA Ala 22	GAA Glu 68	GGA Gly 12
	GUG Val 36	GCG Ala 34	GAG Glu 32	GGG Gly 15

in-frame hexamer count Mark Borodovsky

Gene prediction in bacterial genomes-several conserved sequence motifs often found in the regions around the start of transcription

Such sequence motifs are more elusive in eukaryotes

The approaches-prokaryotes-eukaryotes

Exons-130 nucleotides-reliable peaks in the likelihood ratio plot while analyzing ORFs-do not differ enough from random fluctuations to be detectable. Moreover, codon usage and other statistical parameters probably nothing in common in splicing machinery recognizes exons

Topic # 81

Statistical Approach to Gene Prediction

Biologically oriented

Approach

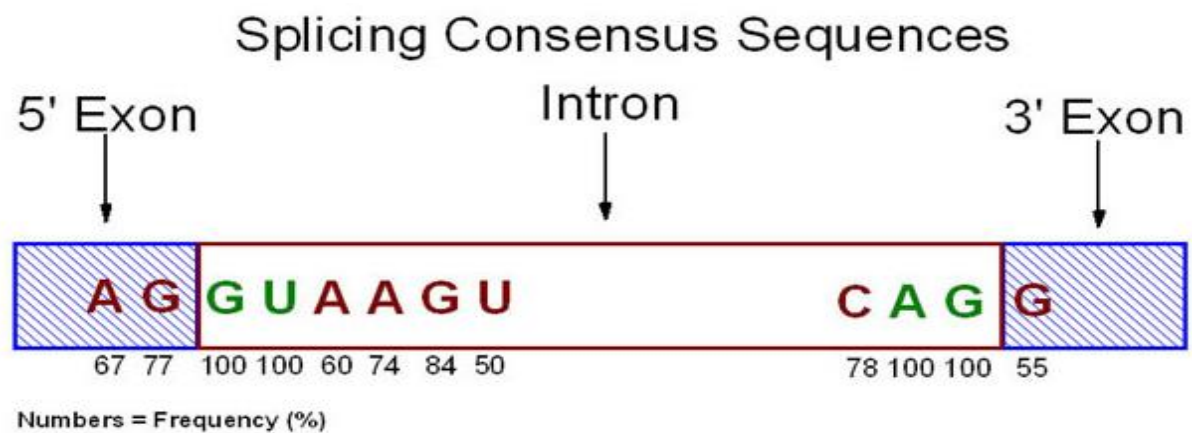
Recognize the locations of splicing signals at exon-intron junctions

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There exists a weakly conserved sequence of eight

nucleotides at the boundary of an exon and an intron (donor splice site) and a sequence of four nucleotides at the boundary of an intron and exon (acceptor splice site)



Profiles for splice sites are weak-limited success Hidden Markov Model (HMM) approaches that capture statistical dependencies between sites GENSCAN developed by Chris Burge and Samuel Karlin. GENSCAN combines coding region and splicing signal predictions into a single framework.

Splice site prediction-coding region appear on one side of the site Such statistics are used in the HMM framework of GENSCAN that merges splicing site statistics, coding region statistics, and motifs near the start of the gene The accuracy of GENSCAN decreases for genes with many short exons or with unusual codon usage

Topic # 82

Similarity Based Approached to Gene Prediction

A similarity-based approach-Previously sequenced genes Unknown genes in newly sequenced DNA fragments Combinatorial puzzle-find a set of substrings (candidate exons) whose splicing best fits the target

Brute force approach-find

all local similarities

Each substring from the genomic sequence that exhibits sufficient similarity to the target protein could be considered a putative exon

Exon-flanking dinucleotides **AG** and **GT**

Overlapping

Model a putative exon with a *weighted interval* in the genomic sequence, parameters (l, r, w)

l is the left-hand position, r is the right-hand position, and w is the weight of the putative exon
“ w ” - local alignment score

Likelihood that this interval is an exon

A chain is any set of non overlapping weighted intervals.
Total weight of a chain

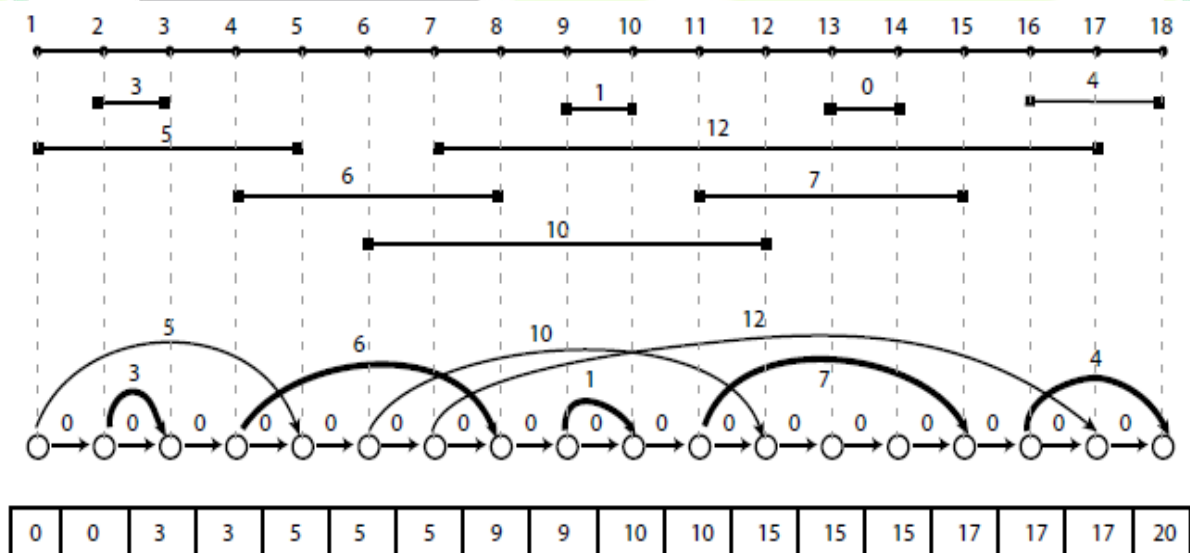
A maximum chain

Weights of all intervals are positive ($w > 0$)

Topic # 83

Similarity Based Approached to Gene Prediction

Model a putative exon with a *weighted interval* in the genomic sequence



Five weighted intervals, $(2, 3, 3)$, $(4, 8, 6)$, $(9, 10, 1)$, $(11, 15, 7)$, and $(16, 18, 4)$, shown by bold edges, form an optimal solution to the Exon Chaining problem. The array at the bottom shows the values $s_1, s_2, \dots, s_{2n}$ generated by the EXONCHANNING algorithm

Topic # 84

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ORF Prediction

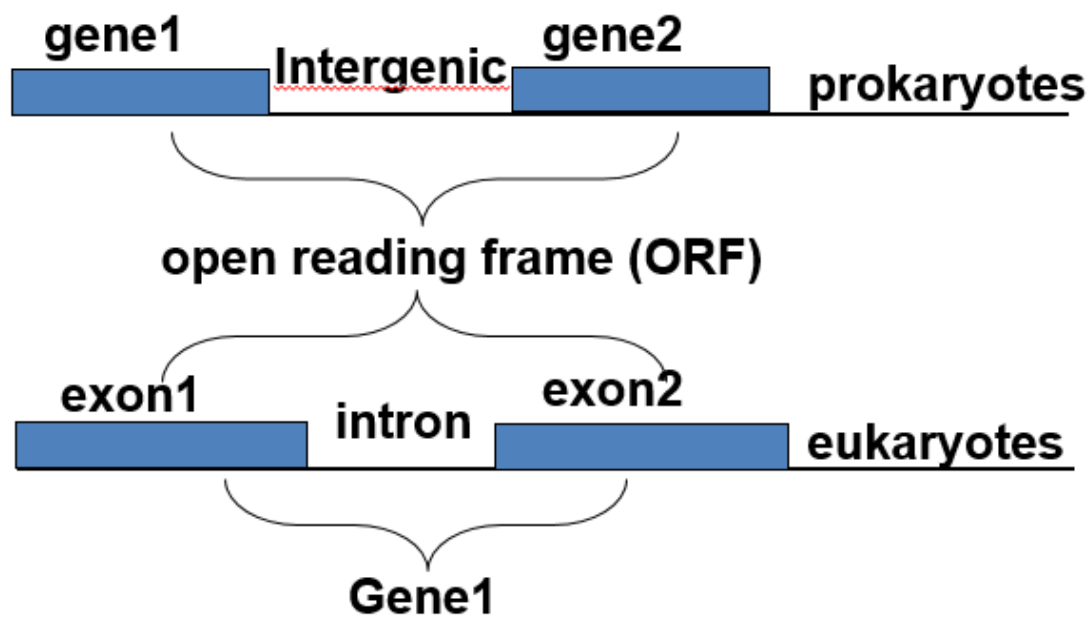
ORF (Open Reading Frame)

Gene finding, specially in prokaryotes starts form searching for an open reading frames (ORF)

ORF (Open Reading Frame)

An ORF is a sequence of DNA that starts with start codon “ATG” (not always) and ends with any of the three termination codons (TAA, TAG, TGA)

Coding regions: exons, genes



ORF and gene finding *sign of bright future*

ORF provide an important evidence in gene finding

Generally longer ORFs are preferred

However presence of ORF not necessarily means the region is translated to a functional product

Reading Frames

Depending on the start point, there are six possible ways of translating any nucleotide sequence into amino acid sequence according to the genetic code

Reading Frames

Three on forward strand and three on complementary strand

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Six Frame translation

+3 ATGGT TTTG GGA →

+2 ATGG TTTG GGA →

+1 ATG GTT TGG GA →

5'-ATGGTTTGGGAACCGAAGTCAATT-3'

3'-TACCAAACCCTTGGCTTCAGTTAA-5'

← GCTTCAGTTAA-1

← GCTTCAGTTAA-2

← GCTTCAGTTAA-3

Conclusions

An ORF is a sequence of DNA that starts with start codon “ATG” (not always) and ends with any of the three termination codons (TAA, TAG, TGA)

Biological Sequence Analysis

R Durbin, S Eddy, A Krogh and G Mitchison

Cambridge University Press, 1998.

Bioinformatics The machine learning approach

P Baldi and S Brunak *sign of bright future*

The MIT Press, 1998

Post-Genome Informatics

M Kanehisa

Oxford University Press, 2000

Topic # 85

ORF Finders

ORF Finding

Long ORF may be a gene

Expected $64/3 \sim 21$ codons before we see a stop codon

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Genes are longer than this

We might scan for ORF longer than a threshold

Codon usage and likelihood ratio

An ORF is more 'reliable' if it has 'likely' codons

We can do sliding window calculations to ORF having 'likely' codon usage

An ORF is more 'reliable' if it has 'likely' codons

However average vertebrate exon length (130 nucleotides) is too small for reliable peaks

An improvement may be;

In-Frame hexamer count

i.e frequencies of pairs

of consecutive codons

ORF Finders

Tools are mainly based on pattern finding algorithms

NCBI's ORF Finder

ORF Investigator

OrfPredictor

ORF Finder (Open Reading Frame Finder)

The ORF Finder (Open Reading Frame Finder) is a graphical analysis tool which finds all open reading frames of a selectable minimum size in a user's sequence or in a sequence already in the database. This tool identifies all open reading frames using the standard or alternative genetic codes. The deduced amino acid sequence can be saved in various formats and searched against the sequence database using the WWW BLAST server. The ORF Finder should be helpful in preparing complete and accurate sequence submissions. It is also packaged with the Sequin sequence submission software.

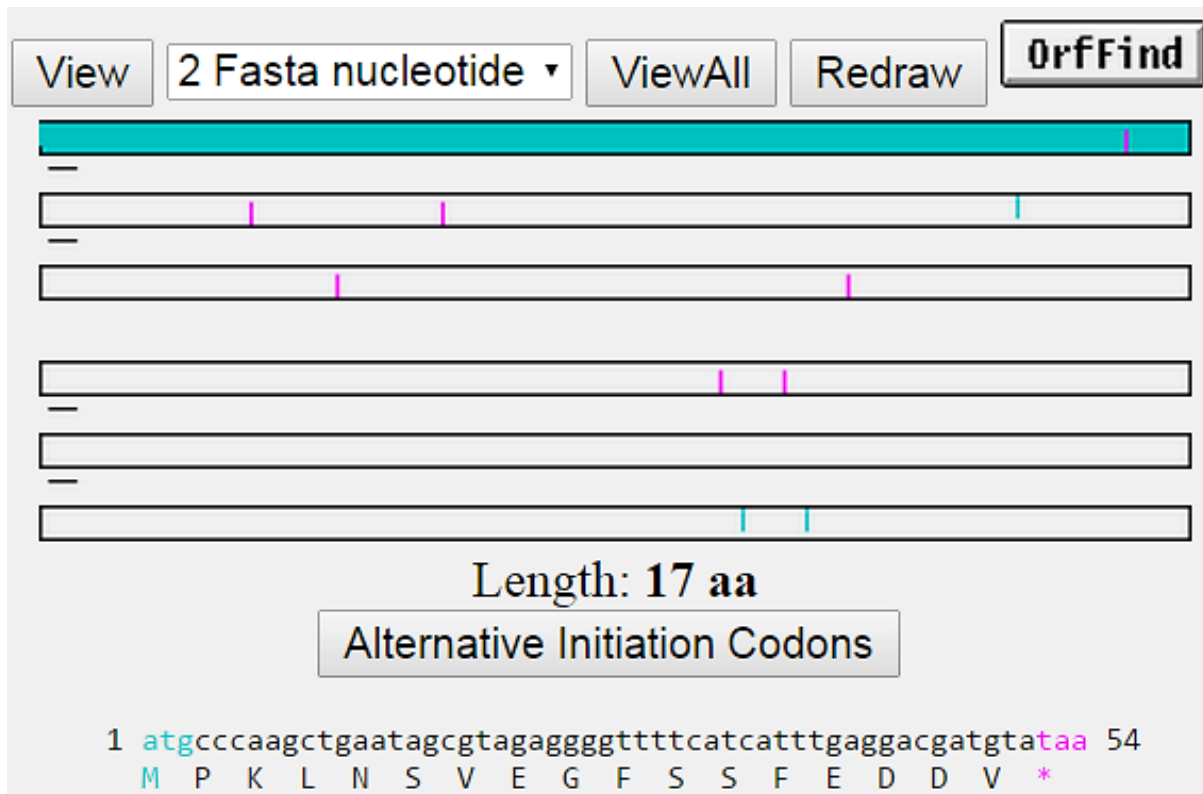
Enter GI or ACCESSION

or sequence in FASTA format

```
atgcccaagctgaatagcgtagaggggttttcattcattt
gaggacgatgtataa
```

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<https://www.facebook.com/groups/vubiomates/>



Conclusions

ORF provide an important evidence in gene finding

Generally longer ORFs are preferred

However presence of ORF not necessarily means the region is translated to a functional product

Biological Sequence Analysis

R Durbin, S Eddy, A Krogh and G Mitchison

Cambridge University Press, 1998.

Bioinformatics The machine learning approach

P Baldi and S Brunak

The MIT Press, 1998

Post-Genome Informatics

M Kanehisa

Oxford University Press, 2000

>TOPIC 86

Translation Start Site (TSS)

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We can call it **Information Content (IC)**

CACC ATA GC

$$\log (58/0.25) + \log (49/0.25) + \log (40/0.25) + \log (50/0.25) + \log (43/0.25) + \log (49/0.25) \\ = 13.69$$

TCGA ATG TT

$$\log (6/0.25) + \log (6/0.25) + \log (15/0.25) + \log (7/0.25) + \log (13/0.25) + \log (14/0.25) \\ = 9.44$$

- Build a mathematical model, based on collected translation start sequence
- For each candidate translation start sequence, apply the model and get a score
- If the score is larger than zero, predict it is a “translation start”; the higher score, the higher the probability the prediction is true

Conclusions

- TSS prediction can be an important step in gene prediction
- TSS can be predicted while using the frequency of neighboring nucleotides
- **Biological Sequence Analysis**
- R Durbin, S Eddy, A Krogh and G Mitchison
- Cambridge University Press, 1998.
- **Bioinformatics The machine learning approach**
- P Baldi and S Brunak
- The MIT Press, 1998
- **Post-Genome Informatics**
- M Kanehisa
- Oxford University Press, 2000

>TOPIC 87

Prediction of splice junctions

Splice Junctions

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Donor site

- Coding region | GT

Acceptor

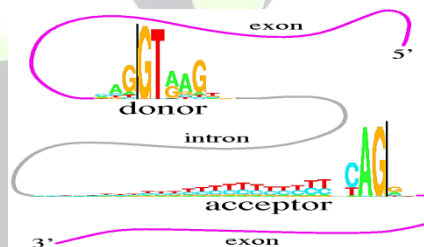
- YAG | coding region

Canonical form

- GT-AG: 99.24%

Splice Junctions

Like TSS, the flanks of splice junctions show “biased” distributions of nucleotides in certain positions. These biased distributions of nucleotides are the basis for prediction of splice junctions.



Sequence LOGOS

A visual representation of a position-specific distribution. Easy for nucleotides, but we need colour to depict up to 20 amino acid proportions.

Sequence LOGOS

Overall height at position is proportional to the information content

Sequence LOGOS

- Proportions of each nucleotide/amino acid are in relation to their observed frequency, with most frequent on top, next most frequent below
- **Non Canonical Splice Junctions**
- In addition to canonical **GT-AG (99.24%)**;
- **GC-AG: 0.69%**
- **AT-AC: 0.05%**
- **Others: 0.02%**
- **Information Content (IC)**
- $S_i = \sum \log (F_i (X)/0.25)$

- If every nucleotide has 0.25 frequency in a position, then the position's information content is ZERO
- **Information Content (IC)**
- $S_i = \sum \log (F_i (X)/0.25)$
- Use “information content as a criterion for determining the length of flanks
- **Algorithm**
- Mathematical model: $F_i (X)$: frequency of X (A, C, G, T) in position I
- Score a segment as a candidate donor/acceptor site by
- $\sum \log (F_i (X)/0.25)$
- For each candidate sequence, apply the model and get a score
- If the score is larger than zero, predict it is “donor/acceptor”; the higher score, the higher the probability the prediction is true

Conclusions

Like TSS, the flanks of splice junctions show “biased” distributions of nucleotides in certain positions. These biased distributions of nucleotides can be used for prediction of splice junctions.

- **Biological Sequence Analysis**
- R Durbin, S Eddy, A Krogh and G Mitchison Cambridge University Press, 1998.
- **Bioinformatics The machine learning approach**
- P Baldi and S Brunak The MIT Press, 1998
- **Post-Genome Informatics**
- M Kanehisa Oxford University Press, 2000

>TOPIC88

Prediction of Exons

Introduction

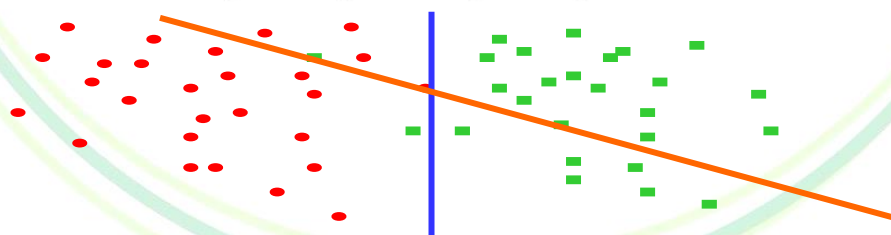
Exons can be predicted within ORF from the information gathered about splice junctions

- **Approach**

- For each segment [acceptor, donor], we get three scores (coding potential, donor score, acceptor score)
- **Various possibilities**
- all three scores are high – probably true exon
- all three scores are low – probably not a real exon
- all in the middle -- ?
- some scores are high and some are low -- ??

Prediction

- Collect a set of exons and non-exons
- Score them using our scoring schemes
- Plot them as follows
- “draw” a separating line between exons and non-exons
- **Linear Discriminate Analysis**
- linear discriminate analysis (LDA) finds an optimal plane surface that best separates points that belong to two classes For example, if there are ten true exons and ten introns, and two feature These samples could be represented by 20 points in a two-dimensional space.LDA would compute a straight line through the space that can best separate the two classes with the minimal classification error



Conclusion

- Collect a set of exons and non-exons
- Score them using our scoring schemes
- Plot them as follows

- “draw” a separating line between exons and non-exons

References

Biological Sequence Analysis

R Durbin, S Eddy, A Krogh and G Mitchison Cambridge University Press, 1998.

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P Baldi and S Brunak The MIT Press, 1998

Post-Genome Informatics

M Kanehisa Oxford University Press, 2000

>TOPIC 89

Annotation of Assembled Genome

A genome sequence is useless without annotation

Three steps in genome annotation:

- Find features not associated with protein-coding genes (e.g. tRNA, rRNA, snRNA, SINE/LINE, miRNA precursors)

A genome sequence is useless without annotation

Three steps in genome annotation:

- Build models for protein-coding genes, including exons, coding regions, regulatory regions

A genome sequence is useless without annotation

Three steps in genome annotation:

- Associate biologically relevant information with the genome features and genes

Ab initio methods

Based on sequence alone

- Gene prediction algorithms (e.g. AUGUSTUS, Glimmer, GeneMark)
- RepeatMasker (repeat families)

Evidence-based Methods

- Require transcriptome data for the target organism (the more the better)

- Align cDNA sequences to assembled genome and generate gene models:
TopHat/Cufflinks, Scripture

Biological Annotations

BLAST of gene models against protein databases

- Sequence similarity to known proteins
- InterProScan of predicted proteins against databases of protein domains
 - Pfam, Prosite, HAMAP, PANTHER, ...
- Mapping against Gene Ontology terms
 - GO terms
 - BLAST2GO
- **Pattern Finding**
- Much of the data processing in bioinformatics involves searching and recognizing certain patterns within DNA, RNA or protein sequences
- **Pattern Finding**
- In Biology it means finding motifs in DNA or proteins while in computational means it is finding a pattern in a string
- **Conclusions**
- After a genome is assembled, genome annotations are performed to identify gene and other features in a genome

>TOPIC 90

Pattern Finding in a Genome

Vocabulary

A **pattern** (keyword) is an ordered sequence of symbols

Vocabulary

Symbols of the pattern and the searched text are chosen from a predetermined finite set, called an **alphabet** (Σ)

Four Cases of Pattern Finding:

Look for a perfect match

- Allow errors due to substitutions

- **CGTA**
- **CGGA**
- **Four Cases of Pattern Finding:**
- Allow errors due to insertions-deletions
- (InDels).
- **C_GTA**
- **CCGGA**
- **Four Cases of Pattern Finding:**
- Rank possible matches according to a weight function and keep matches above a certain threshold

Four Cases of Pattern Finding:

p1 and ***p2*** are two patterns of length 5

- ***W*** is the weight of complete patterns defined via a nucleotide-nucleotide weight function ***w()***
- **CTGTA**
- **CCGGA**

$$W(p1, p2) = \sum_{i=1}^5 w(p1[i], p2[i])$$

Generalized Algorithm

Goal: Finding all occurrences of a pattern in a text

Input:

Pattern ***p*** = [***p1...pn***] of length ***n***

Text ***t*** = [***t1...tm***] of length ***m***

Output:

An indication that pattern ***P*** exist in ***T***

or it does not exist in text ***T***

Conclusion

Pattern searching algorithms search specific sequences in strands of DNA, RNA and proteins having important biological meaning

>TOPIC 91

Pattern Finding Algorithms

Methods Devised for Pattern Finding

- Exact searching methods
- Approximate searching methods
- Position weight matrices
- Suffix trees
- **Exact Pattern Matching**
- Given a pattern p of length m
- and a string or text T of length n ($m \leq n$)
- Find all the occurrences of p in T
- **The matching needs to be exact, which means that the exact word or pattern is found**

Exact Pattern Matching Algorithms

- Naïve Brute Force algorithm
- Boyer-Moore algorithm
- Knuth Morris Pratt algorithm
- **Approximate Pattern Matching**
- Also referred as approximate string matching or matches with k mismatches or differences
- **Approximate Pattern Matching**
- **Given:** a pattern p of length m
- and
- a string or text T of length n ($m \leq n$)

- **Find:** all the occurrences of substring X in T that are similar to p , allowing a limited number, say k different characters in similar matches

Approximate Pattern Matching Algorithms

- Dynamic programming approach
- Automata approach
- Filtering and automation algorithms

Position Weight Matrices

Also known as position specific scoring matrices (PSSM)

- A matrix representing the frequencies of residues observed for a position in multiple alignment

Suffix Trees

- It is a compressed tree containing all the suffixes and allows many problems on strings to be solved quickly

Conclusions

- Exact searching or pattern matching methods
- Approximate searching or pattern matching methods
- Position weight matrices.
- Suffix trees

>TOPIC 92

Methods Devised for Pattern Finding

- Exact searching methods
- Approximate searching methods
- Position weight matrices
- Suffix trees
- **Exact Pattern Matching**
- Given a pattern p of length m
- and a string or text T of length n ($m \leq n$)

- Find all the occurrences of p in T
- **The matching needs to be exact, which means that the exact word or pattern is found**

Exact Pattern Matching Algorithms

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- Knuth Morris Pratt algorithm
- **Approximate Pattern Matching**
- Also referred as approximate string matching or matches with k mismatches or differences
- **Approximate Pattern Matching**
- **Given:** a pattern p of length m and a string or text T of length n ($m \leq n$)
- **Find:** all the occurrences of substring X in T that are similar to p , allowing a limited number, say k different characters in similar matches

Approximate Pattern Matching Algorithms

- Dynamic programming approach
- Automata approach
- Filtering and automation algorithms

Position Weight Matrices

Also known as position specific scoring matrices (PSSM)

- A matrix representing the frequencies of residues observed for a position in multiple alignment
- **Suffix Trees**
- It is a compressed tree containing all the suffixes and allows many problems on strings to be solved quickly

Conclusions

- Exact searching or pattern matching methods
- Approximate searching or pattern matching methods
- Position weight matrices.

- Suffix trees

>TOPIC 93

Knuth-Morris-Pratt Algorithm

Introduction

A linear time algorithm for string matching

- Does not involve backtracking on string s i.e., repetitive comparison of nucleotide residues

Components

- **The Prefix Function**
- **The KMP Matcher**
- **The Prefix Function Π**
- Encapsulates knowledge about how the pattern matches against shifts of itself

The Prefix Function Π

- This information can be used to avoid useless shifts of the pattern ' p '
- This enables avoiding backtracking on the string ' S '
- **The KMP Matcher**
- **Given:** string ' S ', pattern ' p ' and prefix function ' Π '
- **Find:** the occurrence of ' p ' in ' S '
- **Return:** the number of shifts of ' p ' after which occurrence is found

Compute-Prefix-Function (p)

```

1   $m \leftarrow \text{length}[p]$     // ' $p$ ' pattern to be matched
2   $\Pi[1] \leftarrow 0$ 
3   $k \leftarrow 0$ 
4  for  $q \leftarrow 2$  to  $m$ 
5      do while  $k > 0$  and  $p[k+1] \neq p[q]$ 
6  do  $k \leftarrow \Pi[k]$ 
7      If  $p[k+1] = p[q]$ 
8      then  $k \leftarrow k + 1$ 

```

```

9    $\Pi[q] \leftarrow k$ 
10  return  $\Pi$ 

The KMP Matcher (S, p)
1   $n \leftarrow \text{length}[S]$ 
2   $m \leftarrow \text{length}[p]$ 
3   $\Pi \leftarrow \text{Compute-Prefix-Function}(p)$ 
4   $q \leftarrow 0$  //number of characters matched
5  for  $i \leftarrow 1$  to  $n$  //scan S from left to right
6  do while  $q > 0$  and  $p[q+1] \neq S[i]$ 
7      do  $q \leftarrow \Pi[q]$  //next character does not match
8      if  $p[q+1] = S[i]$ 
9          then  $q \leftarrow q + 1$  //next character matches

```

The KMP Matcher (S, p)

```

10     if  $q = m$  //is all of p matched?
11     then print "Pattern occurs at position"
         $i - m + 1$ 
12      $q \leftarrow \Pi[q]$  // look for the next match

```

Conclusions

A linear time algorithm for string matching

- avoid useless shifts of the pattern ' p '
- Does not involve backtracking on string S

>Topic 94

Knuth-Morris-Pratt Algorithm

Components

- The Prefix Function Π

- The KMP Matcher

Compute-Prefix-Function (p)

1 $m \leftarrow \text{length}[p]$

2 $\Pi[1] \leftarrow 0$

3 $k \leftarrow 0$

4 for $q \leftarrow 2$ to m

5 do while $k > 0$ and $p[k+1] \neq p[q]$

6 do $k \leftarrow \Pi[k]$

7 If $p[k+1] = p[q]$

8 then $k \leftarrow k + 1$

9 $\Pi[q] \leftarrow k$

10 return Π

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
Π	0						

Initially:

$m = \text{length}[p] = 7$

$\Pi[1] = 0$

$k = 0$

Compute-Prefix-Function (p)

```
1  $m \leftarrow \text{length}[p]$ 
2  $\Pi[1] \leftarrow 0$ 
3  $k \leftarrow 0$ 
4   for  $q \leftarrow 2$  to  $m$ 
5     do while  $k > 0$  and  $p[k+1] \neq p[q]$ 
6       do  $k \leftarrow \Pi[k]$ 
7     If  $p[k+1] = p[q]$ 
8       then  $k \leftarrow k + 1$ 
9      $\Pi[q] \leftarrow k$ 
10  return  $\Pi$ 
```

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
Π	0	0					

Step 1: $q = 2, k = 0$

$p[0+1] \neq p[q_2]$

$\Pi[2] = 0$

Compute-Prefix-Function (p)

```
1  $m \leftarrow \text{length}[p]$ 
2  $\Pi[1] \leftarrow 0$ 
3  $k \leftarrow 0$ 
4   for  $q \leftarrow 2$  to  $m$ 
5     do while  $k > 0$  and  $p[k+1] \neq p[q]$ 
6       do  $k \leftarrow \Pi[k]$ 
7     If  $p[k+1] = p[q]$ 
8       then  $k \leftarrow k + 1$ 
9      $\Pi[q] \leftarrow k$ 
10  return  $\Pi$ 
```

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
Π	0	0	1				

Step 2: $q = 3, k = 0,$

$p[0+1] = p[q_3]$

$K = 0+1 = 1$

$\Pi[3] = 1$

Compute-Prefix-Function (π)

1 $m \leftarrow \text{length}[p]$

2 $\pi[1] \leftarrow 0$

3 $k \leftarrow 0$

4 for $q \leftarrow 2$ to m

5 do while $k > 0$ and $p[k+1] \neq p[q]$

6 do $k \leftarrow \pi[k]$

7 **If $p[k+1] = p[q]$**

8 **then $k \leftarrow k + 1$**

9 **$\pi[q] \leftarrow k$**

10 return π

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
π	0	0	1	2			

Step 3: $q = 4, k = 1$

$p[1+1] = p[q_4]$

$K = 1+1 = 2$

$\pi[4] = 2$

Compute-Prefix-Function (π)

1 $m \leftarrow \text{length}[p]$

2 $\pi[1] \leftarrow 0$

3 $k \leftarrow 0$

4 for $q \leftarrow 2$ to m

5 do while $k > 0$ and $p[k+1] \neq p[q]$

6 do $k \leftarrow \pi[k]$

7 **If $p[k+1] = p[q]$**

8 **then $k \leftarrow k + 1$**

9 **$\pi[q] \leftarrow k$**

10 return π

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
π	0	0	1	2	3		

Step 4: $q = 5, k = 2$

$p[2+1] = p[q_5]$

$K = 2+1 = 3$

$\pi[5] = 3$

Compute-Prefix-Function (p)

1 $m \leftarrow \text{length}[p]$

2 $\Pi[1] \leftarrow 0$

3 $k \leftarrow 0$

4 for $q \leftarrow 2$ to m

5 do while $k > 0$ and $p[k+1] \neq p[q]$

6 do $k \leftarrow \Pi[k]$

7 If $p[k+1] = p[q]$

8 then $k \leftarrow k + 1$

9 $\Pi[q] \leftarrow k$

10 return Π

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
Π	0	0	1	2	3	1	

Step 5: $q = 6, k = 3$

$p[3+1] \neq p[q_6]$

$K = \Pi[3] = 1$

$\Pi[6] = 1$

Compute-Prefix-Function (p)

1 $m \leftarrow \text{length}[p]$

2 $\Pi[1] \leftarrow 0$

3 $k \leftarrow 0$

4 for $q \leftarrow 2$ to m

5 do while $k > 0$ and $p[k+1] \neq p[q]$

6 do $k \leftarrow \Pi[k]$

7 If $p[k+1] = p[q]$

8 then $k \leftarrow k + 1$

9 $\Pi[q] \leftarrow k$

10 return Π

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
Π	0	0	1	2	3	1	0

Step 6: $q = 7, k = 1$

$p[1+1] \neq p[q_7]$

$K = \Pi[1] = 0$

$\Pi[7] = 0$

Conclusions

- The Prefix Function Π
- The KMP Matcher

Topic # 95

Knuth-Morris-Pratt Algorithm

Components

- The Prefix Function Π
- The KMP Matcher

The KMP Matcher (S, p)

```
1 n  $\leftarrow$  length[S]
2 m  $\leftarrow$  length[p]
3  $\Pi \leftarrow$  Compute-Prefix-Function(p)
4 q  $\leftarrow$  0 //number of characters matched
5 for i  $\leftarrow$  1 to n //scan S from left to right
6     do while q > 0 and p[q+1]  $\neq$  S[i] //if next character does not match
7         do q  $\leftarrow$   $\Pi$ [q]
8     if p[q+1] = S[i] //next character matches
9         then q  $\leftarrow$  q + 1
10    if q = m //is all of p matched?
```

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11 then print "Pattern occurs at position"

$i - m + 1$

11 $q \leftarrow \Pi[q]$ // look for the next match

Let us execute the KMP algorithm to find whether 'p' occurs in 'S'.

For 'p' the prefix function, Π was computed previously and is as follows:

q	1	2	3	4	5	6	7
p	a	t	a	t	a	c	a
Π	0	0	1	2	3	1	0

Initially: $n = \text{size of } S = 15;$

$m = \text{size of } p = 7$

Step 1: $i = 1, q = 0$ comparing $p[1]$ with $S[1]$

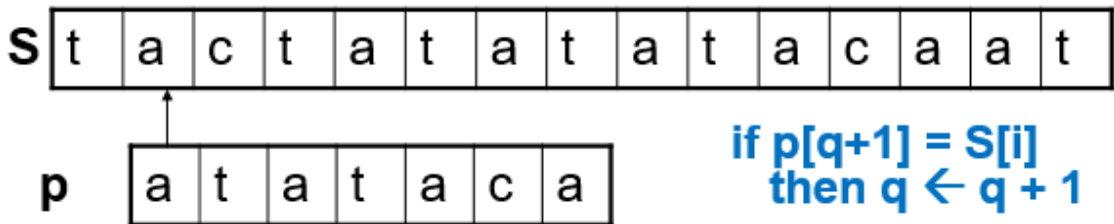
t	a	c	t	a	t	a	t	a	c	a	a	t
a	t	a	t	a	c	a						
p	a	t	a	t	a	c	a					
Π	0	0	1	2	3	1	0					

$P[1]$ does not match with $S[1]$
'p' will be shifted to the right as i increments

Step 2:

$p[q+1] : S[i]$

$i = 2, q = 0$ comparing $p[1]$ with $S[2]$



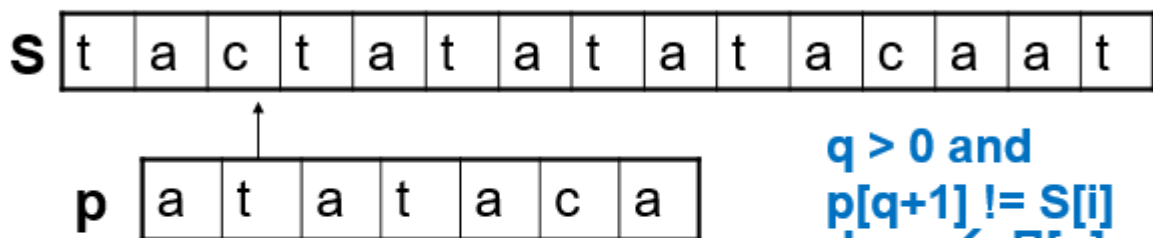
$P[1]$ matches $S[2]$

$q = 0 + 1 = 1$

p	a	t	a	t	a	c	a
Π	0	0	1	2	3	1	0

Step 3: $i = 3, q = 1$ $p[q+1] : S[i]$

$p[2]$ does not match with $S[3]$

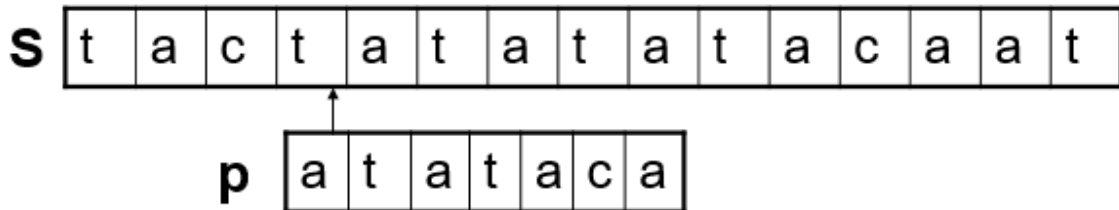


**Backtracking on p, comparing $p[1]$ and $S[3]$
because after mismatch $q = \Pi[1] = 0$**

p	a	t	a	t	a	c	a
Π	0	0	1	2	3	1	0

Step 4: $i = 4, q = 0$

$p[1]$ does not match with $S[4]$



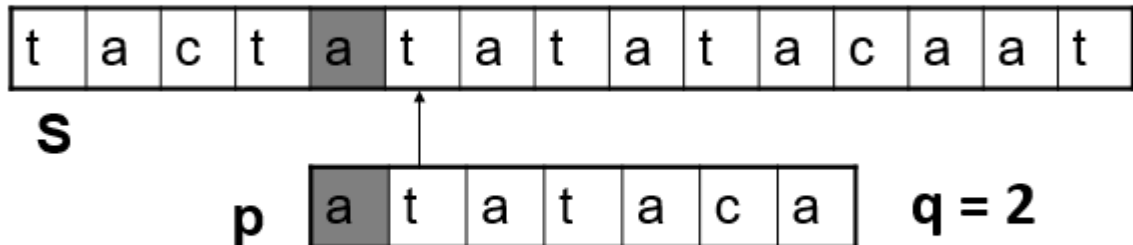
Step 5: $i = 5, q = 0$

$p[1]$ matches with $S[5]$



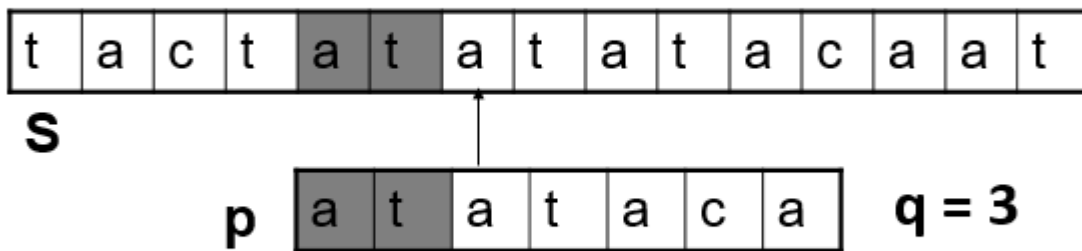
Step 6: $i = 6, q = 1$

$p[2]$ matches with $S[6]$



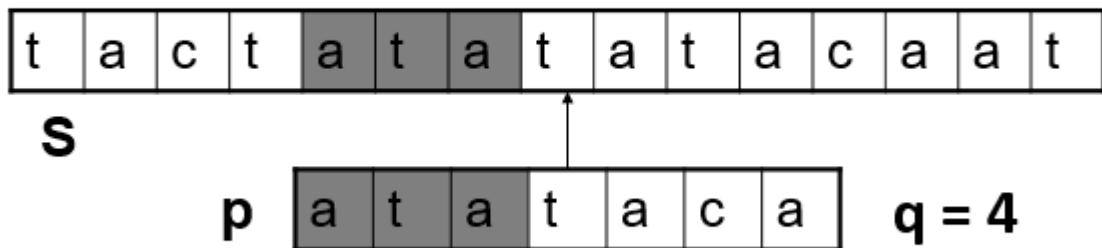
Step 7: $i = 7, q = 2$

p[3] matches with S[7]



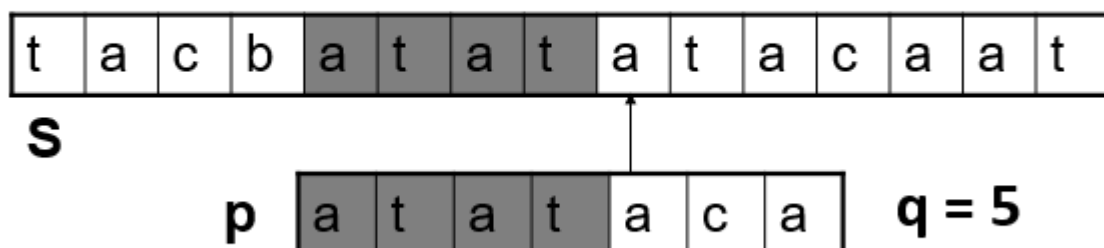
Step 8: $i = 8, q = 3$

p[4] matches with S[8]



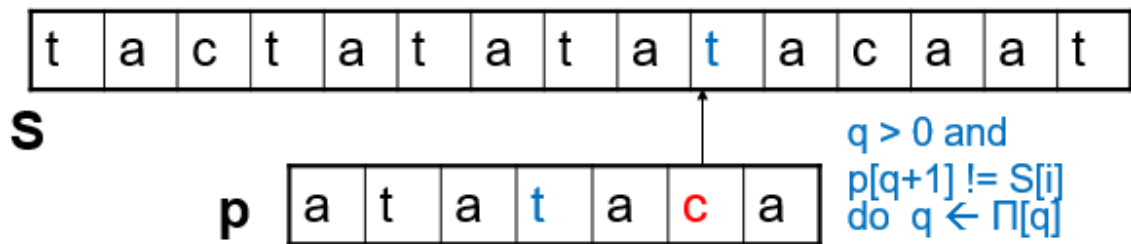
Step 9: $i = 9, q = 4$

p[5] matches with S[9]



Step 10: $i = 10, q = 5$

$p[6]$ does not match with $S[10]$

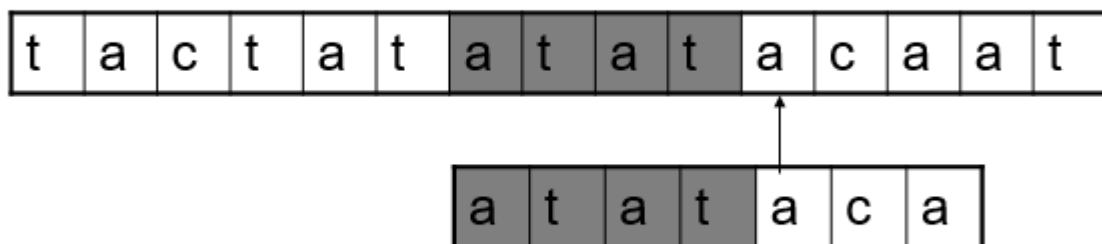


**Backtracking on p, comparing $p[4]$ with $S[10]$
because after mismatch $q = \Pi[5] = 3$**

p	a	t	a	t	a	c	a
Π	0	0	1	2	3	1	0

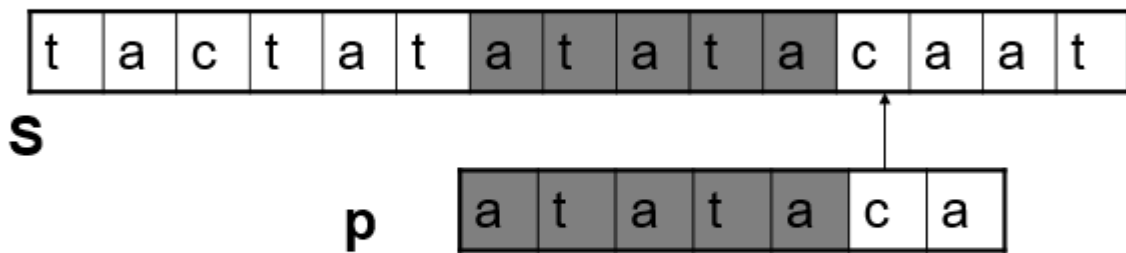
Step 11: $i = 11, q = 4$

$p[5]$ matches with $S[11]$



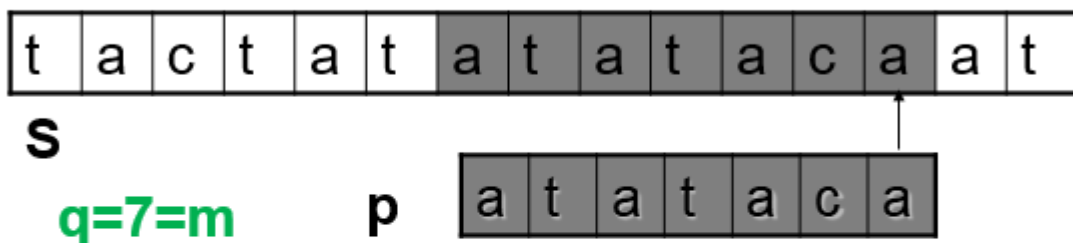
Step 12: $i = 12, q = 5$

$p[6]$ matches with $S[12]$



Step 13: $i = 13, q = 6$

$p[7]$ matches with $S[13]$



Pattern 'p' is found completely in string 'S'
The total number of shifts that took place for the match to be found are:
 $i - m = 13 - 7 = 6$ shifts

Complexity

- $O(m)$ - It is to compute the prefix function values.
- $O(n)$ - It is to compare the pattern to the text.
- Total of $O(m + n)$ run time.

Advantages

- The running time of the KMP algorithm is optimal ($O(m + n)$), which is very fast
- The algorithm never needs to move backwards that makes the algorithm good for processing very large files

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Drawback

- Doesn't work so well as the size of the alphabets increases

Conclusion

- A fast linear time algorithm for string matching

Topic 96-100

Scoring Schemes

Scoring System: Introduction:

- Total score assigned to an alignment is sum of terms for each aligned pair of residues, plus terms for each gap

$$\sum S(x_i, y_j) + d$$

d = linear gap penalty

Simple Alignment Scores:

- A simple way (but not the best) to score an alignment is to count 1 for each match and 0 for each mismatch

Scoring or Substitution Matrices:

- Used for scoring amino acid substitutions in pairwise alignments
- They reflect substitution rates that are originated by evolutionary events

Some of the substitution matrices to compute sequence alignments are:

- PAM: Point Accepted mutations
- BLOSUM: BLOCK Substitution Matrix

Substitution Matrices:

For a set of well known proteins:

- Align the sequences
- Count the mutations at each position
- For each substitution set the score to the log-odds ratio

For each substitution set the score to the log-odds ratio

$$\log \left(\frac{\text{observed}}{\text{expected by chance}} \right)$$

Positive Score: The amino Acids are similar, mutations from one into the other occur more often than expected by chance during evolution

Negative Score:

The amino Acids are dissimilar, mutations from one into the other occur less often than expected by chance during evolution

Conclusion

- Substitution matrices are the log-odds matrices used for scoring amino acid substitutions in pairwise alignments

- **Substitution matrices**

Introduction:

- Substitution scores can be derived from probabilistic model

Notations:

- Let a pair of sequences x and y of length n and m
- X_i be the i th symbol in x , y_j be the j th symbol in y
- Symbols are from alphabet A
- $A = \{A, T, G, C\}$
- $A = \{\text{twenty amino acids}\}$
- Symbols from alphabet be $a, b, \dots$

Objective:

Given a pair of aligned sequences, we want to assign a score to the alignment that gives a measure of the relative likelihood that the sequences are related as opposed to being unrelated

Unrelated or random Model R:

Letter a occurs independently with frequency q_a and the probability of two sequences is the product of probability of each amino acid

$$P(x, y | R) = \prod_i q_{x_i} \prod_j q_{y_j}$$

Match Model M:

- Aligned pairs occur with a joint probability p_{ab} .
- p_{ab} can be thought of as the probability that the residues a and b have been independently derived from some unknown original residue c in their common ancestor.
- The probability of the alignment is

$$P(x, y | M) = \prod_i p_{x_i y_i}$$

- The ratio of two likelihoods can be calculated as;

$$\frac{P(x, y | M)}{P(x, y | R)}$$

$$P(x,y|R)$$

Odds ratio:

- The ratio of two likelihoods can be calculated as;

$$\frac{P(x,y|M)}{P(x,y|R)} = \frac{\prod_i p_{x_i y_i}}{\prod_i q_{x_i} \prod_j q_{y_j}} = \prod_i \frac{p_{x_i y_i}}{q_{x_i} q_{y_i}}$$

Logarithm can be used to have an additive score

$$S = \sum S(x_i, y_j)$$

$$\text{where } S(a,b) = \log(p_{ab}/q_a q_b)$$

log likelihood of (a,b) as aligned vs unaligned pair

1. Dayhoff PAM matrices:

Dayhoff, Schwartz and Orcutt (1978) presented their famous PAM (Point accepted mutations) using substitution data from similar proteins then extrapolating this information to longer evolutionary distances

$$S(a,b) = \log(p_{ab}/q_a q_b)$$

incorporating the evolutionary time

$$S(a,b|t) = \log P(b|a,t)/q_b$$

Since $p_{ab}/q_a = P(b|a)$

Values are rounded to near integer for computational convenience

- PAM250 is scaled by $3/\log 2$ to give scores in third-bits

2. BLOSUM matrices:

- Dayhoff matrices do not capture true difference between short time substitutions and long term ones.
- PAM matrices do not perform well in case of distantly related proteins.
- BLOSUM matrices are derived from set of aligned, ungapped regions from protein families called BLOCKS database (Henikoff&Henikoff 1992).
- Sequences from each block was clustered together with score $>L\%$.
- Matrices with $L= 62$ and $L= 50$ known as **BLOSUM62** and **BLOSUM50** respectively.
- **BLOSUM62** is good for ungapped alignments and **BLOSUM50** is good for gapped alignments

Conclusions

- Substitution scores can be derived from probabilistic models
- PAM and BLOSUM are famous substitution matrices

• Optimal Alignment Methods

Introduction:

Finding the path whose total score is maximal will give the best sequence alignment

- Two methods
 - Local alignment
 - Global alignment

1. **Global Alignment:**

It is an alignment that essentially spans the full extents of input sequences

- Hence it covers the entire length of sequences involved
- The Needleman-Wunsch algorithm finds best global alignment between two sequences

2. **Local Alignment:**

- It only covers parts of the sequences to be aligned
- Smith-Waterman algorithm finds the best local alignment between two sequences

3. **Dynamic Programming:**

Dynamic programming is used to find an optimal alignment of two sequences and its scores. It is a method by which a larger problem may be solved by first solving smaller, partial versions of the problem. **Three steps in dynamic programming:**

- Initialization
 - Matrix filling (scoring)
 - Traceback (alignment)
1. **Initialization:** Create matrix with $M+1$ columns and $N+1$ rows where M and N correspond to the size of sequences to be aligned
 2. **Matrix filling:** Fill the matrix with highest possible score
 3. **Trace back:** Move from the last corner and follow the arrow

Conclusion:

- Dynamic programming is used to find an optimal alignment of two sequences and its scores

• Needleman-Wunsch Algorithm

Introduction:

It performs global alignment on two sequences

- **The algorithm was developed by Saul B. Needleman and Christian D. Wunsch and published in 1970**

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Basic idea is

- to build up the best alignment by using optimal alignments of smaller subsequences
- It was the first application of dynamic programming to compare biological sequences

Steps:

Three steps

- Initialization
- Matrix filling
- Traceback

Creating the Matrix:

Initial matrix is created with $M+1$ columns and $N+1$ rows

- Where M and N correspond to the length of sequences

Initialization:

- The cell of first row and first column of the matrix is initially filled with zero
- Add gap penalty for each shift to the right

Initialization:

- a. $F(0, 0) = 0$
- b. $F(0, j) = -j \times d$
- c. $F(i, 0) = -i \times d$

Matrix Fill:

- Move through the cells row by row, calculating the score for each cell
- Compute three scores:
 - A match score
 - Vertical gap score
 - Horizontal gap score
- The *match score* is the sum of the diagonal cell score and the score for a match
- The *horizontal gap score* is the sum of the cell to the left and the gap score
- The *vertical gap score* is computed analogously

Matrix filling: Filling-in partial alignments

For each $i = 1 \dots M$

For each $j = 1 \dots N$

$$F(i, j) = \max \begin{cases} F(i-1, j-1) + s(x_i, y_j), & [\text{case 1}] \\ F(i-1, j) - d, & [\text{case 2}] \\ F(i, j-1) - d & [\text{case 3}] \end{cases}$$

$$\text{Ptr}(i, j) = \begin{cases} \text{DIAG}, & \text{if } [\text{case 1}] \\ \text{LEFT}, & \text{if } [\text{case 2}] \\ \text{UP}, & \text{if } [\text{case 3}] \end{cases}$$

Traceback:

- The final step in the algorithm is the trace back for the best alignment
- Start at the bottom-right corner
- Follow where maximum value comes from
- **F(M, N) is the optimal score, and from Ptr(M, N) can trace back optimal alignment**

Scoring Scheme:

- Scoring scheme introduced can be user defined
- It contains specific scores for match and mismatch residues as well as gap

Conclusions:

- Needleman and Wunsch Algorithm performs global alignment on two sequences using a dynamic programming approach
- **Smith-Waterman Algorithm**

Introduction:

- Finds the best local alignment between two subsequences

Steps:

- Initialization
- Matrix filling
- Traceback or alignment

Algorithm

1. Initialization:

- a. $F(0, 0) = 0$
- b. $F(0, j) = 0$
- c. $F(i, 0) = 0$

Trace-back:

$F(M, N)$ is the optimal score, and from $\text{Ptr}(M, N)$ can trace back optimal alignment

Creating the Matrix:

- Initial matrix is created with $M+1$ columns and $N+1$ rows
 - Where M and N correspond to the length of sequences

Initialization:

- First row and first column of the matrix is filled with zero

		T	G	G	T	G
	0	0	0	0	0	0
A	0					
T	0					
C	0					
G	0					
T	0					

Matrix Filling:

- Add the match or mismatch scores diagonally
- Add gap penalties vertically and horizontally
- Replace the negative values by zero

Traceback:

- Traceback starts with maximum value in the matrix and then go backwards

Alignment:

_ T G G T
 A T C G T

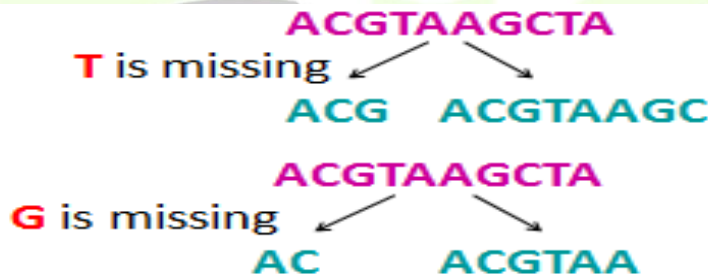
Conclusion:

- Finds the best local alignment between two subsequences

Topic 101-102

- **DNA Sequencing**

Magazine cut into millions of pieces. Problem of fragment assembly in DNA sequencing Short 500 to 700 nucleotide sequences per experiment, Assembling entire genome reassembling the magazine Both problems are complicated by unavoidable experimental error. The data are frequently incomplete. DNA sequencing methods-Fred Sanger and Walter Gilbert. Cells make copies of DNA DNA fragments of different lengths- if one base is missing.



For A and C missing, will also result in DNA fragments by length. Each of four starvation experiments produces a ladder of fragments of varying lengths called the *Sanger ladder*

Sequencing of a 5386- nucleotide virus, DNA sequence data, Human Genome Project , 3 billion- nucleotide sequence, DNA sequencing technology, Sequencing reads, Continuous genome, DNA reading process, DNA sequencing machines, Single DNA fragment.

- **Shotgun sequencing**

Sonicated, Inserts, Vector, Bacterial host, Cloning proce, DNA sequencing- inserts and computational

Topic-103

- **DNA Array**

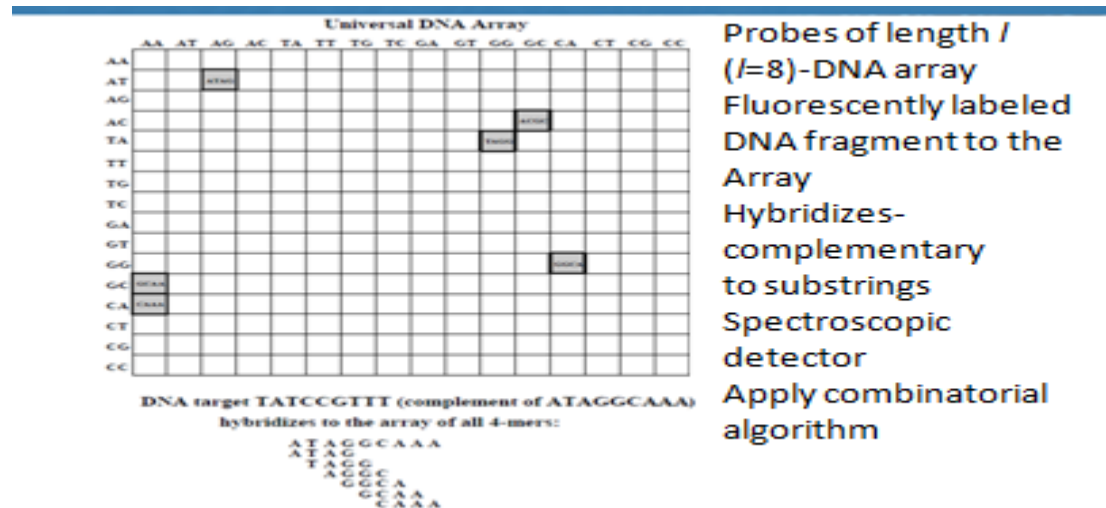
Human Genome Project, Sequencing **by Hybridization**, DNA array (DNA chip) , Probes, Hybridization, Weak chemical bond. DNA probes, Biochemical problem and the combinatorial problem **Science**, SteveFodor and colleagues, Light-directed polyme synthesis –similarities to computer chip manufacturing.



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In Fig. An array with all 4^l probes- $4 \cdot l$ separate reactions Affymetrix-64-kb DNA array in 1994 1-Mb or larger Probes- unknown target DNA- l -mer composition Universal DNA array contains all 4^l probes of length l .



Topic-104

• Sequencing by Hybridization

Unknown DNA sequence- length l , String s of length n , the l -mer composition is the multiset of $n - l + 1$ l -mers in s and is written $Spectrum(s, l)$.

If $l = 3$ and $s = \text{TATGGTGC}$, then

$Spectrum(s, l) = \{\text{TAT}, \text{ATG}, \text{TGG}, \text{GGT}, \text{GTG}, \text{TGC}\}$

Lexicographic order, like $\text{ATG}, \text{GGT}, \text{GTG}, \text{TAT}, \text{TGC}, \text{TGG}$

Sequencing by Hybridization (SBH) Problem:

Reconstruct a string from its l -mer composition

Input: A set, S , representing all l -mers from an (unknown) string s

Output: String s such that $Spectrum(s, l) = S$

Topic-105-106

Fragment Assembly in DNA Sequencing

500- to 700-bp DNA reads Reconstruct the entire genomic DNA sequence Fragment assembly Error rate in DNA reads Sequencing errors, Assignments to read Major problem-repeats in DNA- 20% of human genome Alu sequence-million times Repeats occur at many scales Human T-cell receptor locus-Trypsinogen gene (4 kb).1 million Alu repeats (300 bp) and 200,000 LINE repeats (1000 bp) 3 billion-letter sequence 500-letter reads, large number of repeats.

Strategy for Sequencing

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Clone into BAC, Sequenced each one-minigenome, Simplifies the computational assembly problem, Human Genome project, Whole-genome assembly paradigm (James Weber and Gene Myers) Celera Genomics Mate-pair reads sequencing technique.

Inserts of length L - both ends are sequenced, Mates at distance L - length is larger than most repeats

Most fragment assembly algorithms consist of the following three steps

Overlap: Finding potentially overlapping reads

Layout: Finding the order of reads along DNA

Consensus: Deriving the DNA sequence from layout Best match between the suffix of one read and the prefix of another. Sequencing errors-Dynamic programming algorithm. constructing the layout is the hardest step in fragment assembly. Whole-genome shotgun sequencing. Large number of reads to ensure that experimental errors are reduced to minor noise.

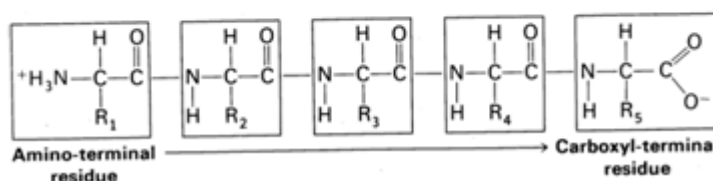
Topic-107-126

Protein Sequencing and Identification

Routinely sequenced proteins-Frederick Sanger-Nobel prize, Computational problem **Edman degradation reaction** to chop off one terminal a.a, Sanger digested insulin with proteases DNA sequencing “break-read the fragments-assemble”

Edman degradation reaction, 1960s protein sequencing machines were on the market DNA sequencing technology Protein-Obtaining reads-problem Assembly- easy

Let $A = \{a_1, a_2, \dots, a_{20}\}$ be the set of amino acids
molecular masses $m(a_i)$
A peptide $P = p_1 \cdots p_n$ is a sequence of amino acids,
Parent mass $m(P) = \sum_{i=1}^n m(p_i)$
Partial N-terminal peptide $p_1, \dots, p_i$ of mass
 $m_i = \sum_{j=1}^i m(p_j)$ P_i and the partial C-terminal peptide
 $p_{i+1}, \dots, p_n$ of mass $m(P) - m_i$ as P_{i+1} , for $1 \leq i \leq n$.



Computational Protein Sequencing

Proteins produced in cell Sequence of previously unknown proteins. Proteins -biological system and Range-Brain cells-Liver cells

De novo protein sequencing Protein identification Gedanken experiment Proteins constitute the DNA polymerase

Protein sequencing and identification, **Spliceosome**, Matthias Mann and colleagues purified the spliceosome complex

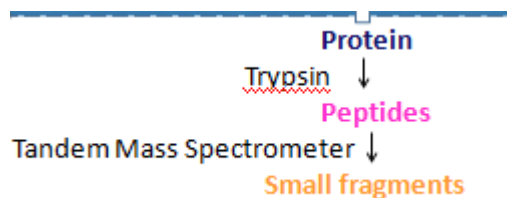
Mass Spectrophotometry

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Programmed cell death, Survival factors, Developing nematode DNA sequence data Nervous system- mutation in several genes, Mass spectrometry.

Mass spectrum of a peptide is a collection of masses of these fragments- Derive the sequence of a peptide given its mass spectrum. For an ideal fragmentation process the peptide sequencing problem is simple. The fragmentation process is not ideal, and mass spectrometers measure mass with some imprecision.



Peptide Sequencing Problem

Amass spectrometer breaks a peptide $p_1 p_2 \dots p_n$
GPFNA **GPFNA**
G, GP, GPF, GPFN **PFNA, FNA, NA, A**
GPFNA into **GP** and **FNA**, lose some small parts of GP and FNA, fragments of a lower mass
GP might lose a water (H_2O), and the peptide **FNA** might lose an ammonia (NH_3). Mass of GP minus the mass of water ($1 + 1 + 16 = 18$ daltons), and the mass of FNA minus the mass of ammonia ($1 + 1 + 1 + 14 = 17$ daltons). Two different ion types

Peptide fragmentation is characterized by a set of numbers $\Delta = \{\delta_1, \dots, \delta_k\}$ types of ions
 The set of ion types. A δ -ion of an N-terminal partial peptide P_i is a modification of P_i that has mass $m_i - \delta$. The most frequent N-terminal ions are called *b*-ions (ion b_i corresponds to P_i with $\delta = -1$) and the most frequent C-terminal ions are called *y*-ions (ion y_i corresponds to P_i^- with $\delta = 19$)

Protein identification via Database Search

De novo protein sequencing algorithms- invaluable for known, unknown proteins. Useful for complete spectra Spectra far from complete, De novo peptide sequencing algorithms. If we had access to a database of all proteins from a genome, then we would no longer need to consider all 20^l peptide sequences to interpret an MS/MS spectrum, but could instead limit our search to peptides present in this database. Database search “the back of a book”, Experimental spectrum, Sequence of the experimental peptide, SEQUEST algorithm – John, Yates and colleagues.

Protein Identification Problem:

Find a protein from a database that best matches the experimental spectrum.

Input: A database of proteins, an experimental spectrum S , a set of ion types, and a parent mass m .

Output: A protein of mass m from the database with the best match to spectrum S

Topic-113

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- **SEQUEST Algorithm**

Exhaustive search approach

Virtual database

Potential modifications

Combinatorial problem

Modified Protein Identification Problem:

Find a peptide from the database that best matches the experimental spectrum with up to k modifications.

Input: A database of proteins, an experimental spectrum S , a set of ion types Δ , a parent mass m , and a parameter k capping the number of modifications

Output: A protein of mass m with the best match to spectrum S that is at most k modifications away from an entry in the database

Modified Protein Identification problem P_1 and P_2 - S_1 and S_2 , Notion of spectral similarity Shared peaks count, Limitations in detecting similarities by database search.

Topic 114-125

- **Protein Structures**

Background

Complex protein structures enable proteins to perform complex functions. We know over a million protein sequences but only about 100,000 protein structures.

Why only 100,000 proteins for over million protein sequences

Estimating exact protein structures is very difficult. Its difficult to crystallize proteins. Even if we manage to get protein's X-Ray, to reconstruct the structure is extremely complex

Introduction

What if we could somehow predict protein structures?

- Since we know so many sequences, they can be used for predicting protein structures. This indeed is possible and helpful.

The Basic Idea

1. Amino acids determine the protein structure
2. We have a large protein sequence dataset (uniprot)

Hence, we can fold protein sequences and predict their structures!

Why predict and why not exact solutions?

A deterministic solution of protein folding is a major unsolved problem in molecular biology! Proteins fold spontaneously or with the help of enzymes or chaperones.

To predict we must first learn!

To computationally predict protein structures, we need to copy or mimic the natural folding! What are the steps in protein folding and structure formation?

To fold we must learn the steps

Step 1: "Collapse"- leading to burial of hydrophobic AA's

Step 2: Fluid globule - helices & sheets form, but are unorganized

Step 3: Compaction, and rearrangement of 2^o structures

Conclusion

- Protein structure prediction involves learning how the amino acids in primary sequence fold.
- Using this information, upon getting a protein sequence, we can try to predict how it folds!

- **Predicting Secondary Structures**

Background

Since the first step in protein folding is the formation of secondary structures, we must evaluate which amino acids in the primary sequence prefer which secondary structures?

- By looking at the structures in PDB, we know that Alanine mostly found in Alpha Helices. So if we have several Alanines in the sequence, then we can anticipate that a helix may be formed by them

Introduction

What if we survey the entire PDB and check the presence of each amino in each type of secondary structure

- **If we know which amino acid is found in which specific secondary structure, then we can use it for prediction!**

Amino Acid	P _α	P _β	P _t
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Conclusion

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- Several algorithms have been designed to predict 2' given an amino acid sequence
- The first such algorithm was the Chou-Fasman Algorithm!
- We will see it in the upcoming modules!

• 2' Structures in Chou Fasman Algorithm

Background

- For a primary sequence, and a tentative 2' structure, propensity table can help us compute the overall propensity
- Product of propensity values is computed for overall propensity for each 2' structure

Introduction

- An important point to note here is that 2' structures are formed due to hydrogen bonding between amino acids
- So, we need to consider the neighboring amino acids as well!

Sequence: E M A V I Y P G

Viable 2' Structures:

$\alpha\alpha\alpha\alpha\beta\beta\beta\beta$
 $\alpha\alpha\alpha\alpha\alpha\alpha\beta\dots$
 $\dots\alpha\beta\beta\beta\beta\beta\alpha\alpha\dots$

	P_{α}	P_{β}	P_t
Glu	1.51	0.37	0.74
Met	1.45	1.05	0.60
Ala	1.42	0.83	0.66
Val	1.06	1.70	0.50
Ile	1.08	1.60	0.50
Tyr	0.69	1.47	1.14
Pro	0.57	0.55	1.52
Gly	0.57	0.75	1.56

Chou & Fasman
(1974 & 1978)

Very small number of 2' structural combinations are left!

Conclusion

- You only need to compute propensities for a small number 2' structures
- The highest net propensity will be the most probably secondary structure that will be formed!

• Chou Fasman Algorithm - I

Only a small number of combinations of secondary structures are possible due to their individual properties. Such as 4 amino acids are needed to start an Alpha Helix and 5 amino acids for Beta Sheet

Note that besides the alpha helix and beta sheets, LOOPS are an other secondary structure.

How can loops be integrated into predicting 2' structures?

- Loops are small ~ 3-4 amino acids

1. Scan through the sequence : **E M A V I Y P G**

2. Identify sequence regions where:

- 4 out of 6 contiguous residues give a $P(\alpha) > 1.0$
- That region is **declared as alpha-helix**
- Extend helix to both sides until 4 out of 6 contiguous residues give a $P(\alpha) < 1.0$
- That is **declared end of the helix**

Conclusion

- For Alpha Helices, 4 contiguous amino acids are required
- Their Alpha-Helix propensity should be more than 1.0
- Once this propensity falls below 1.0, Alpha-Helix stops
- **Chou Fasman Algorithm - II**

Background

Alpha Helices are formed from 4 contiguous amino acids having an Alpha-Helix propensity over 1.0. The Alpha-Helix stops if this propensity falls below 1.0!

Introduction

- Once Alpha Helices are constructed, and concluded, the remaining amino acids can be evaluated for Beta sheets and turns etc
- Let's see how Beta sheets are evaluated using Chou Fasman Algorithm

1. Compute $P(\beta)$ for contiguous regions of 5 Amino Acids

2. From these regions, identify regions where:

- 5 contiguous residues have $P(\alpha) > P(\beta)$
- That region is **finalized as alpha-helix**

Repeat this step for the full amino acid sequence to finalize all possible alpha helical regions in the sequence.

Conclusion

- Alpha Helices can be finalized if their propensity is higher than the propensity for Beta Sheets in regions of 5 amino acids
- For those regions where that is not the case, further evaluation is required
- **Chou Fasman Algorithm - III**

Alpha Helices were finalized if their propensity was higher than the propensity for Beta Sheets in regions of 5 amino acids. For those regions where that are not the case, what should be done?

We can evaluate such regions for Beta Sheets.

Let us see step by step how to find a beta sheet and how to differentiate them from alpha helices

Scan the sequence to identify regions where:

- 3 out of 5 amino acids have $P(\beta) > 1.0$
- That region is declared as beta sheet
- Extend beta sheet to both sides until 4 contiguous residues average $P(\beta) < 1.0$
- That is declared end of the beta sheet
- Those regions are finalized as beta-sheets which have average $P(\beta) > 1.05$ and the average $P(\beta) > P(\alpha)$ for that region.

Regions where overlapping alpha-helices and beta-sheets occur are declared helices if

- the average $P(\alpha\text{-helix}) > P(\beta\text{-sheet})$ for that region

Else, a beta sheet is declared if

- average $P(\beta\text{-sheet}) > P(\alpha\text{-helix})$ for that region

Conclusion

- Using the strategy of higher propensity, alpha helices and beta sheets can be completely resolved
- Assignments for each beta sheet and alpha helix can be finalized
- But what about the loops?

• Chou Fasman Algorithm - IV

Background

- After computing the propensity of alpha helices and beta sheets, we need to settle for loops
- Let's see how can we find out the loops using Chou Fasman Algorithm

• For any j th residue in sequence, we calculate $f(\text{Total}) = f(j) f(j+1) f(j+2) f(j+3)$ (tetrapeptide)

• If

1. $f(\text{Total}) > 0.000075$
2. the average value for $P(\text{turn}) > 1.00$ in the tetrapeptide
3. the averages for the tetrapeptide are such $P(\alpha\text{-helix}) < P(\text{turn}) > P(\beta\text{-sheet})$,

Conclusion

- Chou Fasman Algorithm helps predict Alpha Helices, Beta Sheets and Turns

- The algorithm is based on statistical occurrence of Amino Acids in known structures
- **Chou Fasman Algorithm – Flowchart I, II & III.**
 - Chou Fasman Algorithm helps predict secondary structures such as Alpha Helices, Beta Sheets and Turns. Step by step flowchart of the entire algorithm. Beta sheets can be predicted from primary amino acid sequences
 - Chou Fasman Algorithm helps predict secondary structures from amino acid sequences. Step by step flowchart of the algorithm for extracting Alpha Helices
 - Alpha helices, beta sheets and turns can be predicted using Chou Fasman Algorithm. This algorithm is based on statistical analysis of amino acid occurrences in proteins.
- **Chou Fasman Algorithm – Improvements**
 - Secondary structure propensity values of alpha helix, beta sheet and turns should be recalculated with the latest protein data sets.

Special consideration for:

 - Nucleation regions
 - Membrane proteins
 - Hydrophobic domains
 - Consider variable coil and loop sizes besides the from tetra peptide turns
 - Consider local protein folding environments
 - Solvent accessibility of residues
 - Protein structural class
 - Protein's organism

Conclusion

Chou Fasman can be improved to better predict secondary structures by incorporating biochemical factors and updated statistics!

Summary of Visualization, Classification & Prediction

1. Why do we need to visualize proteins?
2. Which atoms are used to reconstruct proteins?
3. Where are the positions of these atoms stored?

Structure Classification

1. What is the relationship between protein structure and function?
2. What is the need to classify proteins

3. Hierarchy of classification

Structure Prediction

1. Why structure of proteins are important?
2. Why are so few structures reported till date?
3. Benefits of predicting structures

Conclusion

Structure visualization, classification and prediction equip us to perform functional evaluation of proteins! This is important for understanding disease and designing drugs for treating them

Topic-127-151

- **Introduction to Homology Modelling**

Proteins are 3D molecules with their own unique structures. Protein structure is reflective of the protein function. Protein structure includes 1', 2', 3' and 4' structures. 1' structure of proteins is the sequence of proteins and can be obtained by mass spectrometry. 2' structures formed by proteins are the helices, beta sheets, loops and coils. 3' structure of proteins is the combination of 2' structures such that the overall protein structure is formed. 4' protein structure is formed when two or more proteins complex together. X-Ray Crystallography and NMR Spectroscopy are used to find the structures of proteins. However, these methods are difficult and expensive. Solution: Prediction of structures. Protein sequence gives rise to its structure. If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein. So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences.

Conclusions

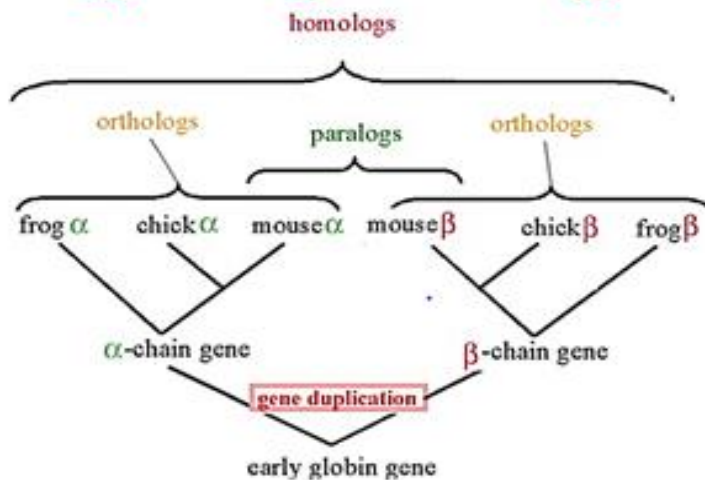
- Sequence Identity
- Alignment Length

Which combination of identity and alignment length is suitable for best for structure prediction?

- **Homology, Paralogy and Orthology**

In homology modelling, proteins with similar 1' sequences are considered. Given that one of them has its 3' structure known, then the 3' structure of other protein can be predicted

Homology: Paralogy vs. Orthology



Conclusions

- Good sequence alignment and identity ensures that homology modelling will give accurate results
- **Workflow of Structural Modelling**

Homology modelling is used to predict structures of proteins having high sequence similarity with other proteins with known structures!

Overall, there are three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling

• **Seven Steps to Homology Modelling – I**

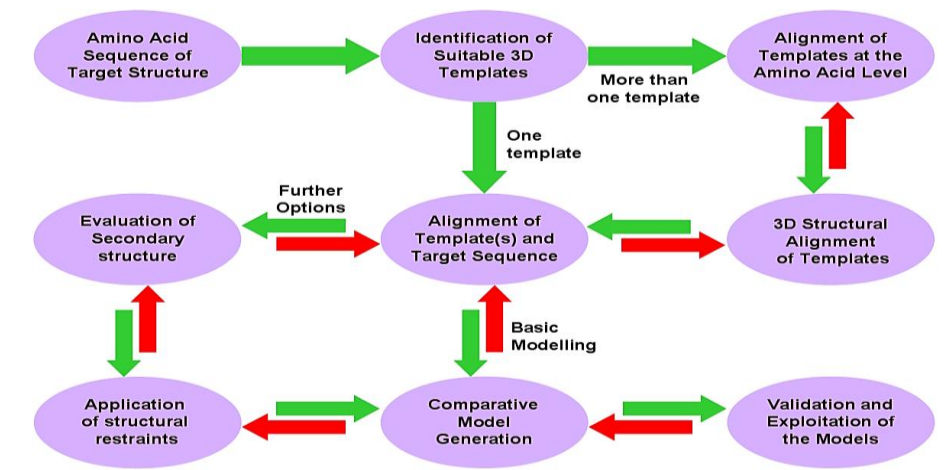
Protein structure can be predicted by 3 methods:

1. Homology Modelling
2. Fold Recognition / Threading
3. Ab Initio Modelling

Let's start by looking at Homology Modelling. There are seven salient steps in any Homology Modelling pipeline. Definition of Template (known) & Target (unknown). Homology modeling of the target structure can be done as follows:

1. Template recognition and initial alignment
2. Alignment correction
3. Backbone generation
4. Loop modeling
5. Side-chain modeling
6. Model optimization

7. Model validation



So, Homology modelling works in seven steps. It is a repetitive process

MODELLER for Homology Modelling

- Template recognition and initial alignment
- Alignment correction
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation

Modeller is a software for homology modelling

salilab.org/modeller

Inputs:

Python script file, Sequence alignment & Template (PDB)

.log : log output from the run.

.B* : model generated in the PDB format.

.D* : progress of optimisation.

.V* : violation profile.

.ini : initial model that is generated.

.rsr : restraints in user format.

.sch : schedule file for the optimisation process.

Automated Modelling Servers

Swiss Model

<http://swissmodel.expasy.org//SWISS-MODEL.html>

Robetta

<http://robetta.bakerlab.org/>

3D Jigsaw

<http://www.bmm.icnet.uk/servers/3djigsaw/>

Phyre

<http://www.sbg.bio.ic.ac.uk/phyre/>

Homology modelling helps predict protein structures by using prior structural information. Several tools are available to perform homology modelling in a programmatic or automated way!

- **Fold Recognition – Threading II**

Fold recognition is also called Threading. Technique for predicting protein structures. Employed when homology modelling cannot predict quality structures. A protein fold is defined by the way the secondary structure elements of the structure are arranged relative to each other in space. Common folds include 4-helix bundle and the TIM barrel. 5,000 stable folds in nature. Fold recognition: Finding the best fit of a sequence to a set of candidate folds. Fold recognition or Threading is a technique for predicting protein structures. It is useful in cases where homology modelling fails to predict quality structures

The process of threading

- In the process of "Threading", we mount an amino acid sequence on to the backbone of template structures in a folds library
- Each step is "drag" along the sequence (MQVKLFY...) through each location of each template fold
- Then, for each fold, we must compute the fitness of sequence matching that fold!

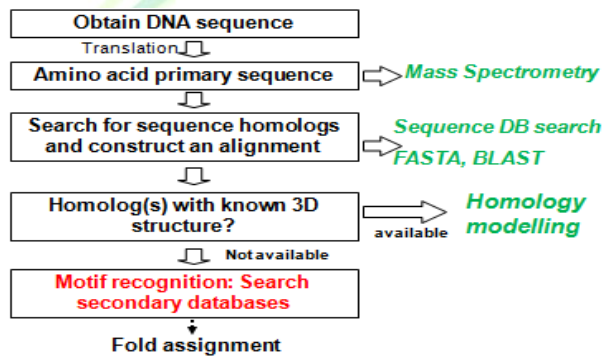
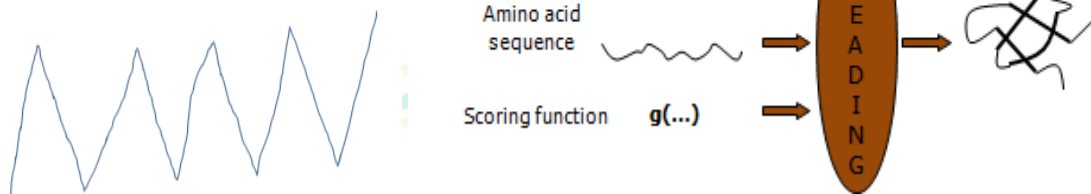
Threading involves "passing" the amino acid sequence through each fold in the database. The best match is computed using a scoring function. Combinations of secondary structures come together to form the best prediction. Scoring typically involves using a Z-Score function based on energy of a molecule.

Find the best way to

Inputs and outputs of threading

"mount" the residue sequence of one protein

onto a known protein structure!



• Online Tools for Threading – iTasser

iTASSER helps thread amino acid sequences on fold and secondary structure databases. It also helps predict function of structures output.

Iterative threading assembly refinement (I-TASSER) server

- Software for automated protein structure & function prediction based on the sequence-to-structure-to-function.
- Steps:
 - Starts from amino acid sequence
 - i-TASSER first generates 3D atomic models from multiple threading alignments and iterative structural assembly simulations.
 - The function of the protein is then inferred by structurally matching the 3D models with other known proteins.
 - Outputs full-length secondary & tertiary structures and functional annotations on ligand-binding sites
 - An estimate of accuracy of the predictions is provided based on the confidence score of the modeling

• Advantages and Disadvantages of Threading

Fold recognition or Threading is a technique for predicting protein structures. It is useful in cases where homology modelling fails to predict quality structures.

Advantages

Threading helps predict secondary structures of proteins towards tertiary structure prediction. For the "Twilight Zone" with low alignment quality and identity, threading is use.

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Disadvantages

Novel proteins cannot be predicted using threading. Fewer than 30% of the predicted first hits are true remote homologues. Validation of each result is necessary.

3D-1D Bowie Algorithm

Homology employed high alignment scores. Threading worked by creating combinations of primary sequences and corresponding secondary structures. Proposed by **Bowie et al in 1991**. It converts 3D structure into a 1-D string profile for each structure in the fold library. Align the target sequence to these profiles. 3D-1D methods convert structure and environment information into “profiles”. Score for each amino acid is computed for each profile.

Inputs and outputs of 3D-1D

- Identify amino acids based on: protein core, side chain positioning, solubility etc. (6 in all)
- Part of secondary structure including α -helix, β -sheet etc (3 in all)
- Total of $3 \times 6 = 18$ distinct states
- $P_{a,j}$ = prob. of finding amino acid (a) in environment (j)
- P_a = probability of finding (a) anywhere
- Maximize sum of scores for the fold:

$$s_{aj} = \log \left(\frac{P_{a,j}}{P_a} \right)$$

Introduction to Ab Initio Modelling

Ab initio methods have Anfinsen's thermodynamic hypothesis at the center. These methods attempt to identify the structure with minimum free energy. Ab initio methods rely on computing the energies of folded proteins. The protein structures with the lowest energy are deemed as plausible predictions.

Need for Ab Initio Modelling

- Applicable to any sequence
- Not very accurate biologically
- Accuracy and applicability are limited by our understanding of the protein folding problem

Limitation

- Computationally expensive
- Suitable for proteins with less than 100 residues

Rationale of Ab Initio Modelling

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Ab initio methods rely on computing the energies of folded proteins. The protein structures with the lowest energy are declared as plausible predictions

Rationale

Sometimes it so happens that even slightly homologous proteins may not be available. This renders homology modelling and threading/fold recognition as futile. Also, newer protein structures continue to be discovered every day. These could not have been identified by methods which only rely on matching with available structures. Lastly, homology / fold recognition predict protein structures without computing fundamental physical/chemical properties of the mechanisms and driving forces in structure formation. Ab initio methods, in contrast, base their predictions on physical models for these mechanisms. Energy released during the folding process is computed for predicting structure.

Strategies for Ab Initio Modelling

Ab initio methods base their predictions on physical models of folding mechanisms. Stabilization is measured by energy released during the folding process. Start with an energy function. Fold structures in order to obtain the most stable structure. This structure will have the minimum energy.

Energy Optimization in Ab Initio Modelling

1. Start with a rough initial model.
2. Define an energy function mapping structures to energy values. We have to minimize this later!!
3. Solve the computational problem of finding the global minimum.

Simulation of the Folding Process

1. Build an accurate initial model (including energy and forces).
2. Accurately simulate the dynamics of the protein folding process.
3. The native structure will steadily emerge.

Energy States of Folded Proteins

Ab initio methods predict protein structures by folding proteins based on each constituent atom's volume, charge, mass etc. The protein structure reporting lowest energy is selected to be the optimal structure.

Energies of Bonded Atoms vs. Nonbonded Atoms

$$V(R) = E_{bonded} + E_{non-bonded}$$

Local versus Global Minima

The protein structure reporting lowest energy is selected to be the optimal structure

Best Case Energy Function

- Clear energy minimum in the native structure

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- Viable path towards this minimum
- Global optimization finds the most stable structure

Optimal Energy Function

- Easier to design and compute
- Native structure not always at the global minimum
- No clear way of choosing among alternative structures that are generated

Pros and Cons of Ab Initio Modelling

Native structure not always at the global minimum. No clear way of choosing among alternative structures that are generated

Advantages

- Ab Initio methods can fold any target sequence using only physical atomic properties
- Predictions are mostly accurate and correctly describe the natural folding process

Disadvantages

- Ab initio methods are the very difficult to design (energy function)
- These methods are slow due to the huge possibilities

Summary of Structural Modelling

Strategies for Structural Modelling

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling

Homology modeling of the target structure can be done as follows:

1. Template recognition and initial alignment
2. Alignment correction
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Energy Optimization in Ab Initio Modelling

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Conclusion

- Homology modelling is performed in cases of high identity and alignment score
- For the “Twilight zone”, other strategies are employed
- For low identity and alignment scores, a “Twilight zone” for structure prediction exists
- Fold recognition / threading is useful in such cases
- For cases where even the fold libraries do not give any high scoring matches, Ab Initio strategies can help model the structure
- However, this is a complex and computationally expensive process

Topic152-160

Conclusion of the Course

1. Review of Sequence Analysis

Important Concepts

How do we sequence:

Genomes, Proteomes

How do we compare sequences:

Pair-wise Sequence Alignment, Multiple Sequence Alignment

Types of Alignments:

Global Alignment , (Needle Wunsch), Local Alignment, (Smith Waterman)

- **Advanced Tools:**

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Fast Alignment (FASTA), Basic Local Alignment Search Tool, (BLAST)

Databases:

GenBank, UniProt

Online Portals:

Ensemble, Expasy, UniProtKB

2. Review of Phylogenetics

• **Important Concepts**

Molecular Evolution

Insertions, Deletions, Substitutions

Phylogenetic Trees

Scaled Trees, Unscaled Trees

Phylogenetic Trees

Rooted Trees, Unrooted Trees

UPGMA: Unweighted Pair – Group Method using arithmetic Averages

Two sequences with the shortest evolutionary distance between them are considered

These sequences will be the last to diverge, and represented by the most recent internal node.

Clustering Vs. Non-clustering Methods:

UPGMA is a clustering method

Maximum Parsimony etc are non-clustering methods (not included in this course).

3. Review of Protein Sequencing

Important Concepts

Techniques of protein sequencing, Edman Degradation, Mass Spectrometry, Protein Ionization, Mass Analysis, Protein Fragmentation, MS1, MS2, Estimating and scoring whole protein mass, Extracting & Scoring Peptide Sequence Tags, Searching Post-translational Modifications

Composite Scoring Schemes

Online tools:

Mascot, Sequest, Prosight PC

4. Review of Protein Structures

Important Concepts

Protein Structures are generally of four types:

Primary, Secondary, Tertiary, Quaternary

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Techniques for determining protein structures

X-Ray Crystallography, NMR Spectroscopy

Types of Protein Secondary Structures

Helices, Beta Sheets, Coils, Loops

- **Foundation of structure prediction algorithms**
Propensities of certain amino acids to form specific secondary structures
- **Algorithm for predicting protein structures**
Chou Fasman Algorithm
- Protein Structure Database – PDB, Online tools for predicting structures by using proteins sequences

5. Review of Homology Modelling

Four Strata of Protein Structures

Primary, Secondary, Tertiary, Quaternary

- **Justification for homology modelling**
Number of known protein sequences is much larger as compared to known proteins structures

Three Strategies for Structure Prediction

Homology Modelling, Fold Recognition, Ab Initio Modelling, Protein Structure Database – PDB, Online tools for predicting structures such as MODELLER and iTASSER

Conclusions from this Course

Definition of Bioinformatics, Need for Bioinformatics, Areas within Bioinformatics, Bioinformatics as an interdisciplinary area, Need to store, process and analyze biological data, Requirement of newer faster algorithms

Specific areas focused were:

Comparing sequences, Comparing structures, Predicting structures

We looked at:

Algorithms, Databases, Online Tools for each topic.

- We studied the basic algorithms for each topic, With evolution and growth of Bioinformatics, newer and better algorithms are now also available!

Advanced Follow-up Courses

We looked into the foundations of Bioinformatics, However, each topic that was studied has a undergone a lot of development.

- For advanced study in Genomics, you may take “Computational Genomics” course **Topics:** Genome Assembly, Gene Finding, Annotation, GWAS etc

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- For advanced study in Proteomics, you may take “Computational Proteomics” course.
Topics: Protein Sequencing, PTM search, Structure Modelling and PPI studies
- **For advanced study in Integrative Biology, you may take “Systems Biology” course.** **Topics:** Metabolomics, Transcriptomics, Network Biology etc

Also, now there are cutting edge courses on:

Nano-Bio-IT, Computational Drug Design, Personalized Medicine

Careers in Bioinformatics

Pakistan as an infrastructure-limited country. The onset of digital revolution. Emergence of data as the most precious commodity, globally. Specifically, health data as a key commodity of the future. Health and disease as the primordial challenge of mankind

- **Unique opportunity for us in Pakistan**

Bioinformatics requires two things

1. **Smart mind**
2. **Internet connected computer**

One man company

You can take public databases and design drugs. One man vs. Roche?

BIGDATA

You can make a startup company which manages and process health BIGDATA. All it needs is basic software development skills coupled with Bioinformatics

The next disruption

The next Google, Facebook and Uber is going to emerge from Health and Bioinformatics. Pharmaceutical companies are investing into bioinformatics human resource development

Jobs Market

Pharmaceutical Giants, Research Centers & Universities, Hospital & Diagnostic IT departments ,
Your own startup company

Topic-161-165

RNA Structure

Outline

RNA folding, dynamic programming for RNA secondary structure prediction, Covariance model for RNA structure prediction

Base Pairing

RNA bases A,C,G,U, Bases can only pair with one other base

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- **Canonical Base Pairs**

A-U, G-C

- **“wobble” pairing**

G-U, I-U, I-A & I-C

RNA Types

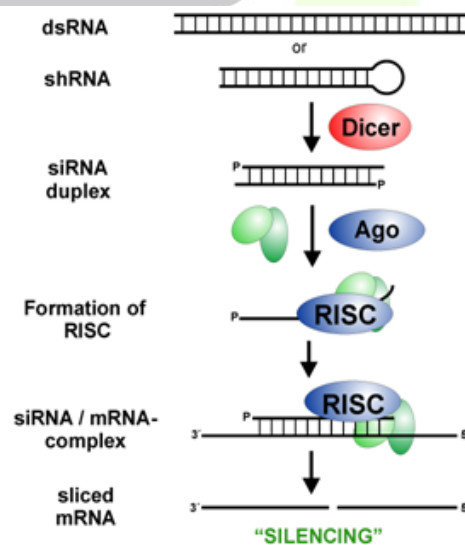
messenger RNA (mRNA)

Non-coding RNA

transfer RNA (tRNA), ribosomal RNA (rRNA), small interfering RNA (siRNA), micro RNA (miRNA), small nucleolar RNA (snoRNA)

Types of RNAs	Primary Function(s)	Types of RNAs	Primary Function(s)
mRNA - messenger	translation (protein synthesis), Coding, regulatory	scRNA - small cytoplasmic	signal recognition particle (SRP), tRNA processing
rRNA - ribosomal	translation (protein synthesis)	snRNA - small nuclear	mRNA processing, poly A addition, rRNA processing/maturation/methylation
tRNA - transfer	translation (protein synthesis)	snoRNA - small nucleolar	mRNA processing, poly A addition, rRNA processing/maturation/methylation
hnRNA - heterogeneous nuclear	precursors & intermediates of mature mRNAs & other RNAs	regulatory RNAs (siRNA, miRNA, etc.)	regulation of transcription and translation, other??

Small RNA



Conclusions

- messenger RNA (mRNA)

Non-coding RNA

- transfer RNA (tRNA)
- ribosomal RNA (rRNA)
- small interfering RNA (siRNA)
- micro RNA (miRNA)
- small nucleolar RNA (snoRNA)

RNA Secondary Structure

Some form of RNA can form secondary structures by “pairing up” with itself. This can change its properties dramatically.

Base Pairing

Pairing of bases helps in determining the secondary structure

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- Aligning bases, based on pairing with each other gives an algorithmic approach to determining the optimal structure

RNA Folding

RNA is produced as a single stranded molecule (unlike DNA)

- Strand folds upon itself to form base pairs & secondary structures
- RNA sequence analysis is different from DNA sequence

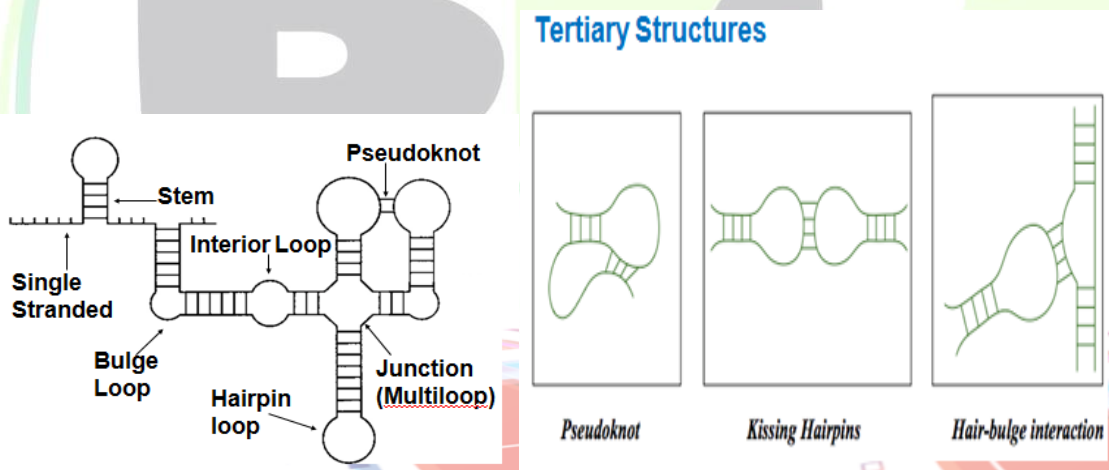
RNA Structure

Structures are more conserved than sequences

- Covariation

Secondary Structure representation

2D, Circle plot, Dot plot, Mountain, Parentheses, Tree model



Pseudoknots

Pseudoknots cause a breakdown in the Dynamic Programming Algorithm. In order to form a pseudoknot, checks must be made to ensure base is not already paired – this breaks down the recurrence relations

Conclusions

Some forms of RNA can form secondary structures by “pairing up” with itself. This can change its properties dramatically.

RNA Secondary Structure Prediction

There are different approaches to predict secondary structure of RNA

- Energy minimization

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- Comparative sequence analysis
- Folding and alignment
- Base-Pair Maximization

1. Energy minimization

Dynamic programming approach

Does not require prior sequence alignment. Require estimation of energy terms contributing to secondary structure

Assumptions;

Energetically most stable structure is more likely structure. Energy associated with any position is only influenced by local sequence and structure. Neglect pseudoknots

Approach

Energy minimization algorithm predicts secondary structure by minimizing the free energy (ΔG). ΔG calculated as sum of individual contributions of:

- Loops
- stacking

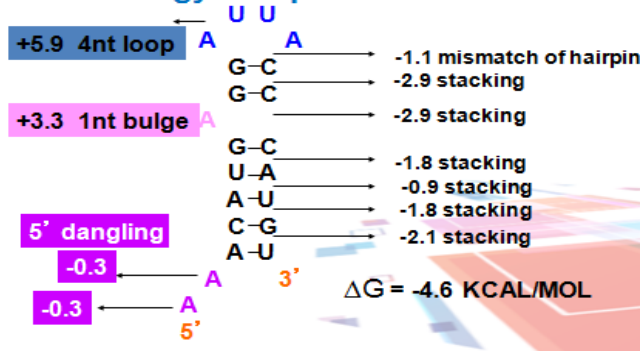
Energy minimization

- Thermodynamic Stability
- Estimated using experimental techniques
- Theory : Most Stable is the Most likely
- No Pseudoknots due to algorithm limitations
- Uses Dynamic Programming alignment technique
- Attempts to maximize the score taking into account thermodynamics
- MFOLD and ViennaRNA

Drawbacks

Compute only one optimal structure. Usual drawbacks of purely mathematical approaches

Free energy computation



2. Comparative sequence Analysis

Need a multiple sequence alignment as input.

Requires sequences be similar enough (so that they can be initially aligned), Sequences should be dissimilar enough for covarying substitutions to be detected, comparative analysis produces accurate structure predictions.

RNASeq

Calculating transcript abundance and prevalence by Ultra high throughput cDNA sequencing (Mortazwi et al, 2008)

The sequence reads are individually mapped to the source genome and counted to obtain the number and density of reads corresponding to RNA from each

- known exon
- splice event
- new candidate gene

Procedures

Isolation of all mRNA, Convert to cDNA using reverse transcriptase, Sequence the cDNA, Map sequences to the genome.

The more times a given sequence is detected, the more abundantly transcribed it is.

If enough sequences are generated (> 40 Million), *a comprehensive and quantitative view* of the entire transcriptome of an organism or tissue can be obtained (Mortazvi et al, 2008)

Data analysis

Mapping reads, Visualization Genome browser , De novo assembly , Quantification, Differential Gene Expression, Functional Analysis, Gene Networks

In RNASeq, transcript abundance and prevalence is calculated using Ultra high throughput cDNA sequencing

RNASeq Normalization

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Sequencing reactions may vary across different sequencing plate-forms as well as within different lanes of the same sequencer.

Transcript lengths also vary

Raw read counts may vary

RNASeq challenges

Uniformity of sequence coverage, Quantity of sequence required to reliably detect RNAs of lower abundance classes, Quantification and conversion of relative quantification to absolute RNA concentrations, Transcriptomes of organisms with large genomes, containing genes with more complicated structure, present some special challenges

Mapping Biases

Read counts will be higher if sequencer produces more reads. Longer genes will have the probability of mapping more reads than smaller ones

RPKM (reads per kilobase of transcript (or exon model) per million mapped reads)

Method for quantification of transcript levels. **RPKM** measure of read density reflects the molar concentration of a transcript in the starting sample by normalizing for RNA length and for the total read number in the measurement. This facilitates transparent comparison of transcript levels both within and between samples

RPKM

Number of reads mapped per gene length in KB per total reads in that sample in millions

C = Count of Mapped Reads

L = Length of transcript

M = Mapped reads of sample

$$\text{RPKM} = \frac{C}{L/1000 * M/1000000}$$

RPKM

$$\text{RPKM} = \frac{C}{L/1000 * M/1000000}$$

$$\text{RPKM} = \frac{C \times 10^9}{L M}$$

Example: What is the RPKM for a transcript of length 2500KB, with 900 alignments in a sample of 10000000 reads out of which 8000000 reads mapped?

$$\text{RPKM} = \frac{900 \times 10^9}{2500 \times 8 \times 10^6} = 4.5$$

RPKM

How many reads are required to map at 1 RPKM with a transcript of 2Kb length from a total of 40 Million Mapped reads?

$$\text{RPKM} = \frac{C \times 10^9}{L \times M}$$

$$1 = \frac{C \times 10^9}{2000 \times 40 \times 10^6}$$

$$C = \frac{2000 \times 40 \times 10^6}{10^9} = 80$$

FPKM (Fragments per kilobase of transcript per million mapped reads)

- Paired end RNASeq experiments produce two reads per fragment
- FPKM counts fragments not the reads
- Both reads might not map
- Counting reads might double count some fragments

(Trapnell et al 2010)

RPM

While comparing the same genes expression across different samples (treatments), normalizing for gene may not be necessary

$$\text{RPM} = \frac{C}{M/1000000}$$

Can compare relatively bigger numbers and get more DEG

RPKM reflects the molar concentration of transcript in starting sample normalized for

- Length of RNA
- Total reads in the sample

It facilitates transcript comparison within and across samples

Topic 166-176

Advanced Bio Informatics

Neural Network

The human brain can be described as a biological neural network an interconnected web of neurons transmitting elaborate patterns of electrical signals A neural network is a “connectionist” computational system. Information is processed collectively in parallel throughout a network of nodes. Complex adaptive system.

- Learning processes in biological systems.
- Learning as an optimization process.
- Learning by modification of synaptic strength.

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Association Rule Mining

It is an important data mining model studied extensively by the database and data mining community. Assume all data are categorical. No good algorithm for numeric data. An association rule has two parts

- an antecedent (if)
- a consequent (then)

Antecedent is an item found in the data. A consequent is an item that is found in combination with the antecedent

Market basket transactions:

t1: {bread, cheese, milk}

t2: {apple, eggs, salt, yogurt}

...

tn: {biscuit, eggs, milk}

Concepts:

An item: an item/article in a basket

I: The set of all items sold in the store

A transaction:

Items purchased in a basket; it may have TID (transaction ID)

A transactional dataset:

A set of transactions

Clustering

Clustering is “a process of organizing objects into groups whose members are similar in some way” A cluster is therefore a collection of objects which are “similar” between them and are “dissimilar” to the objects belonging to other clusters.

- Simplifications
- Pattern detection
- Useful in data concept construction
- Unsupervised learning process

Hierarchical agglomerative general algorithm

- Find the 2 closest objects and merge them into a cluster
- Find and merge the next two closest points, where a point is either an individual object or a cluster of objects
- If more than one cluster remains, return to step 2

Applications

- For administrative purposes
- Hospital activity and performance
- Used for researchers
- Data used by physician
- For laboratory use etc.

Machine Learning

Programming computers to optimize performance criterion using example data or past experience

When to learn? Calculate payroll , Solution needs to be adapted to particular cases (user biometrics)

When To Learn

Human expertise does not exist (navigating on Mars), Humans are unable to explain their expertise (speech recognition), Solution changes in time (routing on a computer network)

Model

Build a model that is a good and useful approximation to the Data

KDD is the non-trivial process of identifying valid, novel, potentially useful, & ultimately understandable patterns in data

Applications

Retail , Finance , Manufacturing , Medicine , Telecommunications , Bioinformatics, Web mining

Retail: Market basket analysis, Customer relationship management (CRM)

Finance: Credit scoring, fraud detection

Manufacturing: Optimization, troubleshooting

Medicine: Medical diagnosis

Telecommunications: Quality of service optimization

Bioinformatics: Motifs, alignment

Web mining: Search engines

Machine Learning

Study of algorithms that Improve performance at some task with exp. Role of Statistics, Role of CS

Applications of ML

Speech recognition, NLP, Computer vision, Medical outcomes analysis/Computational biology, Robot control

ML Concepts

ML Concepts

Association Analysis

Supervised Learning

- * Classification
- * Regression/Prediction

Unsupervised Learning

Reinforcement Learning

Learning Associations

Basket analysis:

$P(Y|X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

Example: $P(\text{chips} | \text{Beer}) = 0.7$

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Classification Apps

FR: Pose, lighting, occlusion (glasses, beard), make-up, hair style

Character recognition:

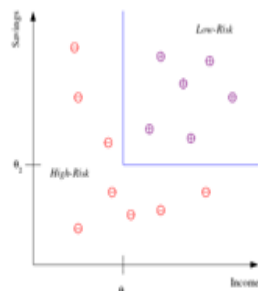
Speech recognition:

Medical diagnosis: From symptoms to illnesses

Web Advertising

Classification

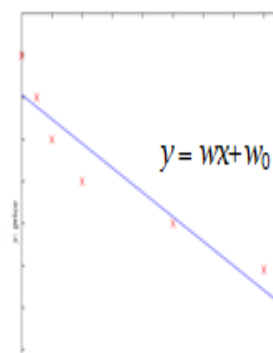
- Example: Credit scoring
- Differentiating between low-risk and high-risk customers from their income and savings



Discriminant: IF $\text{income} > \theta_1$ AND $\text{savings} > \theta_2$ THEN low-risk ELSE high-risk

Prediction: Regression

- Example: Price of a used car
- x : car attributes
- y : price
- $y = g(x | \theta)$
- $g()$ model, θ parameters



Machine Learning

Mates
sign of bright future

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Supervised Learning

Prediction of future cases

Knowledge extraction

Compression

Outlier detection

Un-supervised Learning

Learning "what normally happens"
No output

Clustering: Grouping similar instances

Applications

Customer segmentation in CRM
Image compression
Bioinformatics

Reinforcement Learning

Policies: what actions should an agent take in a particular situation

Utility estimation: how good is a state (→ used by policy)

No supervised output

Delayed reward

Reinforcement Learning

Credit assignment problem (what was responsible for the outcome)

Applications:

Game playing
Robot in a maze
Multiple agents, partial observability, ...

Forensic Science

Forensic science

Study & application of science to matters of law

Associations between people, places, things & events involved in crimes.

Crime investigation & Criminal Identification.

Methods

- Thumb printing.
- Face Recognition.
- Iris Recognition.
- Finger Printing.

Limitations of physical analysis

- Every one has unique iris pattern, Finger prints, Thumb prints.
- We can get the fingerprints and thumb prints from the crime scene.
- Fingerprints can be changed by cutting or burning of finger/thumb.

Limitations of Thumb

Using fingerprints require that you get a print from the finger or the thumb, no other parts of body can be used and criminal can use gloves and no prints are there for investigation.

Limitations of FR

Face Recognition can only be used when the photographs of the crime scene has been taken or suspicious person has been arrested and victim tells the physical appearance of criminal

Genotype Method

- DNA finger printing

DNA Finger Printing / DNA Profiling:

It is a form of forensic identification that is used primarily to identify people.

Advantages

Even though all humans share 99.9% of their DNA sequences, the remaining 0.01% of sequences is unique enough to differentiate people.

Advantages

Because DNA is in every cell of a person's body is present and same, any part of a human's body including dead skin cells, hair, saliva, and more contain DNA sequences.

DNA Fingerprinting

- A forensic scientist will need two pieces of DNA to be compared.
- For example, DNA discovered at crime scene should be compared to a DNA sample taken from a suspect.

Steps

- Comparison of repeating DNA sequences
- Match b/w two samples

DNA extraction from cell

1. Collecting cell from sample:

Two meters of DNA in cell

Collect cells from the sample with buccal swab.

Place the swab into Eppendorf tube.

2. Burst cells open to release DNA:

Add the lysis solution to the tube to separate the cells.

LS breaks Cell membrane & nuclear envelope causing cells to burst open & release DNA . It also removes histones proteins from DNA.

Inside the centrifuge tube spin around and debris & heavy proteins sink in bottom of tube and DNA strands remains distributed throughout liquid.

4. Isolate concentrated DNA:

Add the liquid containing DNA in separate tube Now add isopropyl alcohol to tube.

DNA is not soluble in this alcohol so it comes out and it can be seen with naked eye.

DNA is collected at bottom tube after placing in centrifuge.

3. Separate DNA from proteins and Debris:

- Cells have stayed in warm to for such a time that DNA is freed from cells.
- Salt causes proteins & other cellular debris to clump together.
- Place tube into micro centrifuge.

DNA Profile

DNA Profile

- Encrypted sets of numbers that reflect a person's DNA makeup,
- Variable number tandem repeats(VNTRS).

DNA Profile

- Short term tandem repeats (STR) in making the DNA profile of a person.
- These DNA profiles are the basis of a national DNA databases.

DNA profiling processes

- RFLP analysis.
- PCR analysis.
- STR analysis.
- Amp FLP.
- Y-chromosome analysis.



RFLP

- It analyzes the length of strands of DNA that include repeating base pairs (VNTRs).
- Repeated sequence of human genome can be same but the number of times it is repeated is unique to everyone.
- **Example:** CAT is repeated continuously 13 times in a row. In somebody else, it might be 12 times or 14 or whatever.
- **Limitation:** RFLP analysis requires a fairly large sample of DNA that hasn't been contaminated with dirt.

RFLP analysis requires investigators to dissolve DNA in an enzyme that breaks the strand at specific points.

The number of repeats affects the length of each resulting strand of DNA.

Investigators compare samples by comparing the lengths of the strands.

DNA Profile Methods

Polymerase Chain

Reaction (PCR)

Replicate a small amount of DNA to create a larger sample for analysis. First, a heat-stable DNA polymerase -- a special enzyme that binds to the DNA and allows it to replicate -- is added. Next, the DNA sample is heated to 200 degrees F (93 degrees C) to separate the threads. Then the sample is cooled and reheated. Reheating doubles the number of copies. Process is repeated about 30 times, there is enough DNA for further analysis.

Analyzing STRs

PCR is the first step in analyzing STRs (Short Tandem Repeats), which are very small, specific alleles in a variable number tandem repeat (VNTR).

Short Tandem repeats

Analyzing STRs is more accurate than the RFLP technique because their small size makes them easier to separate. If you want to create a fingerprint, you might look at 20 different STRs at different places in order to create a profile.

STRs

- It is impossible for two persons to have same number of STR repeated in a given sequence.

Y-chromosome Analysis

STRs in Y-chromosome Useful if the sample has mixed DNA. Gender analysis cases. It is processed just like simple STR analysis.

AmpFLP

Amplified fragment length polymorphism, is another technique that uses PCR to replicate DNA. Like RFLP, it first uses a restriction enzyme. Then, the fragments are amplified using PCR and sorted using gel electrophoresis. Can be automated, Doesn't cost very much., DNA sample must be high quality otherwise errors may result, which is the case with most DNA analysis techniques.

Topic 177-

Drug Discovery

Drug: agent used for the psychotic effect by the media or general public. Even the drugs abused have their **activity**. No drug is completely safe. Suitable quantity to cure or excess to be poisonous! E.g. aspirin, paracetamol can be toxic if excesses.

Drug Discovery	Requirements	Drugs: Definition
Primary objective — design & discovery of new compounds that are suitable for use as drugs	(i) Synthesis of the drug (ii) Administration method (iii) Development of tests (iv) Procedures to establish how it operates in the body (v) safety assessment (vi) research into the biological and chemical nature of diseased state.	Chemical substances that are used to prevent or cure diseases in humans, animals and plants
A team of workers — chemistry, biology, biochemistry, pharmacology, mathematics, medicine & computing ...		Activity: Pharmaceutical/ pharmacological effect on the subject, e.g. Analgesic or β -blocker Potency: quantitative nature of the effect

Drug Discovery Applications:-

Areas Influencing DD

- Molecular Biology on Drug Discovery
- High-Throughput Screening
- Combinatorial Chemistry

Molecular Biology Influence

Genetic information

Biochemical and chemical terms.

Cloning and expressing genes that encode therapeutically useful protein

High Throughput Screening

Widely used in the pharmaceutical industry.

Automation to quickly assay the biological or biochemical activity of a large number of drug-like compounds.

Combinatorial Chemistry

Laboratory technique in which millions of molecular constructions can be synthesized and tested for biological activity.

Pharmacogenetics

It is the branch of pharmacology concerned with the effect of genetic factors on reactions to drugs.

How people respond to medicines, Correlating heritable genetic variation to drug response.

Defination:-

Biotechnological science combines techniques of medicine, pharmacology & genomics which developing drug therapies to compensate for genetic differences in patients which cause varied responses to a single therapeutic regimen.

Applications:-

1. Detection of genetic variability of drug effects on the genome level
2. Agent selection
3. Analysis of drug
4. reactions and drug toxicity on gene expression
5. Development of new indications for already approved drugs
6. Discovery of new drug targets
7. Identification of (non) responders in clinical trials of phase I-IV
8. Identification of genotype dependent adverse drug reactions
9. Identification of individuals at risk for severe adverse drug effects

Pharmacogenomics

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How genes affect persons response to drugs. Pharmacology (science of drugs). Genomics (the study of genes and their functions). Develop effective, safe medications & doses tailored to a person's genetic makeup.

Applications:-

- Improve drug safety, Reduce ADRs, Tailor treatments to meet patients unique genetic pre-disposition, Optimal dosing, Improve drug discovery and Improve proof of principle for efficacy trials.

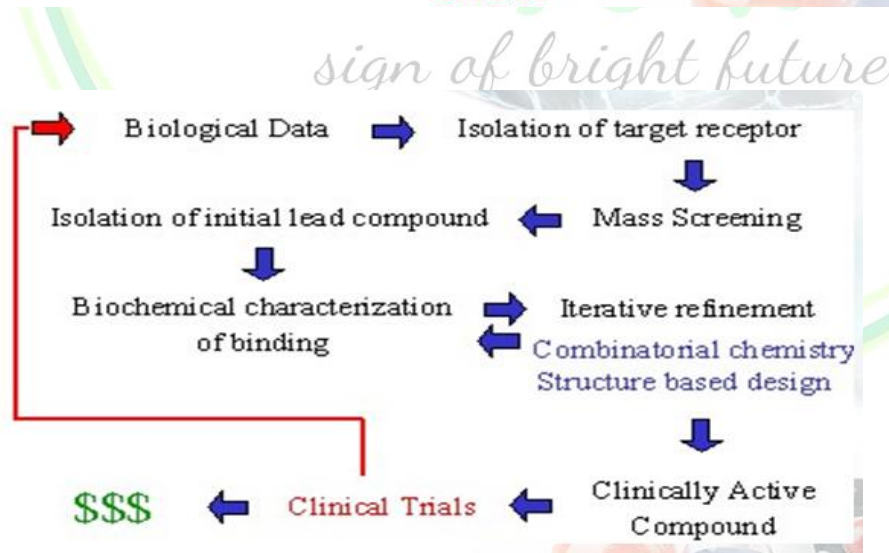
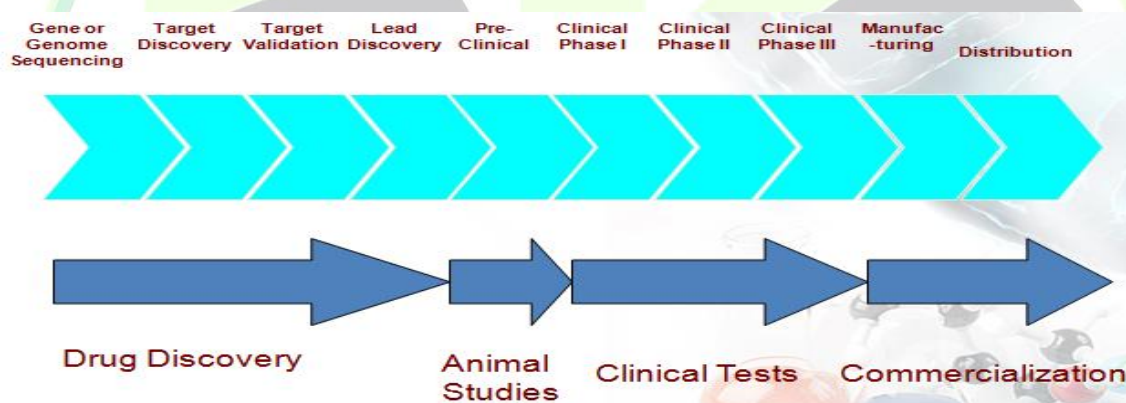
Future

Blessing in research. As a simple example, for nearly a decade the ability to store more information on a hard drive has enabled us to investigate a human genome sequence cheaper.

Drug Discovery – Pipeline

Target Identification, Target Validation, Lead Identification, Lead Optimization, Pre-Clinical

- Pharmacology & Toxicology



Methods of DD

Past:

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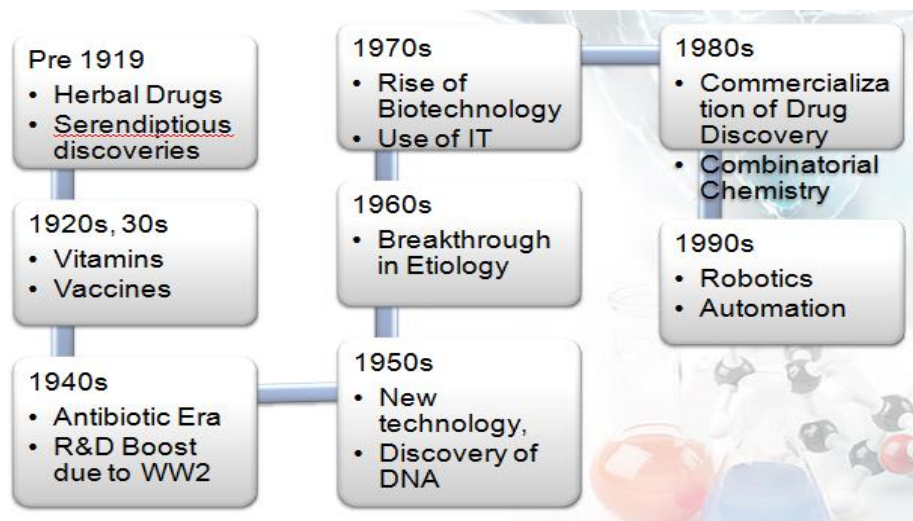
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(i) Identification of active ingredient from traditional remedies

(2) serendipitous discovery.

Current:

Diseases are controlled at molecular & physiological level. Information of Human Genome



Methods for DD

- Random Screening
- Molecular Manipulation
- Molecular Designing
- Drug Metabolites
- Serendipity

Random Screening

Higher/crude plants, opium, senna, reserpine, etc. Penicillin microorganism. Antibacterials with improved therapeutic profiles.

Molecular Manipulation

- Molecular biology is concerned with the handling and manipulation of DNA.
- Standard molecular biology techniques centre around the use of plasmid DNA.
- Plasmids all contain certain basic features.
 - Origin of replication
 - Resistance cassette
 - Promoter
 - Multiple cloning site
- If the plasmid is to be used to express a protein in bacteria a phage promoter will be used.
- If the plasmid is to be used to express a protein in a mammalian cell a viral promoter can be used.

Drug Metabolism

Xenobiotic metabolism

Biochemical modification of pharmaceutical substances or xenobiotics by living organisms, usually through specialized enzymatic systems

Drug Metabolites

Lipophilic chemical compounds into more readily excreted hydrophilic products.

Rate of metabolism determines duration & intensity of a drug's pharmacological action

Serendipity

Prototype psychotropic drugs

Development of psychiatry

Finding of one thing while looking for something else

Biomedical annotated corpora

Biomedical researchers are interested

- (i) Understanding data
- (ii) context information
- (iii) background knowledge
- (iv) curated databases &
- (v) Literature extensive

What is Annotated Corpora?

Dataset for extraction of disease/treatment entities relations. Corpora are usually constructed for training or evaluation purposes during the development of particular system

Annotation Consumers

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The linguistic community typically uses annotation as training data or for specific tasks. An abundance of tools that can *produce* annotations in the specific format of those resources. Biomedical annotation typically used for gene set enrichment analysis

Information deluge

Bio-databases, controlled vocabularies and bio-ontologies encode small fraction of information

Linking text to dbs and ontologies

Curators struggling to process scientific literature. Discovery of facts & events crucial for gaining insights in biosciences: need for text mining

Steps for Creation of Biomedical Corpora

- Nature of data
- Standard datasets
- Formalism
- Users of dataset
- Evaluation measures

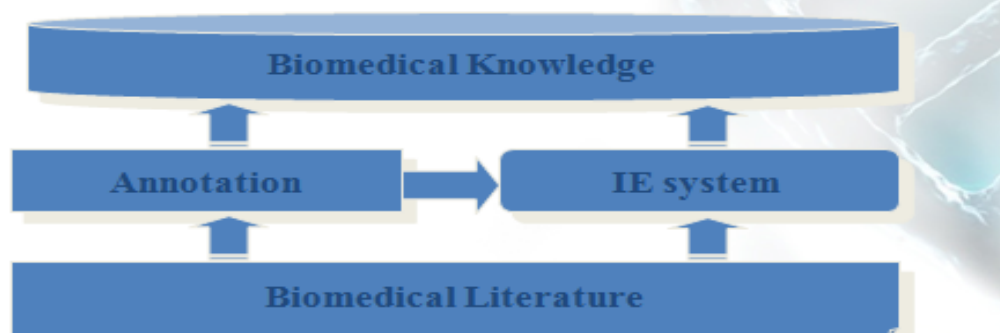
Data Mining

- Discovers unsuspected associations
- Combines & links facts
- and events
- Discovers *new* knowledge, finds new associations

IR: yields all relevant . Corpora; gathers, selects, filters documents that may prove useful , Finds what is known

IE: extracts facts & events of interest to user, Finds relevant concepts, facts about concepts. Finds only what we are looking for

Annotation & Information Extraction



- IE systems can be developed by referencing annotated corpus.
- The performance of IE systems can be evaluated by being compared to the annotated corpus.

Text Mining Pipelines

- ❑ Text documents
- ❑ Retrieval/storage Index access relevant storage
- ❑ Text Processing: word Filters, Pattern filters, Lexicon matching, Ontology, NLP parsing etc, ...

Feature Extractions:

Statistical: Word counts, pattern extraction & counts, etc

Domain-specific: Gene Name counts, etc

NLP-specific: Phrase counts, etc

Data Mining:

Classification, Clustering, Association, Statistical Analysis, Visual Analysis, etc ...

Target Discovery Strategy

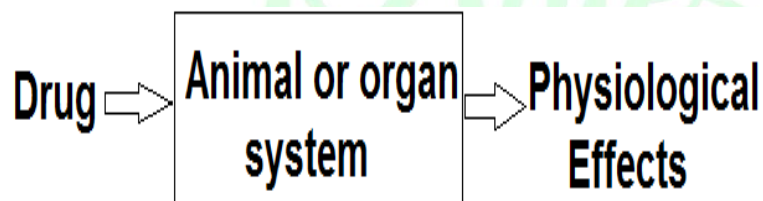
Target discovery to clinical application saga

- Physiology-based approach
- Target-based approach

1. Physiology-based approach

Is a disease-centric approach in which target is not identified, multiple targets are involved. In vivo screening is done by using drugs, siRNA or antisense oligonucleotides.

This process relies on disease, not on the target.



2. Target-centric approach

Target based discovery starts with the identification of genes and their protein products. Aim to develop drugs affecting one gene or a molecular mechanism. The identification of disease-relevant genes in vitro cellular models has been possible due to several tools. Gene-suppression tool used to link the genes with disease.

Target has two types

- Genetic
- Mechanistic

Genetic targets are represented by genes and genes products. Mechanistic targets include mechanism

based targets such as receptors, enzymes or genes, identified on the basis of the disease state.



Strategies To Identify Possible Drug Targets

First step of drug discovery is to identification of disease-associated targets. Genome sequencing and screening have enhanced opportunities for target identification and lead optimization.

Structure-based Target Discovery

It helped in defining the contours of the cognate surfaces of ligands and their protein targets, permitting optimization of their potency and selectivity. Some drugs that are originated from structure-based approach:

- Dorzolamide
- Captopril
- Imatinib
- Zanamivir

The protein structure contributes in the following fields:

- Target identification from sequence structure homolog recognition.
- Structural genomics and drug targets.
- Identification of ligand binding region
- Identification of hits and leads
- Structure-guided design and screening

Kinase drug discovery & Kinome

Target Discovery through Cell-based Genetics

- New drugs are based on target identification.
- It wants the thorough knowledge of the disease processes and characterization of genes.
- Combination of three target discovery will provide the desired result.

Target Discovery Strategies

Target discovery strategies based on

- Expression profiling Proteomic approach to identify disease related genes based on differential EP, homology and post translational modification

- Biochemical and cell biological assays - To identify genes and proteins linked with disease pathways
- Cell-based genetics - Leads to the discovery of targets by disturbing gene function in whole organisms, correlation with phenotypes.

Cell-based Genetics

Cell-based assays may lead to the identification of genes involved in cellular transformation, activation, migration and a host of biological processes relevant to a human disease.

Genetic-based Target Identification

It has some methods:

- Positional Cloning- Laboratory technique used to locate the position of a disease associated gene along the chromosomes.
- **Candidate gene approach**

To identify complex disease-linked genes through SNP markers. 10 million in HGP and 3 million identified.

Target class genetic approach

- Is applied to drug target gene families such as proteases, ion channels and GPCRs.
- 24000 protein coding genes & 2400 DTs
- Best candidate are selected from gene family for genetic analysis.

Target Validation

- A potential target is identified in the context of a specific disease.
- A validated target is the one that can be manipulated with drugs to produce positive clinical effects in humans.

Requirements for TV

☐ **Genetic Approach**

Gene to disease correlation in animal model

- Forward Genetics
- Reverse Genetics

Target Validation Tools

Helps in assessing their genetic association with disease. The following tools are:

- Antisense agents
- Ribozymes
- Peptide nucleic acids
- Transcription factors
- Gene knockouts

Antisense Technology

- Drugs as molecules action cellular proteins (enzymes, receptors).
- Downstream: Block events at nuclear/ribosomal levels.
- Prevent expression of aberrant proteins at earlier stage.
- Oligonucleotide reagents

Ribozymes

- Small RNA mol. cleave other RNA mol. at sequence specific sites.
- Hairpin Ribozyme: GUC

HammerHead: NUH, H is AUC

- Mode of Delivery: Exogenously , Endogenously
 - Peptide NAs: block protein translation
 - Transcription Factors: Zinc finger proteins – high Affinity to bind to correct region of DNA.
- Gene Knockouts:

Lead Identification

- Compound for biological activity on target.
- Potency threshold
- Libraries of molecules

Lead Identification – Technologies

Virtual Screening:

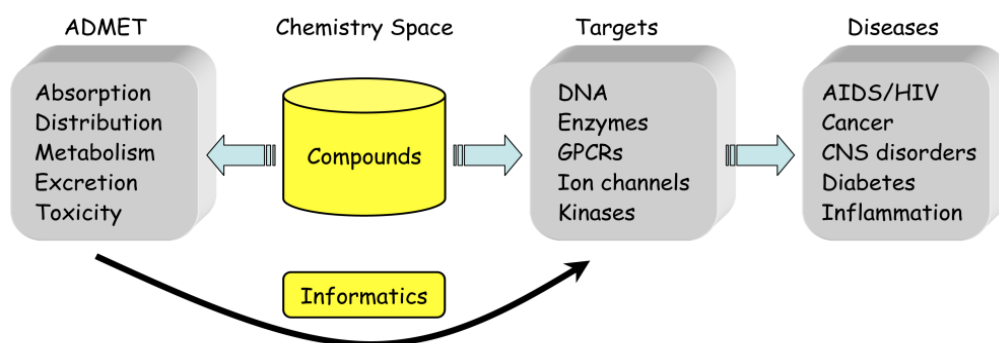
Protein structure , docking Chemical similarity search, Knowledge of compounds against receptor, receptor structure & receptor ligand interactions

Visual Screening

MLCC: Multilevel chemical Compatibility scoring

- Top selling drugs
- Compounds under biological scrutiny
- Anticancer drugs
- Compounds with poor drug like characters

Chemoinformatics



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Pharmacophore Mapping

Identify lead compounds against a desired target

Definition: 3-D arrange...

Usage: interaction of

receptor & legend

DB concept

- **QSAR**

Quantitative structure activity relationship

- **SAR:**

synthesizing & testing a series of structurally related compounds

Least squares & KNNs

High Throughput Docking

- Ligand & protein
- Docking algos
- Force fields, knowledge based & empirical

NMR based screening

- Nuclear Magnetic
- Resonance
- 3-D potential DC & tertiary structure of Proteins
- Need of prior information
- SHAPES
- WaterLogsy

Chemical Genetics

- Gene-product function in cellular or organismal context using exogenous ligands
- Knockouts
- Cell cycle - arresting agents

Lead Optimization

- Optimize the desirable traits of the lead
- Lead should be amenable for chemistry optimization
- Methods from Lead Identification

Lead Optimization – Technologies

- Medicinal chemists conduct extensive SARs to improve potency and selectivity.
- Improve physicochemical and drug-like properties
- Best molecules are advanced to animal Models & preliminary toxicology

LO Methods

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De novo drug desing

Charge distribution, lipophilicity or pK_a of side chains and H-bonds donors and acceptors

SBDD

Structure based drug Design. Effective: 3-D structure of inhibitor with target known. Large no of medicines. Molecular recognition in protein ligand complexes

Drug Like properties

- ADMET in phase 1
- Filters
- Bioavailability, PK, CNS

Pre- Clinical Pharma-cology & Toxicity

- Animals testing
- Xenograft models
- ADME/T testing and validation



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